Almost-Idempotent Stochastic Maps — internal lab-book

(scientific-exploration repo; see PRD.md / CLAUDE.md)

July 28, 2026

The campaign in numbers (snapshot 2026-07-28; see §52)

107	**295**	**800**	**67**	**57**	**2 026**
`af`-validated results (T0)	registry results	`fr` controller cycles	logged work sessions	days, first commit to last	adversarial proof nodes

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1 Orientation and rigour boundary

The north star, and its status

The target of this project is the classical (stochastic) stability statement `op-classical`: there are universal constants $\eta_0, C > 0$, independent of the dimension, such that every row-stochastic Q with $\|Q^2 - Q\|_{\infty\to\infty} \le \eta \le \eta_0$ admits a stochastic idempotent E (`def-stochastic`) with

$$\|Q - E\|_{\infty\to\infty} \;\le\; C\sqrt{\eta}, \tag{1}$$

the exponent $\frac{1}{2}$ being sharp.

This is OPEN. Nothing in this document proves it, and nothing in this document should be read as evidence that it is close to proved. What follows is a rigorous *partial* record: the components of the currently live route that have cleared this repository's top in-repo rigour rung.

Two pictures, one bridge

Two equivalent formulations are in play throughout. In the *stochastic picture* the object is a genuinely positive row-stochastic Q that is only approximately idempotent, with defect η measured in the $\infty\to\infty$ map norm (`def-almost-idempotent`). In the *signed picture* the object is an exact signed affine retraction P — $P\mathbf{1} = \mathbf{1}$ and $P^2 = P$ *exactly* (`def-signed-idempotent`) — whose failure of positivity is quarantined in the single scalar δ, the negative mass (`def-negative-mass`). The two are interchangeable up to universal constants, and the licence to cross between them is `lem-classical-equiv` (§3); a bound must always name the picture it lives in and cite the bridge when it crosses.

What this report contains

Five blocks, in dependency order.

1. **The bridge** (§3). `lem-classical-equiv`, the signed$\leftrightarrow$stochastic equivalence, with explicit constants in both directions.

2. **Positive-retract hardening** (§4). `lem-prh` converts a *positive approximate retract* — a pair of positive unital maps A, M between finite commutative C^*-algebras whose composite MA is ε-close to the identity (`def-positive-approximate-retract`) — into an exact stochastic idempotent at distance $2\sqrt{2\varepsilon}$. It is the step that turns approximate retraction data into the exact object (1) asks for, at the sharp square-root rate.

3. **The COMP tier** (§§5–18, interleaved). This tier builds, over an *extended* ε-C^*-algebra (`def-extended-epsilon-cstar-algebra`), the approximate corner calculus of δ-projections: the compressed corners $\mathcal{S}_{P,Q}$, the compression maps $\mathrm{Co}_{P,Q}$ and the compressed product (`def-compressed-corner`). Its content is that all of this is *matrix-uniform*: every estimate holds at every amplification level n with one constant. The keystones are the amplification identity $1_{M_n} \otimes \mathrm{Co}_{P,Q} = \mathrm{Co}_{I_n\otimes P, I_n\otimes Q}$, the uniform rectangular-product estimate, and the statement that a nonvanishing corner $\mathcal{S}_P$ is itself an *extended* $O(e)$-C^*-algebra — the object the next tier needs.

4. **The H-CB tier** (§§12–30, interleaved). Over a closed H-CB datum (`def-hcb-datum`) this tier establishes the amplified estimates for the map $\mathrm{Ha}^{Q}_{P,R}$ of `def-ha-map`: the column-Hilbert comparison, the variational identity that defines it, its column action, exact adjointness, its

multiplicative defect, the diagonal unit and upper-norm estimates, and — under explicit level-one hypotheses — the diagonal lower modulus and the diagonal inverse bound, all uniform in n.

5. **The H-CB parent contract and EXT tier** (§31 and §§32–37). This tier develops corner arithmetic through a one-dimensional index, the dimension selection $(r, 1)$ of an EXT-CB datum, the four-corner merge, and the EXT parent that adds one dimension to a level-one datum.

The honesty boundary

Three rules govern what may appear here.

Only af-validated results are stated as lemmas. Every numbered lemma below is a registry result whose adversarial proof tree has root node `validated` with clean taint, produced by fresh `codex` provers and *separate* fresh per-node verifiers; the orchestrator never judged a proof. Each shard records the node count and names its workspace. Nothing else in the registry — the inherited `proved-mod-audit` transcriptions, the `conjecture` rows, the `numerical` evidence, the `heuristic` arguments — is restated here as a theorem; those keep their tags in `argument/` and are surveyed, without promotion, in §53.

Every lemma is exactly as strong as its registry contract. Each statement is a typeset transcription of the one-line `contract` of its `argument/` shard, reproduced byte-verbatim beneath it. Where a contract is deliberately conditional or one-sided — the two-sided compressed-unit bound only for *nonvanishing* projections, the level-one hypotheses of the inverse tier — that shape is load-bearing and is preserved. In particular a level-one conclusion is never glossed as a matrix-uniform one.

No definition is restated. Terms resolve to their canonical `definitions/` shards, cited by `def-`identifier. Where a proof tree consumed a byte-matched auxiliary registered inside its own `af` workspace — the ε-C^* norm axioms, the compressed-product display, the one-dimensional nonvanishing bridge — it is named as such, and identified as a workspace external rather than a registry definition.

What is deliberately absent

No claim, direct or implied, that `op-classical` follows. The H-CB tier feeds the parent contract `conj-hcb` (§31), and that in turn feeds the EXT parent `conj-extcb` (§37); both are `proved` with `af: validated`. What remains below the af-validated rung beyond them is the assembly, whose registry carrier `lem-thmainext-conditional` is `proved-mod-audit` with `af: none`, together with the quarantined assembly interface. The route is a route, not a proof. Also absent: any numerical statement (those live in `runs/`, and are never rigorous), and the earlier signed-geometry material, which is off the live route and inventoried in §53.

2 The vocabulary

What this section is, and why it is generated

Every later section names its terms by `def-` identifier and never restates them. This section is where those identifiers resolve. It reproduces the project's Layer-0 definitions database — one shard per term under `definitions/` — as typeset definitions, in a deliberate reading order, each carrying its own provenance.

The material below is *generated*, and that is a load-bearing design decision rather than a convenience. The governing law of this repository is that each term is defined exactly once (`CLAUDE.md` L2): a second, hand-maintained statement of a definition inside the paper is drift, and drift here is fatal, because a lemma proved against one reading of δ or of exposedness and quoted against another is a wrong result that no gate would catch. So the canonical shard stays the single source of truth and the pages below are a deterministic projection of it, produced by `scripts/gen-report-defs.py` into `report/generated/defs/` in the same way that `argument/INDEX.md` and `argument/DAG.md` are produced from the registry shards. The generator's `--check` mode fails the local gate whenever the committed projection differs from a fresh render, so a definition edited in `definitions/` and not re-rendered here cannot be committed. Nothing in this section may be edited in place: corrections belong in the named shard, after which the projection is regenerated.

The statement is the definition; the source text is the second check

Each entry is a typeset `definition` environment, and that environment is the definition. Underneath it sits a small provenance block; underneath that, where the shard carries one, a SOURCE CHECK (BYTE-VERBATIM) block. The order is deliberate. A definition transcribed from the literature is only usable if it can be *read*, and a block of detokenized monospace is not a readable definition; but the byte-verbatim text is what the repository's `cited` criterion actually locks (`L1`), so it cannot simply be dropped. It is therefore kept and demoted: the small print at the end, against which the typeset statement above may be checked, rather than the statement itself.

Where a shard carries a harmonised **Statement** section, that section is what is typeset — it exists in the shard precisely to be the readable form, and it is where a transcription's known corrections are recorded. Where the shard's only content is the pinned source's own TeX, the generator typesets *that*, through an explicit macro-translation table. The table exists because a quoted source is written in that source's private macro dialect (`\eps`, `\calA`, `\Co`, ...), which this document's preamble does not define. Every row of the table maps a source macro to the expansion *the source itself gives it*, taken from that source's own preamble and recorded with the line number that defines it; the table is reproduced in full in `report/generated/defs/MANIFEST.md` for audit, together with its two declared glyph-only deviations. Translation changes spelling, never meaning. Each definition typeset this way says so in its provenance block, names the table version, and lists the macros that fired.

The honesty boundary is mechanical, not editorial. A translated fragment is validated before it is emitted — every surviving control sequence must be one this document defines, braces and math delimiters must balance, no alignment or script character may escape math mode — and a fragment that fails is *not* typeset. It is replaced on the page by a loud **NOT TYPESET** block naming the reason, its bytes are left to the source check below, and it is listed in `report/generated/defs/MANIFEST.md`. Partial success is allowed: what translates is typeset, what does not is flagged, and nothing in between is guessed at. As of this render there are no such flags. The byte-verbatim blocks themselves are set with TeX's detokenizer, which prints a control word followed by one space; apart from that spacing artefact, and from the line breaks the quote block inserts to keep long source lines inside the text block, what is printed is the source's own bytes.

Reading the kind and status of a definition

Each definition is followed by a small provenance block naming its *kind*, its *status*, its source, its locus, and — for literature transcriptions — the SHA256 prefix of the pinned source file. The vocabulary of that block is the repository's rigour ladder applied to definitions, and it is not the same ladder that grades results.

- `cited` — transcribed from the literature and byte-matched to a local source under `refs/` at the recorded locus and hash. The gate `scripts/check-defs.py` recomputes that hash. What this certifies is *provenance*: that the text below is what the source says. It certifies nothing about whether the source's claim is true, and several of the cited shards here carry explicit scope notes recording exactly which neighbouring assertions of the source are *not* part of the definition.
- `consensus` — a formulation this project adopted and agreed on, with no single external source to byte-match against.
- `original` — a notion introduced here, typically to name a configuration or a hypothesis package that the exploration kept re-inlining.

The status is orthogonal. `locked` means usable downstream: byte-verified for a `cited` shard, signed off for a `consensus` or `original` one. `draft` means the shard is pinned but not yet gated — fine as working vocabulary, and flagged whenever a validated result leans on it. A definition is never `proved`: it carries no truth claim, so there is nothing in it to prove. When a statement about a defined object needs to be true, it is a registry result with its own status, not a clause smuggled into the vocabulary.

The two pictures, and which one a term lives in

The single most reliable way to misread this project is to forget which picture a symbol belongs to. The stochastic picture holds genuinely positive row-stochastic maps that are only approximately idempotent (Definition 3, Definition 4). The signed picture holds exactly idempotent signed affine retractions whose failure of positivity is quarantined in a single scalar (Definition 2, Definition 1); everything built on row geometry — exposedness, the visible set, height, invisible mass — is stated there. The licence to cross between the pictures, with explicit constants, is `lem-classical-equiv` (§3), and it is the only such licence.

Two symbol collisions are deliberate in the sources and are preserved rather than silently renamed. The letter δ denotes the negative mass of a signed idempotent in the classical layer and the defect of a δ-projection in the approximate-C^* layer (Definition 21); these are different quantities, and `CONVENTIONS.md` is the registry that keeps them apart. Likewise ε is the ambient approximate-C^* parameter, never the retract defect of Definition 38. When a bound is quoted, the picture and the parameter must be named with it.

Which definitions appear here

This section renders the vocabulary of the *current proof strategy*, not the whole Layer-0 database. The strategy is the Route-F landing chain, and it is not re-described here: the generator imports `scripts/gen-report-dag.py` and reuses the very same subgraph computation that draws the atlas of §51, so the map and the vocabulary cannot drift apart. A definition is rendered when it is imported by a registry result in that subgraph, or by a registry result reproduced in this document, or when it is reached from one of those by following the `[[def-...]]` crosslinks that the rendered statements themselves use. Anything outside that closure is not printed, and a generated paragraph at the end of the section says how many such definitions there are and names them; they remain canonical in `definitions/` and in `definitions/INDEX.md`, they are simply not part of what this document needs. A crosslink pointing at one of them degrades to a bare identifier rather than to a link this document could not honour.

The three groups below are computed from that set, not curated. The first is the connected component of the crosslink graph that the classical signed and stochastic vocabulary forms: those definitions genuinely define each other, and they are ordered so that a term appears after everything its statement uses, with mutually recursive clusters (negative mass, signed idempotent, stochastic idempotent; exposedness and the visible set) kept together. The second group is the approximate-C^* vocabulary transcribed from the pinned source, which is the language of the tiers reproduced in §§5–37. The third is the project-internal packaging built over that language: the datum bundles that fix the hypotheses of a tier, and the notation the proofs of that tier share. A group with no member in scope would not be printed at all.

One honest asymmetry survives the scoping. Being *needed by* the strategy is a much weaker condition than being *proved on* it: much of the classical group is the vocabulary of the signed-geometry trunk, whose results are ancestors of the target but are recorded in `argument/` and whitelisted in `report/UNWIRED.md` rather than reproduced here. Those definitions belong in this section — the target `op-classical` of §1 is stated in their terms — but the reader should not read their presence as a claim that the surrounding argument is in this document. Each definition's pointer line says how many registry results import it and which of those, if any, appear here; a term with no reproduced consumer says so instead of leaving the reader to guess.

Citing a definition

Every definition carries `\label{def:`*slug*`}` and a matching `\hypertarget`, where *slug* is the shard identifier with its first hyphen replaced by a colon — `def-co-top` becomes `def:co-top`. That is the same identifier-to-label transform the provenance gate uses for results, so a lemma section may write `\ref{def:co-top}` and the knowledge-graph page may anchor at the same target. Inside the generated statements, a crosslink is a live link exactly when its target is in this document — a definition rendered in this section, or a registry result reproduced in a later one — and otherwise is printed as a bare identifier, because a link into material this document does not contain would be a promise it cannot keep. Each definition also names its canonical shard path, so the reader can go from the page to the source of truth in one step, and from there to `definitions/INDEX.md` and `argument/DAG.md`.

The classical picture: the signed and stochastic vocabulary

Definition 1 (negative mass (`def-negative-mass`)). For an exact signed idempotent $P \in \mathbb{R}^{n\times n}$ (or any signed matrix), the *negative mass* (equivalently *negativity*) is

$$\delta(P) := \max_{1\le i\le n} \sum_{j=1}^{n} \max\{-P_{ij}, 0\},$$

the largest total negative mass in any single row. The *row polytope* is

$$K := \operatorname{conv}\{p_1, \dots, p_n\},$$

the convex hull of the rows $p_i \in \mathbb{R}^n$. A stochastic idempotent is exactly a signed idempotent with $\delta(P) = 0$.

Kind. original (introduced by this project); `status: locked`. **Aliases.** negativity; row polytope; row polytope K.
Provenance. `source: internal`; locus `quoted from classical-portfolio/report/kernel-conjecture.tex §Setting (def:esi,` δ`(P)); row polytope from ../almost-idempotent-positive-maps/definitions/def-stochastic.md`; no source hash (not a literature transcription); record: adapted from kernel-conjecture.tex §Setting (negativity δ(P)) and ../almost-idempotent-positive-maps definitions/def-stochastic.md (row polytope K).
Used in this report by `lem-classical-equiv` (164 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-negative-mass.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. $\delta(P)$ quoted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (Definition: *Exact signed idempotent*); the row polytope K adopted from `../almost-idempotent-positive-maps/definitions/def-stochastic.md`. This scalar is the single knob that controls the geometry: the three derived scales $\tau = \sqrt{\delta}$, $\rho = 4\tau$, $\kappa = \tau/4$ live in `def-visible-set`. The δ-positivity of `def-near-positive-projection` is the same quantity: a unital idempotent R is δ-positive iff $\delta(R) \le \delta$.

Definition 2 (exact signed idempotent (`def-signed-idempotent`)). Let $n \in \mathbb{N}$. A matrix $P \in \mathbb{R}^{n\times n}$ is an *exact signed idempotent* if

$$P\mathbf{1} = \mathbf{1} \qquad \text{and} \qquad P^2 = P,$$

where $\mathbf{1} = (1, \dots, 1)^{\mathsf{T}}$. Equivalently, P is a linear self-map of $\mathbb{R}^n$ fixing the all-ones vector with $P^2 = P$ *exactly*, whose rows are signed measures of total mass 1 ($\sum_j P_{ij} = 1$ for each i). A row-stochastic idempotent — a stochastic idempotent — is exactly an exact signed idempotent whose negative mass $\delta(P) = 0$.

Row geometry. Writing $p_i \in \mathbb{R}^n$ for the i-th row with the ℓ^1 metric, each row satisfies $\sum_j p_{ij} = 1$ and $\sum_j |p_{ij}| \le 1 + 2\delta$; hence all rows lie in a common affine hyperplane and pairwise ℓ^1 distances are at most $2 + 4\delta$ (with $\delta = \delta(P)$).

Kind. original (introduced by this project); `status: locked`. **Aliases.** signed idempotent; signed affine retraction; signed stochastic retraction.
Provenance. `source: internal`; locus `quoted from classical-portfolio/report/kernel-conjecture.tex §Setting (Definition, exact signed idempotent)`; no source hash (not a literature transcription); record: adapted from ../almost-idempotent-positive-maps/agent-A/explorations/classical-portfolio/report/kernel-conjecture.tex §Setting (def:esi); exact-idempotence form adopted for this repo.
Used in this report by `lem-classical-equiv` (186 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-signed-idempotent.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Quoted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (Definition: *Exact signed idempotent*, and the row-geometry remark). The exact-idempotence form is the working object: $P^2 = P$ holds exactly while non-positivity is quarantined into the scalar negative mass $\delta(P)$. This is the matrix/geometry framing of `def-near-positive-projection` (map/positivity framing). Its exposed row vertices, visible set, height and invisible mass are `def-exposed`, `def-visible-set`, `def-height`, `def-invisible-mass`.

Definition 3 (stochastic idempotent (`def-stochastic`)). A *unital positive map* of $\ell_n^\infty = \mathbb{R}^n$ is a *row-stochastic matrix* Q ($Q \geq 0$ entrywise, $Q\mathbf{1} = \mathbf{1}$) — an affine self-map of the probability simplex Δ_n. A *stochastic idempotent* E is a row-stochastic E with $E^2 = E$: an affine retraction of Δ_n onto a sub-polytope.

Dropping positivity but keeping *exact* idempotence gives the *signed* generalization — a signed affine retraction P with $P\mathbf{1} = \mathbf{1}$ and $P^2 = P$ exactly, whose rows are signed measures of total mass 1. Its failure of positivity is quantified by the negative mass $\delta(P)$; a stochastic idempotent is exactly a signed idempotent with $\delta(P) = 0$.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** row-stochastic idempotent; stochastic retraction; unital positive map on ℓ∞_n.
Provenance. `source: internal`; locus `adopted from ../almost-idempotent-positive-maps/definitions/def-stochastic.md`; no source hash (not a literature transcription); record: adopted from ../almost-idempotent-positive-maps definitions/def-stochastic.md (B's classical formulation, report sec:classical def:stochastic).
Used in this report by `lem-classical-equiv`, `lem-prh`, `lem-routef-f0-defect-identity`, `lem-routef-f0-ucp-lift`, `lem-routef-f2-positive-unital-compression`, `lem-routef-f3-retract-defect` and 1 further reproduced result (19 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-stochastic.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Adopted from `../almost-idempotent-positive-maps/definitions/def-stochastic.md` (B's classical formulation, report sec:classical `def:stochastic`); re-tagged `consensus / source: internal` for this repo. The signed form is the working object because $P^2 = P$ holds *exactly* (non-positivity quarantined in δ, see `def-negative-mass`). This is the commutative specialization of `def-near-positive-projection`; the approximate-idempotence sibling is `def-almost-idempotent` (a genuinely positive Q with $\|Q^2 - Q\|$ small). Exposed row vertices and the visible set are `def-exposed` / `def-visible-set`.

Definition 4 (almost idempotent (`def-almost-idempotent`)). A row-stochastic matrix Q on ℓ_n^∞ is *almost idempotent* (with defect η) if

$$\|Q^2 - Q\|_{\infty\to\infty} \leq \eta, \qquad \eta \in [0, \tfrac{1}{4}),$$

where $\|\cdot\|_{\infty\to\infty}$ is the operator norm induced by ℓ_n^∞ — i.e. the maximum over rows of the ℓ^1-norm of the corresponding row of $Q^2 - Q$. The range $\eta < \frac{1}{4}$ is what makes the binomial series producing the exact idempotent converge.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** η-idempotent; eta-idempotent; almost-idempotent stochastic matrix.
Provenance. `source: internal`; locus `adopted from ../almost-idempotent-positive-maps/definitions/def-almost-idempotent.md (stochastic specialization)`; no source hash (not a literature transcription); record: adopted from ../almost-idempotent-positive-maps definitions/def-almost-idempotent.md (operator-norm formulation, report rem:cb-norm); $\infty \to \infty$ specialization per kernel-conjecture.tex thm:chain(c).
Used in this report by `lem-classical-equiv`, `lem-routef-ai-defect-linearization`, `lem-routef-f0-defect-identity`, `lem-routef-functional-calculus-closeness` (5 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-almost-idempotent.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Adopted from `../almost-idempotent-positive-maps/definitions/def-almost-idempotent.md` (the operator-norm — *not* cb-norm — formulation, report rem:cb-norm), specialized to the $\|\cdot\|_{\infty\to\infty}$ map norm on stochastic matrices as in `classical-portfolio/report/kernel-conjecture.tex` Theorem (chain)(c). Measuring non-idempotence in the map operator norm (rather than the cb-norm) is why the bridge exponent is $\sqrt{\eta}$ rather than η. The classical stability question (`op-classical`) asks whether every such Q lies within $C\sqrt{\eta}$ (max-row-ℓ^1) of an exactly idempotent row-stochastic matrix; its parent object is `def-near-positive-projection`. Distinguish η (this defect) from the negative mass δ of the associated `def-signed-idempotent`, with $\delta = O(\eta)$.

Definition 5 (exposed vertex (`def-exposed`)). Fix an exact signed idempotent $P \in \mathbb{R}^{n\times n}$ and write $p_i \in \mathbb{R}^n$ for its i-th row, with the ℓ^1 metric $\|x - y\|_1 = \sum_j |x_j - y_j|$.

A row p_v is a *row vertex* of P if $p_v \notin \operatorname{conv}\{\, p_j : p_j \neq p_v \text{ as vectors of } \mathbb{R}^n \,\}$ (geometrically coincident duplicate rows count as a single point; a repeated point is still a vertex if it lies outside the hull of the *other* points).

An *admissible exposer* for a row vertex v is an affine function $h : \mathbb{R}^n \to \mathbb{R}$ with

$$h(p_v) = 0 \qquad \text{and} \qquad 0 \le h(p_j) \le 1 \quad \text{for every row } p_j.$$

The *exposedness margin* of v is

$$t^*(v) := \sup_{h \text{ admissible}} \min\{\, h(p_j) : \|p_j - p_v\|_1 \ge \rho \,\},$$

with the convention $t^*(v) = +\infty$ when no row is at ℓ^1-distance $\ge \rho$ from p_v. The vertex v is (ρ, κ)-*exposed* if $t^*(v) \ge \kappa$, and *hidden* otherwise. The scales $\rho = 4\tau$, $\kappa = \tau/4$ (with $\tau = \sqrt{\delta}$) and the resulting *visible set* $\mathcal{W}(P)$ of exposed vertices are recorded in `def-visible-set`.

Informally: an exposed vertex can be linearly separated, with margin $\ge \kappa$, from every row that is genuinely far ($\ge \rho$) from it; rows inside the ρ-ball are exempt.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** (ρ,κ)-exposed; (rho,kappa)-exposed; exposedness margin; admissible exposer; hidden row vertex; well-exposed vertex.
Provenance. `source: internal`; locus `adapted from classical-portfolio/report/kernel-conjecture.tex §Setting (def:vertex, def:exposed); ../almost-idempotent-positive-maps/definitions/def-exposed.md`; no source hash (not a literature transcription); record: adopted from ../almost-idempotent-positive-maps definitions/def-exposed.md (B's exposed-circuit machinery, report sec:classical); reconciled with kernel-conjecture. tex def:exposed.
Used by 129 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-exposed.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Adapted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (def:vertex, def:exposed) and `../almost-idempotent-positive-maps/definitions/def-exposed.md`. The latter states the equivalent *exposedness modulus* $e_v(\rho) = \sup_h \min_{\|p_i - v\|_1 \ge \rho} h(p_i)$ and calls a vertex *well-exposed* at $\sqrt{\cdot}$-scale if $e_v(\rho) \ge c\sqrt{\delta}$ for some $\rho = O(\sqrt{\delta})$, where δ is the negative mass. Central to the proved classical special cases (well-exposed $\Rightarrow$ simplex) and to the open global exposed-hull lemma (`op-exposed-hull`). A pointwise exposed-or-redundant dichotomy is provably *insufficient* (dense regular polygons) — the gap must be stated globally (see `def-height`).

Definition 6 (visible set (`def-visible-set`))**.** Given an exact signed idempotent P with $\delta = \delta(P)$ (the negative mass), the *three scales* are

$$\tau := \sqrt{\delta}, \qquad \rho := 4\tau, \qquad \kappa := \tau/4.$$

The *visible set* is

$$\mathcal{W} = \mathcal{W}(P) := \{\, v : \ p_v \text{ is a } (\rho, \kappa)\text{-exposed row vertex of } P \,\},$$

the set of row vertices that are (ρ, κ)-exposed at these scales; all other row vertices are *hidden.* At $\delta = 0$ the visible vertices are exactly the distinct rows of the recurrent (equal-input) blocks of a stochastic idempotent.

Kind. original (introduced by this project); `status: locked`. **Aliases.** visible set W; W(P); (ρ,κ)-exposed set; three scales.
Provenance. `source: internal`; locus `quoted from classical-portfolio/report/kernel-conjecture.tex §Setting (def:scales, def:exposed)`; no source hash (not a literature transcription); record: adapted from kernel-conjecture.tex §Setting (Definition: three scales; Definition: (ρ,κ)-exposedness and the visible set W).
Used by 137 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-visible-set.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Adapted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (Definition: *The three scales*; Definition: (ρ, κ)*-exposedness and the visible set* $\mathcal{W}$). The scales couple exposedness (`def-exposed`) to the negative mass (`def-negative-mass`). The convex hull $C_{\mathcal{W}} := \mathrm{conv}\{p_w : w \in \mathcal{W}\}$ is consumed by `def-height` (max ℓ^1-distance of a row from $C_{\mathcal{W}}$) and `def-invisible-mass` (positive mass a row places outside $C_{\mathcal{W}}$).

Definition 7 (height (`def-height`))**.** Let $C_{\mathcal{W}} := \mathrm{conv}\{p_w : w \in \mathcal{W}\}$, where $\mathcal{W} = \mathcal{W}(P)$ is the `def-visible-set`, and let $\mathrm{dist}_1(x, S) = \inf_{s \in \mathrm{conv}\, S} \|x - s\|_1$. If $\mathcal{W} \neq \emptyset$, the *height* of a row p_i is $\mathrm{dist}_1(p_i, C_{\mathcal{W}})$, and the *height of* P is

$$H = H(P) := \max_{1 \le i \le n} \mathrm{dist}_1\big(p_i,\ \mathrm{conv}\{p_w : w \in \mathcal{W}\}\big).$$

Because $\mathrm{dist}_1(\cdot, C_{\mathcal{W}})$ is convex and every row is a convex combination of the geometrically distinct row vertices, the maximum is attained at a row vertex; if $H > 0$, any maximizing vertex is necessarily hidden. Such a vertex is called a *hidden top vertex*.

Kind. original (introduced by this project); `status: locked`. **Aliases.** height H; H(P); hidden top vertex.
Provenance. `source: internal`; locus `quoted from classical-portfolio/report/kernel-conjecture.tex §Setting (Definition: Height)`; no source hash (not a literature transcription); record: adapted from kernel-conjecture.tex §Setting (Definition: Height, def:height).
Used by 112 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-height.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Quoted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (Definition: *Height*). H measures how far the rows spill outside the hull of the visible (`def-visible-set`, `def-exposed`) vertices; the linear hull bound / HLC target is $H(P) \le C\sqrt{\delta}$ in the negative mass δ. The invisible mass `def-invisible-mass` of a hidden top vertex is the hypothesis quantity that governs whether H can be large.

Definition 8 (always-tight hull (`def-actor-hull`))**.** Fix an exact signed idempotent P and a hidden geometrically distinct row vertex u with $t^*(u) > 0$. Let $T(u)$, $O(u)$, $Z(u)$ be the always-tight ρ-far / upper-box / zero-face constraint families of the exposedness LP at u (`lem-always-tight-dual-support`). The **always-tight hulls** (or *actor hulls*) are the displacement convex bodies

$$K_T(u) := \mathrm{conv}\{p_f - p_u : f \in T(u)\}, \qquad K_O(u) := t^*(u) \cdot \mathrm{conv}\{p_i - p_u : i \in O(u)\}.$$

They are **disjoint** when $g := \mathrm{dist}_1(K_T(u), K_O(u)) > 0$ (the huddle / Branch-I predicate), and **intersect** otherwise (Branch II, the alpha-free optimal display of `lem-optimal-face-conic-reduction`). Boundary ownership: tangency / any common point counts as intersecting.

Kind. original (introduced by this project); `status: draft`. **Aliases.** actor hull; K_T; K_O; K_T(u); K_O(u); always-tight hulls; disjoint always-tight hulls; displacement hull.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-downhill-cotop-conic-mass.md and lem-always-tight-dual-support.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-always-tight-dual-support (the T/O families) + lem-downhill-cotop-conic-mass (the hulls K_T, K_O).
Used by 27 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-actor-hull.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; "actor hull" and "always-tight hull(s)" name the same displacement bodies K_T, K_O built from the actors (far / upper-box rows) the dual witness charges. The disjoint-vs-intersect dichotomy on K_T, K_O is the root split $S1$ of the W54 huddle-charge tree. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-zero-face`, `def-dual-witness`, `def-co-top`.

Definition 9 (hiddenness dual witness (`def-dual-witness`))**.** Fix an exact signed idempotent P and a hidden row vertex v ($\rho = 4\tau$, $\kappa = \tau/4$, $\tau = \sqrt{\delta(P)}$), with $F_v = \{j : \|p_j - p_v\|_1 \ge \rho\}$ the (nonempty) ρ-far row set.

A **hiddenness dual witness** of v is a triple (λ, α, β) of nonnegative coefficients — λ_f ($f \in F_v$) with $\sum_f \lambda_f = 1$ (a probability on the far rows), and $\alpha_i, \beta_i \ge 0$ over all rows with $\sum_i \beta_i = t^*(v) < \kappa$ — satisfying the *witness balance*

$$\sum_{f \in F_v} \lambda_f\,(p_f - p_v) \;+\; \sum_i \alpha_i\,(p_i - p_v) \;=\; \sum_i \beta_i\,(p_i - p_v).$$

It is the LP dual of the exposedness program (see `def-exposed`); the *small-beta* property is $\sum_i \beta_i = t^*(v) < \kappa$. A witness is **reduced** when redundant centered-zero constraints are deleted; by `lem-always-tight-dual-support` a reduced optimal witness has $\mathrm{supp}(\lambda) \subseteq T$, $\mathrm{supp}(\beta) \subseteq O$, and $\mathrm{supp}(\alpha) \subseteq Z$ (the always-tight far / upper-box / zero-face families).

Kind. original (introduced by this project); `status: draft`. **Aliases.** dual witness; dual certificate; small-beta witness; reduced optimal witness; witness balance; (lambda, alpha, beta).
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-hiddenness-dual-witness.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-hiddenness-dual-witness (the LP-dual object) + lem-always-tight-dual-support (support localization).
Used by 1 registry result, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-dual-witness.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; from `lem-hiddenness-dual-witness` (existence via finite LP duality). The witness converts the scalar hiddenness fact $t^*(v) < \kappa$ into a geometric object: a convex combination of ρ-far rows reproducing p_v up to controlled signed slack. "dual certificate" is used interchangeably in the shards. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-zero-face`, `def-actor-hull`, `def-co-top`.

Definition 10 (zero-face row (`def-zero-face`))**.** Fix an exact signed idempotent P and a hidden (not (ρ,κ)-exposed) geometrically distinct row vertex u, with the exposedness LP at u (the program defining $t^*(u)$, see `def-exposed`) and an *optimal exposer* h^* (an admissible affine h attaining $t^*(u)$).

A row z is a **zero-face row** of u if $h^*(p_z) = 0$, i.e. z lies on the zero face $\{x : h^*(x) = 0\}$ of the optimal exposer. The **always-tight zero-face family** $Z(u)$ is the set of rows whose lower-box constraint $h \geq 0$ is tight on the *whole* primal optimal face of the LP (equivalently, $\mathrm{supp}(\alpha) \subseteq Z(u)$ for every reduced optimal hiddenness dual witness (λ, α, β), by `lem-always-tight-dual-support`). The **zero-face conic mass** is $\sum_{z \in Z(u)} a_z$ for the reduced optimal display's zero-face coefficients $a_z \geq 0$.

By `lem-zero-face-localization` every zero-face row is ρ-near u ($\|p_z - p_u\|_1 < 4\tau$, $\tau = \sqrt{\delta}$); if u is within 4τ of a hidden top v of height H then additionally $\|p_z - p_v\|_1 < 8\tau$ and z has depth $> H - 8\tau$.

Kind. original (introduced by this project); `status: draft`. **Aliases.** zero face; zero-face family; always-tight zero-face family; Z(u); h-zero row; zero-face conic mass.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-zero-face-localization.md and lem-always-tight-dual-support.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-zero-face-localization (rho-nearness) + lem-always-tight-dual-support (the alpha-carrier family Z).
Used by 2 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-zero-face.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; the vocabulary in which the huddle-charge terminal node is posed (kill or bound the zero-face conic mass). Distinguish two uses in the shards, kept consistent here: the *pointwise* zero-face row ($h^*(p_z) = 0$ for one fixed optimal exposer) and the *always-tight family* $Z(u)$ (tight on the entire optimal face). $Z(u)$ is the always-tight version; statements that fix one optimal exposer h^* use the pointwise version, and the two agree on the relative interior optimal exposer. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-actor-hull` (the always-tight far/upper-box hulls K_T, K_O), `def-dual-witness`.

Definition 11 (co-top (`def-co-top`))**.** Fix an exact signed idempotent P with nonempty visible set $\mathcal{W}$ and a hidden top vertex v of height H; write $d_j = \mathrm{dist}_1(p_j, \mathrm{conv}\{p_w : w \in \mathcal{W}\})$, $\tau = \sqrt{\delta(P)}$.

A row f is **co-top** (relative to v, at slack c) if it sits nearly as deep as the top:

$$d_f \; > \; H - c\,\tau \qquad (c > 0 \text{ a small universal constant, typically } c \in \{4, 8\}).$$

The **co-top far web** of v is the set of rows that are simultaneously co-top **and** ρ-far from v ($\|p_f - p_v\|_1 \geq 4\tau$) — the **starved set** $\{j : \|p_j - p_v\|_1 \geq 4\tau,\ d_j > H - 8\tau\}$. By `lem-cotop-`

`witness-pinning` more than 13/16 of a hidden top's dual-witness λ-mass sits in this starved set (at $c = 4$, $\delta \le 1/4$), and by `lem-downhill-cotop-conic-mass` (disjoint always-tight hulls) a definite share of the zero-face conic mass lands on nonclone, ρ-near, co-top rows.

Kind. original (introduced by this project); `status: draft`. **Aliases.** co-top row; co-top vertex; co-top web; co-top far web; co-top zero-face mass; starved set.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-cotop-witness-pinning.md and lem-downhill-cotop-conic-mass.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-cotop-witness-pinning + lem-downhill-cotop-conic-mass.
Used by 40 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-co-top.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; "co-top" names the deep-far actors that a hidden top's dual witness is forced to charge yet its positive coefficient mass need not reach (the dual-required / primal-starved tension at the heart of the huddle charge and `conj-cotop-web-coupling`). 'status: draft' — A+B sign-off pending (Rule 7). Related: `def-dual-witness`, `def-actor-hull`, `def-near-cluster`.

Definition 12 (invisible mass (`def-invisible-mass`))**.** Let $C_{\mathcal{W}} := \operatorname{conv}\{p_w : w \in \mathcal{W}\}$ (`def-visible-set`) and let p_v be a row vertex. The *invisible (positive) mass* of v is

$$\widetilde{\sigma}_v := \sum_{j\,:\,\operatorname{dist}_1(p_j,\,C_{\mathcal{W}})>0} \max\{P_{vj},\, 0\},$$

the total positive coefficient mass that row v places on rows lying strictly outside $C_{\mathcal{W}}$ — *including* the index $j = v$ itself when $p_v \notin C_{\mathcal{W}}$ and $P_{vv} > 0$.

Halo robustness. Replacing the condition $\operatorname{dist}_1(p_j, C_{\mathcal{W}}) > 0$ by $\operatorname{dist}_1(p_j, C_{\mathcal{W}}) > \varepsilon$ changes $\widetilde{\sigma}_v$ only by mass sitting in the ε-halo of $C_{\mathcal{W}}$, which is absorbable as an ε-loss in the height; the $\varepsilon = 0$ form is used throughout.

Kind. original (introduced by this project); `status: locked`. **Aliases.** invisible positive mass; $\widetilde{\sigma}$_v; invisible mass $\widetilde{\sigma}$_v.
Provenance. `source: internal`; locus `quoted from classical-portfolio/report/kernel-conjecture.tex §Setting (Definition: Invisible mass` $\widetilde{\sigma}$`_v; Remark: Halo robustness)`; no source hash (not a literature transcription); record: adapted from kernel-conjecture.tex §Setting (Definition: Invisible mass $\widetilde{\sigma}$_v, def:sigt).
Used by 35 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-invisible-mass.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Quoted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (Definition: *Invisible mass* $\widetilde{\sigma}_v$) with the *Halo robustness* remark. $\widetilde{\sigma}_v$ is the hypothesis quantity of the Kernel Conjecture: a hidden top vertex (`def-height`) with $\widetilde{\sigma}_v > \tau = \sqrt{\delta}$ is the missing branch, while the $\widetilde{\sigma}_v \le \tau$ branch is the proved height cap. Uses `def-visible-set`, `def-negative-mass`.

Definition 13 (near-cluster (`def-near-cluster`))**.** Fix an exact signed idempotent P with nonempty visible set $\mathcal{W}$, a hidden top vertex v of height H, $\tau = \sqrt{\delta(P)}$, $\rho = 4\tau$, and a width constant $a \ge 4$. Write $d_j = \operatorname{dist}_1(p_j, \operatorname{conv}\{p_w : w \in \mathcal{W}\})$.

The **near-cluster** (or ρ-near deep cluster) of v is

$$C(v) \;:=\; \{\, j : \; \|p_j - p_v\|_1 < 4\tau \;\; \text{and} \;\; d_j > a\,\tau \,\}$$

— rows simultaneously ρ-near v **and** deep (both conditions required). The **cluster mass** is $\sum_{j\in C(v)} \max(P_{vj}, 0)$; a top is **heavy** when this is $\ge 1 - \theta_0$. A **cluster vertex** is a geometrically distinct row vertex u carrying positive disintegrated weight from $C(v)$-rows (or, in the re-rooted assembly, $u := v$ itself).

Kind. original (introduced by this project); `status: draft`. **Aliases.** rho-near deep cluster; near-deep cluster; cluster C(v); C(v); huddle cluster; cluster mass.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/conj-near-cluster-absorption.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by conj-near-cluster-absorption (the summand set C(v)) and the W54 huddle-charge tree.
Used by 3 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-near-cluster.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; the exact summand set of `conj-near-cluster-absorption` and the pinned target of the W54 huddle-charge decomposition tree. The width a is a calibration constant (any universal $a \geq 4$; $a = 16$ in the W54 assembly), not load-bearing downstream. NOTE ON DIVERGENCE: the bare word "cluster" is also used generically for an arbitrary row-index subset C (e.g. `lem-cluster-return-flow`'s return flow over any subset) — that generic set-theoretic use is common knowledge and is NOT this term; this shard defines only the huddle near-cluster $C(v)$. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-co-top`, `def-slab`.

Definition 14 (near-positive projection (`def-near-positive-projection`))**.** A *near-positive projection* is a linear map $R\colon \ell_n^\infty \to \ell_n^\infty$ that is an exact unital idempotent, $R^2 = R$ and $R(\mathbf{1}) = \mathbf{1}$, and is *$\delta$-positive*: for every $0 \leq x$ (entrywise) with $\|x\|_\infty \leq 1$ one has $Rx \geq -\delta\,\mathbf{1}$, for a small parameter $\delta \in [0, \delta_0)$. As a matrix this is exactly an exact signed idempotent whose negative mass satisfies $\delta(R) \leq \delta$ (and $\|R\| \leq 1 + O(\delta)$).

The motivating operator-algebra example is the spectral idempotent $P = \theta(2\Phi - \mathbf{1})$ built from an `def-almost-idempotent` Φ, with $\delta = O(\eta)$; such a P need **not** be a genuinely positive map. *Near-positive-projection stability* (the open hypothesis parenting `op-classical`) asks whether every near-positive projection is within $C\sqrt{\delta}$ (operator norm) of a genuine *positive* unital idempotent — a stochastic idempotent E; the $\sqrt{}$ exponent is sharp (Hume's 3×3 family).

Kind. original (introduced by this project); `status: locked`. **Aliases.** δ-positive unital idempotent; delta-positive unital idempotent; near-idempotent stochastic map; near-positive idempotent.
Provenance. `source: internal`; locus `adopted from ../almost-idempotent-positive-maps/definitions/def-near-positive-projection.md (commutative shadow)`; no source hash (not a literature transcription); record: adopted from ../almost-idempotent-positive-maps definitions/def-near-positive-projection.md (B's factorization program, op:npps); commutative shadow agreed for this repo.
Used by 2 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-near-positive-projection.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Adopted from `../almost-idempotent-positive-maps/definitions/def-near-positive-projection.md` (B's factorization program, `op:npps`), taken here as the commutative shadow / general parent of the classical object. This is the map/positivity framing of the same object as `def-signed-idempotent` (matrix/geometry framing): R is δ-positive iff $\delta(R) \leq \delta$. The *stability* statement is an open registry problem, not part of this definition.

Definition 15 (top support functional (`def-top-support-functional`))**.** Fix an exact signed idempotent P with nonempty visible set $\mathcal{W}$, write $C_\mathcal{W} = \mathrm{conv}\{p_w : w \in \mathcal{W}\}$, and let v be a hidden top vertex of height H.

A **top support functional** at v is an affine $\phi : \mathbb{R}^n \to \mathbb{R}$ with

$$\phi(p_v) = H, \qquad \phi \leq 0 \text{ on } C_\mathcal{W}, \qquad \phi \text{ is 1-Lipschitz for } \ell^1.$$

The set of all such is Φ_v; it is nonempty (`lem-top-deficit-price`), convex and compact, and by `lem-top-support-dual-face` equals $\{\phi_y(x) = y \cdot x - h_C(y) : y \in Y_v\}$ where $h_C(y) = \sup_{c \in C_\mathcal{W}} y \cdot c$ and $Y_v = \{y : \|y\|_\infty \leq 1,\ y \cdot p_v - h_C(y) = H\}$. A finite convex average of top support functionals (an **averaged** $\bar{\phi}$) is again a top support functional (the three defining conditions are convex).

Kind. original (introduced by this project); `status: draft`. **Aliases.** top support functional phi; Phi_v; support functional at the top; averaged phi-bar.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-top-deficit-price.md and lem-top-support-dual-face.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-top-deficit-price (existence) + lem-top-support-dual-face (the dual face Phi_v).
Used by 39 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-top-support-functional.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; the affine functional that realizes the height of the top and pins the top-deficit $z_j = H - \phi(p_j) \geq 0$. It is the charging channel of `lem-top-deficit-price`. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-top-deficit`, `def-height`, `def-visible-set`.

Definition 16 (selected-corner configuration (`def-selected-corner`))**.** Fix a finite exact signed idempotent P with $0 < \delta(P)$, $\tau = \sqrt{\delta(P)}$, $D = 2+4\delta(P)$, nonempty visible set $\mathcal{W}$, $C_{\mathcal{W}} = \operatorname{conv}\{p_w : w \in \mathcal{W}\}$, and a hidden top v of height $H > 16\tau$. A **selected-corner configuration** is such (P, v) together with:

- an affine ϕ with $\phi(p_v) = H$, $\phi \leq 0$ on $C_{\mathcal{W}}$, and $|\phi(a) - \phi(b)| \leq \|a - b\|_1$ (a top support functional), an admissible exposer h at v, and a **selected corner row** f with $\|p_f - p_v\|_1 \geq 4\tau$, $\operatorname{dist}_1(p_f, C_{\mathcal{W}}) > H - 4\tau$ (co-top and ρ-far) satisfying the corner inequality $2(H - \phi(p_f))/D + h(p_f) \leq 12\tau/13$;
- a **disintegration kernel** $\xi_x(u)$ from row points to geometrically distinct row vertices, constant on clone fibers, with $p_x = \sum_u \xi_x(u) p_u$ and Dirac at vertex points.

Write $P^+_{fx} = \sum_{j:p_j=p_x} \max(P_{fj}, 0)$, $\Gamma_f(x,u) = P^+_{fx}\,\xi_x(u)$, $C_f = \{(x,u) : H - \phi(p_x) < 4\tau,\ h(p_x) < 4\tau,\ H - \phi(p_u) < 4\tau,\ h(p_u) < 4\tau\}$, $B_F = C_f \cap \{\|p_u - p_v\|_1 \geq 4\tau\}$, $B_N = C_f \cap \{\|p_u - p_v\|_1 < 4\tau\}$. For a block $B \in \{B_F, B_N\}$ the three **corner masses** are

$$M_X(B) = \Gamma_f\{(x,u) \in B : p_x \neq p_u\}, \quad M_I(B) = \Gamma_f\{(x,u) \in B : p_x = p_u,\ K_T(u) \cap K_O(u) \neq \emptyset\},$$
$$M_D(B) = \Gamma_f\{(x,u) \in B : p_x = p_u,\ K_T(u) \cap K_O(u) = \emptyset\},$$

with $K_T(u), K_O(u)$ the always-tight hulls at u (off-diagonal X = distinct $p_x \neq p_u$; intersection I = diagonal with intersecting hulls; deep D = diagonal with disjoint hulls).

Kind. original (introduced by this project); `status: draft`. **Aliases.** selected corner; corner configuration; disintegration kernel xi; Gamma_f; M_X; M_I; M_D; B_F; B_N; C_f.
Provenance. `source: internal`; locus `internal; first pinned (inlined verbatim) in argument/lemmas/lem-sl1a-three-cell-reduction.md and the three conj-sl1a-*-cell shards`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-sl1a-three-cell-reduction (the SL1a three-cell surface).
Used by 37 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-selected-corner.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; the single home of the SL1a "corner" machinery inlined verbatim in `lem-sl1a-three-cell-reduction` and the three cell conjectures. The three cells partition the selected corner's mass into the $X/I/D$ diagonal cells. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-actor-hull`, `def-co-top`, `def-top-support-functional`, `def-top-deficit`.

Definition 17 (top-deficit (`def-top-deficit`))**.** In the setting of `def-top-support-functional` — an exact signed idempotent P, nonempty visible set, hidden top v of height H, and a top support functional $\phi \in \Phi_v$ — the **top-deficit** of a row j under ϕ is

$$z_j \;:=\; H - \phi(p_j) \;\geq\; 0,$$

nonnegative because $\phi(p_j) \le \mathrm{dist}_1(p_j, C_{\mathcal{W}}) \le H$ (and bounded above, $z_j \le 2+4\delta$). The **top-deficit supremum** of a row f is $Z_v(f) := \sup_{\phi\in\Phi_v} \big(H - \phi(p_f)\big) = \sup_{y\in Y_v} y\cdot(p_v - p_f)$ (`lem-top-support-dual-face`). A **top-deficit slab** is a band of rows cut by a threshold on z, e.g. $\{j : z_j \ge L\}$.

Kind. original (introduced by this project); `status: draft`. **Aliases.** top deficit; z_j; deficit; top-deficit slab; Z_v(f); top-deficit supremum.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-top-deficit-price.md and lem-top-support-dual-face.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-top-deficit-price (z_j) + lem-top-support-dual-face (the supremum Z_v(f)).
Used by no registry result yet (vocabulary registered ahead of the argument).
Canonical shard. `definitions/def-top-deficit.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; "deficit" alone always means this top-deficit in the huddle-charge shards. The charging identity (`lem-top-deficit-price`, from $P^2 = P$ and $P\mathbf{1} = \mathbf{1}$) is $\sum_j a_j^+ z_j = \sum_j a_j^- z_j \le \nu_v(2+4\delta)$ with $a_j = P_{vj}$; its structural blind spot is that rows in the ρ-ball of v carry $z_j < 4\tau$. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-top-support-functional`, `def-slab`, `def-co-top`.

Definition 18 (slab (`def-slab`))**.** Fix an exact signed idempotent P, $\tau = \sqrt{\delta(P)}$, nonempty visible set $\mathcal{W}$. A **slab** is the set of rows on which a fixed affine functional lies in a prescribed threshold band. Two families recur, kept distinct here:

- **Low slab** (exposer-value): for an optimal exposer h^* at a vertex and a threshold s, the set $\{j : h^*(p_j) < s\}$. The **deep / very-low slab** takes $s = \kappa = \tau/4$ or $s = \tau/8$; combined with depth ($d_j > a\tau$) it is the summand set of `conj-low-slab-cap` and `conj-far-low-slab-cap`. The pincer `lem-cs-low-slab-pincer` controls its complementary shell $\{s \le h^* < \kappa\}$.
- **Top slab** (depth / top-deficit): the near-top band $\{j : d_j > H - c\tau\}$ (equivalently small top-deficit $z_j < c'\tau$); the **far slab** additionally imposes ρ-farness. `lem-top-slab-companion` forces a ρ-far row in the top slab.

Kind. original (introduced by this project); `status: draft`. **Aliases.** low slab; low-slab; deep low slab; very-low slab; top slab; top-slab; far slab; slab leakage.
Provenance. `source: internal`; locus `internal; first pinned across argument/lemmas/conj-low-slab-cap.md, lem-cs-low-slab-pincer.md, lem-top-slab-companion.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by conj-low-slab-cap (exposer-value slab) and lem-top-slab-companion (depth slab).
Used by 2 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-slab.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; "slab" is overloaded in the shards between the exposer-value cut (h^*) and the depth cut (d / z) — this shard records BOTH canonical instances and which functional each uses, so downstream statements can name the functional explicitly and avoid drift. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-top-deficit`, `def-near-cluster`, `def-exposed`.

The approximate C^*-algebra vocabulary (transcribed from the pinned source)

Definition 19 (level-one and amplified column-Hilbert corner (`def-column-hilbert-corner`))**.** Let P and Q be δ-projections in an ε-C^* algebra. If Q is one-dimensional, then $\mathcal{S}_{P,Q}$ is a Hilbert space with the inner product $\langle\cdot,\cdot\rangle$ defined by the equation

$$Y^\dagger \cdot X = \langle Y, X\rangle\, \widetilde{Q} \qquad (X, Y \in \mathcal{S}_{P,Q}),$$

where $Y^\dagger \cdot X = \mathrm{C}_Q(Y^\dagger X)$ and $\widetilde{Q} = \mathrm{C}_Q(Q)$.
One-dimensional projections and the corresponding Hilbert spaces. Lemma [`lem_PQ_Hilb`] says that if $P, Q \in \mathcal{A}$ are δ-projections and Q is one-dimensional, then the space $\mathcal{S}_{P,Q}$ is equipped

with the Hermitian inner product given by [1dQ_ip]. The extended version of this space, $\mathbb{C}^n \otimes \mathcal{S}_{P,Q} = \mathbf{M}_{n,1} \otimes \mathcal{S}_{P,Q}$, satisfies essentially the same equation:

$$Y^\dagger \cdot X = \langle Y, X\rangle\, \widetilde{Q} \qquad (X, Y \in \mathbf{M}_{n,1} \otimes \mathcal{S}_{P,Q}),$$

where $\langle Y, X\rangle = \sum_l \langle [Y]_{l1}, [X]_{l1}\rangle$.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** column-Hilbert corner; amplified column space.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1123-1128,1546-1550`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\CC`, `\Co`, `\Ma`, `\braket`, `\calA`, `\calS`, `\dag`, `\eps`, `\wt`; display environments unnumbered (labels are stripped); source cross-reference keys shown in brackets; source theorem-environment delimiters dropped; source-internal \labels dropped. The byte-verbatim source is reproduced below as the second check.
Used in this report by `lem-compcb-row-column-product`, `lem-hcb-column-hilbert-squared` (2 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-column-hilbert-corner.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft transcription; ratification is required before locking. The known false unsquared display at source lines 1551-1555 is deliberately not part of this definition.
Provenance. Byte-verbatim from the pinned source at the recorded loci and SHA256 prefix; the exclusion is required by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and is consistent with `VERDICT-W74F-H-STAGE1.md`.

Source check (byte-verbatim) — `kitaev-2405.02434, approximate_algebras.tex:1123-1128,1546-1550`

```
\begin {Lemma}\label {lem_PQ_Hilb}
Let $P$ and $Q$ be $\delta $-projections in an $\eps $-$C^*$ algebra. If $Q$ is one-dimensional, then $\calS
_{P,Q}$ is a Hilbert space with the inner product $\braket {\cdot }{\cdot }$ defined by the equation
\begin {equation}\label {1dQ_ip}
Y^\dag \cdot X=\braket {Y}{X}\,\wt {Q}\qquad (X,Y\in \calS _{P,Q}),
\end {equation}
where $Y^\dag \cdot X=\Co _{Q}(Y^{\dag }X)$ and $\wt {Q}=\Co _Q(Q)$.
```

```
\noindent \textbf {One-dimensional projections and the corresponding Hilbert spaces.} Lemma~\ref {lem_PQ_Hilb}
says that if $P,Q\in \calA $ are $\delta $-projections and $Q$ is one-dimensional, then the space $\calS
_{P,Q}$ is equipped with the Hermitian inner product given by \eqref {1dQ_ip}. The extended version of this
space, $\CC ^n\otimes \calS _{P,Q}=\Ma {n,1}\otimes \calS _{P,Q}$, satisfies essentially the same equation:
\begin {equation}\label {1dQ_ip_ext}
Y^\dag \cdot X=\braket {Y}{X}\,\wt {Q}\qquad (X,Y\in \Ma {n,1}\otimes \calS _{P,Q}),
\end {equation}
where $\braket {Y}{X}=\sum _{l}\braket {[Y]_{l1}}{[X]_{l1}}$.
```

Definition 20 (compressed corner (`def-compressed-corner`))**.** To each pair of projections (P, Q) in a C^* algebra $\mathcal{B}$, one assigns the vector space $\mathcal{S}_{P,Q} = \{PXQ : X \in \mathcal{B}\}$. In this section, we discuss an approximate version of this construction. Recall that a δ-projection in an ε-C^* algebra $\mathcal{A}$ is a Hermitian element P such that $\|P^2 - P\| \le \delta$. For each pair (P, Q) of δ-projections, there are two slightly different maps that could be called "compressions": $\mathrm{L}_P\mathrm{R}_Q : X \mapsto P(XQ)$ and $\mathrm{R}_Q\mathrm{L}_P : X \mapsto (PX)Q$. We will use their symmetric combination, $\frac{1}{2}(\mathrm{L}_P\mathrm{R}_Q + \mathrm{R}_Q\mathrm{L}_P)$. It is an $O(\delta + \varepsilon)$-idempotent element of the algebra of operators acting on $\mathcal{A}$, and therefore, can be approximated by an idempotent using Proposition [prop_P]. The *compression map* thus defined,

$$\mathrm{C}_{P,Q} : \mathcal{A} \to \mathcal{A}, \qquad \mathrm{C}_{P,Q} = \theta(\mathrm{L}_P\mathrm{R}_Q + \mathrm{R}_Q\mathrm{L}_P - 1),$$

satisfies the equations

$$\mathrm{C}_{P,Q}^2 = \mathrm{C}_{P,Q}, \qquad \mathrm{C}_{P,Q}(X)^\dagger = \mathrm{C}_{Q,P}(X^\dagger) \quad (X \in \mathcal{A}),$$

and both $\|\mathrm{L}_P\mathrm{R}_Q - \mathrm{C}_{P,Q}\|$ and $\|\mathrm{R}_Q\mathrm{L}_P - \mathrm{C}_{P,Q}\|$ are bounded by $O(\delta+\varepsilon)$. The image of this map, $\mathcal{S}_{P,Q} = \operatorname{Im}\mathrm{C}_{P,Q} = \operatorname{Ker}(1-\mathrm{C}_{P,Q})$, is a closed linear subspace of $\mathcal{A}$. There is a variant of this construction where only L_P or only R_Q is applied, and one writes $\mathrm{C}_{P,1}, \mathcal{S}_{P,1}$ or $\mathrm{C}_{1,Q}, \mathcal{S}_{1,Q}$, respectively. (This is the same as $\mathrm{C}_{P,I}$, etc. if the unit is exact.)

When $P = Q$, the abbreviations C_P, $\mathcal{S}_P$ are used.

For any triple (P, Q, R) of δ-projections, one defines the *compressed product* between elements of the corresponding subspaces:

$$(X,Y) \mapsto X \cdot Y : \mathcal{S}_{P,Q} \times \mathcal{S}_{Q,R} \to \mathcal{S}_{P,R}, \qquad X \cdot Y = \mathrm{C}_{P,R}(XY).$$

It is close to the ambient product in $\mathcal{A}$, namely, $\|X \cdot Y - XY\| \le O(\delta+\varepsilon)\|X\|\|Y\|$. If P is a nonvanishing projection, the compressed product turns the Banach space $\mathcal{S}_P$ (which is closed under the involution $X \mapsto X^\dagger$) into an $O(\delta+\varepsilon)$-C^* algebra with unit $\widetilde{P} = \mathrm{C}_P(P)$.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** compression map; compressed product; compressed unit.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1054-1066,1077-1082`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\Co`, `\Img`, `\Ker`, `\La`, `\Ra`, `\calA`, `\calB`, `\calS`, `\dag`, `\eps`, `\ts`, `\wt`; display environments unnumbered (labels are stripped); source cross-reference keys shown in brackets; source-internal \labels dropped. The byte-verbatim source is reproduced below as the second check.
Used in this report by `lem-compcb-amplified-almost-containment`, `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-identities`, `lem-compcb-compressed-unit-action`, `lem-compcb-compressed-unit-norm`, `lem-compcb-corner-algebra` and 11 further reproduced results (17 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-compressed-corner.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft transcription; ratification is required before locking.
Provenance. Byte-verbatim from the pinned source at the recorded loci and SHA256 prefix; provisioned by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and admitted only for the safe subset by `VERDICT-FUDW-DECOMP-V3.md` §D.

SOURCE CHECK (BYTE-VERBATIM) — KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:1054-1066,1077-1082

```
To each pair of projections $(P,Q)$ in a $C^*$ algebra $\calB $, one assigns the vector space $\calS
_{P,Q}=\{PXQ:\,X\in \calB \}$. In this section, we discuss an approximate version of this construction. Recall
that a $\delta $-projection in an $\eps $-$C^*$ algebra $\calA $ is a Hermitian element $P$ such that
$\|P^2-P\|\le \delta $. For each pair $(P,Q)$ of $\delta $-projections, there are two slightly different maps
that could be called ‘‘compressions’’: $\La _P\Ra _Q\colon X\mapsto P(XQ)$ and $\Ra _Q\La _P\colon X\mapsto
(PX)Q$. We will use their symmetric combination, $\frac {1}{2}(\La _P\Ra _Q+\Ra _Q\La _P)$. It is an $O(\delta
+\eps )$-idempotent element of the algebra of operators acting on $\calA $, and therefore, can be approximated
by an idempotent using Proposition~\ref {prop_P}. The \emph {compression map} thus defined,
\begin {equation}
\Co _{P,Q}\colon \calA \to \calA ,\qquad
\Co _{P,Q}=\theta (\La _P\Ra _Q+\Ra _Q\La _P-1),
\end {equation}
satisfies the equations
\begin {equation}
\Co _{P,Q}^2=\Co _{P,Q},\qquad \:
\Co _{P,Q}(X)^\dag =\Co _{Q,P}(X^\dag )\quad \: (X\in \calA ),
\end {equation}
and both $\|\La _P\Ra _Q-\Co _{P,Q}\|$ and $\|\Ra _Q\La _P-\Co _{P,Q}\|$ are bounded by $O(\delta +\eps )$.
The image of this map, $\calS _{P,Q}=\Img \Co _{P,Q}=\Ker (1-\Co _{P,Q})$, is a closed linear subspace of
$\calA $. There is a variant of this construction where only $\La _P$ or only $\Ra _Q$ is applied, and one
```

```
writes $\Co _{P,1},\calS _{P,1}$ or $\Co _{1,Q},\calS _{1,Q}$, respectively. (This is the same as $\Co
_{P,I}$, etc.\ if the unit is exact.)
When $P=Q$, the abbreviations $\Co _{P}$, $\calS _P$ are used.
```

```
For any triple $(P,Q,R)$ of $\delta $-projections, one defines the \emph {compressed product} between elements
of the corresponding subspaces:
\begin {equation}\label {compr_prod}
(X,Y)\mapsto X\cdot Y\,\colon \,
\calS _{P,Q}\times \calS _{Q,R}\to \calS _{P,R},\qquad X\cdot Y=\Co _{P,R}(XY).
\end {equation}
It is close to the ambient product in $\calA $, namely, $\|X\cdot Y-XY\|\le O(\delta +\eps )\|X\|\ts \|Y\|$.
If $P$ is a nonvanishing projection, the compressed product turns the Banach space $\calS _P$ (which is closed
under the involution $X\mapsto X^\dag $) into an $O(\delta +\eps )$-$C^*$ algebra with unit $\wt {P}=\Co
_{P}(P)$.\medskip
```

Definition 21 (δ-projection and nonvanishing alternative (`def-delta-projection`)). A *projection* in an ε-C^* algebra $\mathcal{A}$ is a Hermitian element P such that $P^2 = P$. Such elements are generally hard to come by, see Remark [`rem_X2`], so will content ourselves with *δ-projections* for sufficiently small δ. They are defined by the conditions

$$P^\dagger = P, \qquad \|P^2 - P\| \le \delta.$$

$$\|P\| \le O(\delta) \quad \text{or} \quad \big|\|P\| - 1\big| \le O(\delta + \varepsilon).$$

A δ-projection is called *nonvanishing* if the second alternative holds. If P is a δ-projection, then $I-P$ is a δ'-projection, where $\delta' = \delta$ if $\mathcal{A}$ has exact unit and $\delta' = \delta + O(\varepsilon)$ in general. A δ-projection P is called *nontrivial* if both P and $I - P$ are nonvanishing.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** delta-projection; nonvanishing delta-projection.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:917-920,926-929`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: \calA, \dag, \eps; display environments unnumbered (labels are stripped); source cross-reference keys shown in brackets; source-internal \labels dropped. The byte-verbatim source is reproduced below as the second check.
Used in this report by `conj-extcb`, `conj-hcb`, `lem-compcb-amplified-almost-containment`, `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-identities`, `lem-compcb-compressed-unit-action` and 9 further reproduced results (15 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-delta-projection.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft transcription; ratification is required before locking.
Provenance. Byte-verbatim from the pinned source at the recorded loci and SHA256 prefix; provisioned by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and admitted only for the safe subset by `VERDICT-FUDW-DECOMP-V3.md` §D.

Source check (byte-verbatim) — `kitaev-2405.02434, approximate_algebras.tex:917-920,926-929`

```
A \emph {projection} in an $\eps $-$C^*$ algebra $\calA $ is a Hermitian element $P$ such that $P^2=P$. Such
elements are generally hard to come by, see Remark~\ref {rem_X2}, so will content ourselves with \emph
{$\delta $-projections} for sufficiently small $\delta $. They are defined by the conditions
\begin {equation}\label {delta_P}
P^\dag =P,\qquad \|P^2-P\|\le \delta .
\end {equation}
```

```
\begin {equation}\label {P_alternatives}
```

```
\|P\|\le O(\delta )\quad \text {or}\quad \bigl |\|P\|-1\bigr |\le O(\delta +\eps ).
\end {equation}
A $\delta $-projection is called \emph {nonvanishing} if the second alternative holds. If $P$ is a $\delta
$-projection, then $I-P$ is a $\delta '$-projection, where $\delta '=\delta $ if $\calA $ has exact unit and
$\delta '=\delta +O(\eps )$ in general. A $\delta $-projection $P$ is called \emph {nontrivial} if both $P$
and $I-P$ are nonvanishing.
```

Definition 22 (ε-C^*-algebra (`def-epsilon-cstar-algebra`))**.** An *ε-Banach algebra* is a Banach space $\mathcal{A}$ endowed with a bilinear multiplication map $\mathcal{A}\times\mathcal{A}\to\mathcal{A}$ such that

$$\begin{aligned} \|XY\| &\le (1+\varepsilon)\|X\|\|Y\| && (X,Y\in\mathcal{A}),\\ \|(XY)Z-X(YZ)\| &\le \varepsilon\|X\|\|Y\|\|Z\| && (X,Y,Z\in\mathcal{A}). \end{aligned}$$

Such an algebra is called *ε'-commutative* if

$$\|XY-YX\|\le\varepsilon'\|X\|\|Y\| \qquad (X,Y\in\mathcal{A}).$$

A *$*\varepsilon$-Banach algebra* is a complex ε-Banach algebra with a conjugate linear involution $X\mapsto X^\dagger$ satisfying the equations

$$\|X^\dagger\|=\|X\|,\qquad (XY)^\dagger=Y^\dagger X^\dagger \qquad (X,Y\in\mathcal{A}).$$

An *ε-C^* algebra* is one with this additional property:

$$\|X^\dagger X\|\ge(1-\varepsilon)\|X\|^2 \qquad (X\in\mathcal{A}).$$

(A bound from the other side, $\|X^\dagger X\|\le(1+\varepsilon)\|X\|^2$, follows from [`ax_prodnorm`] and [`ax_*`].) We assume that all algebras are unital. The unit element $I\in\mathcal{A}$ should satisfy the approximate or exact conditions

$$\begin{aligned} &\|XI-X\|\le\varepsilon\|X\|, && \|IX-X\|\le\varepsilon\|X\|, && \bigl|\|I\|-1\bigr|\le\varepsilon && \text{(by default)}, \quad \text{or}\\ &XI=X, && IX=X, && \|I\|=1 && \text{(if specified)}. \end{aligned}$$

If the involution is defined, one also requires that $I^\dagger=I$.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** epsilon-C^*-algebra; approximate C^*-algebra.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:407-440`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\calA`, `\dag`, `\eps`, `\ts`; display environments unnumbered (labels are stripped); source cross-reference keys shown in brackets; source theorem-environment delimiters dropped; source-internal \labels dropped. The byte-verbatim source is reproduced below as the second check.
Used in this report by `lem-extcb-corner-dimension-additivity`, `lem-extcb-one-dimensional-corner-dimension`, `lem-extcb-one-dimensional-product`, `lem-stage1-approximate-group-laws`, `lem-stage1-exact-unit-rectification`, `lem-stage1-group-closeness` and 16 further reproduced results (29 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-epsilon-cstar-algebra.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft transcription; ratification is required before locking.
Provenance. Byte-verbatim from the pinned source at the recorded locus and SHA256 prefix; provisioned by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and admitted only for the safe subset by `VERDICT-FUDW-DECOMP-V3.md` §D.

```
\begin {Definition}
An \emph {$\eps $-Banach algebra} is a Banach space $\calA $ endowed with a bilinear multiplication map $\calA
\times \calA \to \calA $ such that
\begin {alignat}{2}
\label {ax_prodnorm}
\|XY\| &\le (1+\eps )\ts \|X\|\ts \|Y\|\qquad &&(X,Y\in \calA ),\\[2pt]
\label {ax_assoc}
\|(XY)Z-X(YZ)\| &\le \eps \ts \|X\|\ts \|Y\|\ts \|Z\|\qquad &&(X,Y,Z\in \calA ).
\end {alignat}
Such an algebra is called \emph {$\eps '$-commutative} if
\begin {equation}
\label {ax_comm}
\|XY-YX\|\le \eps '\ts \|X\|\ts \|Y\|\qquad (X,Y\in \calA ).
\end {equation}
A \emph {$*\eps $-Banach algebra} is a complex $\eps $-Banach algebra with a conjugate linear involution
$X\mapsto X^\dag $ satisfying the equations
\begin {equation}
\label {ax_*}
\|X^\dag \|=\|X\|,\qquad (XY)^\dag =Y^\dag X^\dag \qquad (X,Y\in \calA ).
\end {equation}
An \emph {$\eps $-$C^*$ algebra} is one with this additional property:
\begin {equation}
\label {ax_C*}
\|X^{\dag }X\|\ge (1-\eps )\ts \|X\|^{2}\qquad (X\in \calA ).
\end {equation}
(A bound from the other side, $\|X^{\dag }X\|\le (1+\eps )\ts \|X\|^{2}$, follows from \eqref {ax_prodnorm}
and \eqref {ax_*}.) We assume that all algebras are unital. The unit element $I\in \calA $ should satisfy the
approximate or exact conditions
\begin {alignat}{4}
\label {ax_eps_unit}
\|XI&-X\|\le \eps \ts \|X\|,\qquad &\|IX&-X\|\le \eps \ts \|X\|,\qquad
&\bigl |\|I\|&-1\bigr |\le \eps \qquad &&\text {(by default)},\quad \text {or}\\[2pt]
\label {ax_exact_unit}
XI&=X,\qquad &IX&=X,\qquad
&\|I\|&=1\qquad &&\text {(if specified)}.
\end {alignat}
If the involution is defined, one also requires that $I^\dag =I$.
\end {Definition}
```

Definition 23 (extended ε-C^*-algebra (`def-extended-epsilon-cstar-algebra`)). An *extended ε-C^* algebra* is a complete self-adjoint operator space $\mathcal{A}$ with a multiplication and a unit that make each space $\mathbf{M}_n \otimes \mathcal{A}$ into an ε-C^* algebra. An *extended δ-homomorphism* is a linear map $v\colon \mathcal{A}' \to \mathcal{A}''$ such that for each n, the map $1_{\mathbf{M}_n} \otimes v\colon \mathbf{M}_n \otimes \mathcal{A}' \to \mathbf{M}_n \otimes \mathcal{A}''$ is a δ-homomorphism.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** extended epsilon-C^*-algebra; extended approximate C^*-algebra.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1477-1479`; SHA256 prefix `e7eb512a2ec2438d`.
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\Ma`, `\calA`, `\eps`. The byte-verbatim source is reproduced below as the second check.
Used in this report by `conj-extcb`, `conj-hcb`, `lem-compcb-amplified-almost-containment`, `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-identities`, `lem-compcb-compressed-unit-action` and 12 further reproduced results (23 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-extended-epsilon-cstar-algebra.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notation. Registry contracts write the source's `\(\eps\)` as `\(\varepsilon\)` and `\(\Ma n\)` as `\(M_n\)`.
Provenance. Byte-verbatim from the pinned source at the recorded locus and SHA256 prefix. User-ratified and locked 2026-07-24 (tobiasosborne, in-session sign-off).

SOURCE CHECK (BYTE-VERBATIM) — KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:1477-1479

```
An \emph {extended $\eps $-$C^*$ algebra} is a complete self-adjoint operator space $\calA $ with a
multiplication and a unit that make each space $\Ma {n}\otimes \calA $ into an $\eps $-$C^*$ algebra. An \emph
{extended $\delta $-homomorphism} is a linear map $v\colon \calA '\to \calA ''$ such that for each $n$, the
map $1_{\Ma {n}}\otimes v\colon \Ma {n}\otimes \calA '\to \Ma {n}\otimes \calA ''$ is a $\delta
$-homomorphism.
```

Definition 24 (diagonal of a finite-dimensional C^*-algebra (`def-fd-cstar-diagonal`)). For arbitrary $\tilde{D} = \sum_j A_j \otimes B_j \in \mathcal{B} \otimes \mathcal{B}$ and $X \in \mathcal{B}$, we write

$$X\tilde{D} = \sum_j XA_j \otimes B_j, \qquad \tilde{D}X = \sum_j A_j \otimes B_j X, \qquad \pi(\tilde{D}) = \sum_j A_j B_j.$$

These operations extend from $\mathcal{B} \otimes \mathcal{B}$ to $\mathcal{B} \hat{\otimes} \mathcal{B}$. An element $D \in \mathcal{B} \hat{\otimes} \mathcal{B}$ is called a *diagonal* if

$$XD = DX \quad \text{for all } \, X \in \mathcal{B}, \qquad \pi(D) = I_{\mathcal{B}}.$$

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** finite-dimensional C^*-algebra diagonal; algebra diagonal.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1233-1242`; SHA256 prefix `e7eb512a2ec2438d`.
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\calB`, `\hotimes`; display environments unnumbered (labels are stripped). The byte-verbatim source is reproduced below as the second check.
Used by 4 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-fd-cstar-diagonal.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Specialization. The registry uses this definition only when `\(\mathcal B\)` is a finite-dimensional `\(C^*\)`-algebra.
Provenance. Byte-verbatim from the pinned source at the recorded locus and SHA256 prefix. User-ratified and locked 2026-07-24 (tobiasosborne, in-session sign-off).

SOURCE CHECK (BYTE-VERBATIM) — KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:1233-1242

```
For arbitrary $\tilde {D}=\sum _{j}A_j\otimes B_j\in \calB \otimes \calB $ and $X\in \calB $, we write
\begin {equation}
X\tilde {D}=\sum _{j}XA_j\otimes B_j,\qquad
\tilde {D}X=\sum _{j}A_j\otimes B_jX,\qquad
\pi (\tilde {D})=\sum _{j}A_jB_j.
\end {equation}
These operations extend from $\calB \otimes \calB $ to $\calB \hotimes \calB $. An element $D\in \calB
\hotimes \calB $ is called a \emph {diagonal} if
\begin {equation}
XD=DX\quad \text {for all }\, X\in \calB ,\qquad \pi (D)=I_\calB .
\end {equation}
```

Definition 25 (H-space and left inversion (`def-h-space-left-inversion`)). An *H-space* is a topological space M with basepoint e and a continuous multiplication $\mu\colon M \times M \to M$ with $\mu(e,e) = e$ such that $x \mapsto \mu(x,e)$ and $x \mapsto \mu(e,x)$ are each basepoint-preserving homotopic to the identity. It is *associative* if the two triple-product maps are homotopic. An *inversion map* is a continuous basepoint-preserving $\sigma\colon M \to M$ with both $x \mapsto \mu(\sigma(x),x)$ and $x \mapsto \mu(x,\sigma(x))$ homotopic to the constant map e; when only the first homotopy is required, σ is a *left inversion map*.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** H-space; associative H-space; inversion map; left inversion map.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:895-912`; SHA256 prefix `e7eb512a2ec2438d`; record: byte-match lock per definitions/README.md (cited criterion); provisioned by DESIGN-FUDW-DECOMP-v4.1.md §4.1 (kind "cited / lock after byte-match").
Used by 2 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-h-space-left-inversion.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Definition only (per the DESIGN-FUDW-DECOMP-v4.1 §4.1 register row: "pinned TeX 895–912, definition only") — no claim about $\mathcal{U}$ or any Stage-1 object is carried by this shard. Source typo note: the source writes $\mu\colon M \to M$ for the multiplication map's type where $\mu\colon M \times M \to M$ is meant (its uses $\mu(x, y)$ are two-argument throughout); the harmonised statement records the corrected type. Elisions `[...]` in the verbatim block skip source sentences about the specific space $\mathcal{U}$ (claims, not definition content). Used by the Stage-1 topology rows `lem-topology-hopf-structure` and (via fixed-point data) the Lefschetz rows; see `def-lefschetz-fixed-point-data`.

Source check (byte-verbatim) — `kitaev-2405.02434, approximate_algebras.tex:895-912`

```
By definition, an \emph {H-space} is a topological space $M$ with a basepoint $e$ and a continuous
multiplication map $\mu \colon M\to M$ such that $\mu (e,e)=e$ and
\begin {equation}\label {H-unit}
\bigl (x\mapsto \mu (x,e)\bigr )
\,\sim \,\bigl (x\mapsto x\bigr )
\,\sim \,\bigl (x\mapsto \mu (e,x)\bigr ),
\end {equation}
where ‘‘$f\sim g$’’ means that $f$ and $g$ are homotopic through basepoint-preserving maps. [...] An H-space
$M$ is called \emph {associative} if
\begin {equation}
\bigl ((x,y,z)\mapsto \mu (\mu (x,y),z)\bigr )
\,\sim \,\bigl ((x,y,z)\mapsto \mu (x,\mu (y,z))\bigr ).
\end {equation}
[...] An \emph {inversion map} is a continuous basepoint-preserving map $\sigma \colon M\to M$ such that
\begin {equation}\label {homo_inverse}
\bigl (x\mapsto \mu (\sigma (x),x)\bigr )
\,\sim \,\bigl (x\mapsto e\bigr )
\,\sim \,\bigl (x\mapsto \mu (x,\sigma (x))\bigr ).
\end {equation}
[...] When only the existence of that homotopy is required, $\sigma $ is called a \emph {left inversion map}.
```

Definition 26 (Ha-map (`def-ha-map`))**.** Let Q be a one-dimensional δ-projection in an ε-C^* algebra $\mathcal{A}$. For any δ-projections $P, R \in \mathcal{A}$, we define a map $\mathrm{H}^Q_{P,R}\colon \mathcal{S}_{P,R} \to \mathbf{B}(\mathcal{S}_{R,Q}, \mathcal{S}_{P,Q})$. (Here $\mathbf{B}(\mathcal{S}_{R,Q}, \mathcal{S}_{P,Q})$ is the space of bounded linear maps from $\mathcal{S}_{R,Q}$ to $\mathcal{S}_{P,Q}$ with the norm induced by the Euclidean norm $\|X\|_{\mathrm{E}} = \sqrt{\langle X, X\rangle}$. Due to inequality [`1dQ_ip_norm`], the latter is equal to $\|X\|$ up to a $1 \pm O(\varepsilon + \delta)$ factor.) For each $Z \in \mathcal{S}_{P,R}$ and $X \in \mathcal{S}_{R,Q}$, the element $\mathrm{H}^Q_{P,R}(Z)(X) \in \mathcal{S}_{P,Q}$ is defined by the condition

$$(Y^\dagger \cdot Z) \cdot X + Y^\dagger \cdot (Z \cdot X) = 2 \left\langle Y, \mathrm{H}^Q_{P,R}(Z)(X)\right\rangle \widetilde{Q} \quad \text{for all } Y \in \mathcal{S}_{P,Q}.$$

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** Ha map; Kitaev Ha-map.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1146-1149`; SHA256 prefix `e7eb512a2ec2438d`.
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\Bo`, `\Euc`, `\Ha`, `\bbraket`, `\braket`, `\calA`, `\calS`, `\dag`, `\eps`, `\wt`; display environments unnumbered (labels are stripped); source cross-reference keys shown in brackets; source-internal \labels dropped. The byte-verbatim source is reproduced below as the second check.

Used in this report by `conj-extcb`, `conj-hcb`, `lem-hcb1-column-action`, `lem-hcb1-variational-identity`, `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect` and 7 further reproduced results (13 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-ha-map.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notation. Registry contracts use `Ha` for the source macro `\(\Ha\)`.
Provenance. Byte-verbatim from the pinned source at the recorded locus and SHA256 prefix. User-ratified and locked 2026-07-24 (tobiasosborne, in-session sign-off).

Source check (byte-verbatim) — `kitaev-2405.02434, approximate_algebras.tex:1146-1149`

```
Let $Q$ be a one-dimensional $\delta $-projection in an $\eps $-$C^*$ algebra $\calA $. For any $\delta
$-projections $P,R\in \calA $, we define a map $\Ha ^{Q}_{P,R}\colon \calS _{P,R}\to \Bo (\calS _{R,Q},\calS
_{P,Q})$. (Here $\Bo (\calS _{R,Q},\calS _{P,Q})$ is the space of bounded linear maps from $\calS _{R,Q}$ to
$\calS _{P,Q}$ with the norm induced by the Euclidean norm $\|X\|_\Euc =\sqrt {\braket {X}{X}}$. Due to
inequality \eqref {1dQ_ip_norm}, the latter is equal to $\|X\|$ up to a $1\pm O(\eps +\delta )$ factor.) For
each $Z\in \calS _{P,R}$ and $X\in \calS _{R,Q}$, the element $\Ha ^{Q}_{P,R}(Z)(X)\in \calS _{P,Q}$ is
defined by the condition
\begin {equation}\label {Ha_def}
(Y^\dag \cdot Z)\cdot X+Y^\dag \cdot (Z\cdot X)
=2\,\bbraket {Y}{\Ha ^{Q}_{P,R}(Z)(X)}\,\wt {Q}\quad \text {for all }\,Y\in \calS _{P,Q}.
\end {equation}
```

Definition 27 (Lefschetz number and local fixed-point index (`def-lefschetz-fixed-point-data`))**.** For a smooth self-map f of a compact orientable manifold M: at a fixed point x with $\det(1-f'(x)) \neq 0$ (where $f'(x)\colon T_xM \to T_xM$ is the differential), the *fixed point index* is $\operatorname{ind}(f,x) = \operatorname{sgn}\det(1-f'(x))$; the *Lefschetz number* is $\Lambda(f) = \sum_k (-1)^k \operatorname{Tr} f^{*k}$, where $f^{*k}\colon H^k(M;\mathbb{R}) \to H^k(M;\mathbb{R})$ is the induced map on real cohomology.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** Lefschetz number; fixed point index; fixed-point index; ind(f,x); Lambda(f).
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:961-967`; SHA256 prefix `e7eb512a2ec2438d`; record: byte-match lock per definitions/README.md (cited criterion); provisioned by DESIGN-FUDW-DECOMP-v4.1.md §4.1 (kind "cited / lock after byte-match").
Used by 3 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-lefschetz-fixed-point-data.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Definitions only (per the DESIGN-FUDW-DECOMP-v4.1 §4.1 register row: "pinned TeX 957–967, definitions only"): this shard carries the two displayed definitions and NO theorem content — in particular it does not assert the Lefschetz–Hopf identity $\sum_{x\in \mathrm{Fix}(f)} \operatorname{ind}(f,x) = \Lambda(f)$ (that is the separate result row `lem-topology-lefschetz-hopf`, to be `cited` from Arkowitz–Brown), nor the index-sign theorem (`lem-topology-local-index-sign`, Granas–Dugundji). The source's index formula is stated in the nondegenerate ($\det \neq 0$) smooth case; the general topological index the theorem rows use is the standard one in their own sources. See `def-h-space-left-inversion`.

Source check (byte-verbatim) — `kitaev-2405.02434, approximate_algebras.tex:961-967`

```
The \emph {fixed point index} $\ind (f,x)$ and the \emph {Lefschetz number} $\Lambda (f)$ are defined as
follows:
\begin {alignat}{2}
\ind (f,x)&=\sgn \det (1-f'(x)),\qquad
& f'(x)\colon \Ta _xM&\to \Ta _xM,\\[2pt]
\Lambda (f)&=\sum _{k}(-1)^k\Tr f^{*k},\qquad
& f^{*k}\colon H^k(M;\RR )&\to H^k(M;\RR ),
\end {alignat}
```

Definition 28 (one-dimensional δ-projection and equivalence (`def-one-dimensional-delta-projection`))**.** When $P = Q$, the abbreviations C_P, $\mathcal{S}_P$ are used. It is clear that $\mathcal{S}_P = 0$ if and only if P is sufficiently close to 0 as described by the first alternative in [`P_alternatives`]. (Recall that in the opposite case, we call P "nonvanishing".) Similarly, $\mathrm{C}_P = 1$ and $\mathcal{S}_P = \mathcal{A}$ if and only if P is close to I. If $\dim \mathcal{S}_P = 1$, we say that P is a *one-dimensional δ-projection.*

One-dimensional δ-projections P and Q are called *equivalent* if $\dim \mathcal{S}_{P,Q} = 1$. This is indeed an equivalence relation; in particular, it is transitive due to Lemma [`lem_PQR`].

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** one-dimensional delta-projection; equivalent delta-projections.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1066,1187`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\Co`, `\calA`, `\calS`; source cross-reference keys shown in brackets. The byte-verbatim source is reproduced below as the second check.
Used in this report by `conj-hcb`, `lem-extcb-one-dimensional-corner-dimension`, `lem-extcb-one-dimensional-product` (3 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-one-dimensional-delta-projection.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Locked after the cited-definition byte-match criterion was user-ratified on 2026-07-24.
Provenance. Byte-verbatim from the pinned source at the recorded loci and SHA256 prefix; provisioned by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and admitted only for the safe subset by `VERDICT-FUDW-DECOMP-V3.md` §D.

SOURCE CHECK (BYTE-VERBATIM) — KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:1066,1187

```
When $P=Q$, the abbreviations $\Co _{P}$, $\calS _P$ are used. It is clear that $\calS _P=0$ if and only if
$P$ is sufficiently close to $0$ as described by the first alternative in \eqref {P_alternatives}. (Recall
that in the opposite case, we call $P$ ‘‘nonvanishing’’.) Similarly, $\Co _P=1$ and $\calS _P=\calA $ if and
only if $P$ is close to $I$. If $\dim \calS _P=1$, we say that $P$ is a \emph {one-dimensional $\delta
$-projection}.
```

```
One-dimensional $\delta $-projections $P$ and $Q$ are called \emph {equivalent} if $\dim \calS _{P,Q}=1$. This
is indeed an equivalence relation; in particular, it is transitive due to Lemma~\ref {lem_PQR}.
```

Definition 29 (projection basis of a finite-dimensional commutative C^*-algebra (`def-projection-basis`))**.** The second application of Lemma [`lem_merging`] involves an auxiliary statement about δ-projections $P_1, \ldots, P_p, Q_1, \ldots, Q_q$ satisfying certain conditions, which will be formulated in terms of inclusions of finite-dimensional commutative C^* algebras. Such an algebra $\mathcal{B}$ is described by a *projection basis* $\{\Pi_1, \ldots, \Pi_n\} \subseteq \mathcal{B}$ such that $\Pi_j^\dagger = \Pi_j,\ \Pi_j \Pi_k = \delta_{jk} \Pi_k$, and $\sum_{j=1}^n \Pi_j = I$.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** projection basis.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1361`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\calB`, `\dag`; source cross-reference keys shown in brackets. The byte-verbatim source is reproduced below as the second check.
Used in this report by `lem-extcb-corner-dimension-additivity`, `lem-extcb1-cross-corner-dimension`, `lem-routef-f2-positive-unital-compression` (3 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-projection-basis.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Locked on the recorded user ratification of 2026-07-24 (see frontmatter `consensus:`; lock commit b9270ef4). An earlier draft-transcription note stood here until 2026-07-27 (stale-body drift caught by `AUDIT-F2-TYPING.md` correction 3).
Provenance. Byte-verbatim from the pinned source at the recorded locus and SHA256 prefix; provisioned by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and admitted only for the safe subset by `VERDICT-FUDW-DECOMP-V3.md` §D.

Source check (byte-verbatim) — kitaev-2405.02434, approximate_algebras.tex:1361

```
The second application of Lemma~\ref {lem_merging} involves an auxiliary statement about $\delta $-projections
$P_1,\dots ,P_p,Q_1,\dots ,Q_q$ satisfying certain conditions, which will be formulated in terms of inclusions
of finite-dimensional commutative $C^*$ algebras. Such an algebra $\calB $ is described by a \emph {projection
basis} $\{\Pi _1,\dots ,\Pi _n\}\subseteq \calB $ such that $\Pi _j^\dag =\Pi _j$,\, $\Pi _j\Pi _k=\delta
_{jk}\Pi _k$, and $\sum _{j=1}^{n}\Pi _j=I$.
```

Definition 30 (theta idempotent-approximation map (`def-theta-idempotent-approximation`)). Power-series functional calculus (tex:505-510):

$$f(X) = a_0 I + \sum_{n=1}^{\infty} a_n (X - x_0 I)^n \quad \text{if } \|X - x_0 I\| < r,$$
$$\|f(X) - f(x_0 I)\| \leq \sum_{n=1}^{\infty} |a_n| \|X - x_0 I\|^n.$$

Absolute value and sign (tex:517-520):

$$|X| = (X^2)^{1/2}, \qquad \operatorname{sgn}(X) = X(X^2)^{-1/2} \qquad\qquad \big(\|X^2 - x_0^2 I\| < x_0^2\big).$$

The theta map (tex:527-528, the definitional display of Proposition `prop_P`):

$$\widetilde{P} = \theta(2P - I), \qquad \text{where} \quad \theta(X) = \frac{1}{2}\big(I + \operatorname{sgn}(X)\big).$$

Kind. cited (byte-matched to a pinned source under `refs/`); `status: draft`. **Aliases.** theta map; functional-calculus idempotent approximation.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:505-528 (displays Taylor_simple, Taylor_simple_bound, abs_sgn, and the theta display inside Proposition prop_P --- ONLY the definitional displays are cited; the Proposition's claims are NOT citable and must be re-proved)`; SHA256 prefix `e7eb512a2ec2438d`.
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\sgn`, `\ts`, `\wt`; the shard quotes the naked body of a source display; restored to a display environment. The byte-verbatim source is reproduced below as the second check.
Used in this report by `lem-compcb-amplification-naturality`, `lem-compcb-entrywise-compression-naturality` (2 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-theta-idempotent-approximation.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Scope note. This is the map `theta` referenced by the compression-map display `Co_{P, Q}=\theta(\La_P\Ra_Q+\Ra_Q\La_P-1)` in `def-compressed-corner` (tex:1057). The neighbouring Proposition's conclusions (idempotence of `wtP`, commutation, distance bound) are claims, not definitions — they enter the registry only as re-proved results.
Ratification. Draft pending recorded sign-off; expected to lock under the user's recorded byte-match criterion (2026-07-24: "if they match the source then they are valid").

Source check (byte-verbatim) — kitaev-2405.02434, approximate_algebras.tex:505-528 (displays Taylor_simple, Taylor_simple_bound, abs_sgn, and the theta display inside Proposition prop_P --- ONLY the definitional displays are cited; the Proposition's claims are NOT citable and must be re-proved)

```
f(X)=a_0I+\sum _{n=1}^{\infty }a_n(X-x_0I)^n\quad \:
\text {if }\, \|X-x_0I\|<r,\\
```

```
\|f(X)-f(x_0I)\|\le \sum _{n=1}^{\infty }|a_n|\ts \|X-x_0I\|^n.
```

```
|X|=(X^2)^{1/2},\qquad \sgn (X)=X(X^2)^{-1/2}\qquad \quad
\bigl (\|X^2-x_0^2I\|<x_0^2\bigr ).
```

```
\wt {P}=\theta (2P-I),\qquad
\text {where}\quad \theta (X)=\frac {1}{2}\bigl (I+\sgn (X)\bigr ).
```

Project-internal packaging: hypothesis data and derived notation

Definition 31 (approximate unitary space (`def-approximate-unitary-space`))**.** Let $(\mathcal{X}, J, \cdot, \dagger)$ be an exact-unit ε-C^*-algebra (`def-epsilon-cstar-algebra`). Define:

- $\mathcal{H} = \{X \in \mathcal{X} : X^\dagger = X\}$ (Hermitian) and $i\mathcal{H} = \{X \in \mathcal{X} : X^\dagger = -X\}$ (anti-Hermitian);
- $\overline{\mathcal{U}}_\delta = \{X \in \mathcal{X} : \ \|X^\dagger \cdot X - J\| \le 2\delta$ and X has a right inverse$\}$, and $\mathcal{U}_\delta$ the analogous set with the strict inequality $\|X^\dagger \cdot X - J\| < 2\delta$;
- the *unitaries* $\mathcal{U} = \overline{\mathcal{U}}_0$;
- $\mathcal{U}_e$ = the connected component of $\mathcal{U}$ containing J, and the scalar action $U \equiv cU$ for $c \in U(1)$ on it.

For a point V for which the left-multiplier L_V is invertible, reserve the *coordinate notation* $\phi_V(X) = L_V^{-1} \cdot (X - V)$ and its components $\phi_V^{\parallel}$ (anti-Hermitian part) and $\phi_V^{\perp}$ (Hermitian part). Reserve the symbols u, h (polar inverse components) and μ, σ (group operations) ONLY as partial notation on domains supplied by result rows.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** approximate unitaries; unitary set of an epsilon-C^*-algebra.
Provenance. `source: internal`; locus `adopted from refs/kitaev-2405.02434 approximate_algebras.tex: 692-706 (sets), 845-859 (u,h and group maps), 945 (U_e and the scalar quotient)`; no source hash (not a literature transcription); record: user-ratified 2026-07-27 (W79 decision D2, docs/plans/2026-07-27-W78-ratification-package.md; content per DESIGN-S1-POLAR-v6.md sect 8, audited AUDIT-S1-POLAR-v4/v5/v6).
Used in this report by `lem-stage1-approximate-group-laws`, `lem-stage1-group-closeness`, `lem-stage1-group-domain-membership`, `lem-stage1-inversion-derivative-control`, `lem-stage1-inversion-derivative-transport`, `lem-stage1-maurer-cartan-transport` and 11 further reproduced results (24 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-approximate-unitary-space.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Vocabulary/typing only — this shard asserts NO chart, inverse, estimate, smoothness, compactness, orientation, or isolation theorem (all of those are the contracts of the Stage-1 polar result rows, `DESIGN-S1-POLAR-v6.md` §3). Adopted from the local source's approximate- unitary discussion with harmonised notation; the source loci are recorded above for orientation, but this is a `consensus` adoption, not a byte-cited definition. Related: `def-epsilon-cstar-algebra`, `def-stage1-polar-witness-data`, `def-h-space-left-inversion`, `def-lefschetz-fixed-point-data`.

Definition 32 (canonical corner identifications (`def-canonical-corner-identifications`))**.** After identifying the one-dimensional corner $\mathcal{S}_{Q,Q}$ with $\mathbb{C}$ by the normalized vector q_0, the canonical column identification is

$$[J_{P,Q,n}(Z)c]_i = \sum_j Z_{ij} c_j,$$

and the canonical row identification is

$$J_{Q,P,n}(Z) = J_{P,Q,n}(Z^\dagger)^\dagger.$$

Kind. original (introduced by this project); `status: locked`. **Aliases.** canonical J maps; canonical column and row identifications.
Provenance. `source: internal`; locus `PROOF-W74F-E-HCB.md §8.1; DESIGN-FUDW-DECOMP-v3.md §4.1`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers).
Used in this report by `conj-hcb`, `lem-hcb4-canonical-closeness`, `lem-hcb4-canonical-gram`, `lem-hcb4-canonical-inverse` (4 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-canonical-corner-identifications.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft packaging of notation only; no norm or inverse estimate is included, and ratification is required before locking.
Provenance. Transcribed from `PROOF-W74F-E-HCB.md` §8.1 and the provisioning row of `DESIGN-FUDW-DECOMP-v3.md` §4.1; admitted by `VERDICT-FUDW-DECOMP-V3.md` §D.

Definition 33 (compressed associator (`def-compressed-associator`))**.** For compatible compressed rectangular products, the *compressed associator* is

$$(A \cdot B) \cdot C - A \cdot (B \cdot C).$$

Kind. original (introduced by this project); `status: locked`. **Aliases.** rectangular compressed associator.
Provenance. `source: internal`; locus `PROOF-W74F-E-HCB.md §3; DESIGN-FUDW-DECOMP-v3.md §4.1`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers).
Used in this report by `lem-hcb0-compressed-associator`, `lem-hcb1-column-action` (2 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-compressed-associator.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft project name for the displayed source expression; it asserts no estimate and requires ratification before locking.
Provenance. Transcribed from `PROOF-W74F-E-HCB.md` §3 and the provisioning row of `DESIGN-FUDW-DECOMP-v3.md` §4.1; admitted by `VERDICT-FUDW-DECOMP-V3.md` §D.

Definition 34 (closed EXT-CB datum (`def-extcb-datum`))**.** A *closed EXT-CB datum* consists of an extended ε-C^*-algebra $\mathcal{A}$; δ-projections P, Q with $\|P + Q - I\| \le \delta$; an extended δ-isomorphism $v : M_r \to S_P$; $\dim S_Q = 1$; and $S_{P,Q} \neq 0$, with $e = \delta + \varepsilon$.

Kind. original (introduced by this project); `status: locked`. **Aliases.** EXT-CB datum.
Provenance. `source: internal`; locus `DESIGN-FUDW-DECOMP-v3.md §4.1 exact proposed datum package`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers).
Used in this report by `conj-extcb`, `lem-extcb1-close-corner-dimension`, `lem-extcb1-cross-corner-dimension` (3 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-extcb-datum.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft theorem-free hypothesis package; it includes no extension conclusion and requires ratification before locking.
Provenance. Byte-faithful transcription of the exact datum package in `DESIGN-FUDW-DECOMP-v3.md` §4.1, approved as definition provisioning by `VERDICT-FUDW-DECOMP-V3.md` §§6,D.

Definition 35 (extended δ-inclusion and isomorphism (`def-extended-delta-inclusion`))**.** An *extended δ-inclusion* is a linear map v for which every amplification $1_{M_n} \otimes v$ is a δ-inclusion: it is a δ-homomorphism and satisfies the two-sided $(1 \pm \delta)$ norm bounds. It is an *extended δ-isomorphism* when v is bijective.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** extended delta-inclusion; extended delta-isomorphism.
Provenance. `source: internal`; locus `harmonization of approximate_algebras.tex:443-456,1477-1484 prescribed by DESIGN-FUDW-DECOMP-v3.md §4.1`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers).
Used in this report by `conj-extcb`, `lem-compcb-single-compression-transfer`, `lem-extcb-corner-dimension-additivity`, `lem-extcb-four-corner-merge` (5 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-extended-delta-inclusion.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft harmonization only; it contains no theorem and requires ratification before locking.
Provenance. Transcribed from the exact provisioning decision in `DESIGN-FUDW-DECOMP-v3.md` §4.1, which harmonizes the two pinned-source loci; `VERDICT-FUDW-DECOMP-V3.md` §§6,D admits it only for the safe subset.

Definition 36 (amplified four-corner merging datum (`def-four-corner-merging-datum`)). An *amplified four-corner merging datum* consists of two complementary source projections, two approximately complementary target projections, and four fixed level-one corner maps whose amplifications share one defect bound for: involution compatibility, compressed-product compatibility, diagonal-unit approximation, and two-sided norm control. The four maps are the same maps at every amplification. **Quantitative complementarity (amendment 2026-07-25, per the cited locus):** the target projections are delta-projections and satisfy $\|P_1 + P_2 - I\| \le \rho$ with ρ the same common defect bound — byte-faithful to the pinned source at `approximate_algebras.tex:1326` ("let P_1, P_2 be δ-projections in an ε-C^* algebra $\mathcal{A}$ such that $\|P_1 + P_2 - I\| \le \delta$").

Kind. original (introduced by this project); `status: locked`. **Aliases.** four-corner merging datum.
Provenance. `source: internal`; locus `project packaging of approximate_algebras.tex:1325-1345 prescribed by DESIGN-FUDW-DECOMP-v3.md §4.1`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers); quantitative-complementarity amendment user-ratified 2026-07-25 (tobiasosborne, in-session sign-off — transcription-fidelity correction to the cited locus; af challenge ch-4dd98c4d).
Used in this report by `lem-extcb-four-corner-merge`, `lem-extcb-four-corner-norm` (2 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-four-corner-merging-datum.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft project packaging of the four source hypotheses; it asserts no merge conclusion and requires ratification before locking.
Provenance. Transcribed from `DESIGN-FUDW-DECOMP-v3.md` §4.1 and the four conditions at the pinned-source locus; admitted only for the safe EXT front end by `VERDICT-FUDW-DECOMP-V3.md` §D.

Definition 37 (closed H-CB datum (`def-hcb-datum`)). A *closed H-CB datum* consists of an extended ε-C^*-algebra $\mathcal{A}$; one level-one one-dimensional δ-projection Q; δ-projections P, R, S; the corresponding compressed corners, amplified column spaces, and their defined sesquilinear forms; and $e = \delta + \varepsilon$.

Kind. original (introduced by this project); `status: locked`. **Aliases.** H-CB datum.
Provenance. `source: internal`; locus `DESIGN-FUDW-DECOMP-v3.md §4.1 exact proposed datum package`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers).
Used in this report by `lem-hcb-column-hilbert-squared`, `lem-hcb0-compressed-associator`, `lem-hcb1-column-action`, `lem-hcb1-variational-identity`, `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect` and 9 further reproduced results (15 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-hcb-datum.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Scope. This is data and notation only. It includes no column-norm comparison, Ha estimate, inverse, or parent H-CB conclusion.
Status. Draft theorem-free package; ratification is required before locking.
Provenance. Byte-faithful transcription of the exact datum package in `DESIGN-FUDW-DECOMP-v3.md` §4.1, approved as definition provisioning by `VERDICT-FUDW-DECOMP-V3.md` §§6,D.

Definition 38 (positive approximate retract (`def-positive-approximate-retract`)). A *positive approximate retract* between finite commutative C^*-algebras is a pair of positive unital maps

$$A : \ell^\infty(k) \longrightarrow \ell^\infty(n), \qquad M : \ell^\infty(n) \longrightarrow \ell^\infty(k)$$

for which MA is close to I_k in the $\ell^\infty \to \ell^\infty$ operator norm. Its retract defect is

$$\varepsilon_{\rm ret} := \|MA - I_k\|_{\infty\to\infty}.$$

Equivalently, the matrices of A and M have probability-vector rows.

Kind. original (introduced by this project); `status: locked`. **Aliases.** positive retract pair; PRH datum.
Provenance. `source: internal`; locus `project formulation transcribed from docs/plans/2026-07-23-W74F-artifacts/PROOF-W74F-A-PRH.md §§1,4`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off).
Used in this report by `lem-prh`, `lem-routef-prh-finish` (3 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-positive-approximate-retract.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Scope. This project term packages the datum hardened by `lem-prh`; it does not assert that such a pair exists for any particular almost-idempotent map.
Provenance. Project formulation extracted from the W74F-A PRH artifact. User-ratified and locked 2026-07-24 (tobiasosborne, in-session sign-off).

Definition 39 (Stage-1 polar witness data (`def-stage1-polar-witness-data`)).
NOT TYPESET. This paragraph could not be typeset: math uses \rm, which report/main.tex does not define and the translation table does not cover. Its bytes follow, unaltered.

```
A *Stage-1 polar witness datum* is a
tuple $W$ of fourteen named scalar fields:
$$W=(C_{\rm rect},\,C_{\rm ch},\,C_{\rm pol},\,C_{\rm grp},\,C_{\rm path},\,C_{\rm der},\,
e_{\rm rect},\,\kappa _{\rm ch},\,\kappa _{\rm pol},\,\kappa _{\rm der},\,
\delta _*,\,\varepsilon _*^r,\,e_{\rm S1},\,r_{\rm iso}).$$
The first six are called coefficients, the next four margins, the last four
derived scales.
```

Kind. original (introduced by this project); `status: locked`. **Aliases.** polar witness tuple; Stage-1 polar constants tuple.
Provenance. `source: internal`; locus `DESIGN-S1-POLAR-v6.md sect 8 (unchanged from v2 sect 8; audited AUDIT-S1-POLAR-v3/v4/v5/v6)`; no source hash (not a literature transcription); record: user-ratified 2026-07-27 (W79 decision D2, docs/plans/2026-07-27-W78-ratification-package.md; verbatim from the audited polar design).
Used in this report by `lem-stage1-inversion-derivative-transport`, `lem-stage1-maurer-cartan-transport`, `lem-stage1-polar-path-transport`, `lem-stage1-polar-retraction-transport`, `lem-stage1-polar-scalar-arithmetic`, `lem-stage1-rectified-cstar-transport` and 1 further reproduced result (9 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-stage1-polar-witness-data.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Pure data: this shard contains NO positivity, inequality, existence, uniqueness, estimate, map, regularity, admissibility, or topological assertion (R35; `AUDIT-S1-POLAR-v4.md` §7: "VALID AS DATA"). The analytic-witness relation — that one such tuple simultaneously witnesses the seven Stage-1 polar producer rows and satisfies the scalar arithmetic — is exported ONLY by the result rows `lem-stage1-polar-scalar-arithmetic` and `lem-stage1-polar-constant-ledger` and their seven parameterized transport helpers (`DESIGN-S1-POLAR-v6.md` §§2–3), never by this definition. Related: `def-approximate-unitary-space`, `def-epsilon-cstar-algebra`.
Rendering note. 1 block(s) of this shard are flagged above; see `report/generated/defs/MANIFEST.md`.

Definition 40 (unital completely positive map (`def-ucp-map`))**.** Let A and B be unital C^*-algebras (in this repo's Route-F usage: finite-dimensional, typically ℓ^n_∞, M_n, or a finite-dimensional unital C^*-algebra $\mathcal{B}$). A linear map $\varphi : A \to B$ is *positive* if $\varphi(a) \geq 0$ whenever $a \geq 0$; *completely positive* (CP) if for every $r \geq 1$ the amplification $\mathrm{id}_{M_r} \otimes \varphi : M_r(A) \to M_r(B)$ is positive; and *unital* if $\varphi(1_A) = 1_B$. A *UCP map* is a unital completely positive map. Standard facts used silently at the BSc/MSc level: a UCP map is contractive; a positive map out of a commutative C^*-algebra is automatically CP; compositions of UCP maps are UCP.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** UCP map; UCP; completely positive unital map.
Provenance. `source: internal`; locus `standard operator-algebra textbook notion (e.g. Paulsen, Completely Bounded Maps and Operator Algebras, ch. 2-3); adopted for the Route-F rows per AUDIT-F0-ASSEMBLY.md sect 1.1`; no source hash (not a literature transcription); record: user-ratified 2026-07-27 (W79 decision D3, docs/plans/2026-07-27-W78-ratification-package.md; def shard chosen over an L2 textbook exemption because the F0 lift row's contract concludes "Phi is UCP").
Used in this report by `lem-routef-f0-defect-identity`, `lem-routef-f0-ucp-lift`, `lem-routef-f2-positive-unital-compression` (3 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-ucp-map.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Provisioned per the F0 assembly audit (`AUDIT-F0-ASSEMBLY.md` §1.1: the acronym UCP appeared in proposed contracts with no canonical anchor in the definitions layer). Consensus adoption of the universal textbook notion — no project-specific content. Consumers: the F0 lift rows (`lem-routef-f0-ucp-lift`), the K-ledger factorization rows (UCP Δ, Υ), F2. Related: `def-stochastic` (row-stochastic = unital positive on ℓ^n_∞), `def-fd-cstar-diagonal`.

5 further canonical definitions in `definitions/` are outside the current proof strategy's dependency closure and are not rendered here. They remain canonical — the shard is the single source of truth either way — and reachable at the shard path and through `definitions/INDEX.md`:

`def-maincb-partition-state`, `def-maincb-raw-call`, `def-maincb-reset-state`, `def-operator-space`, `def-pivot`

3 The signed-stochastic bridge

Lemma 41 (lem-classical-equiv). *The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants. There is a universal $C > 0$ such that:*

1. *(stochastic $\to$ signed) if Q is row-stochastic with $\|Q^2 - Q\|_{\infty\to\infty} \le \eta$, then $P = \theta(2Q-1)$ is a signed affine retraction (def-signed-idempotent) satisfying $\|P - Q\|_{\infty\to\infty} \le C\eta$ and negative mass $\delta \le C\eta$;*
2. *(signed $\to$ stochastic) conversely, row-normalising the positive parts p_i^+ of a signed idempotent P yields a row-stochastic Q with $\|P - Q\|_{\infty\to\infty} \le 2\delta$ and $\|Q^2 - Q\|_{\infty\to\infty} \le 6\delta + 4\delta^2$.*

Transcription note. *The contract writes the map norm as $\|\cdot\|$ without a subscript; per CONVENTIONS.md (b) it is the $\infty\to\infty$ map norm $\|\cdot\|_{\infty\to\infty}$. The operator θ is carried verbatim from the contract, which does not expand it; the validated tree instantiates it as the functional calculus $\theta(X) = \frac{1}{2}(I + \operatorname{sgn} X)$ of def-theta-idempotent-approximation.*

Registry contract (verbatim).

```
The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent
up to universal constants: Q row-stochastic with ||Q^2-Q|| <= eta gives P=theta(2Q-1) signed affine
retraction with ||P-Q|| <= C eta and neg mass delta <= C eta, and conversely row-normalising p_i^+
gives Q with ||P-Q|| <= 2 delta, ||Q^2-Q|| <= 6 delta+4 delta^2.
```

Proof (prose account of the af-validated tree). *Forward direction.* Put $X = 2Q - I$. Row-stochasticity gives $X\mathbf{1} = \mathbf{1}$ and $\|X\|_{\infty\to\infty} \le 3$, while the algebraic identity

$$X^2 - I = 4\,(Q^2 - Q) \tag{2}$$

converts the almost-idempotence hypothesis into $\|X^2 - I\|_{\infty\to\infty} \le 4\eta$. Write $Y = X^2 - I$. Two properties of Y drive everything: $Y\mathbf{1} = 0$ (because $X\mathbf{1} = \mathbf{1}$ forces $X^2\mathbf{1} = \mathbf{1}$), and $\|Y\|_{\infty\to\infty} \le 4\eta \le \frac{1}{2}$ once $\eta \le \frac{1}{8}$.

The smallness of Y puts the inverse square root inside its convergence radius: the binomial series

$$B = (I+Y)^{-1/2} = \sum_{m\ge 0} \binom{-\frac{1}{2}}{m} Y^m$$

converges absolutely in the map norm. Being a norm limit of polynomials in $Y = X^2 - I$, the operator B commutes with X; since $Y\mathbf{1} = 0$ kills every positive power, $B\mathbf{1} = \mathbf{1}$; and the absolute coefficient sum gives the quantitative bound $\|B - I\|_{\infty\to\infty} \le (1-t)^{-1/2} - 1 \le 2t$ at $t = \|Y\|_{\infty\to\infty} \le \frac{1}{2}$, hence $\|B - I\|_{\infty\to\infty} \le 8\eta$. The scalar identity $\big((1+z)^{-1/2}\big)^2(1+z) = 1$ transfers to the absolutely convergent series in Y, so $B^2X^2 = I$; commutation then makes $S = XB$ an exact involution,

$$S^2 = XBXB = X^2B^2 = I, \qquad S\mathbf{1} = XB\mathbf{1} = X\mathbf{1} = \mathbf{1},$$

with $S - X = X(B - I)$ and therefore $\|S - X\|_{\infty\to\infty} \le 3\cdot 8\eta = 24\eta$.

Setting $P = \theta(X) = \frac{1}{2}(I + S)$ now gives $P\mathbf{1} = \mathbf{1}$ and $P^2 = \frac{1}{4}(I + 2S + S^2) = \frac{1}{2}(I + S) = P$, so P is an exact signed idempotent in the sense of def-signed-idempotent. Since $Q = \frac{1}{2}(I + X)$, the difference is $P - Q = \frac{1}{2}(S - X)$ and hence $\|P - Q\|_{\infty\to\infty} \le 12\eta$. Finally the negative mass: row-stochastic Q is entrywise nonnegative (`def-stochastic`), so for every entry $\max\{-P_{ij}, 0\} \le |P_{ij} - Q_{ij}|$ — trivially where $P_{ij} \ge 0$, and because $-P_{ij} \le Q_{ij} - P_{ij}$ where $P_{ij} < 0$. Summing over j and maximising over i gives, with δ as in `def-negative-mass`,

$$\delta(P) \;\le\; \|P - Q\|_{\infty\to\infty} \;\le\; 12\eta. \tag{3}$$

So $C = 12$ serves. The export records an instructive scope correction here: the negative-mass step was first blocked because `def-negative-mass` — although permitted by the registry shard — had not been registered inside the workspace, and the tree carried an explicit gap certificate rather than the estimate; registering the definition discharged it.

Converse direction. Let P be an exact signed idempotent with negative mass δ. Split each row into disjointly supported nonnegative parts, $p_i = p_i^+ - p_i^-$, and set $a_i = \sum_k (P_{ik})_- \le \delta$. Total mass one (`def-signed-idempotent`) gives $\sum_k p_{ik}^+ = 1 + a_i$, so

$$q_i = \frac{p_i^+}{1 + a_i}$$

is a probability vector and the matrix Q with rows q_i is row-stochastic. The correction $e_i = q_i - p_i$ is

$$e_i = p_i^- - \frac{a_i}{1 + a_i}\, p_i^+,$$

a difference of two disjointly supported vectors of ℓ^1-mass a_i each; hence $\|e_i\|_1 = 2a_i \le 2\delta$ and $\|P - Q\|_{\infty\to\infty} \le 2\delta$.

For the idempotence defect, write $D = Q - P$. The row-geometry clause of `def-signed-idempotent` gives $\|P\|_{\infty\to\infty} \le 1 + 2\delta$, hence $\|P - I\|_{\infty\to\infty} \le 2 + 2\delta$. Using $P^2 = P$ exactly, each row satisfies the exact decomposition

$$q_i Q - q_i = e_i(P - I) + q_i D. \tag{4}$$

Since q_i is a probability vector, $\|q_i D\|_1 \le \|D\|_{\infty\to\infty} \le 2\delta$, while $\|e_i(P - I)\|_1 \le 2\delta\,(2 + 2\delta)$. Adding, $\|Q^2 - Q\|_{\infty\to\infty} \le 6\delta + 4\delta^2$, exactly as contracted.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-classical-equiv`: a 29-node tree (21 live nodes validated, 8 superseded nodes archived), root `validated`, taint clean.

Role in the chain. This is the licence to move between the two pictures of §1, and the only such licence in the registry. It has no dependencies. Everything stated in the signed picture reaches the stochastic formulation of `op-classical` only through it, and only up to the universal constants above; `op-classical` itself remains open.

4 Positive-retract hardening

Lemma 42 (`lem-prh`). *Let $k, n \geq 1$ and let*

$$A : \ell^\infty(k) \longrightarrow \ell^\infty(n), \qquad M : \ell^\infty(n) \longrightarrow \ell^\infty(k)$$

be positive unital maps — equivalently, maps whose matrices have probability-vector rows. If

$$\|MA - I_k\|_{\infty\to\infty} \;\leq\; \varepsilon, \qquad 0 \leq \varepsilon < \tfrac{1}{2},$$

then there is a stochastic idempotent $E : \ell^\infty(n) \to \ell^\infty(n)$ (`def-stochastic`) with

$$\|AM - E\|_{\infty\to\infty} \;\leq\; 2\sqrt{2\varepsilon}. \tag{5}$$

Transcription note. *The pair (A, M) is the datum of `def-positive-approximate-retract`, with retract defect $\varepsilon_{\mathrm{ret}} = \|MA - I_k\|_{\infty\to\infty}$. Following the validated tree, A is presented by rows $a_i \in \Delta_k$ indexed by $i \in \{1, \ldots, n\}$ and M by rows μ_s indexed by $s \in \{1, \ldots, k\}$, so that the s-th row of MA is the probability vector $\sum_i \mu_s(i)\, a_i$.*

Registry contract (verbatim).

```
Positive-retract hardening (PRH): let k,n >= 1 and let A:l-infinity(k)->l-infinity(n) and
M:l-infinity(n)->l-infinity(k) be positive unital maps (equivalently, have probability-vector
rows); if ||MA-I_k||_{infinity->infinity} <= epsilon with 0 <= epsilon < 1/2, then there is a
stochastic idempotent E:l-infinity(n)->l-infinity(n) with ||AM-E||_{infinity->infinity} <=
2*sqrt(2*epsilon).
```

Proof (prose account of the af-validated tree). The tree splits on the defect. If $\varepsilon = 0$ the hypothesis reads $MA = I_k$ exactly, and then $E := AM$ is already row-stochastic with $E^2 = A(MA)M = AM = E$, so (5) holds with error 0. The substance is the case $0 < \varepsilon < \frac{1}{2}$.

The scale and the cores. Fix

$$\lambda = \sqrt{\varepsilon/2}, \qquad C_s = \{\, i : \; a_{is} > 1 - \lambda \,\}, \qquad \beta_s = \mu_s(C_s^c).$$

The single quantitative input is a coordinate defect identity. The s-th row of MA is the probability vector $\sum_i \mu_s(i) a_i$, and for probability vectors the ℓ^1 distance to the basis vector e_s is governed by one coordinate:

$$\|(MA)_{s,*} - e_s\|_1 = 2\big(1 - (MA)_{ss}\big) = 2 \sum_i \mu_s(i)\,(1 - a_{is}) =: 2D_s, \tag{6}$$

so the norm hypothesis is exactly $D_s \leq \varepsilon/2$ for every s. Two consequences follow. First, the cores are pairwise disjoint: if $i \in C_s \cap C_t$ with $s \neq t$ then $a_{is} + a_{it} > 2(1 - \lambda) > 1$ (using $\lambda < \frac{1}{2}$, which is where $\varepsilon < \frac{1}{2}$ enters), contradicting that a_i is a probability vector. Second, off the core one has $1 - a_{is} \geq \lambda$, so dropping the on-core terms of the nonnegative sum D_s gives $\lambda\beta_s \leq D_s \leq \varepsilon/2$, whence

$$\beta_s \;\leq\; \frac{\varepsilon}{2\lambda} \;=\; \lambda \;<\; \tfrac{1}{2}. \tag{7}$$

The tree records a correction at this point: an earlier node asserted the strict inequality $\beta_s < \lambda$, which the threshold $a_{is} \leq 1 - \lambda$ off the core does not give; the weak form (7) is what holds and all that is used.

The exact construction. Condition each decoder row on its core and replace each encoder row inside a core by a basis vector:

$$\nu_s(i) = \frac{\mu_s(i)\,\mathbf{1}_{C_s}(i)}{1-\beta_s}, \qquad \widehat{A} \text{ has row } e_s \text{ on } C_s \text{ and row } a_i \text{ off } \bigcup_s C_s.$$

Both are well defined: $\beta_s < 1$ makes ν_s a probability vector, and disjointness of the cores makes $\widehat{A}$ unambiguous. The construction is engineered so that the retraction becomes *exact*: the s-th row of $N\widehat{A}$ is $\sum_{i\in C_s} \mu_s(i)e_s/(1-\beta_s) = e_s$, i.e.

$$N\widehat{A} = I_k, \qquad \text{hence} \qquad E := \widehat{A}N \text{ satisfies } E^2 = \widehat{A}(N\widehat{A})N = \widehat{A}N = E. \tag{8}$$

As a product of positive unital maps, E is row-stochastic, so it is a stochastic idempotent in the sense of `def-stochastic`.

The two error terms. Write $AM - E = A(M-N) + (A-\widehat{A})N$ and bound the summands separately.

For the conditioning term, ν_s is μ_s conditioned on C_s, so the two rows differ by mass β_s removed from C_s^c and mass β_s redistributed on C_s; hence $\|\mu_s - \nu_s\|_1 = 2\beta_s \le \varepsilon/\lambda$ by (7) and $\|M-N\|_{\infty\to\infty} \le \varepsilon/\lambda$. Because A has probability rows it is an $\infty\to\infty$ contraction — concretely, $\sum_j |(AB)_{ij}| \le \sum_s a_{is} \sum_j |B_{sj}| \le \|B\|_{\infty\to\infty}$ — so $\|A(M-N)\|_{\infty\to\infty} \le \varepsilon/\lambda$.

For the core-replacement term, the rows of $(A-\widehat{A})N$ vanish outside $\bigcup_s C_s$. For $i \in C_s$ the corresponding row is $\sum_t a_{it}\nu_t - \nu_s$, and since every ν_t is a probability vector its ℓ^1 norm is at most $(1-a_{is}) + \sum_{t\neq s} a_{it} = 2(1-a_{is}) < 2\lambda$. Hence $\|(A-\widehat{A})N\|_{\infty\to\infty} \le 2\lambda$.

Optimisation. Adding the two bounds,

$$\|AM - E\|_{\infty\to\infty} \;\le\; \frac{\varepsilon}{\lambda} + 2\lambda,$$

and the choice $\lambda = \sqrt{\varepsilon/2}$ balances the two terms at the common value $\sqrt{2\varepsilon}$, giving (5). The square-root rate is intrinsic to this balance: conditioning costs ε/λ and core replacement costs 2λ, and no choice of threshold does better than their geometric mean.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-prh`: 14 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. `lem-prh` has no registry dependencies. It is the finishing step of the live route: it installs the reduction of `op-classical` to the *existence* of a positive approximate retract for a given almost-idempotent Q — stochastic A, M with $\|AM-Q\|_{\infty\to\infty} = O(\eta)$ and $\|MA-I\|_{\infty\to\infty} = O(\eta)$ — after which (5) delivers the exact idempotent at rate $\sqrt{\eta}$. Producing such a pair is what the COMP and H-CB tiers below are ultimately for, and that production is *not* established: `lem-prh` is a conditional finisher, and `op-classical` remains open. Downstream registry consumers are `lem-routef-prh-finish` and `lem-routef-k-ledger`, both non-rigorous.

5 Amplification naturality of the functional calculus

Lemma 43 (`lem-compcb-amplification-naturality`)**.** *Let $\mathcal{B}$ be a unital Banach algebra and let $\iota_n : \mathcal{B} \to M_n(\mathcal{B})$, $X \mapsto 1_{M_n} \otimes X$, be the unital amplification, an isometric unital homomorphism. Let f be given on $\|X - x_0 I\| < r$ by the power series*

$$f(X) = a_0 I + \sum_{m \geq 1} a_m \,(X - x_0 I)^m. \tag{9}$$

Then for every $X \in \mathcal{B}$ with $\|X - x_0 I\| < r$ the element $f(\iota_n(X))$ is defined and

$$f(\iota_n(X)) = \iota_n(f(X)). \tag{10}$$

In particular, whenever $\|(2P - I)^2 - I\| < 1$,

$$\theta\big(\iota_n(2P - I)\big) = \iota_n\big(\theta(2P - I)\big), \qquad \theta(X) = \tfrac{1}{2}\big(I + \operatorname{sgn} X\big).$$

Transcription note. *θ and* sgn *are the power-series functional calculus of* ***`def-theta-idempotent-approximation`****; only its definitional displays are in scope here. The norm is the ambient Banach-algebra norm.*

Registry contract (verbatim).

```
Amplification naturality of the power-series functional calculus: let B be a unital Banach algebra,
iota_n: B -> M_n(B) the unital amplification X -> 1_{M_n} tensor X (an isometric unital
homomorphism), and f given on ||X - x_0 I|| < r by the power series f(X) = a_0 I + sum_{m>=1} a_m
(X - x_0 I)^m; then for every X in B with ||X - x_0 I|| < r, f(iota_n(X)) is defined and
f(iota_n(X)) = iota_n(f(X)); in particular, whenever ||(2P-I)^2 - I|| < 1, theta(iota_n(2P-I)) =
iota_n(theta(2P-I)) with theta(X) = (I + sgn(X))/2.
```

Proof (prose account of the af-validated tree). *The general statement.* Three structural facts about ι_n do all the work. Unitality gives $\iota_n(I_{\mathcal{B}}) = I_{M_n(\mathcal{B})}$, hence $\iota_n(X) - x_0 I = \iota_n(X - x_0 I)$; isometry then transports the convergence hypothesis verbatim, $\|\iota_n(X) - x_0 I\| = \|X - x_0 I\| < r$, so the series (9) evaluated at $\iota_n(X)$ is inside its domain; and multiplicativity gives $(\iota_n(X) - x_0 I)^m = \iota_n\big((X - x_0 I)^m\big)$ for every $m \geq 0$. Consequently the N-th partial sums correspond exactly,

$$T_N = a_0 I + \sum_{m=1}^{N} a_m (\iota_n(X) - x_0 I)^m = \iota_n(S_N), \qquad S_N = a_0 I + \sum_{m=1}^{N} a_m (X - x_0 I)^m.$$

Since $S_N \to f(X)$ in $\mathcal{B}$ and ι_n is isometric, hence continuous, $T_N \to \iota_n(f(X))$. The power-series definition identifies the limit of the same sequence T_N with $f(\iota_n(X))$, so uniqueness of limits yields both definedness and (10). No property of ι_n beyond isometry, unitality and multiplicativity is used.

The θ specialization. Here the functional calculus is not a single series but a composite built from the inverse square root, so the tree first pins that branch down. With $c_m = \binom{-1/2}{m}$ one has $c_0 = 1$ and $c_m / c_{m-1} = -(2m - 1)/(2m)$, whose modulus tends to 1; the ratio test therefore gives radius exactly one for $h(w) = \sum_{m \geq 0} c_m w^m$. On $|w| < 1$ termwise differentiation is legitimate and the coefficient recurrence is precisely the differential equation $2(1 + w)h'(w) + h(w) = 0$; hence $\frac{d}{dw}\big[(1+w)h(w)^2\big] = 0$, and $h(0) = 1$ gives $(1+w)h(w)^2 = 1$. So h *is* the analytic inverse-square-root branch normalised at 1.

Transporting this to a unital Banach algebra: for $\|Z-I\| < 1$ the series $H = \sum_{m\geq 0} c_m(Z-I)^m$ converges absolutely in norm, and absolute convergence licenses the same Cauchy products as in the scalar identity, so $ZH^2 = I$; the normalisation $H = I +$ (terms of positive degree) identifies H with the branch $Z^{-1/2}$ used by the registered calculus. Applying the general statement to this series with $Z = Y^2$, $Y = 2P - I$, gives $\big(\iota_n(Y)^2\big)^{-1/2} = \iota_n\big((Y^2)^{-1/2}\big)$ whenever $\|Y^2 - I\| < 1$; multiplicativity then propagates the identity through $\operatorname{sgn}(Y) = Y(Y^2)^{-1/2}$ and finally through $\theta = \frac{1}{2}(I + \operatorname{sgn})$.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-amplification-naturality`: 12 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. This lemma exists because of a stall, and the shape of the campaign record is worth stating: three separate blocking challenges in the first orchestration of `lem-compcb-amplified-compression` named this one identity, so it was factored out into its own shard and validated first, bottom-up. It has no dependencies. It is imported by `lem-compcb-amplified-compression` (§6) and by `lem-compcb-entrywise-compression-naturality` (§8), and it is the reason every compression identity in the COMP tier holds at *every* amplification level rather than only at level one.

6 The amplified compression identity

Lemma 44 (`lem-compcb-amplified-compression`). *There is a universal $e_{\mathrm{cmp}} > 0$ such that, whenever $e = \delta + \varepsilon \le e_{\mathrm{cmp}}$, every pair of δ-projections P, Q in an extended ε-C^*-algebra and every $n \ge 1$ satisfy*

$$1_{M_n} \otimes \mathrm{Co}_{P,Q} \;=\; \mathrm{Co}_{I_n \otimes P, I_n \otimes Q} \tag{11}$$

and

$$M_n \otimes \mathcal{S}_{P,Q} \;=\; \mathcal{S}_{I_n \otimes P, I_n \otimes Q}. \tag{12}$$

Transcription note. *$\mathrm{Co}_{P,Q}$ and $\mathcal{S}_{P,Q}$ are the compression map and compressed corner of* ***`def-compressed-corner`****; δ-projections are as in* ***`def-delta-projection`****, and "extended" is as in* ***`def-extended-epsilon-cstar-algebra`****, so that every $M_n \otimes \mathcal{A}$ is itself an ε-C^*-algebra. On the left of (11) $1_{M_n} \otimes (\cdot)$ denotes the entrywise amplification of an* operator *on $\mathcal{A}$; on the right $I_n \otimes P$ is the amplified* element.

Registry contract (verbatim).

```
Amplified compression identity: there is a universal e_cmp > 0 such that, whenever e=delta+epsilon
<= e_cmp, every pair of delta-projections P,Q in an extended epsilon-C*-algebra and every n >= 1
satisfy 1_{M_n} tensor Co_{P,Q}=Co_{I_n tensor P,I_n tensor Q} and M_n tensor S_{P,Q}=S_{I_n tensor
P,I_n tensor Q}.
```

Proof (prose account of the af-validated tree). *A uniform threshold.* Write $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$. Because $\mathcal{A}$ is *extended*, each $M_n \otimes \mathcal{A}$ is an ε-C^*-algebra with compatible matrix multiplication, involution, unit and amplification norm; since $P_n^\dagger = P_n$ and $\|P_n^2 - P_n\| = \|I_n \otimes (P^2 - P)\| = \|P^2 - P\| \le \delta$, the amplified elements are δ-projections *with the same δ, uniformly in n.* Next, the compression is the value of θ at a fixed operator: writing $\mathrm{A}_{P,Q} = \frac{1}{2}(L_P R_Q + R_Q L_P)$ and $T_{P,Q} = 2\mathrm{A}_{P,Q} - I$, `def-compressed-corner` supplies a single universal C with $\|\mathrm{A}_{P,Q}^2 - \mathrm{A}_{P,Q}\| \le Ce$ in *every* ε-C^*-algebra, and the exact operator identity

$$T_{P,Q}^2 - I = 4\big(\mathrm{A}_{P,Q}^2 - \mathrm{A}_{P,Q}\big) \tag{13}$$

converts this into $\|T_{P,Q}^2 - I\| \le 4Ce$. Since the same C applies inside every $M_n \otimes \mathcal{A}$ to the pair (P_n, Q_n), choosing e_{cmp} with $4Ce_{\mathrm{cmp}} < 1$ puts $T_{P,Q}$ and every T_{P_n,Q_n} simultaneously inside the θ/sgn domain of `def-theta-idempotent-approximation`. Uniformity of the threshold in n is exactly what (13) buys.

The amplification is entrywise, not diagonal. Left and right multiplication by the block-diagonal P_n, Q_n act slotwise: for $X = [x_{ij}]$ one has $L_{P_n} R_{Q_n}(X) = [P(x_{ij}Q)]$ and $R_{Q_n} L_{P_n}(X) = [(Px_{ij})Q]$, whence $T_{P_n,Q_n} = 1_{M_n} \otimes T_{P,Q}$. Here the tree records a correction that is genuinely load-bearing. The relevant map is *not* the diagonal embedding of $B(\mathcal{A})$ into $M_n\big(B(\mathcal{A})\big)$ but the entrywise operator amplification

$$\Gamma_n : B(\mathcal{A}) \to B(M_n \otimes \mathcal{A}), \qquad \Gamma_n(U)\big([x_{ij}]\big) = [U(x_{ij})],$$

and the tree proves exactly the properties Γ_n has and no more: coordinate compression gives $\|x_{ij}\| \le \|X\|$ and $\|E_{ij} \otimes y\| = \|y\|$, so $\|\Gamma_n(U)\| \le n^2 \|U\|$ — bounded at each fixed n, with *no* claim of isometry and *no* uniform-in-n operator bound; and entrywise evaluation gives $\Gamma_n(I) = I$ and $\Gamma_n(UV) = \Gamma_n(U)\Gamma_n(V)$. Boundedness at fixed n is all that is needed, because the θ series is a norm limit for each fixed n separately.

Transporting θ. Put $T = T_{P,Q}$ and $R = T^2 - I$. On $\|R\| < 1$ the registered calculus writes $\theta(T) = \frac{1}{2}\big(I + T\sum_{k\geq 0} b_k R^k\big)$ with b_k the Taylor coefficients of $(1+z)^{-1/2}$ — the branch identified in `lem-compcb-amplification-naturality`. Since Γ_n is a unital algebra homomorphism it commutes with every partial sum and satisfies $\Gamma_n(R) = \Gamma_n(T)^2 - I$; boundedness lets one pass to the norm limit; and the threshold above guarantees that the limit at $\Gamma_n(T) = T_{P_n,Q_n}$ is the value $\theta(T_{P_n,Q_n})$. Hence $\Gamma_n(\theta(T_{P,Q})) = \theta(T_{P_n,Q_n})$, which by $\mathrm{Co}_{P,Q} = \theta(T_{P,Q})$ is precisely (11).

Ranges. For any linear $C : \mathcal{A} \to \mathcal{A}$ and finite n one has $\mathrm{Im}(1_{M_n} \otimes C) = M_n \otimes \mathrm{Im}\, C$: every output has entries $C(x_{ij})$, and conversely finitely many entrywise preimages assemble into a single matrix preimage. Since $\mathcal{S}_{P,Q} = \mathrm{Im}\,\mathrm{Co}_{P,Q}$ by `def-compressed-corner`, applying this to (11) gives (12).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-amplified-compression`: 13 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplification-naturality`. This is the structural keystone of the COMP tier: every later result that says "uniformly in n" does so because (11)–(12) let a level-one fact be read at level n with no loss. It is imported by `lem-compcb-amplified-compression-identities`, `lem-compcb-entrywise-compression-naturality`, `lem-compcb-rectangular-product`, `lem-compcb-amplifie` `lem-compcb-compressed-unit-norm`, `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-product` `lem-compcb-single-compression-transfer` and `lem-hcb2-product-defect`.

7 Idempotence and adjoint symmetry of amplified compressions

Lemma 45 (`lem-compcb-amplified-compression-identities`). *There is a universal $e_{\mathrm{cmp}} > 0$ such that, whenever $\mathcal{A}$ is an extended ε-C^*-algebra, $e = \delta + \varepsilon \le e_{\mathrm{cmp}}$, P, Q are δ-projections in $\mathcal{A}$, $n \ge 1$, and $X \in M_n \otimes \mathcal{A}$, one has*

$$\mathrm{Co}_{P_n,Q_n}^2 = \mathrm{Co}_{P_n,Q_n}, \qquad \mathrm{Co}_{P_n,Q_n}(X)^\dagger = \mathrm{Co}_{Q_n,P_n}\big(X^\dagger\big), \tag{14}$$

where $P_n = I_n \otimes P$ and $Q_n = I_n \otimes Q$.

Registry contract (verbatim).

```
Amplified compression identities: there is a universal e_cmp > 0 such that, whenever A is an
extended epsilon-C*-algebra, e=delta+epsilon <= e_cmp, P,Q are delta-projections in A, n >= 1, and
X is in M_n tensor A, one has Co_{P_n,Q_n}^2=Co_{P_n,Q_n} and
Co_{P_n,Q_n}(X)^dagger=Co_{Q_n,P_n}(X^dagger), where P_n=I_n tensor P and Q_n=I_n tensor Q.
```

Proof (prose account of the af-validated tree). The argument is short because the amplification lemma has already done the structural work; what remains is to check that two *exact* level-one identities survive the passage to matrices.

Identification. Take e_{cmp} to be the universal constant supplied by `lem-compcb-amplified-compression`. For $e \le e_{\mathrm{cmp}}$ that lemma gives $\mathrm{Co}_{P_n,Q_n} = 1_{M_n} \otimes \mathrm{Co}_{P,Q}$, and — applied a second time to the *ordered* pair (Q, P), which is a distinct instance — also $\mathrm{Co}_{Q_n,P_n} = 1_{M_n} \otimes \mathrm{Co}_{Q,P}$. Both amplified compressions are therefore canonical amplifications of level-one maps.

The level-one identities. Writing $T = \mathrm{Co}_{P,Q}$ and $U = \mathrm{Co}_{Q,P}$, `def-compressed-corner` states exactly $T^2 = T$ and $T(x)^\dagger = U(x^\dagger)$ for all $x \in \mathcal{A}$. These are equalities, not estimates; no smallness is consumed by them, and the threshold above is needed only for the identification step.

Entrywise transfer. Let T, U be any linear maps on a self-adjoint operator space satisfying those two identities, and let $T_n = 1_{M_n} \otimes T$, $U_n = 1_{M_n} \otimes U$. For $X = [x_{ij}]$, idempotence is checked slotwise,

$$(T_n^2 X)_{ij} = T\big(T(x_{ij})\big) = (T^2)(x_{ij}) = T(x_{ij}) = (T_n X)_{ij},$$

so $T_n^2 = T_n$. Adjoint symmetry combines the matrix involution with the slotwise action: the (i, j) entry of $T_n(X)^\dagger$ is $T(x_{ji})^\dagger = U\big(x_{ji}^\dagger\big)$, while the (i, j) entry of $U_n(X^\dagger)$ is $U\big((X^\dagger)_{ij}\big) = U\big(x_{ji}^\dagger\big)$. The transposition of indices is what makes the *swap* $\mathrm{Co}_{P_n,Q_n} \mapsto \mathrm{Co}_{Q_n,P_n}$ in (14) the right statement. Combining the identification with the transfer gives the lemma.

A recorded contract amendment. The export's final node is a contract-reconciliation step, and it documents why the statement is shaped as it is. The originally codified contract left the ambient algebra unbound: a challenge observed that the proof used, but the contract never asserted, that $\mathcal{A}$ is an extended ε-C^*-algebra with P, Q in $\mathcal{A}$. The contract was amended to the prover's corrected root text — the version reproduced above — as a mechanical, verdict-driven typing repair. The added clauses are the intended and already-used hypotheses of the source argument, so nothing was strengthened or weakened.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-amplified-compression-identities`: 7 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression`. These two identities are the algebraic skeleton on which every quantitative COMP estimate is hung: idempotence is what makes "X lies in the corner" equivalent to "X is fixed by its compression", and the dagger swap is what lets an estimate proved on one side of a rectangular corner be read on the other. It is imported by `lem-compcb-rectangular-product`, `lem-compcb-amplified-almost-containment`, `lem-compcb-compressed-unit-norm`, `lem-compcb-compressed-unit-action`, `lem-compcb-corner-algebra`, `lem-compcb-single-compression-transfer`, `lem-hcb2-amplified-adjointness` and `lem-hcb1-variational-`

8 Entrywise compression naturality at a matrix unit

Lemma 46 (`lem-compcb-entrywise-compression-naturality`). *There is a universal $e_{\mathrm{nat}} > 0$ such that, whenever $e = \delta + \varepsilon \le e_{\mathrm{nat}}$, Q is a δ-projection in an extended ε-C^*-algebra $\mathcal{A}$, $n \ge 1$, and E_{11} is the $(1,1)$ matrix unit of M_n, every $Z \in \mathcal{A}$ satisfies*

$$\mathrm{Co}_{I_n \otimes Q, I_n \otimes Q}\big(E_{11} \otimes Z\big) \;=\; E_{11} \otimes \mathrm{Co}_Q(Z). \tag{15}$$

Transcription note. Co_Q *is the $P = Q$ abbreviation* $\mathrm{Co}_{Q,Q}$ *of* `def-compressed-corner`. *The point of the statement is that the amplified compression does not merely preserve the corner but acts* in the single occupied slot*, leaving the zero blocks zero.*

Registry contract (verbatim).

```
Entrywise compression naturality: there is a universal e_nat > 0 such that, whenever e =
delta+epsilon <= e_nat, Q is a delta-projection in an extended epsilon-C*-algebra A, n >= 1, and
E_{11} is the (1,1) matrix unit of M_n, every Z in A satisfies Co_{I_n tensor Q, I_n tensor
Q}(E_{11} tensor Z) = E_{11} tensor Co_Q(Z).
```

Proof (prose account of the af-validated tree). Set $e_{\mathrm{nat}} := e_{\mathrm{cmp}}$, the universal threshold supplied by `lem-compcb-amplified-compression`. That lemma is an equality of *linear maps* on $M_n(\mathcal{A})$, and the proof consists of specializing it and then evaluating.

Specializing the imported identity to the pair $(P, Q) = (Q, Q)$ gives

$$\mathrm{Co}_{I_n \otimes Q, I_n \otimes Q} \;=\; 1_{M_n} \otimes \mathrm{Co}_{Q,Q} \;=\; 1_{M_n} \otimes \mathrm{Co}_Q,$$

the last equality being nothing but the abbreviation $\mathrm{Co}_{Q,Q} = \mathrm{Co}_Q$ recorded in `def-compressed-corner`. This is where the whole quantitative content of the statement sits: the threshold, the θ-domain control and the amplification argument are all inherited from `lem-compcb-amplified-compression`, which in turn rests on `lem-compcb-amplification-naturality`.

What remains is a one-line evaluation. For any linear $T : \mathcal{A} \to \mathcal{A}$ the tensor amplification acts on elementary tensors by $(1_{M_n} \otimes T)(X \otimes Z) = X \otimes T(Z)$; taking $T = \mathrm{Co}_Q$ and $X = E_{11}$ gives (15).

Status. `proved`; `af:  validated` (T0). Workspace `proofs/lem-compcb-entrywise-compression-naturality`: 4 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplification-naturality`. Like §5, this shard exists because of a recorded stall: the column-Hilbert workspace (§17) reached its third consecutive stall on exactly the identity (15), whose statement was then extracted mechanically from the blocking challenge and validated on its own. It is an explicit single-slot specialization of the $1_{M_n}\otimes$-clause: that clause identifies two maps, while (15) records its value on the single-slot element needed downstream. It has one registry consumer, `lem-hcb-column-hilbert-squared`.

9 The uniform rectangular compressed-product estimate

Lemma 47 (`lem-compcb-rectangular-product`)**.** *There are universal $C_{\mathrm{co}} < \infty$ and $e_{\mathrm{co}} > 0$ such that, for $e = \delta + \varepsilon \le e_{\mathrm{co}}$, every compatible amplified rectangular pair A, B satisfies*

$$\|A \cdot B - AB\| \;\le\; C_{\mathrm{co}}\, e\, \|A\|\, \|B\|. \tag{16}$$

Transcription note. *The tree fixes "compatible amplified rectangular pair" as an explicit abbreviation, and the quantifier in* (16) *ranges over exactly these data and no others: an extended ε-C^*-algebra $\mathcal{A}$; an integer $n \ge 1$; δ-projections P, Q, R in $\mathcal{A}$; the amplifications $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$, $R_n = I_n \otimes R$; elements $A \in \mathcal{S}_{P_n,Q_n}$ and $B \in \mathcal{S}_{Q_n,R_n}$; and the compressed product $A \cdot B = \mathrm{Co}_{P_n,R_n}(AB)$ formed as in* `def-compressed-corner` *inside $M_n \otimes \mathcal{A}$. In particular both factors live in one* square *amplification $M_n \otimes \mathcal{A}$ — a restriction that later forces a separate row/column statement (§14).*

Registry contract (verbatim).

```
Uniform rectangular compressed-product estimate: there are universal C_co < infinity and e_co > 0
such that, for e=delta+epsilon <= e_co, every compatible amplified rectangular pair satisfies ||A
dot B-AB|| <= C_co*e*||A||||B||.
```

Proof (prose account of the af-validated tree). The shape of the argument is: unpack a level-one universal estimate into named constants, verify that the amplified data satisfy its hypotheses, and apply it inside $M_n \otimes \mathcal{A}$. No new analysis is done at level n; the whole point is that none is needed.

Unpacking the level-one estimate. `def-compressed-corner` states, for any triple U, V, W of δ-projections in an ε-C^*-algebra, that $X \cdot Y = \mathrm{Co}_{U,W}(XY)$ on $\mathcal{S}_{U,V} \times \mathcal{S}_{V,W}$ and that $\|X \cdot Y - XY\| \le O(\delta+\varepsilon)\|X\|\|Y\|$. The tree makes the big-O explicit: there are constants $C_{\mathrm{corner}} < \infty$ and $e_{\mathrm{corner}} > 0$, independent of the algebra, of the projections and of the elements, such that the bound reads $C_{\mathrm{corner}} e\|X\|\|Y\|$ whenever $e \le e_{\mathrm{corner}}$. This independence is the reason the final constant is uniform in n: the level-one estimate is a statement about *every* ε-C^*-algebra, and $M_n \otimes \mathcal{A}$ is one of them.

The amplified data are admissible. Because $\mathcal{A}$ is extended, $M_n \otimes \mathcal{A}$ is an ε-C^*-algebra for every n; and P_n is a δ-projection whenever P is, since $P_n^\dagger = P_n$, $P_n^2 - P_n = I_n \otimes (P^2 - P)$ and the block-diagonal norm satisfies $\|I_n \otimes Z\| = \|Z\|$ — likewise for Q_n, R_n. Applying `lem-compcb-amplified-compression` to the three pairs (P, Q), (Q, R), (P, R) identifies the amplified corners with $M_n \otimes \mathcal{S}_{P,Q}$, $M_n \otimes \mathcal{S}_{Q,R}$, $M_n \otimes \mathcal{S}_{P,R}$ and the amplified compressions with the entrywise amplifications; `lem-compcb-amplified-compression` supplies the idempotence and dagger identities for those maps. Hence a compatible amplified rectangular pair really is an ordinary rectangular pair in $M_n \otimes \mathcal{A}$, and its amplified compressed product is literally the ordinary compressed product there, $A \cdot B = \mathrm{Co}_{P_n,R_n}(AB)$, which lands in $\mathcal{S}_{P_n,R_n}$.

Assembly. Take

$$C_{\mathrm{co}} = C_{\mathrm{corner}}, \qquad e_{\mathrm{co}} = \min\bigl(e_{\mathrm{corner}}, e_{\mathrm{cmp}}\bigr),$$

with e_{cmp} the threshold of the amplification dependencies; both are universal, finite and positive. For $e \le e_{\mathrm{co}}$ the level-one estimate applies inside $M_n \otimes \mathcal{A}$ with $U = P_n$, $V = Q_n$, $W = R_n$, $X = A$, $Y = B$, and gives (16) with constants independent of n and of all the data.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-rectangular-product`: 12 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplified-compression-i` This is the workhorse estimate of the COMP tier — the licence to replace a compressed product by an ambient product at cost $O(e)$ — and it is invoked by almost everything downstream: `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-product`, `lem-hcb0-compressed-associator`, `lem-compcb-corner-algebra`, `lem-hcb-column-hilbert-squared`, `lem-compcb-single-compression-transfe` and `lem-hcb2-product-defect`. Its square-amplification compatibility clause is a real restriction, not a formality: §14 exists precisely because a row/column pair fails it.

10 Amplified almost-containment of corners

Lemma 48 (`lem-compcb-amplified-almost-containment`). *There are universal $C_{\mathrm{ac}} < \infty$ and $e_{\mathrm{ac}} > 0$ such that, whenever $e = \delta + \varepsilon \le e_{\mathrm{ac}}$, the elements P_1, P, Q_1, Q are δ-projections with*

$$\|P_1P - P_1\| \le \delta, \qquad \|Q_1Q - Q_1\| \le \delta, \tag{17}$$

and $n \ge 1$, every $X \in M_n \otimes \mathcal{S}_{P_1,Q_1}$ satisfies

$$\|\mathrm{Co}_{P_n,Q_n}(X) - X\| \ \le\ C_{\mathrm{ac}}\, e\, \|X\|, \tag{18}$$

with $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$. Transcription note. *Hypothesis* (17) *is the approximate form of "$P_1 \le P$ and $Q_1 \le Q$"; the conclusion says that the inner corner $\mathcal{S}_{P_1,Q_1}$ sits almost inside the outer one, uniformly at every amplification level.*

Registry contract (verbatim).

```
Amplified almost-containment: there are universal C_ac < infinity and e_ac > 0 such that, whenever
e=delta+epsilon <= e_ac, P_1,P,Q_1,Q are delta-projections with ||P_1 P-P_1||,||Q_1 Q-Q_1|| <=
delta, every n >= 1, and X in M_n tensor S_{P_1,Q_1}, one has ||Co_{P_n,Q_n}(X)-X|| <=
C_ac*e*||X||.
```

Proof (prose account of the af-validated tree). *A uniform ledger.* All four elements amplify to δ-projections in $M_n \otimes \mathcal{A}$, and the containment hypotheses amplify too, with the involution converting one side into the other: $P_nP_{1,n} - P_{1,n} = I_n \otimes (PP_1 - P_1) = I_n \otimes (P_1P - P_1)^\dagger$ has norm at most δ, and $Q_{1,n}Q_n - Q_{1,n} = I_n \otimes (Q_1Q - Q_1)$ likewise. A crude but uniform norm bound is also available: for $\varepsilon, \delta \le \frac{1}{2}$ the ε-C^* axioms and `def-delta-projection` give $(1-\varepsilon)\|R_n\|^2 \le \|R_n^2\| \le \|R_n\| + \delta$, hence $\|R_n\| \le B := 3$ for each of the four. Finally `def-compressed-corner` supplies one universal k_{co} with

$$\|\mathrm{Co}_{R,S}(Z) - R(ZS)\| \le k_{\mathrm{co}} e\|Z\|, \qquad \|\mathrm{Co}_{R,S}(Z) - (RZ)S\| \le k_{\mathrm{co}} e\|Z\|, \tag{19}$$

in every $M_n \otimes \mathcal{A}$. The threshold e_0 is the minimum of $\frac{1}{2}$, the threshold behind (19), and those of the two amplification dependencies.

Control in the source corner. By `lem-compcb-amplified-compression`, $M_n \otimes \mathcal{S}_{P_1,Q_1} = \mathcal{S}_{P_{1,n},Q_{1,n}}$, so X lies in the image of $\mathrm{Co}_{P_{1,n},Q_{1,n}}$; idempotence (`lem-compcb-amplified-compression-identities`) then upgrades membership to the fixed-point identity $\mathrm{Co}_{P_{1,n},Q_{1,n}}(X) = X$. Feeding this into (19) with $R = P_{1,n}$, $S = Q_{1,n}$, $Z = X$ replaces the compression by X itself and yields the two bracketings

$$\|X - P_{1,n}(XQ_{1,n})\| \le k_{\mathrm{co}} e\|X\|, \qquad \|X - (P_{1,n}X)Q_{1,n}\| \le k_{\mathrm{co}} e\|X\|. \tag{20}$$

Transfer to the outer projections. The claim is that X is almost invariant under P_n on the left and Q_n on the right, with a universal $L := k_{\mathrm{co}}(2B+1) + 2B(B^2+2)$. For the left estimate put $D = P_{1,n}(XQ_{1,n})$ and $Y = XQ_{1,n}$, and telescope

$$P_nX - X = P_n(X - D) + \big(P_nD - D\big) + (D - X).$$

The outer terms are controlled by (20) and submultiplicativity. The middle term is where the containment hypothesis and the *approximate* associativity of an ε-C^*-algebra both enter: $P_nD - D = P_n(P_{1,n}Y) - P_{1,n}Y$ is bounded by $\big[\varepsilon B^2 + (1+\varepsilon)\delta\big]\|Y\|$, since re-associating costs $\varepsilon\|P_n\|\|P_{1,n}\|\|Y\|$

and $(P_nP_{1,n} - P_{1,n})Y$ costs $(1+\varepsilon)\delta\|Y\|$. With $\|Y\| \le 2B\|X\|$ and $\varepsilon, \delta \le e$ the three terms sum to at most $Le\|X\|$. The right estimate is the mirror image, run with $D = (P_{1,n}X)Q_{1,n}$, $Y = P_{1,n}X$ and the hypothesis $\|Q_{1,n}Q_n - Q_{1,n}\| \le \delta$.

Assembly. Insert $P_n(XQ_n)$ and P_nX:

$$\|\mathrm{Co}_{P_n,Q_n}(X) - X\| \le \|\mathrm{Co}_{P_n,Q_n}(X) - P_n(XQ_n)\| + \|P_n(XQ_n - X)\| + \|P_nX - X\|.$$

The first term is $\le k_{\mathrm{co}}e\|X\|$ by (19); the second is $\le (1+\varepsilon)\|P_n\|\, Le\|X\| \le 2BLe\|X\|$; the third is $\le Le\|X\|$. Hence (18) holds with $C_{\mathrm{ac}} = k_{\mathrm{co}} + (2B+1)L$ and $e_{\mathrm{ac}} = e_0$, both finite, positive and universal.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-amplified-almost-containment`: 13 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplified-compression-i`
It is the corner-inclusion primitive of the COMP tier: it says that once a corner is approximately contained in another, the outer compression is a near-identity on it, again with one constant at every level. Its registry consumer is `lem-compcb-single-compression-transfer` (§18).

11 The compressed-unit norm estimate

Lemma 49 (`lem-compcb-compressed-unit-norm`). *There are universal $C_{\rm co} < \infty$ and $e_{\rm co} > 0$ such that, for $e = \delta + \varepsilon \le e_{\rm co}$,*

$$\|u_T\| \ \le\ 1 + C_{\rm co}\, e \qquad \textit{for every } \delta\textit{-projection } T, \tag{21}$$

and

$$\big|\,\|u_T\| - 1\,\big| \ \le\ C_{\rm co}\, e \qquad \textit{for every } \text{nonvanishing } T. \tag{22}$$

Transcription note. *$u_T = \mathrm{Co}_T(T)$ is the compressed unit of* ***`def-compressed-corner`****; "nonvanishing" is the second branch of the alternative in* ***`def-delta-projection`****. The asymmetry is deliberate and load-bearing: the two-sided bound is* false *in general, since a δ-projection may fall in the small branch $\|T\| = O(\delta)$, in which case u_T is small rather than close to a unit.*

Registry contract (verbatim).

```
Compressed-unit norm estimate: there are universal C_co < infinity and e_co > 0 such that, for
e=delta+epsilon <= e_co, every delta-projection T satisfies ||u_T|| <= 1+C_co*e, and every
nonvanishing T satisfies abs(||u_T||-1) <= C_co*e.
```

Proof (prose account of the af-validated tree). This shard's tree is instructive twice over, because its first attempt was audited and found unsound, and the audit — not the estimate — is what is validated in the early nodes.

The blocked first attempt. The originally registered inputs (`def-extended-epsilon-cstar-algebra` together with `def-compressed-corner`) yield only the δ-projection dichotomy, the identity $u_T = \mathrm{Co}_T(T)$, and the comparison $\|\mathrm{Co}_T - L_T R_T\| \le ke$. They do *not* yield the quantitative multiplication bound $\|XY\| \le (1+\varepsilon)\|X\|\|Y\|$ — the extended definition says only that each $M_n \otimes \mathcal{A}$ *is* an ε-C^*-algebra, without supplying that term's quantitative axioms — nor any unit-norm axiom that would give (22) from "$\mathcal{S}_T$ is an $O(e)$-C^*-algebra with unit u_T". A closure audit node records this verdict explicitly and refuses the root: without those axioms there is no sound bridge from the permitted premises to the contract.

The repair. The missing quantitative content was then registered in the workspace as byte-matched externals — the ε-C^* norm axioms (`epsilon-banach-cstar-norm-axioms`) and the nonvanishing alternative — and each $O(\cdot)$ in them unpacked into a named universal constant after shrinking the threshold. That gives universal $a, b, k < \infty$ and $e_0 > 0$ such that for $e \le e_0$: (i) every δ-projection T satisfies $\|T\| \le a\delta$ or is nonvanishing; (ii) $\|XY\| \le (1+\varepsilon)\|X\|\|Y\|$; (iii) $u_T = \mathrm{Co}_T(T)$ with $\|\mathrm{Co}_T - L_T R_T\| \le ke$; and (iv) nonvanishing T obeys $|\|T\| - 1| \le be$.

Small branch. Suppose $\|T\| \le a\delta$ and $e \le \min(e_0, 1)$. Since $(L_T R_T)(T) = T(T^2)$, two applications of (ii) give $\|(L_T R_T)(T)\| \le (1+\varepsilon)^2\|T\|^3 \le 4a^3e^3 \le 4a^3e$, while (iii) gives $\|u_T - (L_T R_T)(T)\| \le ke\|T\| \le kae^2 \le kae$. Hence $\|u_T\| \le De$ with $D = 4a^3 + ka$, and in particular $\|u_T\| \le 1 + De$. Note that here the compressed unit is *small*, which is why no two-sided statement can hold.

Nonvanishing branch. Shrink e_0 so that $e_0 \le 1$ and $be_0 \le 1$; then (iv) gives $\|T\| \le 2$. The compressed unit is compared directly with T rather than through any unit axiom: bilinearity gives $T(T^2) - T = T(T^2 - T) + (T^2 - T)$, so by (iii), (ii) and $\|T^2 - T\| \le \delta$,

$$\|u_T - T\| \le \|u_T - (L_T R_T)(T)\| + \|T(T^2) - T\| \le ke\|T\| + (1+\varepsilon)\|T\|\delta + \delta \le (2k+5)e.$$

The reverse triangle inequality together with (iv) then gives $\big|\|u_T\| - 1\big| \le (2k+5+b)e$, hence also the upper bound.

Assembly. With e_{co} the shrunken threshold and $C_{\mathrm{co}} = \max\bigl(4a^3+ka,\ 2k+5+b\bigr)$, the dichotomy routes every δ-projection to one of the two branches: the first gives (21), the second gives (21) and (22). The final assembly node states expressly that it uses neither the superseded audit node nor any compressed-unit norm axiom.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-compressed-unit-norm`: 15 nodes (12 `validated`, 3 superseded nodes archived), root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplified-compression-ic`
It supplies the unit-size input to `lem-compcb-corner-algebra` (§16) and to `lem-hcb-column-hilbert-squared` (§17); in the latter it is applied to the one-dimensional projection Q of an H-CB datum, which is why the nonvanishing hypothesis of (22) has to be discharged there by a separate bridge rather than assumed.

12 Exact adjointness of the amplified Ha map

Lemma 50 (`lem-hcb2-amplified-adjointness`). *There is a universal $e_{\mathrm{adj}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{adj}}$, every $n \ge 1$ and every $Z \in M_n \otimes \mathcal{S}_{P,R}$ satisfy*

$$\big(\mathrm{Ha}^Q_{P,R}\big)_n(Z)^\dagger \;=\; \big(\mathrm{Ha}^Q_{R,P}\big)_n(Z^\dagger). \tag{23}$$

Transcription note. *The datum is that of* ***`def-hcb-datum`*** *and $\mathrm{Ha}^Q_{P,R}$ is the map of* ***`def-ha-map`****; $e = \delta + \varepsilon$. The two daggers in (23) are different operations: on the left it is the Hilbert-space adjoint of the operator $\big(\mathrm{Ha}^Q_{P,R}\big)_n(Z)$ between amplified column spaces, on the right it is the involution of the ambient algebra applied to Z. The identity is* exact *— no $O(e)$ error — and the threshold is needed only so that the sesquilinear forms involved are genuine inner products.*

Registry contract (verbatim).

```
Exact amplified Ha adjointness: there is a universal e_adj > 0 such that every H-CB datum with e <=
e_adj, every n >= 1, and every Z in M_n tensor S_{P,R} satisfy
(Ha^Q_{P,R})_n(Z)^dagger=(Ha^Q_{R,P})_n(Z^dagger).
```

Proof (prose account of the af-validated tree). *Threshold.* Set $e_{\mathrm{adj}} = \min(e_{\mathrm{cmp}}, e_{\mathrm{ip}})$, where e_{cmp} comes from `lem-compcb-amplified-compression-identities` and e_{ip} is a universal smallness threshold below which the Euclidean sesquilinear forms of `def-ha-map` and of the column-Hilbert displays are genuine inner products — such a threshold exists because those forms agree with the norms up to a $1 \pm O(e)$ factor. Fix an arbitrary datum, n and Z.

Corner symmetries. Applying `lem-compcb-amplified-compression-identities` at $n = 1$: if $a \in \mathcal{S}_{A,B} = \mathrm{Im}\,\mathrm{Co}_{A,B}$ then $a = \mathrm{Co}_{A,B}(a)$, so $a^\dagger = \mathrm{Co}_{B,A}(a^\dagger)$ and $a^\dagger \in \mathcal{S}_{B,A}$; and since $Q^\dagger = Q$, the compressed unit $\widetilde{Q} = \mathrm{Co}_Q(Q)$ is Hermitian.

Exact product reversal. The pivotal algebraic fact is

$$(a \cdot b)^\dagger = b^\dagger \cdot a^\dagger \qquad (a \in \mathcal{S}_{A,B},\ b \in \mathcal{S}_{B,C}), \tag{24}$$

and it is *exact*. The proof is three substitutions: the compressed-product display gives $a \cdot b = \mathrm{Co}_{A,C}(ab)$; the imported identities at $n = 1$ give $\mathrm{Co}_{A,C}(ab)^\dagger = \mathrm{Co}_{C,A}\big((ab)^\dagger\big)$; the involution axiom gives $(ab)^\dagger = b^\dagger a^\dagger$; and a second use of the display, legitimate because the reversed corner memberships were just established, gives $\mathrm{Co}_{C,A}(b^\dagger a^\dagger) = b^\dagger \cdot a^\dagger$. No associativity estimate and no approximation enter. The tree records a correction here worth one sentence: an earlier audit node concluded that (24) was *not* derivable, because the formula $a \cdot b = \mathrm{Co}_{A,C}(ab)$ had not been registered as an allowed input in the workspace; after the compressed-product display was registered, that negative conclusion — and the associated demand for further definition provisioning — was explicitly withdrawn.

Level one. Fix $z \in \mathcal{S}_{P,R}$ and put $T = \mathrm{Ha}^Q_{P,R}(z)$. The defining relation of `def-ha-map` reads, for all $x \in \mathcal{S}_{R,Q}$ and $y \in \mathcal{S}_{P,Q}$,

$$(y^\dagger \cdot z) \cdot x + y^\dagger \cdot (z \cdot x) = 2\,\langle y, Tx\rangle\,\widetilde{Q}. \tag{25}$$

Apply † to (25). On the left, (24) reverses each compressed product, turning the two summands into $x^\dagger \cdot (z^\dagger \cdot y)$ and $(x^\dagger \cdot z^\dagger) \cdot y$. On the right, $\widetilde{Q}$ is Hermitian and the scalar is conjugated, and conjugate symmetry together with the definition of the Hilbert-space adjoint gives $\overline{\langle y, Tx\rangle} = \langle Tx, y\rangle = \langle x, T^\dagger y\rangle$. Thus for all x, y,

$$(x^\dagger \cdot z^\dagger) \cdot y + x^\dagger \cdot (z^\dagger \cdot y) = 2\,\langle x, T^\dagger y\rangle\,\widetilde{Q}.$$

This is precisely the defining relation of `def-ha-map` for the pair (R, P), the symbol $z^\dagger$, the input y and the test vector x; and that relation determines its solution uniquely. Hence $\mathrm{Ha}^Q_{R,P}(z^\dagger)y = T^\dagger y$ for every y, i.e. $\mathrm{Ha}^Q_{R,P}(z^\dagger) = \mathrm{Ha}^Q_{P,R}(z)^\dagger$.

Amplification. Write $Z = [Z_{ij}]$. By the amplification convention of the contract, $\big(\mathrm{Ha}^Q_{P,R}\big)_n(Z)$ is the block operator $[\mathrm{Ha}^Q_{P,R}(Z_{ij})]$ from the n-fold Hilbert direct sum of $\mathcal{S}_{R,Q}$ to that of $\mathcal{S}_{P,Q}$; its Hilbert-space adjoint therefore has (i,j) block $\mathrm{Ha}^Q_{P,R}(Z_{ji})^\dagger$. On the other side, $(Z^\dagger)_{ij} = Z^\dagger_{ji}$, which lies in $\mathcal{S}_{R,P}$ by the corner symmetry above, so the (i,j) block of $\big(\mathrm{Ha}^Q_{R,P}\big)_n(Z^\dagger)$ is $\mathrm{Ha}^Q_{R,P}(Z^\dagger_{ji})$. The level-one identity equates the two blockwise, and equality of all blocks is (23). Since n and Z were arbitrary, the lemma follows.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb2-amplified-adjointness`: 21 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression-identities`. Being exact rather than approximate, it is what allows a C^*-identity argument to be run on the amplified Ha map: it is used in `lem-hcb3-diagonal-upper-norm` (§23) to identify $\big(\mathrm{Ha}^Q_{P,P}\big)_n(Z)^\dagger$ with $\big(\mathrm{Ha}^Q_{P,P}\big)_n(Z^\dagger)$ inside $\|T\|^2 = \|T^\dagger T\|$, and the same role in `lem-hcb3-diagonal-lower-modulus` (§25). It is also a declared dependency of `lem-hcb3-offdiagonal-inverse` (§27) and `lem-hcb4-canonical-close` (§29), both now `af: validated`.

13 The uniform compressed-unit action

Lemma 51 (`lem-compcb-compressed-unit-action`). *There are universal $C_{\mathrm{co}} < \infty$ and $e_{\mathrm{co}} > 0$ such that, for $e = \delta + \varepsilon \le e_{\mathrm{co}}$, every compatible amplified rectangular corner satisfies*

$$\|u_T \cdot A - A\| \le C_{\mathrm{co}}\, e\, \|A\|, \qquad \|A \cdot u_R - A\| \le C_{\mathrm{co}}\, e\, \|A\|. \tag{26}$$

Transcription note. *As spelled out in the tree, the datum is an extended ε-C^*-algebra $\mathcal{A}$, an $n \ge 1$, δ-projections T, R with $T_n = I_n \otimes T$, $R_n = I_n \otimes R$, and an element $A \in \mathcal{S}_{T_n,R_n}$; the compressed units are $u_{T,n} = \mathrm{Co}_{T_n,T_n}(T_n)$ and $u_{R,n} = \mathrm{Co}_{R_n,R_n}(R_n)$. By `lem-compcb-amplified-compression` these equal $I_n \otimes \mathrm{Co}_T(T)$ and $I_n \otimes \mathrm{Co}_R(R)$, which is the notation u_T, u_R of the contract.*

Registry contract (verbatim).

```
Uniform compressed-unit action: there are universal C_co < infinity and e_co > 0 such that, for
e=delta+epsilon <= e_co, every compatible amplified rectangular corner satisfies ||u_T dot A-A|| <=
C_co*e*||A|| and ||A dot u_R-A|| <= C_co*e*||A||.
```

Proof (prose account of the af-validated tree). The estimate is assembled in three layers: replace the compressed units by the projections, act with the projections in the *ambient* product, then replace the ambient product by the compressed one.

Primitive bounds. Since $\mathcal{A}$ is extended, $M_n \otimes \mathcal{A}$ is an ε-C^*-algebra and T_n, R_n are δ-projections there (adjoints and products amplify entrywise and $\|I_n \otimes Z\| = \|Z\|$). Unpacking the $O(\cdot)$ terms of `def-delta-projection` and `def-compressed-corner` gives universal $a, k < \infty$ and $e_0 \le 1$ such that every δ-projection S obeys $\|S\| \le a\delta$ or $\|S\| \le 1 + ae$ — so $\|T_n\|, \|R_n\| \le M := \max(a, 1+a)$ in either branch — and such that each compression is within ke of both $L \cdot R$ orderings. Idempotence of the amplified compression (`lem-compcb-amplified-compression-identities`) also gives the fixed-point identity $A = \mathrm{Co}_{T_n,R_n}(A)$.

Compressed unit versus projection. For $S \in \{T_n, R_n\}$ and $u_S = \mathrm{Co}_{S,S}(S)$ one has $(L_S R_S)(S) = S(S^2)$, and bilinearity gives $S(S^2) - S = S(S^2 - S) + (S^2 - S)$. With $\|S^2 - S\| \le \delta$, the multiplication bound, and $\|\mathrm{Co}_{S,S} - L_S R_S\| \le ke$,

$$\|u_S - S\| \;\le\; ke\|S\| + (1+\varepsilon)\|S\|\delta + \delta \;\le\; K_u e, \qquad K_u := kM + 2M + 1, \tag{27}$$

and hence $\|u_S\| \le M + K_u$, uniformly in n.

Ambient action of the projections. Put $D = T_n(AR_n)$. The fixed-point identity and the compression comparison give $\|A - D\| \le ke\|A\|$, and one telescopes

$$T_n A - A = T_n(A - D) + \big(T_n D - D\big) + (D - A).$$

The outer terms cost $2Mke\|A\|$ and $ke\|A\|$. For the middle term write $Y = AR_n$, so that $T_n D - D = T_n(T_n Y) - T_n Y$; approximate associativity contributes $\varepsilon M^2 \|Y\|$ and $\|T_n^2 - T_n\| \le \delta$ contributes $2\delta\|Y\|$, and $\|Y\| \le 2M\|A\|$. Altogether $\|T_n A - A\| \le K_T e\|A\|$ with $K_T = 2Mk + 2M(M^2+2) + k$; the right-hand estimate $\|AR_n - A\| \le K_R e\|A\|$ is the mirror argument with $D = (T_n A)R_n$ and the same constant. Substituting the units via (27) — $\|(u_{T,n} - T_n)A\| \le 2K_u e\|A\|$ and symmetrically — gives the ambient form

$$\|u_{T,n} A - A\| \le K_a e\|A\|, \quad \|A\, u_{R,n} - A\| \le K_a e\|A\|, \qquad K_a := \max(K_T, K_R) + 2K_u. \tag{28}$$

From ambient to compressed. The pairs $(u_{T,n}, A)$ and $(A, u_{R,n})$ are compatible amplified rectangular pairs in the exact sense required by `lem-compcb-rectangular-product`, which therefore gives universal C_p with $\|u_{T,n} \cdot A - u_{T,n}A\| \leq C_p e \|u_{T,n}\| \|A\|$ and its mirror. Combining with (28) and $\|u_S\| \leq M + K_u$ yields (26) with

$$C_{\mathrm{co}} = K_a + C_p\,(M + K_u), \qquad e_{\mathrm{co}} = \min\bigl(e_0, e_p, e_{\mathrm{cmp}}, e_{\mathrm{id}}, 1\bigr),$$

all universal and independent of n, $\mathcal{A}$, T, R, A.

A bookkeeping remark: the later nodes of the export re-derive (27)–(28) in "dependency-closed" form, so that the final assembly rests only on validated premises and not on any node still pending at the time. The mathematics is unchanged; the restructuring is what let the tree close bottom-up.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-compressed-unit-action`: 20 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-iden` and `lem-compcb-rectangular-product`. It supplies the unit axiom needed by `lem-compcb-corner-algebra` (§16) — a corner cannot be an approximate C^*-algebra without an approximate unit — and it is the input that drives the diagonal unit estimate `lem-hcb3-diagonal-unit` (§22).

14 The row-column compressed-product estimate

Lemma 52 (`lem-compcb-row-column-product`). *There are universal $C_{\mathrm{rc}} < \infty$ and $e_{\mathrm{rc}} > 0$ such that, whenever $e = \delta + \varepsilon \le e_{\mathrm{rc}}$, P, Q are δ-projections in an extended ε-C^*-algebra $\mathcal{A}$, $n \ge 1$, and $X, Y \in M_{n,1} \otimes \mathcal{S}_{P,Q}$, one has*

$$\|\mathrm{Co}_Q\big(Y^\dagger X\big) - Y^\dagger X\| \;\le\; C_{\mathrm{rc}}\, e\, \|Y\|\, \|X\|, \tag{29}$$

where $Y^\dagger X$ is the ambient product of the $1 \times n$ row $Y^\dagger$ with the $n \times 1$ column X. Transcription note. *The product $Y^\dagger X$ is a single element of $\mathcal{A}$, not a matrix: the row and column contract. This is exactly why* ***`lem-compcb-rectangular-product`*** *does not apply directly — its compatibility clause requires both factors inside one* square *amplification.*

Registry contract (verbatim).

```
Row-column compressed-product estimate: there are universal C_rc < infinity and e_rc > 0 such that,
whenever e = delta+epsilon <= e_rc, P,Q are delta-projections in an extended epsilon-C*-algebra A,
n >= 1, and X,Y are in M_{n,1} tensor S_{P,Q}, one has ||Co_Q(Y^dagger X) - Y^dagger X|| <=
C_rc*e*||Y||*||X||, where Y^dagger X is the ambient product of the 1-by-n row Y^dagger with the
n-by-1 column X.
```

Proof (prose account of the af-validated tree). Set $C_{\mathrm{rc}} = C_{\mathrm{co}}$ and $e_{\mathrm{rc}} = \min(e_{\mathrm{co}}, e_{\mathrm{cmp}})$ with the constants of `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`, and fix arbitrary admissible data. The whole proof is a change of ambient size: embed the row/column pair as off-diagonal blocks of one square amplification, apply the validated square estimate there, and read the result off the single occupied entry.

The embedding. Use the decomposition $\mathbb{C}^{n+1} = \mathbb{C}^n \oplus \mathbb{C}$ and let

$$C_X = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \qquad R_Y = \begin{pmatrix} 0 & 0 \\ Y^\dagger & 0 \end{pmatrix}$$

in $M_{n+1} \otimes \mathcal{A}$. Writing $X = (X_i)$ and $Y = (Y_i)$ with entries in $\mathcal{S}_{P,Q}$, the dagger identity of `def-compressed-corner` gives $Y_i^\dagger \in \mathcal{S}_{Q,P}$, so X has entries in $\mathcal{S}_{P,Q}$ and $Y^\dagger$ has entries in $\mathcal{S}_{Q,P}$ with the indicated orientations. Since $e \le e_{\mathrm{cmp}}$, `lem-compcb-amplified-compression` at level $n+1$ identifies $M_{n+1} \otimes \mathcal{S}_{P,Q} = \mathcal{S}_{I_{n+1}\otimes P, I_{n+1}\otimes Q}$ and the transposed statement, so zero-padding puts C_X and R_Y in exactly those compatible corners. The padding is isometric: in the operator-space matrix norms, insertion by a scalar coordinate isometry and extraction by its adjoint are both contractive, hence equalities; and involution is isometric because $\mathcal{A}$ is self-adjoint. Therefore

$$\|C_X\| = \|X\|, \qquad \|R_Y\| = \|Y^\dagger\| = \|Y\|, \tag{30}$$

and (R_Y, C_X) is a compatible *square* amplified rectangular pair — the right projection of the first factor and the left projection of the second are both $I_{n+1} \otimes P$.

One occupied entry. Ordinary block multiplication gives

$$R_Y C_X = \mathrm{diag}\big(0_{n\times n},\, Y^\dagger X\big),$$

every other block containing a zero factor. On the compressed side, `def-compressed-corner` applied to the triple $(I_{n+1}\otimes Q, I_{n+1}\otimes P, I_{n+1}\otimes Q)$ gives $R_Y \cdot C_X = \mathrm{Co}_{I_{n+1}\otimes Q}(R_Y C_X)$, using the abbreviation $\mathrm{Co}_{S,S} = \mathrm{Co}_S$; and `lem-compcb-amplified-compression` applied to the pair (Q, Q) at level $n+1$ says $\mathrm{Co}_{I_{n+1}\otimes Q} = 1_{M_{n+1}} \otimes \mathrm{Co}_Q$, an entrywise map. Applying it to the diagonal block matrix above,

$$R_Y \cdot C_X = \mathrm{diag}\big(0_{n\times n},\, \mathrm{Co}_Q(Y^\dagger X)\big). \tag{31}$$

Conclusion. Since $e \le e_{\mathrm{co}}$ and (R_Y, C_X) is compatible, `lem-compcb-rectangular-product` yields $\|R_Y \cdot C_X - R_Y C_X\| \le C_{\mathrm{co}} e \|R_Y\| \|C_X\|$. By (31) the left-hand difference is the zero-padded element with single entry $\mathrm{Co}_Q(Y^\dagger X) - Y^\dagger X$, whose norm equals the norm of that entry by the same corner isometry as in (30); and (30) converts the right-hand side. This is (29) with $C_{\mathrm{rc}} = C_{\mathrm{co}}$. Both thresholds are positive universal constants, so $e_{\mathrm{rc}} > 0$, and universal generalisation over the arbitrary data completes the proof.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-row-column-product`: 20 nodes (18 `validated`, 2 superseded nodes archived), root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`. This shard was factored out of the column-Hilbert workspace at a recorded third stall: the blocking challenge observed that the validated rectangular-product contract quantifies only over square amplifications, so the pair $(Y^\dagger, X)$ — a $1\times n$ row against an $n\times 1$ column — simply does not satisfy its hypotheses, and the gap had to be closed by a statement of its own rather than by an instantiation. It has one registry consumer, `lem-hcb-column-hilbert-squared` (§17), where $Y = X$ and the estimate becomes the bridge between the column inner product and the ambient norm.

15 The uniform compressed associator estimate

Lemma 53 (`lem-hcb0-compressed-associator`)**.** *There are universal $C_{\mathrm{as}} < \infty$ and $e_{\mathrm{as}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{as}}$ and all compatible amplified rectangular A, B, C satisfy*

$$\|(A \cdot B) \cdot C - A \cdot (B \cdot C)\| \;\le\; C_{\mathrm{as}}\, e\, \|A\|\, \|B\|\, \|C\|. \tag{32}$$

Transcription note. *The left-hand side is the compressed associator of* ***`def-compressed-associator`****; "compatible amplified rectangular" is the datum fixed in §9, here for a composable triple. Both the compressed product and the ambient product are only* approximately *associative — the ambient one by the ε-associator axiom of an ε-C^*-algebra — so the content is that the two failures add up to a single $O(e)$.*

Registry contract (verbatim).

```
Uniform compressed associator: there are universal C_as < infinity and e_as > 0 such that every
H-CB datum with e <= e_as and all compatible amplified rectangular A,B,C satisfy ||(A dot B) dot
C-A dot (B dot C)|| <= C_as*e*||A||||B||||C||.
```

Proof (prose account of the af-validated tree). Let $C_{\mathrm{co}}, e_{\mathrm{co}}$ be the universal witnesses of `lem-compcb-rectangular-product` and set

$$e_{\mathrm{as}} = \min(e_{\mathrm{co}}, 1), \qquad C_{\mathrm{as}} = 2C_{\mathrm{co}}(2 + C_{\mathrm{co}}) + 4C_{\mathrm{co}} + 1.$$

A product-norm bound. For a compatible pair, the rectangular estimate plus the ambient bound $\|AB\| \le (1+\varepsilon)\|A\|\|B\|$ from the ε-C^* norm axioms give

$$\|A \cdot B\| \le \big(1 + \varepsilon + C_{\mathrm{co}} e\big)\|A\|\|B\|, \tag{33}$$

and likewise for $B \cdot C$. This is what keeps the outer terms of the telescope below from degrading.

The telescope. The compressed associator has the exact five-term expansion

$$\begin{aligned}(A \cdot B) \cdot C - A \cdot (B \cdot C) = {}& \big[(A \cdot B) \cdot C - (A \cdot B)C\big] + \big[(A \cdot B)C - (AB)C\big] \\ &+ \big[(AB)C - A(BC)\big] + \big[A(BC) - A(B \cdot C)\big] + \big[A(B \cdot C) - A \cdot (B \cdot C)\big],\end{aligned} \tag{34}$$

which routes both associations through the ambient product and isolates the ambient associator in the middle.

Write $N = \|A\|\|B\|\|C\|$. The first and fifth terms are rectangular-product errors for the compatible pairs $(A \cdot B, C)$ and $(A, B \cdot C)$, so by (33) each is at most $C_{\mathrm{co}} e(1+\varepsilon+C_{\mathrm{co}} e)N$. The second and fourth are the rectangular-product errors for (A, B) and (B, C) multiplied by one ambient factor — bilinearity gives $(A \cdot B)C - (AB)C = (A \cdot B - AB)C$ and $A(BC) - A(B \cdot C) = A(BC - B \cdot C)$ — so each is at most $(1+\varepsilon)C_{\mathrm{co}} eN$. The third is precisely the ε-associator axiom, $\|(AB)C - A(BC)\| \le \varepsilon N$.

Summation. Adding and using $0 \le \varepsilon \le e \le 1$, so that $1 + \varepsilon + C_{\mathrm{co}} e \le 2 + C_{\mathrm{co}}$ and $1 + \varepsilon \le 2$, gives (32) with the stated C_{as}.

Two pieces of hygiene in the tree are worth recording, since they shape the constant. The summation is written so that the ambient-associator term is bounded directly by eN rather than as $(\varepsilon/e) \cdot eN$, because at the endpoint $e = 0$ one has $\varepsilon = 0$ and every error vanishes — the quotient ε/e is never formed. And the upstream witness is normalised to $C_{\mathrm{co}} := \max\{C_{\mathrm{co}}, 0\}$, which cannot weaken the imported estimate since $e\|A\|\|B\| \geq 0$, but does make the sign-sensitive arithmetic above valid.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb0-compressed-associator`: 20 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-rectangular-product`. It is the base of the H-CB tier: the defining relation of the Ha map (`def-ha-map`) pairs an element against precisely the difference of the two associations $(Y^\dagger \cdot Z) \cdot X$ and $Y^\dagger \cdot (Z \cdot X)$, so (32) is exactly the estimate that turns that relation into a quantitative bound. It feeds `lem-hcb1-column-action` (§20) and `lem-hcb2-product-defect` (§21).

16 The compressed corner as an extended approximate C^*-algebra

Lemma 54 (`lem-compcb-corner-algebra`). *There are universal $C_{\mathrm{ca}} < \infty$ and $e_{\mathrm{ca}} > 0$ such that, whenever $e = \delta + \varepsilon \le e_{\mathrm{ca}}$ and P is a nonvanishing δ-projection, the corner $\mathcal{S}_P$ — equipped with the compressed product, the inherited involution and the compressed unit $u_P = \mathrm{Co}_P(P)$ — is an extended $C_{\mathrm{ca}}e$-C^*-algebra.* Transcription note. *"Extended" is as in* `def-extended-epsilon-cstar-algebra`*: not merely that $\mathcal{S}_P$ satisfies the approximate C^* axioms, but that* every *$M_n \otimes \mathcal{S}_P$ does, with the* same *parameter $C_{\mathrm{ca}}e$, independent of n. That uniformity is the whole content.*

Registry contract (verbatim).

```
Uniform compressed-corner algebra: there are universal C_ca < infinity and e_ca > 0 such that,
whenever e=delta+epsilon <= e_ca and P is a nonvanishing delta-projection, S_P with the compressed
product, inherited involution, and compressed unit u_P=Co_P(P) is an extended C_ca*e-C*-algebra.
```

Proof (prose account of the af-validated tree). The tree is organised as one node per axiom family, over a shared constant ledger, with a preliminary step that moves everything to level n at once.

Constant ledger and amplified setup. Let C_r, e_r come from `lem-compcb-rectangular-product`; C_u, e_u from `lem-compcb-compressed-unit-action`; C_N, e_N from `lem-compcb-compressed-unit-norm`; and e_i from `lem-compcb-amplified-compression-identities`. Put

$$K_p = C_r+1, \quad K_a = 2C_r(K_p+3)+1, \quad C_{\mathrm{ca}} = \max\{K_p, K_a, C_u, C_N\}, \quad e_{\mathrm{ca}} = \min\{e_r, e_u, e_N, e_i, 1\}.$$

With $P_n = I_n \otimes P$, the tree then *derives inside itself* — from `def-compressed-corner` and the registered θ calculus, rather than importing it — the naturality $\mathrm{Co}_{P_n} = \mathrm{id}_{M_n} \otimes \mathrm{Co}_P$: writing $T_P = L_PR_P+R_PL_P-1$, multiplication by the block-diagonal P_n acts entrywise so $T_{P_n} = \mathrm{id}_{M_n}\otimes T_P$; powers of $T_{P_n} - x_0 I$ amplify by induction; so every partial sum of the θ series is the amplification of the corresponding level-one partial sum, and at fixed finite n the norm limits agree. Consequently $\mathcal{S}_{P_n} = M_n\otimes\mathcal{S}_P$, the P_n-compressed product of $[X_{ij}]$, $[Y_{jk}]$ has entries $\mathrm{Co}_P(\sum_j X_{ij}Y_{jk}) = \sum_j X_{ij}\cdot Y_{jk}$ — exactly the matrix amplification of the compressed product — and $\mathrm{Co}_{P_n}(P_n) = I_n \otimes \mathrm{Co}_P(P)$. Also P_n is a δ-projection, so all four imported estimates apply at every level.

Operator space and completeness. `lem-compcb-amplified-compression-identities` gives $\mathrm{Co}_{P_n}^2 = \mathrm{Co}_{P_n}$, so $\mathcal{S}_{P_n} = \mathrm{Im}\,\mathrm{Co}_{P_n}$ is a closed subspace of the complete space $M_n\otimes\mathcal{A}$ and hence complete in the inherited norm. With the inherited matrix norms $\mathcal{S}_P$ is therefore a complete operator space.

Multiplication and associator. By `lem-compcb-rectangular-product` and the ambient bound $\|AB\| \le (1+\varepsilon)\|A\|\|B\|$, one gets $\|A\cdot B\| \le (1+K_pe)\|A\|\|B\|$. For the associator, insert successively $(A\cdot B)C$, $(AB)C$, $A(BC)$ and $A(B\cdot C)$ between the two associations; the two outer rectangular errors cost $C_re(1+K_pe)N$ each, the two errors obtained by multiplying $A\cdot B - AB$ or $BC - B\cdot C$ by an ambient factor cost $(1+\varepsilon)C_reN$ each, and the ambient associator costs εN, where $N = \|A\|\|B\|\|C\|$. With $\varepsilon \le e \le 1$ the total is at most K_aeN.

Involution. If $A \in \mathcal{S}_{P_n}$ then $A = \mathrm{Co}_{P_n}(A)$, so the imported dagger identity gives $A^\dagger = \mathrm{Co}_{P_n}(A^\dagger)$ and the inherited isometric involution preserves the corner. Anti-multiplicativity is exact: $(A\cdot B)^\dagger = \mathrm{Co}_{P_n}\big((AB)^\dagger\big) = \mathrm{Co}_{P_n}(B^\dagger A^\dagger) = B^\dagger\cdot A^\dagger$. The unit is fixed, $u_{P,n}^\dagger = \mathrm{Co}_{P_n}(P_n^\dagger) = u_{P,n}$, since $P_n^\dagger = P_n$. Finally the lower C^*-norm bound: the rectangular estimate applied to $A^\dagger$ and A, isometry of the involution, and the ambient lower estimate give

$$\|A^\dagger\cdot A\| \;\ge\; \|A^\dagger A\| - C_re\|A\|^2 \;\ge\; \big(1-\varepsilon-C_re\big)\|A\|^2 \;\ge\; \big(1-(C_r+1)e\big)\|A\|^2.$$

Unit. `lem-compcb-compressed-unit-action` applied to the amplified square corner gives $\|u_{P,n} \cdot A - A\| \leq C_u e\|A\|$ and its right-hand mirror at every n. For the unit norm, each P_n is a δ-projection with $\|P_n\| = \|P\|$, so the *nonvanishing* alternative passes from P to P_n; that is exactly the hypothesis under which `lem-compcb-compressed-unit-norm` gives the two-sided $\big|\|u_{P,n}\| - 1\big| \leq C_N e$, uniformly in n. This is where the nonvanishing hypothesis of the contract is used, and why it cannot be dropped.

Collecting the four families under the ledger constants gives the extended $C_{\mathrm{ca}}e$-C^* structure.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-corner-algebra`: 14 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression-identities`, `lem-compcb-rectangular-pro` `lem-compcb-compressed-unit-action` and `lem-compcb-compressed-unit-norm`. It is the output of the COMP tier: it converts a bundle of separate estimates into a single *structure* that later tiers may use as a black box. Among its registry consumers are `lem-compcb-single-compression-transfer` (§18), `lem-hcb3-diagonal-upper-norm` (§23) — where the matrix-uniform product axiom is precisely the input that a level-one statement could not supply — `lem-hcb3-uniform-square-lower` (§24), which uses the matching lower axiom, and `lem-hcb3-diagonal-lower-modulus` (§25).

17 The corrected amplified column-Hilbert estimate

Lemma 55 (`lem-hcb-column-hilbert-squared`). *There are universal $C_{\mathrm{col}} < \infty$ and $e_{\mathrm{col}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{col}}$, every $n \ge 1$ and every $X \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ satisfy*

$$\Big| \langle X, X\rangle_n - \|X\|_{n,1}^2 \Big| \;\le\; C_{\mathrm{col}}\, e\, \|X\|_{n,1}^2. \tag{35}$$

Transcription note. *$\langle\cdot,\cdot\rangle_n$ is the amplified column inner product of `def-column-hilbert-corner`, defined by $Y^\dagger \cdot X = \langle Y, X\rangle_n\, u_Q$ with $u_Q = \mathrm{Co}_Q(Q)$, and $\|\cdot\|_{n,1}$ is the ambient norm on the column space. The word "corrected" in the contract refers to the fact that this is the* squared *comparison: the unsquared display at the corresponding place in the pinned source is known to be false and is deliberately excluded from `def-column-hilbert-corner`.*

Registry contract (verbatim).

```
Corrected amplified column-Hilbert estimate: there are universal C_col < infinity and e_col > 0
such that every H-CB datum with e <= e_col, every n >= 1, and every X in M_{n,1} tensor S_{P,Q}
satisfy abs(<X,X>_n-||X||_{n,1}^2) <= C_col*e*||X||_{n,1}^2.
```

Proof (prose account of the af-validated tree). *Scalarisation.* Fix an admissible datum, n and X, and abbreviate

$$r = \|X\|_{n,1}, \qquad h = \langle X, X\rangle_n, \qquad a = \|u_Q\|, \qquad b = \|X^\dagger X\|.$$

The column space is a Hilbert space for its defined inner product, so $h \ge 0$ is real, and the defining amplified display of `def-column-hilbert-corner` at $Y = X$ reads

$$X^\dagger \cdot X = h\, u_Q. \tag{36}$$

The proof is then the interaction of three scalar estimates for h, a, b, r.

(i) The compressed-versus-ambient comparison. Applying `lem-compcb-row-column-product` with $Y = X$ gives $\|\mathrm{Co}_Q(X^\dagger X) - X^\dagger X\| \le C_{\mathrm{rc}} e\, r^2$; and $\mathrm{Co}_Q(X^\dagger X) = X^\dagger \cdot X = h\, u_Q$ by (36). Absolute homogeneity and the reverse triangle inequality therefore give

$$|ha - b| \;=\; \big| \|h\, u_Q\| - \|X^\dagger X\| \big| \;\le\; \|h\, u_Q - X^\dagger X\| \;\le\; C_{\mathrm{rc}} e\, r^2. \tag{37}$$

(ii) The C^* comparison. The ε-C^* lower norm axiom at the amplified rectangular level gives $b \ge (1-\varepsilon) r^2$, and submultiplicativity together with isometry of the involution gives $b \le (1+\varepsilon)\|X^\dagger\|\, r = (1+\varepsilon) r^2$. Hence

$$|b - r^2| \;\le\; \varepsilon\, r^2 \;\le\; e\, r^2. \tag{38}$$

(iii) The compressed unit is close to a unit. Applying `lem-compcb-compressed-unit-norm` with $T = Q$ gives $|a - 1| \le Ce$, and with $Ce \le \frac{1}{2}$ also $a \ge \frac{1}{2}$. This step needs Q *nonvanishing*, which the datum does not literally state — `def-hcb-datum` says only that Q is a *one-dimensional* δ-projection. The tree first records this as a genuine gap, then closes it by a registered bridge: one-dimensionality means $\dim \mathcal{S}_Q = 1$, so $\mathcal{S}_Q \ne 0$; and $\mathcal{S}_Q = 0$ characterises the first (vanishing) alternative of `def-delta-projection`; hence Q lies in the second alternative, i.e. is nonvanishing.

Synthesis. Enlarge the constant and shrink the threshold so that one pair $C \ge 1$, $e_0 \le 1/(2C)$ serves all three estimates. Then (37) and (38) give $|ha - r^2| \le (C+1)er^2$. The exact scalar identity

$$a\,(h - r^2) = (ha - r^2) + r^2(1 - a) \tag{39}$$

combined with $|a-1| \le Ce$ then yields $a|h-r^2| \le (2C+1)er^2$, and dividing by $a \ge \frac{1}{2}$ gives (35) with $C_{\mathrm{col}} = 4C+2$ and $e_{\mathrm{col}} = e_0$.

The withdrawn route. A substantial part of this tree is a *negative* audit, and it is the reason two of the four dependencies exist. The natural first attempt was to square-pad the column, $X_c = Xe_1^*$ and $X_r = X_c^\dagger$, and apply `lem-compcb-rectangular-product` to the pair (X_r, X_c). The audit found that under this shard's allowed inputs none of the three needed identifications was available: the corner memberships $M_n \otimes \mathcal{S}_{P,Q} = \mathcal{S}_{P_n,Q_n}$ and $\big(\mathcal{S}_{P,Q}\big)^\dagger \subseteq \mathcal{S}_{Q,P}$, the compressed-product identity $X_r \cdot X_c = E_{11} \otimes (X^\dagger \cdot X)$, and the matrix-norm identifications $\|X_c\| = r$ and $\|E_{11} \otimes z\| = \|z\|$. Each affirmative claim was withdrawn rather than repaired in place; a further over-claim — that the failure of this one construction proved an *exhaustive* impossibility — was also withdrawn when challenged. The actual repair was structural: register `lem-compcb-row-column-product` and `lem-compcb-entrywise-compression-naturality` as registry dependencies, after which step (i) above applies directly, with no square embedding and no E_{11} identification needed at this level.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb-column-hilbert-squared`: 50 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-rectangular-product`, `lem-compcb-compressed-unit-norm`, `lem-compcb-entrywise-compression-naturality` and `lem-compcb-row-column-product`. It is the analytic hinge of the H-CB tier: it makes the column-Hilbert norm q_P and the ambient norm interchangeable up to $1 \pm O(e)$, which is what lets estimates proved in one be read in the other. It feeds `lem-hcb1-variational-identity` (§19), `lem-hcb1-column-action` (§20) and `lem-hcb4-canonical-closeness` (§29), now `af: validated`, which consumes it at $n = 1$ to normalise the compressed unit.

18 Single-compression transfer of an extended inclusion

Lemma 56 (`lem-compcb-single-compression-transfer`). *There are universal $C_{\mathrm{co}} < \infty$ and $e_{\mathrm{co}} > 0$ such that restricting an extended α-inclusion to one ideal and following it by one compatible amplified compression produces an extended $C_{\mathrm{co}}(\alpha+\varepsilon)$-inclusion whenever $\alpha + \varepsilon \le e_{\mathrm{co}}$.*
Transcription note. *The tree fixes the datum: $\mathcal{A}$ an extended ε-C^*-algebra, $\mathcal{B}$ an exact unital C^*-algebra, $\mathcal{J} \subseteq \mathcal{B}$ a unital direct-summand ideal with unit q, and $v : \mathcal{B} \to \mathcal{A}$ an extended α-inclusion (`def-extended-delta-inclusion`). The "compatible single compression" is*

$$P := v(q), \qquad T := \mathrm{Co}_P \circ v|_{\mathcal{J}},$$

and throughout $e = \alpha + \varepsilon$.

Registry contract (verbatim).

```
Single-compression transfer: there are universal C_co < infinity and e_co > 0 such that restricting
an extended alpha-inclusion to one ideal and following it by one compatible amplified compression
produces an extended C_co*(alpha+epsilon)-inclusion whenever alpha+epsilon <= e_co.
```

Proof (prose account of the af-validated tree). *The projection and the ledger.* Since $q = q^\dagger = q^2$ with $\|q\| = 1$ in the exact C^*-algebra $\mathcal{B}$, the inclusion clauses give $P^\dagger = v(q^\dagger) = P$, $\|P^2 - P\| = \|v(q)v(q) - v(q^2)\| \le \alpha$ and $1 - \alpha \le \|P\| \le 1 + \alpha$; so P is an α-projection and, for small e, a *nonvanishing* one, its norm lying in the second alternative of `def-delta-projection`. That is what licenses `lem-compcb-corner-algebra` to make $\mathcal{S}_P$ an extended $C_{\mathrm{ca}}e$-C^*-algebra — the target structure. The remaining ledger constants are k_{cmp} from the compression comparison $\|\mathrm{Co}_{P_n}(Z) - (P_nZ)P_n\| \le k_{\mathrm{cmp}}e\|Z\|$, then $D = 2k_{\mathrm{cmp}} + 5$ and $K_N = D + 1$, together with C_r, e_r from `lem-compcb-rectangular-product`.

The exact clauses. By tensor functoriality $T_n = \mathrm{id}_{M_n} \otimes T = (\mathrm{id}_{M_n} \otimes \mathrm{Co}_P) \circ v_n$, and `lem-compcb-amplified-comp` rewrites this as $T_n = \mathrm{Co}_{P_n} \circ v_n$ with range $\mathrm{Im}\,\mathrm{Co}_{P_n} = \mathcal{S}_{P_n} = M_n \otimes \mathcal{S}_P$; linearity is inherited from v_n and Co_{P_n}. Two clauses hold *exactly*, with no error term. The star clause of the extended inclusion gives $v_n(x^\dagger) = v_n(x)^\dagger$, and `lem-compcb-amplified-compression-identities` for the square pair (P_n, P_n) gives $\mathrm{Co}_{P_n}(Z^\dagger) = \mathrm{Co}_{P_n}(Z)^\dagger$, so $T_n(x^\dagger) = T_n(x)^\dagger$. And the unit of $M_n \otimes \mathcal{J}$ is $q_n = I_n \otimes q$ with $v_n(q_n) = P_n$, so

$$T_n(q_n) = \mathrm{Co}_{P_n}(P_n) = I_n \otimes \mathrm{Co}_P(P) = I_n \otimes u_P :$$

the unit defect is zero.

Compression closeness. Because q is the unit of the ideal, $q_nx = xq_n = x$ for $x \in M_n \otimes \mathcal{J}$, so with $Z = v_n(x)$ the multiplicative clause of the inclusion gives $\|P_nZ - Z\| \le \alpha\|x\|$ and $\|ZP_n - Z\| \le \alpha\|x\|$, while its norm clause gives $\|P_n\| \le 1 + \alpha$ and $\|Z\| \le (1+\alpha)\|x\|$. Combining with the compression comparison and $e \le \frac{1}{2}$,

$$\|T_n(x) - v_n(x)\| \le \|\mathrm{Co}_{P_n}(Z) - (P_nZ)P_n\| + \|(P_nZ)P_n - Z\| \le D\,e\,\|x\|, \tag{40}$$

uniformly in n; the second term is bounded by $\big[(1+\varepsilon)(1+\alpha)+1\big]\alpha\|x\| \le 5e\|x\|$.

Norm distortion. The inclusion gives $(1-\alpha)\|x\| \le \|v_n(x)\| \le (1+\alpha)\|x\|$; with (40) the triangle and reverse-triangle inequalities give

$$(1 - K_Ne)\|x\| \;\le\; \|T_n(x)\| \;\le\; (1 + K_Ne)\|x\|, \qquad K_N = D + 1. \tag{41}$$

Multiplicativity. Insert $v_n(xy)$, $v_n(x)v_n(y)$ and $T_n(x)T_n(y)$ between $T_n(xy)$ and $T_n(x) \cdot T_n(y)$. The first term is $\leq De\|x\|\|y\|$ by (40), using that $M_n \otimes \mathcal{J}$ is an exact C^*-algebra — so its norm is genuinely submultiplicative, $\|xy\| \leq \|x\|\|y\|$. The second is $\leq \alpha\|x\|\|y\|$ by the multiplicative clause. The third, after inserting $T_n(x)v_n(y)$, is $\leq 2D(K_N + 3)e\|x\|\|y\|$ by (40), (41) and the ε-product bound. The fourth needs a naturality observation: for $A, B \in M_n \otimes \mathcal{S}_P$ the amplified compressed product has entries $\sum_k \mathrm{Co}_P(a_{ik}b_{kj}) = \mathrm{Co}_P\big(\sum_k a_{ik}b_{kj}\big)$, i.e. it equals $\mathrm{Co}_{P_n}(AB)$ — so it *is* the rectangular compressed product for the compatible triple (P_n, P_n, P_n), not merely an element of the same space, and `lem-compcb-rectangular-product` applies, giving $\leq C_r(1 + K_N)^2 e\|x\|\|y\|$. Summing, $K_M := D + 1 + 2D(K_N + 3) + C_r(1 + K_N)^2$ works.

Assembly. With $C_{\mathrm{co}} = \max\{1, C_{\mathrm{ca}}, K_N, K_M\}$ and e_{co} the positive minimum of all thresholds used, `def-extended-delta-inclusion` makes each T_n a $C_{\mathrm{co}}e$-inclusion; since n was arbitrary and every constant is universal, T is an extended $C_{\mathrm{co}}(\alpha + \varepsilon)$-inclusion, with no dependence on the amplification level or the ideal's dimension.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-single-compression-transfer`: 33 nodes (32 `validated`, 1 superseded node archived), root `validated`, taint clean.

Role in the chain. This is the single-compression transfer primitive aimed at compression-based EXT and Stage-1 assembly, where inclusions have to be pushed through compressions repeatedly. The EXT parent `conj-extcb` is now `proved` with `af: validated`; the broader assembly carrier `lem-thmainext-conditional` remains `proved-mod-audit` with `af: none`. This lemma has no registry consumers.

19 The amplified Ha variational identity

Lemma 57 (`lem-hcb1-variational-identity`). *There is a universal $e_{\mathrm{var}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{var}}$, every $n \ge 1$, every $Z \in M_n \otimes \mathcal{S}_{P,R}$, $X \in M_{n,1} \otimes \mathcal{S}_{R,Q}$ and $Y \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ satisfy*

$$2\,\langle Y, \big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z)X - Z \cdot X\rangle_n\, \mathrm{Co}_Q(Q) \;=\; \big(Y^\dagger \cdot Z\big)\cdot X - Y^\dagger \cdot \big(Z \cdot X\big). \tag{42}$$

Transcription note. *The identity is* exact*; the threshold only guarantees that the objects involved are defined. Its content is that the "Ha defect" — the difference between letting $\mathrm{Ha}^{Q}_{P,R}(Z)$ act on a column and simply multiplying by Z — is, after pairing, literally the compressed associator of `def-compressed-associator` for the triple $(Y^\dagger, Z, X)$.* Contract amendment. *The codified contract originally wrote the normalising factor as u_Q; the ratified `def-ha-map` identity and the column-Hilbert displays write it as $\widetilde{Q} = \mathrm{Co}_Q(Q)$. The contract now spells out $\mathrm{Co}_Q(Q)$. These are definitionally the same element — the compressed unit is $u_Q = \mathrm{Co}_Q(Q)$ — so this was pure notation typing, neither a strengthening nor a weakening.*

Registry contract (verbatim).

```
Amplified Ha variational identity: there is a universal e_var > 0 such that every H-CB datum with e
<= e_var, every n >= 1, Z in M_n tensor S_{P,R}, X in M_{n,1} tensor S_{R,Q}, and Y in M_{n,1}
tensor S_{P,Q} satisfy 2*<Y,(Ha^Q_{P,R})_n(Z)X-Z dot X>*Co_Q(Q)=(Y^dagger dot Z) dot X-Y^dagger dot
(Z dot X).
```

Proof (prose account of the af-validated tree). *Threshold.* Take $e_{\mathrm{var}} = \min(e_{\mathrm{cmp}}, e_{\mathrm{col}})$, the thresholds of `lem-compcb-amplified-compression-identities` and `lem-hcb-column-hilbert-squared`, and fix an arbitrary datum and arbitrary n, Z, X, Y.

Entrywise dictionary. Write $Z = (Z_{ij})_{1\le i,j\le n}$, $X = (X_j)_j$, $Y = (Y_i)_i$. Matrix amplification, the matrix compressed product and the amplified sesquilinear form unfold as

$$\big(\big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z)X\big)_i = \sum_j \mathrm{Ha}^{Q}_{P,R}(Z_{ij})(X_j), \quad (Z\cdot X)_i = \sum_j Z_{ij}\cdot X_j,$$

$$\big(Y^\dagger \cdot Z\big)_j = \sum_i Y_i^\dagger \cdot Z_{ij}, \qquad \langle Y, W\rangle_n = \sum_i \langle Y_i, W_i\rangle.$$

Summing the level-one relation. For each pair (i,j), the defining relation of `def-ha-map`, applied to $Z_{ij} \in \mathcal{S}_{P,R}$, $X_j \in \mathcal{S}_{R,Q}$ and $Y_i \in \mathcal{S}_{P,Q}$, gives

$$2\,\langle Y_i, \mathrm{Ha}^{Q}_{P,R}(Z_{ij})(X_j)\rangle\, \mathrm{Co}_Q(Q) = \big(Y_i^\dagger \cdot Z_{ij}\big)\cdot X_j + Y_i^\dagger \cdot \big(Z_{ij}\cdot X_j\big).$$

Summing over i, j and using the dictionary together with sesquilinearity,

$$2\,\langle Y, \big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z)X\rangle_n\, \mathrm{Co}_Q(Q) = \sum_{i,j}\Big[\big(Y_i^\dagger \cdot Z_{ij}\big)\cdot X_j + Y_i^\dagger \cdot \big(Z_{ij}\cdot X_j\big)\Big]. \tag{43}$$

The column display. The registered amplified column-Hilbert display $Y^\dagger \cdot W = \langle Y, W\rangle_n \mathrm{Co}_Q(Q)$, applied to $W = Z\cdot X$ and expanded entrywise, gives

$$\langle Y, Z\cdot X\rangle_n\, \mathrm{Co}_Q(Q) = Y^\dagger \cdot (Z\cdot X) = \sum_{i,j} Y_i^\dagger \cdot \big(Z_{ij}\cdot X_j\big). \tag{44}$$

Cancellation. Associating only as displayed — this is the point where the two sides differ — ordinary matrix multiplication gives $\big(Y^\dagger \cdot Z\big) \cdot X = \sum_{i,j} \big(Y_i^\dagger \cdot Z_{ij}\big) \cdot X_j$, whereas $Y^\dagger \cdot (Z \cdot X) = \sum_{i,j} Y_i^\dagger \cdot \big(Z_{ij} \cdot X_j\big)$. Subtracting twice (44) from (43) and using sesquilinearity in the second variable, the doubled $Y_i^\dagger \cdot (Z_{ij} \cdot X_j)$ sum cancels once and reverses sign, leaving exactly (42). Since the datum and n, Z, X, Y were arbitrary and e_{var} is universal and positive, the lemma follows.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb1-variational-identity`: 8 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression-identities` and `lem-hcb-column-hilbert-s` It is the conversion device of the H-CB tier: on its own it proves no estimate, but it turns every bound on the compressed associator into a bound on the Ha defect. Its registry consumers are `lem-hcb1-column-action` (§20), where it is combined with `lem-hcb1-combo-bound`, and the parent `conj-hcb` (§31).

20 The uniform Ha column action

Lemma 58 (`lem-hcb1-column-action`). *There are universal $C_{\mathrm{act}} < \infty$ and $e_{\mathrm{act}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{act}}$, every $n \ge 1$, every $Z \in M_n \otimes \mathcal{S}_{P,R}$ and every $X \in M_{n,1} \otimes \mathcal{S}_{R,Q}$ satisfy*

$$q_P\big(\big(\mathrm{Ha}^Q_{P,R}\big)_n(Z)X - Z \cdot X\big) \;\le\; C_{\mathrm{act}}\, e\, \|Z\|\, q_R(X). \tag{45}$$

Transcription note. *$q_P(U) = \langle U, U\rangle_n^{1/2}$ is the column-Hilbert norm of* ***`def-column-hilbert-corner`*** *on $M_{n,1} \otimes \mathcal{S}_{P,Q}$, and likewise q_R; $\|Z\|$ is the ambient norm.*

Registry contract (verbatim).

```
Uniform Ha column action: there are universal C_act < infinity and e_act > 0 such that every H-CB
datum with e <= e_act, every n >= 1, Z in M_n tensor S_{P,R}, and X in M_{n,1} tensor S_{R,Q}
satisfy q_P((Ha^Q_{P,R})_n(Z)X-Z dot X) <= C_act*e*||Z||*q_R(X).
```

Proof (prose account of the af-validated tree). Throughout put $D := \big(\mathrm{Ha}^Q_{P,R}\big)_n(Z)X - Z\cdot X$, the object to be bounded.

Nondegeneracy of the compressed unit. Everything is divided by $\|u_Q\|$ at the end, so a lower bound is needed. Two ingredients combine. First, $\|u_Q - Q\| \le K_1 e$. This is proved *inside the associative operator algebra $B(\mathcal{A})$*, not in the ambient ε-C^*-algebra, which is only approximately associative — a distinction the tree makes explicitly. With $L = L_Q$, $R = R_Q$ and $\mathrm{A} = LR + RL - I$ one has $\mathrm{Co}_Q = \theta(\mathrm{A})$; the δ-projection dichotomy bounds $\|Q\|$, hence $\|L\|, \|R\|$, and the ε-associator and δ-projection bounds give $\|LR - RL\| = O(e)$, $\|L^2 - L\| = O(e)$, $\|R^2 - R\| = O(e)$; from these, $T = \frac{1}{2}(LR + RL)$ satisfies $\|T^2 - T\| \le C_T e$ and so $\|\mathrm{A}^2 - I\| = 4\|T^2 - T\| \le 4C_T e$. Writing $r = Q^2 - Q$ and keeping the displayed parentheses, $(\mathrm{A} - I)Q = 2r + Qr + rQ$, so $\|(\mathrm{A} - I)Q\| \le C_Q e$. Finally, below the threshold making $\|\mathrm{A}^2 - I\| < \frac{1}{2}$, the series $B = (\mathrm{A}^2)^{-1/2}$ converges and factors as $B - I = (\mathrm{A}^2 - I)K(\mathrm{A}^2)$; since A commutes with B and $K(\mathrm{A}^2)$, one gets the clean factorisation $\theta(\mathrm{A}) - I = (\mathrm{A} - I)H(\mathrm{A})$ with $\|H(\mathrm{A})\| \le C_H$, whence $\|u_Q - Q\| = \|(\theta(\mathrm{A}) - I)Q\| \le C_H C_Q e$. Second, $\|Q\| \ge 1 - K_0 e$, which holds *provided Q is nonvanishing*. The tree runs this branch strictly conditionally — it refuses to infer nonvanishing from one-dimensionality — and discharges the antecedent only later, from the registered bridge $\dim \mathcal{S}_Q = 1 \Rightarrow \mathcal{S}_Q \ne 0 \Rightarrow Q$ nonvanishing. Combining the two by the reverse triangle inequality and shrinking the threshold gives

$$\|u_Q\| \;\ge\; c_u := \tfrac{1}{2}. \tag{46}$$

Column-norm comparison. By `lem-hcb-column-hilbert-squared`, $\big|q_T(U)^2 - \|U\|_{n,1}^2\big| \le C_{\mathrm{col}} e \|U\|_{n,1}^2$ for $T = P$ and $T = R$; with $C_{\mathrm{col}} e \le \frac{1}{2}$ all terms are nonnegative and taking square roots gives $(1 - C_{\mathrm{col}} e)^{1/2}\|U\|_{n,1} \le q_T(U) \le (1 + C_{\mathrm{col}} e)^{1/2}\|U\|_{n,1}$, in particular $\|U\|_{n,1} \le \sqrt{2}\, q_T(U)$.

Defect pairing. By `lem-hcb1-variational-identity`, $2\,\langle Y, D\rangle_n\, u_Q = \big(Y^\dagger \cdot Z\big)\cdot X - Y^\dagger \cdot \big(Z \cdot X\big)$ for every compatible column Y. The right-hand side is precisely the compressed associator of $A = Y^\dagger$, $B = Z$, $C = X$ (`def-compressed-associator`), so `lem-hcb0-compressed-associator` bounds it by $C_{\mathrm{as}} e \|Y^\dagger\|\|Z\|\|X\|_{n,1}$, and the involution is isometric, so $\|Y^\dagger\| = \|Y\|_{n,1}$. Taking norms on the left and using absolute homogeneity,

$$2\,\big|\langle Y, D\rangle_n\big|\, \|u_Q\| \;\le\; C_{\mathrm{as}}\, e\, \|Y\|_{n,1}\, \|Z\|\, \|X\|_{n,1}. \tag{47}$$

Assembly by self-pairing. Insert (46) and the two norm comparisons into (47): for every compatible Y,

$$\left|\langle Y, D\rangle_n\right| \;\leq\; 2C_{\mathrm{as}}\, e\, q_P(Y)\,\|Z\|\, q_R(X).$$

Now the finishing move avoids any duality argument: since both $\left(\mathrm{Ha}^{Q}_{P,R}\right)_n(Z)X$ and $Z \cdot X$ lie in $M_{n,1} \otimes \mathcal{S}_{P,Q}$, so does D, and $Y = D$ is an admissible test vector. Substituting it gives $q_P(D)^2 = \langle D, D\rangle_n \leq 2C_{\mathrm{as}}e\|Z\|q_R(X)\, q_P(D)$; if $q_P(D) = 0$ the conclusion is immediate, and otherwise one divides. Hence (45) with $C_{\mathrm{act}} = 2C_{\mathrm{as}}$.

Two recorded refinements. The tree first proved the terminal cancellation only *conditionally*, explicitly declining to divide by an unavailable c_u and carrying a node that says so; the unconditional root appears only after the nonvanishing bridge is registered, and the earlier negative conclusion is marked stale rather than deleted. Separately, the threshold is set as $e_{\mathrm{act}} = t/2$ for $t = \min(e_u, e_{\mathrm{col}}, e_{\mathrm{var}}, e_{\mathrm{as}}, 1/(2\max(C_{\mathrm{col}}, 1)))$ so that the strict inequalities required upstream hold even at the endpoint, and the degenerate case $C_{\mathrm{col}} = 0$ is split off so that the reciprocal $1/(2C_{\mathrm{col}})$ is never formed.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb1-column-action`: 38 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-hcb-column-hilbert-squared`, `lem-hcb0-compressed-associator` and `lem-hcb1-variational-identity`. This is the first genuinely quantitative statement about the Ha map: it says that $\left(\mathrm{Ha}^{Q}_{P,R}\right)_n(Z)$ is, to within $O(e)$, left multiplication by Z. Everything later in the tier is built on it — `lem-hcb2-product-defect` (§21), `lem-hcb3-diagonal-unit` (§22) — and it is a declared dependency of `lem-hcb4-canonical-closeness` (§29), now `af: validated`.

21 The uniform amplified Ha product defect

Lemma 59 (`lem-hcb2-product-defect`)**.** *There are universal $C_{\mathrm{prod}} < \infty$ and $e_{\mathrm{prod}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{prod}}$, every $n \ge 1$, every $Z \in M_n \otimes \mathcal{S}_{P,S}$ and every $W \in M_n \otimes \mathcal{S}_{S,R}$ satisfy*

$$\big\| \big(\mathrm{Ha}^Q_{P,R}\big)_n(Z \cdot W) - \big(\mathrm{Ha}^Q_{P,S}\big)_n(Z)\, \big(\mathrm{Ha}^Q_{S,R}\big)_n(W) \big\| \;\le\; C_{\mathrm{prod}}\, e\, \|Z\|\, \|W\|. \tag{48}$$

Transcription note. *On the left the juxtaposition of the two amplified Ha maps is composition of operators between amplified column spaces, whereas $Z \cdot W$ is the compressed product in the ambient corner; the norm is the operator norm of* `def-ha-map`*. So the statement is that $\mathrm{Ha}^Q_{\cdot,\cdot}$ is multiplicative up to $O(e)$.*

Registry contract (verbatim).

```
Uniform amplified Ha product defect: there are universal C_prod < infinity and e_prod > 0 such that
every H-CB datum with e <= e_prod, every n >= 1, Z in M_n tensor S_{P,S}, and W in M_n tensor
S_{S,R} satisfy ||(Ha^Q_{P,R})_n(Z dot W)-(Ha^Q_{P,S})_n(Z)(Ha^Q_{S,R})_n(W)|| <=
C_prod*e*||Z||||W||.
```

Proof (prose account of the af-validated tree). *Calibration.* Three uniform facts are set up first. The whole-column identity $X^\dagger \cdot X = q_T(X)^2 u_Q$ of `def-column-hilbert-corner`, combined with `lem-compcb-rectangular-product` and the ε-C^* norm axioms, gives $\big|\, \|X^\dagger \cdot X\| - \|X\|^2 \big| \le (C_{\mathrm{co}} + 1) e \|X\|^2$; since Q is one-dimensional it is nonvanishing by the registered bridge, so u_Q is the unit of an $O(e)$-C^* corner and its norm is within $O(e)$ of 1. Shrinking the threshold, this calibrates the column norm against the ambient norm,

$$\tfrac{1}{2}\|X\| \;\le\; q_T(X) \;\le\; 2\|X\|. \tag{49}$$

Next, the rectangular estimate and $\|AB\| \le (1+\varepsilon)\|A\|\|B\|$ give $\|A \cdot B\| \le K_0 \|A\|\|B\|$ with $K_0 = 2 + C_{\mathrm{co}}$; and for a square-by-column pair, (49) converts this into the multiplier bound $q_P(A \cdot X) \le K_{\mathrm{mul}} \|A\| q_R(X)$ with $K_{\mathrm{mul}} = 4K_0$. Finally, `lem-hcb1-column-action` together with the multiplier bound gives a uniform operator bound for the Ha map itself,

$$\big\| \big(\mathrm{Ha}^Q_{A,B}\big)_n(U) \big\| \;\le\; K_H \|U\|, \qquad K_H := K_{\mathrm{mul}} + C_{\mathrm{act}}. \tag{50}$$

The telescope. Abbreviate $H_{A,B}(U) := \big(\mathrm{Ha}^Q_{A,B}\big)_n(U)$. For every compatible column X the following identity is exact, obtained by adding and subtracting compressed multiplications:

$$\begin{aligned} \big[H_{P,R}(Z \cdot W) - H_{P,S}(Z) H_{S,R}(W)\big] X = {} & \big[H_{P,R}(Z \cdot W) X - (Z \cdot W) \cdot X\big] \\ & + \big[(Z \cdot W) \cdot X - Z \cdot (W \cdot X)\big] \\ & + Z \cdot \big[W \cdot X - H_{S,R}(W) X\big] \\ & + \big[Z \cdot H_{S,R}(W) X - H_{P,S}(Z) H_{S,R}(W) X\big]. \end{aligned} \tag{51}$$

Each bracket is one of the two estimates already available. The first is a column-action defect for the element $Z \cdot W$, so `lem-hcb1-column-action` and $\|Z \cdot W\| \le K_0 \|Z\|\|W\|$ bound it by $C_{\mathrm{act}} K_0 e \|Z\|\|W\| q_R(X)$. The second is the compressed associator, bounded by `lem-hcb0-compressed-associator` in the ambient norm by $C_{\mathrm{as}} e \|Z\|\|W\|\|X\|$; converting both ends through (49) costs two factors of 2, giving $4C_{\mathrm{as}} e \|Z\|\|W\| q_R(X)$. The third is the column-action defect for W, pushed through the square-by-column multiplier bound for Z, giving $K_{\mathrm{mul}} C_{\mathrm{act}} e \|Z\| \|W\| q_R(X)$. The fourth is the column-action defect for Z evaluated on the column $H_{S,R}(W)X$, whose column norm is controlled by the operator bound (50), giving $C_{\mathrm{act}} K_H e \|Z\| \|W\| q_R(X)$.

Conclusion. Applying the q_P triangle inequality to (51) and adding the four bounds gives the pointwise estimate with

$$C_* := C_{\mathrm{act}} K_0 + 4C_{\mathrm{as}} + C_{\mathrm{act}} K_{\mathrm{mul}} + C_{\mathrm{act}} K_H.$$

Taking the supremum over columns with $q_R(X) = 1$ converts it into the operator-norm statement (48), with $C_{\mathrm{prod}} = C_*$ and $e_{\mathrm{prod}} = \min\{e_{\mathrm{cmp}}, e_{\mathrm{co}}, e_{\mathrm{as}}, e_{\mathrm{act}}, e_q, 1\}$; `lem-compcb-amplified-compression` is what guarantees that all the corners and compressed products appearing at level n are the compatible amplified objects the imported estimates quantify over.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb2-product-defect`: 12 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-hcb0-compressed-associator`, `lem-hcb1-column-action`, `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`. Together with the exact adjointness of §12 it makes $\mathrm{Ha}^{Q}_{\cdot,\cdot}$ an approximate $*$-homomorphism at every amplification level, which is exactly the input the norm estimates of the next tier consume: it feeds `lem-hcb3-diagonal-upper-norm` (§23) and `lem-hcb3-diagonal-lower-modulus` (§25), and it is a declared dependency of `lem-hcb3-offdiagonal-inverse` (§27), now `af: validated`.

22 The uniform diagonal Ha unit estimate

Lemma 60 (`lem-hcb3-diagonal-unit`). *There are universal $C_{\mathrm{unit}} < \infty$ and $e_{\mathrm{unit}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{unit}}$ and every $n \ge 1$ satisfy*

$$\big\|\big(\mathrm{Ha}^{Q}_{P,P}\big)_n\big(I_n \otimes u_P\big) - I\big\| \;\le\; C_{\mathrm{unit}}\, e. \tag{52}$$

Transcription note. *$u_P = \mathrm{Co}_P(P)$ is the compressed unit of the corner $\mathcal{S}_P$, $I_n \otimes u_P$ its diagonal amplification, and I the identity operator on the amplified column space $M_{n,1} \otimes \mathcal{S}_{P,Q}$; the norm is the operator norm.*

Registry contract (verbatim).

```
Uniform diagonal Ha unit estimate: there are universal C_unit < infinity and e_unit > 0 such that
every H-CB datum with e <= e_unit and every n >= 1 satisfy ||(Ha^Q_{P,P})_n(I_n tensor u_P)-I|| <=
C_unit*e.
```

Proof (prose account of the af-validated tree). Write $U = I_n \otimes u_P$. The estimate is pointwise on columns, then a supremum.

A uniform bound on the amplified unit. The obvious route — bound $\|u_P\|$ at level one and appeal to isometry of the diagonal amplification — is *not* taken, and the tree says why: no registered input supplies that isometry. Instead $\|U\|$ is bounded directly at level n by a C^*-style argument. Put $t = \|U\|$. The ε-C^* norm axioms at level n give $\|U^\dagger U\| \ge (1-\varepsilon)t^2 \ge (1-e)t^2$ and $\|U^\dagger\| = t$; the compressed-product display in the amplified P, P corner gives $\|U^\dagger U - U^\dagger \cdot U\| \le C_{\mathrm{pr}} e t^2$; and `lem-compcb-compressed-unit-action`, in its right-unit form with $A = U^\dagger$, gives $\|U^\dagger \cdot U - U^\dagger\| \le C_{\mathrm{co}} e t$. Chaining,

$$(1-e)t^2 \;\le\; \|U^\dagger U\| \;\le\; C_{\mathrm{pr}} e t^2 + (1 + C_{\mathrm{co}} e)t.$$

If $t = 0$ there is nothing to prove; otherwise divide by t to get $\big[1 - (1 + C_{\mathrm{pr}})e\big]t \le 1 + C_{\mathrm{co}} e$, and choosing $e_u \le 1/\big[2(1 + C_{\mathrm{pr}})\big]$ makes the bracket at least $\frac{1}{2}$, so

$$\|U\| \;\le\; K_u := 2\big(1 + C_{\mathrm{co}} e_u\big), \tag{53}$$

uniformly in n.

Two defects. For a column $X \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ split

$$\big(\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(U) - I\big)X = \big(\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(U)X - U \cdot X\big) + \big(U \cdot X - X\big). \tag{54}$$

The first bracket is a column-action defect: `lem-hcb1-column-action` with $R = P$ and $Z = U$ gives $q_P\big(\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(U)X - U \cdot X\big) \le C_{\mathrm{act}} e \|U\| q_P(X)$, which (53) turns into $C_{\mathrm{act}} K_u e q_P(X)$.

The second bracket says that the amplified compressed unit is an approximate identity on columns, and it is assembled coordinatewise. Multiplication by the block-diagonal $U = I_n \otimes u_P$ acts entrywise, so writing $X = (X_1, \ldots, X_n)^\mathsf{T}$ and applying `lem-compcb-compressed-unit-action` in each level-one P, Q corner gives $\|u_P \cdot X_j - X_j\| \le C_{\mathrm{co}} e \|X_j\|$ for every j. The level-one norm/inner-product comparison registered with `def-ha-map` and the column-Hilbert displays supplies a universal κ with $q_P(Y) \le \kappa\|Y\|$ and $\|Y\| \le \kappa q_P(Y)$, so each coordinate estimate becomes $q_P(u_P \cdot X_j - X_j) \le \kappa^2 C_{\mathrm{co}} e\, q_P(X_j)$. Finally the coordinate-sum display $q_P\big((Y_j)_j\big)^2 = \sum_j q_P(Y_j)^2$ lets the squared coordinate estimates be summed, giving $q_P(U \cdot X - X) \le \kappa^2 C_{\mathrm{co}} e\, q_P(X)$ uniformly in n; the existential constant C_{co} is then simply replaced by the larger universal $\kappa^2 C_{\mathrm{co}}$ and the notation retained.

Conclusion. Setting $e_0 = \frac{1}{2}\min(e_u, e_{\mathrm{act}}, e_{\mathrm{co}})$ and applying the q_P triangle inequality to (54),

$$q_P\Big(\big((\mathrm{Ha}^Q_{P,P})_n(U) - I\big)X\Big) \ \le\ \big(C_{\mathrm{act}}K_u + C_{\mathrm{co}}\big)e\, q_P(X),$$

and passing to the operator norm gives (52) with $C_{\mathrm{unit}} = C_{\mathrm{act}}K_u + C_{\mathrm{co}}$ and $e_{\mathrm{unit}} = e_0$.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-diagonal-unit`: 18 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-hcb1-column-action` and `lem-compcb-compressed-unit-action`. It is one of the estimates the parent contract `conj-hcb` collects — the clause asserting that the amplified Ha map is approximately unital — and its registry consumer is the parent contract itself (§31), now `proved` with `af: validated`; neither it nor this lemma bears on the status of `op-classical`, which remains open.

23 The uniform diagonal Ha upper norm

Lemma 61 (`lem-hcb3-diagonal-upper-norm`). *There are universal $C_{\mathrm{up}} < \infty$ and $e_{\mathrm{up}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{up}}$, every $n \ge 1$ and every $Z \in M_n \otimes \mathcal{S}_P$ satisfy*

$$\big\|\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(Z)\big\| \;\le\; \big(1 + C_{\mathrm{up}}\, e\big)\|Z\|. \tag{55}$$

Transcription note. *The left norm is the operator norm on maps of the amplified column space; $\|Z\|$ is the ambient norm of Z in $M_n \otimes \mathcal{S}_P$. The estimate is uniform in n, which is the whole difficulty: the level-one statement is a routine consequence of the corner structure, the matrix-uniform one is not.*

Registry contract (verbatim).

```
Uniform diagonal Ha upper norm: there are universal C_up < infinity and e_up > 0 such that every
H-CB datum with e <= e_up, every n >= 1, and every Z in M_n tensor S_P satisfy
||(Ha^Q_{P,P})_n(Z)|| <= (1+C_up*e)||Z||.
```

Proof (prose account of the af-validated tree). Write $h_n = \big(\mathrm{Ha}^{Q}_{P,P}\big)_n$ and $b_n = \|h_n\|$ for its operator norm; the goal is $b_n \le 1 + C_{\mathrm{up}}e$ for every n.

The matrix-uniform square estimate. The first ingredient is

$$\|Z^{\dagger} \cdot Z\| \;\le\; (1 + \kappa e)\|Z\|^2, \qquad Z \in M_n \otimes \mathcal{S}_P, \tag{56}$$

with $\kappa = C_{\mathrm{ca}}$ and threshold $e_{\mathrm{sq}} = e_{\mathrm{ca}}$. If $\mathcal{S}_P = 0$ then $M_n \otimes \mathcal{S}_P = 0$ for every n, so $Z = 0$ and (56) is trivial. Otherwise $\mathcal{S}_P \neq 0$ puts P in the nonvanishing branch of the δ-projection dichotomy, and `lem-compcb-corner-algebra` makes $\mathcal{S}_P$ an *extended* $C_{\mathrm{ca}}e$-C^*-algebra — so, by the meaning of "extended", every $M_n \otimes \mathcal{S}_P$ is a $C_{\mathrm{ca}}e$-C^*-algebra with the *same* parameter, independent of n. Its product upper axiom together with isometry of the inherited involution then gives (56).

This is the shard's instructive correction, and it explains the shape of the dependency list. An earlier audit of the tree found that the permitted inputs at that time — the compressed-product display, the level-one ε-C^* axioms, `def-hcb-datum` (data and notation only), and the two Ha lemmas — gave (56) at $n = 1$ and *nowhere else*: passing to arbitrary n was not an inference from any allowed premise but a genuinely missing amplified-corner theorem. Once `lem-compcb-corner-algebra` became an allowed dependency, the negative audit nodes became obsolete and are marked as such.

A quadratic recurrence for b_n. Finiteness of b_n is established without any finite-dimensionality assumption on $\mathcal{S}_P$. From the defining Ha relation and the level-one bounds — $\|A{\cdot}B\| \le K_{\mathrm{dot}}\|A\|\|B\|$, the two-sided comparison $c_q^{-1}\|X\| \le q(X) \le c_q\|X\|$, and $\|u_Q\| \ge c_u > 0$ (with Q nonvanishing because it is one-dimensional) — one gets $2|\langle Y, h(Z)X\rangle|\,\|u_Q\| \le 2K_{\mathrm{dot}}^2\|Y\|\|Z\|\|X\|$, hence $|\langle Y, h(Z)X\rangle| \le K_1 q(Y)\|Z\|q(X)$ with $K_1 = K_{\mathrm{dot}}^2 c_q^2/c_u$, and Hilbert-space duality in the q norm gives $\|h(Z)\| \le K_1\|Z\|$ at level one. For fixed finite n, $h_n(Z) = [h(Z_{ij})]$ on the Hilbert direct sum, so $\|h_n(Z)\| \le \big(\sum_{i,j}\|h(Z_{ij})\|^2\big)^{1/2} \le nK_1\|Z\|$, using contractivity of matrix-coordinate compressions. Crude in n, but finite — which is all that is needed to *form* b_n and take a supremum.

Now let $e \le \min(e_{\mathrm{sq}}, e_{\mathrm{prod}}, e_{\mathrm{adj}})$ and $Z \in E_n := M_n \otimes \mathcal{S}_P$. Put $T = h_n(Z)$. The operator C^*-identity gives $\|T\|^2 = \|T^{\dagger}T\|$; `lem-hcb2-amplified-adjointness` identifies $T^{\dagger} = h_n(Z^{\dagger})$ — exactly, which is what makes this move legitimate; and `lem-hcb2-product-defect` with all three corner indices equal to P gives $\|h_n(Z^\dagger)h_n(Z) - h_n(Z^\dagger \cdot Z)\| \le C_{\mathrm{prod}} e\|Z\|^2$. Chaining with boundedness of h_n and (56),

$$\|h_n(Z)\|^2 \;\le\; b_n\|Z^\dagger \cdot Z\| + C_{\mathrm{prod}} e\|Z\|^2 \;\le\; \big[(1+\kappa e)b_n + C_{\mathrm{prod}} e\big]\|Z\|^2.$$

If $E_n = \{0\}$ then $b_n = 0$ and the recurrence below holds trivially; otherwise the unit sphere of E_n is nonempty, b_n is a finite supremum over it, and $\sup\|h_n(Z)\|^2 = \big(\sup\|h_n(Z)\|\big)^2 = b_n^2$, so

$$b_n^2 \;\le\; (1+\kappa e)\, b_n + C_{\mathrm{prod}} e. \tag{57}$$

The root argument. It remains to solve (57). For nonnegative b, e, κ, C with $b^2 \le ab + Ce$ and $a = 1{+}\kappa e$, set $x = 1{+}(\kappa{+}C)e$ and $f(t) = t^2{-}at{-}Ce$. Direct expansion gives $f(x) = C(\kappa{+}C)e^2 \ge 0$, and f is strictly increasing on $[x,\infty)$ because $2x{-}a = 1{+}(\kappa{+}2C)e > 0$. The hypothesis is $f(b) \le 0$; if $b > x$ then monotonicity would give $f(b) > f(x) \ge 0$, a contradiction. Hence $b \le x$.

Applying this with $b = b_n$ and $C = C_{\mathrm{prod}}$ gives $b_n \le 1 + (\kappa + C_{\mathrm{prod}})e$ for every n, which is (55) with $C_{\mathrm{up}} = \kappa + C_{\mathrm{prod}}$ and $e_{\mathrm{up}} = \min(e_{\mathrm{sq}}, e_{\mathrm{prod}}, e_{\mathrm{adj}})$; both are universal.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-diagonal-upper-norm`: 30 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect` and `lem-compcb-corner-algebra`. It is the upper half of the norm control that the parent contract `conj-hcb` requires; the matching lower half is `lem-hcb3-diagonal-lower-modulus` (§25), which is validated but only *conditionally* — it assumes a level-one lower modulus of at least $\frac{1}{4}$, a hypothesis supplied only inside §37, and the diagonal inverse estimate `lem-hcb3-diagonal-inverse` (§26) inherits it. The off-diagonal inverse (§27), the canonical tier (§§28–30) and `conj-hcb` itself (§31) are now all `af: validated`; the conditional hypotheses named above are supplied only inside §37, and nowhere else here. Nothing here bears on the status of `op-classical`.

24 The uniform square lower estimate

Lemma 62 (lem-hcb3-uniform-square-lower)**.** *There are universal $K_{\mathrm{sq}} < \infty$ and $e_{\mathrm{sq}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{sq}}$, every $n \ge 1$ and every $Z \in M_n \otimes \mathcal{S}_P$ satisfy*

$$\|Z^\dagger \cdot Z\| \;\ge\; \big(1 - K_{\mathrm{sq}}\, e\big)\|Z\|^2. \tag{58}$$

Transcription note. *The datum is that of* `def-hcb-datum` *and $e = \delta + \varepsilon$; $\cdot$ is the compressed product on the corner $\mathcal{S}_P$ of* `def-compressed-corner`*, and † the inherited involution. No hypothesis is placed on P beyond its being one of the datum's δ-projections — in particular* no *nonvanishing assumption — which is what makes the statement usable as a black box downstream.*

Registry contract (verbatim).

```
Uniform square lower estimate: there are universal K_sq < infinity and e_sq > 0 such that every
H-CB datum with e <= e_sq, every n >= 1, and every Z in M_n tensor S_P satisfy ||Z^dagger dot Z||
>= (1-K_sq*e)||Z||^2.
```

Proof (prose account of the af-validated tree). The tree splits on the δ-projection dichotomy of `def-delta-projection`. The nonvanishing branch is a citation; the vanishing branch is the work, and it is settled by showing that the corner is then *literally zero*.

The vanishing branch. Suppose P is not nonvanishing, so the registered dichotomy puts it in the small alternative $\|P\| \le C_v\delta$ for a universal C_v and e below a universal threshold. Put $T = L_P R_P + R_P L_P$. The ε-Banach product bound gives

$$\|T\| \;\le\; 2(1+\varepsilon)^2\|P\|^2 \;\le\; 2(1+e)^2 C_v^2 e^2,$$

so after a further universal shrink of the threshold one may assume $\|T\| < \frac{1}{3}$.

At that point the θ calculus of `def-theta-idempotent-approximation` evaluates in closed form. Write $U = I - T$ and $X = T - I = -U$. Since $\|T\| < \frac{1}{3}$ the Neumann series $\sum_{k\ge 0} T^k$ converges absolutely to U^{-1}; and $X^2 - I = -2T + T^2$ has norm at most $2\|T\| + \|T\|^2 < \frac{7}{9} < 1$, so the registered binomial series defining $(X^2)^{-1/2}$ converges absolutely. The two absolutely convergent series are then identified through their scalar counterparts: on the disc $|z| < \frac{1}{3}$ one has $\mathrm{Re}(1-z) > 0$, so the branch of the square root normalised to 1 at $z = 0$ satisfies $\big((1-z)^2\big)^{-1/2} = (1-z)^{-1} = \sum_{k\ge 0} z^k$; the two sides therefore have identical Taylor coefficients at 0, and evaluating both at the single Banach-algebra element T gives $(X^2)^{-1/2} = U^{-1}$. Consequently

$$\operatorname{sgn}(X) = X\,(X^2)^{-1/2} = (-U)U^{-1} = -I, \qquad \theta(X) = \tfrac{1}{2}\big(I + \operatorname{sgn} X\big) = 0. \tag{59}$$

Since the registered compression construction is $\mathrm{Co}_P = \theta\big(L_P R_P + R_P L_P - I\big) = \theta(T - I)$, (59) gives $\mathrm{Co}_P = 0$ and hence $\mathcal{S}_P = \operatorname{Im} \mathrm{Co}_P = \{0\}$.

The nonvanishing branch. If P *is* nonvanishing and $e \le e_{\mathrm{ca}}$, then `lem-compcb-corner-algebra` makes $\mathcal{S}_P$, with the compressed product, the inherited involution and the compressed unit, an *extended* $C_{\mathrm{ca}}e$-C^*-algebra. By the meaning of “extended” in `def-extended-epsilon-cstar-algebra`, every $M_n \otimes \mathcal{S}_P$ is then a $C_{\mathrm{ca}}e$-C^*-algebra with the *same* parameter, independently of n, and its lower C^*-axiom applied to Z reads exactly $\|Z^\dagger \cdot Z\| \ge (1 - C_{\mathrm{ca}}e)\|Z\|^2$. This — and not any level-one estimate — is where the uniformity in n comes from.

Exhaustion. Set

$$K_{\mathrm{sq}} = C_{\mathrm{ca}}, \qquad e_{\mathrm{sq}} = \min\Big\{e_{\mathrm{ca}},\, e_v,\, \tfrac{1}{2\max\{1,C_{\mathrm{ca}}\}}\Big\} > 0.$$

The two branches exhaust the possibilities for P. In the nonvanishing branch the displayed axiom is (58); in the vanishing branch $M_n \otimes \mathcal{S}_P = \{0\}$, so the only admissible Z is 0 and (58) reads $0 \geq 0$. The tree also records an explicit endpoint check: since $e_{\mathrm{sq}} \leq e_{\mathrm{ca}}$, the nonvanishing branch applies even at $e = e_{\mathrm{sq}} = e_{\mathrm{ca}}$, so no boundary point is omitted.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-uniform-square-lower`: 18 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-corner-algebra`. This shard exists because of a recorded balloon abort: the first orchestration of `lem-hcb3-diagonal-lower-modulus` tripped the node cap on exactly this claim, so it was factored out of that tree — its statement extracted mechanically from the blocking challenges — and validated on its own, bottom-up. Its value to the parent is that it carries the vanishing case internally, so the downstream argument may use (58) without re-splitting on the dichotomy. Its registry consumers include `lem-hcb3-diagonal-lower-modulus` (§25).

25 Diagonal Ha lower-modulus propagation

Lemma 63 (`lem-hcb3-diagonal-lower-modulus`). *There are universal $C_{\text{diag}} < \infty$ and $e_{\text{diag}} > 0$ such that, for every H-CB datum with $e \le e_{\text{diag}}$,* if the level-one lower modulus of $\mathrm{Ha}^{Q}_{P,P}$ is at least $\frac{1}{4}$*, then*

$$\big\|\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(Z)\big\| \;\ge\; \big(1 - C_{\text{diag}}\, e\big)\|Z\| \qquad \textit{for every } n \ge 1 \textit{ and } Z \in M_n \otimes \mathcal{S}_P. \tag{60}$$

Transcription note. *The hypothesis is load-bearing and is* not *discharged here: the conclusion is asserted only for those data whose level-one map already has lower modulus $\ge \frac{1}{4}$. (This conditional shape is the corrected one; the corresponding clause of the parent contract* `conj-hcb` *was made conditional after an unconditional inverse claim was refuted by an exact $\mathbb{C} \oplus \mathbb{C}$ counterexample.) The "lower modulus" of a bounded linear map is the largest c with $\|\cdot\| \ge c\|\cdot\|$, i.e. the infimum of the norm ratios over nonzero arguments.*

Registry contract (verbatim).

```
Diagonal Ha lower-modulus propagation: there are universal C_diag < infinity and e_diag > 0 such
that, for every H-CB datum with e <= e_diag, if the level-one lower modulus of Ha^Q_{P,P} is at
least 1/4, then ||(Ha^Q_{P,P})_n(Z)|| >= (1-C_diag*e)||Z|| for every n >= 1 and Z in M_n tensor
S_P.
```

Proof (prose account of the af-validated tree). Write $T_n = \big(\mathrm{Ha}^{Q}_{P,P}\big)_n$ throughout.

The root gap. Let $C_{\text{prod}}, e_{\text{prod}}$ come from `lem-hcb2-product-defect` — normalised to $C_{\text{prod}} \ge 0$, which cannot weaken that estimate since $e\|Z\|\|W\| \ge 0$ — let $K_{\text{sq}}, e_{\text{sq}}$ come from `lem-hcb3-uniform-square-lo` and put

$$C_{\text{diag}} := K_{\text{sq}} + 4C_{\text{prod}}, \qquad A := 1 - K_{\text{sq}} e, \qquad B := C_{\text{prod}} e.$$

Choose $e_{\text{diag}} > 0$ below e_{prod}, e_{sq} and e_{adj} (from `lem-hcb2-amplified-adjointness`) and small enough that $A \ge \frac{3}{4}$, $A^2 - 4B \ge 0$, $4B < \frac{1}{8}$ and $C_{\text{diag}} e \le \frac{1}{2}$; each requirement is a continuity condition at $e = 0$, where $A = 1$ and $A^2 - 4B = 1$, and is vacuous if the relevant constant vanishes, so the intersection is a positive universal interval. Let $r_{\mp} = \frac{1}{2}\big(A \mp \sqrt{A^2 - 4B}\big)$ be the roots of $x^2 - Ax + B$. Rationalising the smaller root — legitimate because the denominator is at least $A > 0$, including when $B = 0$ — gives

$$0 \le r_- = \frac{2B}{A + \sqrt{A^2 - 4B}} \le \frac{2B}{A} \le \tfrac{8}{3}B \le 4B < \tfrac{1}{8}, \qquad r_+ = A - r_- \;\ge\; 1 - C_{\text{diag}} e \;\ge\; \tfrac{1}{2}. \tag{61}$$

The gap between $r_- < \frac{1}{8}$ and $r_+ \ge \frac{1}{2}$ is the whole mechanism: any quantity known to satisfy the quadratic inequality and to exceed $\frac{1}{8}$ is forced up to r_+.

The zero-corner branch. If $\mathcal{S}_P = \{0\}$ then $M_m \otimes \mathcal{S}_P = \{0\}$ for every m, so the only admissible Z is 0; linearity gives $T_m(0) = 0$ and $(1 - C_{\text{diag}} e)\|Z\| = 0$, so (60) holds at every level. The tree keeps this branch strictly separate, and *no* lower modulus a_m is defined in it — the infimum below would be over an empty set of ratios.

The dichotomy on the nonzero branch. Assume $\mathcal{S}_P \ne \{0\}$ and pick $0 \ne x \in \mathcal{S}_P$. For each n the element $E_{11} \otimes x$ is a nonzero member of $M_n \otimes \mathcal{S}_P$ — the operator-space direct-sum axiom gives $\|\mathrm{diag}(x, 0, \dots, 0)\| = \|x\| > 0$ — so the ratio set $R_n = \big\{\|T_n(Z)\|/\|Z\| : 0 \ne Z \in M_n \otimes \mathcal{S}_P\big\}$ is nonempty and $a_n := \inf R_n$ is a finite nonnegative real.

Fix Z. The C^*-identity in the bounded-operator target gives $\|T_n(Z)\|^2 = \|T_n(Z)^\dagger T_n(Z)\|$; `lem-hcb2-amplified-adjointness` identifies $T_n(Z)^\dagger = T_n(Z^\dagger)$ *exactly*; and `lem-hcb2-product-defect`, applied with all three corner indices equal to P and with the factors $Z^\dagger, Z$, gives

$$\|T_n(Z)\|^2 \;\geq\; \big\|T_n\big(Z^\dagger \cdot Z\big)\big\| - C_{\mathrm{prod}} e \|Z\|^2, \tag{62}$$

the involution being isometric. Applying the defining property $\|T_n(X)\| \geq a_n \|X\|$ at $X = Z^\dagger \cdot Z$ and then `lem-hcb3-uniform-square-lower` — multiplication by $a_n \geq 0$ preserving the inequality — turns (62) into $\|T_n(Z)\|^2 \geq \big[A\, a_n - B\big]\|Z\|^2$. Dividing by $\|Z\|^2$ and taking the infimum over nonzero Z, using that $\inf\{r^2 : r \in R\} = (\inf R)^2$ for a nonempty $R \subseteq [0,\infty)$ with finite infimum, gives

$$a_n^2 \;\geq\; A\, a_n - B, \qquad \text{i.e.} \qquad (a_n - r_-)(a_n - r_+) \geq 0, \tag{63}$$

so $a_n \leq r_-$ or $a_n \geq r_+$: the level-n lower modulus cannot sit in the gap.

Propagation upward. Two operator-space facts move the moduli between levels. First a halving estimate: writing $Z \in M_{2n} \otimes \mathcal{S}_P$ as a 2×2 matrix of $n \times n$ blocks and factoring $Z = \mathrm{A} \cdot \mathrm{diag}(Z_{11}, Z_{12}, Z_{21}, Z_{22}) \cdot \mathrm{B}$ with the explicit scalar matrices $\mathrm{A} = \left[\begin{smallmatrix} I & I & 0 & 0 \\ 0 & 0 & I & I \end{smallmatrix}\right]$ and $\mathrm{B} = \left[\begin{smallmatrix} I & 0 \\ 0 & I \\ I & 0 \\ 0 & I \end{smallmatrix}\right]$ of norm $\sqrt{2}$, the Ruan axioms give $\|Z\| \leq 2\max_{i,j}\|Z_{ij}\|$; and since T_{2n} is the entrywise amplification of T_n, compression to the (i,j) target block gives $\|T_{2n}(Z)\| \geq \max_{i,j}\|T_n(Z_{ij})\|$. Hence $\|T_{2n}(Z)\| \geq a_n \max_{i,j}\|Z_{ij}\| \geq \frac{1}{2}a_n\|Z\|$, i.e. $a_{2n} \geq a_n/2$. Second a monotonicity: the zero-corner inclusion $Z \mapsto \mathrm{diag}(Z, 0_k)$ is isometric by the direct-sum axiom and is carried by T_m to $\mathrm{diag}(T_n(Z), 0_k)$ of the same norm, so $a_n \geq a_m$ whenever $n \leq m$.

Now the bootstrap. The hypothesis of the lemma is exactly $a_1 \geq \frac{1}{4}$, while (61) gives $r_- < \frac{1}{8}$; so $a_1 > r_-$ and (63) at $n = 1$ forces $a_1 \geq r_+$. Inductively, if $a_{2^k} \geq r_+$ then the halving estimate gives

$$a_{2^{k+1}} \;\geq\; \tfrac{1}{2}a_{2^k} \;\geq\; \tfrac{1}{2}r_+ \;\geq\; \tfrac{1}{2}\big(1 - C_{\mathrm{diag}}e\big) \;\geq\; \tfrac{1}{4} \;>\; r_-,$$

using $C_{\mathrm{diag}}e \leq \frac{1}{2}$; so the dichotomy at level 2^{k+1} again excludes the lower alternative and $a_{2^{k+1}} \geq r_+$. Ordinary induction gives $a_{2^k} \geq r_+$ for every $k \geq 0$. Note that the halving alone would degrade the bound geometrically; it is the *gap* that repairs it at each step.

Conclusion. For arbitrary n choose a power of two $m \geq n$. Monotonicity gives $a_n \geq a_m \geq r_+$, and (61) gives $r_+ \geq 1 - C_{\mathrm{diag}}e$. Hence $\|T_n(Z)\| \geq a_n\|Z\| \geq (1 - C_{\mathrm{diag}}e)\|Z\|$ for every Z, which together with the zero-corner branch is (60); the constants are universal and $e_{\mathrm{diag}} > 0$.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-diagonal-lower-modulus`: 29 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`. Together with `lem-hcb3-diagonal-upper` (§23) it pins the amplified diagonal Ha map between $1 \pm O(e)$ — but only under the level-one hypothesis, which nothing here supplies. It is the input to `lem-hcb3-diagonal-inverse` (§26), which inherits that hypothesis, and to `lem-hcb3-offdiagonal-inverse` (§27), which consumes it as its diagonal anchor. The parent `conj-hcb` (§31) is now `proved` with `af: validated`, and `op-classical` remains open.

26 Diagonal Ha inverse propagation

Lemma 64 (`lem-hcb3-diagonal-inverse`)**.** *There are universal $C_{\mathrm{inv}} < \infty$ and $e_{\mathrm{inv}} > 0$ such that, for every H-CB datum with $e \le e_{\mathrm{inv}}$,* if $\mathrm{Ha}^{Q}_{P,P}$ has level-one lower modulus at least $\frac{1}{4}$ and is bijective at level one*, then every amplification is bijective and*

$$\big\| \big((\mathrm{Ha}^{Q}_{P,P})_n \big)^{-1} \big\| \;\le\; 1 + C_{\mathrm{inv}}\, e. \tag{64}$$

Transcription note. ***Both** level-one hypotheses are load-bearing and neither is discharged here: the conclusion is asserted only for those data whose level-one map is* already *bijective* and *already has lower modulus at least $\frac{1}{4}$. The two are used at genuinely different points of the proof — bijectivity supplies the inverse, the lower modulus supplies its norm — and neither implies the other in this setting. The conditional shape is the corrected one, inherited from the same verdict that made the lower-modulus clause of `conj-hcb` conditional after an unconditional inverse claim was refuted by an exact $\mathbb{C} \oplus \mathbb{C}$ counterexample.*

Registry contract (verbatim).

```
Diagonal Ha inverse propagation: there are universal C_inv < infinity and e_inv > 0 such that, for
every H-CB datum with e <= e_inv, if Ha^Q_{P,P} has level-one lower modulus at least 1/4 and is
bijective at level one, then every amplification is bijective and ||((Ha^Q_{P,P})_n)^(-1)|| <=
1+C_inv*e.
```

Proof (prose account of the af-validated tree). The tree is short — five nodes — because the analytic work has already been done by `lem-hcb3-diagonal-lower-modulus` (§25); what remains is an exact algebraic step and one scalar inequality. Write $T := \mathrm{Ha}^{Q}_{P,P}$ and $T_n := \big(\mathrm{Ha}^{Q}_{P,P}\big)_n$.

Constants. Let $C_{\mathrm{diag}}, e_{\mathrm{diag}}$ be universal witnesses supplied by `lem-hcb3-diagonal-lower-modulus`. Enlarging C_{diag} only weakens that external conclusion, so one may take $C_{\mathrm{diag}} > 0$ and set

$$e_{\mathrm{inv}} := \min\Big\{ e_{\mathrm{diag}},\ \tfrac{1}{2C_{\mathrm{diag}}} \Big\}, \qquad C_{\mathrm{inv}} := 2C_{\mathrm{diag}}.$$

Both are universal, $e_{\mathrm{inv}} > 0$, and the choice is made precisely so that

$$0 \le C_{\mathrm{diag}} e \le \tfrac{1}{2} \qquad \text{whenever } 0 \le e \le e_{\mathrm{inv}}. \tag{65}$$

Bijectivity amplifies exactly. Fix an admissible datum and an arbitrary $n \ge 1$. This step uses the *bijectivity* hypothesis and nothing else. By the meaning of matrix amplification, $T_n\big([Z_{ij}]\big) = [T(Z_{ij})]$ acts slotwise. Surjectivity therefore passes upward: for any target $[Y_{ij}]$ the matrix $[T^{-1}(Y_{ij})]$ is a preimage. So does injectivity: $T_n([Z_{ij}]) = 0$ forces $T(Z_{ij}) = 0$ for every i, j, hence $Z_{ij} = 0$ for every i, j and $[Z_{ij}] = 0$. Thus T_n is bijective and, moreover,

$$(T_n)^{-1} \text{ is the entrywise amplification of } T^{-1}. \tag{66}$$

Note what is *not* claimed: nothing here bounds $\|(T_n)^{-1}\|$ — entrywise inversion alone gives no uniform norm control, which is why the second hypothesis is needed.

The imported lower bound. Since $e \le e_{\mathrm{inv}} \le e_{\mathrm{diag}}$ and the level-one lower-modulus hypothesis is at least $\frac{1}{4}$, `lem-hcb3-diagonal-lower-modulus` applies verbatim and gives

$$\|T_n(Z)\| \;\ge\; \big(1 - C_{\mathrm{diag}} e\big)\|Z\| \qquad \text{for every } Z \in M_n \otimes \mathcal{S}_P. \tag{67}$$

This is the only place the $\frac{1}{4}$ hypothesis enters, and it enters through the imported lemma rather than being re-proved.

The reciprocal estimate. Combine the two. For an arbitrary target element Y, put $Z = (T_n)^{-1}Y$, which exists by (66); then (67) reads $\|Y\| = \|T_n(Z)\| \geq (1 - C_{\mathrm{diag}}e)\|Z\|$. By (65) the quantity $x := C_{\mathrm{diag}}e$ lies in $[0, \frac{1}{2}]$, and on that interval the elementary identity

$$\frac{1}{1-x} = 1 + \frac{x}{1-x} \;\leq\; 1 + 2x$$

holds because $1 - x \geq \frac{1}{2}$. Therefore

$$\big\|(T_n)^{-1}Y\big\| \;\leq\; \big(1 - C_{\mathrm{diag}}e\big)^{-1}\|Y\| \;\leq\; \big(1 + 2C_{\mathrm{diag}}e\big)\|Y\| \;=\; \big(1 + C_{\mathrm{inv}}e\big)\|Y\|.$$

Since the datum and n were arbitrary, every amplification is bijective with the inverse bound (64), and the constants are universal. The restriction to $x \leq \frac{1}{2}$ is not cosmetic: it is what converts the reciprocal into a linear bound in e, and it is exactly what the choice of e_{inv} secures.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-diagonal-inverse`: 5 nodes, all `validated`, root `validated`, taint clean; first pass, two rounds, no challenges surviving.
Role in the chain. Imports `lem-hcb3-diagonal-lower-modulus`, `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`. It completes the diagonal block of the H-CB tier: with §23 and §25 it says that, *under the two level-one hypotheses*, the amplified diagonal Ha map is a bijection whose norm and inverse norm are both within $O(e)$ of 1. The matching off-diagonal statement `lem-hcb3-offdiagonal-inverse` (§27), the hcb4 canonical tier (§§28–30) and the parent `conj-hcb` (§31) are now all `proved` with `af: validated`. Nothing here bears on the status of `op-classical`, which is open.

27 Off-diagonal Ha inverse propagation

Lemma 65 (`lem-hcb3-offdiagonal-inverse`). *There are universal $C_{\mathrm{rect}} < \infty$ and $e_{\mathrm{rect}} > 0$ such that, for every H-CB datum with $e \le e_{\mathrm{rect}}$,* if $\mathrm{Ha}^Q_{P,R}$ is bijective at level one and $\mathrm{Ha}^Q_{R,R}$ has level-one lower modulus at least $\frac{1}{4}$, *then every amplification of* $\mathrm{Ha}^Q_{P,R}$ *is bijective with*

$$\left\| \left(\left(\mathrm{Ha}^Q_{P,R} \right)_n \right)^{-1} \right\| \;\le\; 1 + C_{\mathrm{rect}}\, e. \tag{68}$$

Transcription note. *The two level-one hypotheses are placed on* different *maps: bijectivity is assumed of the rectangular map* $\mathrm{Ha}^Q_{P,R}$ *itself, the lower-modulus hypothesis of the* diagonal *map* $\mathrm{Ha}^Q_{R,R}$ *at the target index. Neither is discharged here, and neither implies the other. This is the same conditional shape as §26, inherited from the verdict that made the corresponding clause of* `conj-hcb` *conditional.*

Registry contract (verbatim).

```
Off-diagonal Ha inverse propagation: there are universal C_rect < infinity and e_rect > 0 such
that, for every H-CB datum with e <= e_rect, if Ha^Q_{P,R} is bijective at level one and Ha^Q_{R,R}
has level-one lower modulus at least 1/4, then every amplification of Ha^Q_{P,R} is bijective with
inverse norm at most 1+C_rect*e.
```

Proof (prose account of the af-validated tree). Write $T_n := \left(\mathrm{Ha}^Q_{P,R}\right)_n$ and $D_n := \left(\mathrm{Ha}^Q_{R,R}\right)_n$. The tree has four movements: a constant ledger, a rectangular square estimate, the passage to a lower modulus for T_n, and a reciprocal bound. Only the third movement uses the diagonal hypothesis, and it uses it through `lem-hcb3-diagonal-lower-modulus` rather than re-proving it.

Constants. Let $C_{\mathrm{prod}}, e_{\mathrm{prod}}$ come from `lem-hcb2-product-defect`, let $C_{\mathrm{diag}}, e_{\mathrm{diag}}$ come from `lem-hcb3-diagonal-lower-modulus`, and let e_{adj} be the threshold of `lem-hcb2-amplified-adjointness`. The universal $O(e)$ constant of the compressed-product display (a workspace external, not a registry lemma) is unpacked as a finite C_{cp} valid below a positive e_{cp}. Put

$$B := 1 + C_{\mathrm{cp}}, \qquad A := \max\{1,\ C_{\mathrm{diag}} + B + C_{\mathrm{prod}}\}, \qquad C_{\mathrm{rect}} := 2A,$$

$$e_{\mathrm{rect}} := \min\left\{ e_{\mathrm{adj}}, e_{\mathrm{prod}}, e_{\mathrm{diag}}, e_{\mathrm{cp}}, \tfrac{1}{2A} \right\} > 0.$$

The last entry is what makes $Ae \le \frac{1}{2}$ throughout, and every subsequent coefficient nonnegative.

The rectangular square estimate. Fix an admissible datum, an arbitrary $n \ge 1$ and $Z \in M_n \otimes \mathcal{S}_{P,R}$. Left and right multiplication by $I_n \otimes P$ and $I_n \otimes R$ act entrywise, so the operator to which the θ calculus is applied amplifies as $1_{M_n} \otimes (\cdot)$ and the registered power series gives $\mathrm{Co}_{I_n \otimes P, I_n \otimes R} = 1_{M_n} \otimes \mathrm{Co}_{P,R}$; hence Z lies in the rectangular corner of the amplified algebra, $Z^\dagger$ in the reversed one, and $Z^\dagger \cdot Z$ is the compressed product formed *inside* $M_n \otimes \mathcal{A}$. Comparing it with the ambient product and using the amplified lower C^* axiom,

$$\|Z^\dagger \cdot Z\| \;\ge\; \left(1 - \varepsilon - C_{\mathrm{cp}} e\right)\|Z\|^2 \;\ge\; \left(1 - Be\right)\|Z\|^2, \tag{69}$$

uniformly in n. Note that this is a *rectangular* statement and is derived here from the compressed-product display and the norm axioms; the diagonal statement `lem-hcb3-uniform-square-lower` is not the vehicle.

Lower modulus for the rectangular map. Instantiate `lem-hcb2-product-defect` at corner indices (R, P, R) with first factor $Z^\dagger$ and second factor Z; by `lem-hcb2-amplified-adjointness` the first image is $(\mathrm{Ha}^Q_{R,P})_n(Z^\dagger) = T_n(Z)^\dagger$, and the C^* identity for bounded operators between the amplified column spaces gives $\|T_n(Z)^\dagger T_n(Z)\| = \|T_n(Z)\|^2$. Hence

$$\|T_n(Z)\|^2 \;\geq\; \|D_n(Z^\dagger \cdot Z)\| - C_{\mathrm{prod}} e\|Z\|^2. \tag{70}$$

The diagonal hypothesis now enters exactly once: applying `lem-hcb3-diagonal-lower-modulus` at the index R to $X = Z^\dagger \cdot Z$ gives $\|D_n(X)\| \geq (1 - C_{\mathrm{diag}} e)\|X\|$, and $1 - C_{\mathrm{diag}} e \geq 1 - Ae \geq \frac{1}{2} > 0$ licenses multiplying (69) by it. Substituting into (70) and discarding the nonnegative quadratic remainder,

$$\|T_n(Z)\|^2 \;\geq\; \big[(1-C_{\mathrm{diag}} e)(1-Be)-C_{\mathrm{prod}} e\big]\|Z\|^2 \;\geq\; \big[1-(C_{\mathrm{diag}}+B+C_{\mathrm{prod}})e\big]\|Z\|^2 \;\geq\; (1-Ae)\|Z\|^2.$$

Since $0 \leq Ae \leq \frac{1}{2}$ and $\sqrt{1-x} \geq 1 - x$ on $[0, 1]$,

$$\|T_n(Z)\| \;\geq\; (1 - Ae)\|Z\|. \tag{71}$$

Bijectivity and the reciprocal bound. Bijectivity amplifies algebraically and with no estimate: $\mathrm{Ha}^Q_{P,R}$ is linear with a two-sided linear inverse S by the first hypothesis, $T_n = 1_{M_n} \otimes \mathrm{Ha}^Q_{P,R}$ acts slotwise, and $1_{M_n} \otimes S$ is a two-sided inverse for it. For a target Y put $Z = T_n^{-1} Y$; then (71) reads $\|Y\| \geq (1 - Ae)\|Z\|$, and $1/(1 - x) \leq 1 + 2x$ for $x = Ae \in [0, \frac{1}{2}]$ gives $\|T_n^{-1} Y\| \leq (1 + 2Ae)\|Y\| = (1 + C_{\mathrm{rect}} e)\|Y\|$. The datum and n were arbitrary and the constants are universal, which is (68).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-offdiagonal-inverse`: 16 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Registry imports: `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-hcb3-diagonal-lower-modulus`, `lem-compcb-corner-algebra`, `lem-hcb3-uniform-square-lower` and `lem-hcb3-diagonal-inverse`; the exported tree visibly consumes the first three, the remaining three being available but unused in the validated derivation. Together with §§22–26 this completes the third layer of the H-CB tier: the diagonal maps are controlled on both sides, and the rectangular maps inherit an inverse bound from the diagonal anchor at their target index. It is the last clause consumed by the parent `conj-hcb` (§31). Nothing here bears on `op-classical`, which is open.

28 The canonical corner Gram estimate

Lemma 66 (`lem-hcb4-canonical-gram`). *There are universal $C_J < \infty$ and $e_J > 0$ such that every H-CB datum with $e \le e_J$, every $n \ge 1$ and every Z in either special P, Q corner satisfy*

$$(1 - C_J e)\,\|Z\| \;\le\; \big\|J_n(Z)\big\| \;\le\; (1 + C_J e)\,\|Z\|. \tag{72}$$

Transcription note. *The* special *corners are the two involving the one-dimensional projection of the datum: $Z \in M_n \otimes \mathcal{S}_{P,Q}$ with $J_n = J_{P,Q,n}$ the canonical column identification, or $Z \in M_n \otimes \mathcal{S}_{Q,P}$ with $J_n = J_{Q,P,n}$ the canonical row identification, both fixed by* `def-canonical-corner-identifications`*. The two sides of* (72) *carry* different *norms: $\|Z\|$ is the ambient operator-space norm of the amplified corner, while $\|J_n(Z)\|$ is the operator norm of the resulting map between amplified column spaces — the Euclidean norm of* `def-ha-map`*. The content of the lemma is therefore that the canonical identification, which is pure notation and carries no estimate of its own, is a $1 \pm O(e)$ isometry between those two norms, uniformly in n.*

Registry contract (verbatim).

```
Canonical corner Gram estimate: there are universal C_J < infinity and e_J > 0 such that every H-CB
datum with e <= e_J, every n >= 1, and every Z in either special P,Q corner satisfy (1-C_J*e)||Z||
<= ||J_n(Z)|| <= (1+C_J*e)||Z||.
```

Proof (prose account of the af-validated tree). The tree proves the column case and transfers the row case to it. The column case is one exact identity fed by two uniform inputs, followed by a scalar computation; there is no analysis beyond the two imports.

Exact Gram scalarisation. Fix an admissible datum, $n \ge 1$ and $Z \in M_n \otimes \mathcal{S}_{P,Q}$, and put $G := J_{P,Q,n}(Z)^\dagger J_{P,Q,n}(Z) \in M_n$. Reading the defining formula of `def-canonical-corner-identifications` against the column-Hilbert inner-product displays of `def-column-hilbert-corner` and the compressed-product display, the compressed square of Z is computed entrywise:

$$\big[Z^\dagger \cdot Z\big]_{ij} \;=\; \sum_k \langle Z_{ki}, Z_{kj}\rangle\, u_Q \;=\; G_{ij}\, u_Q, \qquad \text{that is} \qquad Z^\dagger \cdot Z \;=\; G \otimes u_Q. \tag{73}$$

This is an identity, not an estimate. Two norm facts turn it into one. The Ruan scalar-tensor invariance among the operator-space matrix-norm axioms gives $\|G \otimes u_Q\| = \|G\|\,\|u_Q\|$, and the Hilbert-space Gram identity gives $\|G\| = \|J_{P,Q,n}(Z)\|^2$. Hence

$$\big\|Z^\dagger \cdot Z\big\| \;=\; \big\|J_{P,Q,n}(Z)\big\|^2\, \|u_Q\|. \tag{74}$$

Everything after this point is bookkeeping on two scalars: the compressed square on the left and the compressed unit on the right.

The two uniform inputs. There are universal $A, B < \infty$, valid below a universal threshold, with

$$\big|\, \|Z^\dagger \cdot Z\| - \|Z\|^2 \,\big| \le A\, e\, \|Z\|^2, \qquad \big|\, \|u_Q\| - 1 \,\big| \le B\, e. \tag{75}$$

For the first, the ε-C^* norm axioms of `def-extended-epsilon-cstar-algebra` place the *ambient* square $\|Z^\dagger Z\|$ between $(1-\varepsilon)\|Z\|^2$ and $(1+\varepsilon)\|Z\|^2$ at level n, while `lem-compcb-rectangular-product` (§9) applied to the compatible amplified pair $(Z^\dagger, Z)$ gives $\|Z^\dagger \cdot Z - Z^\dagger Z\| \le C_r e \|Z\|^2$; since $\varepsilon \le e$, the reverse triangle inequality yields the first bound with $A = C_r + 1$, uniformly in n. For the second, `lem-compcb-compressed-unit-norm` (§11) gives $|\|u_Q\| - 1| \le C_u e$ once Q is known to be *nonvanishing*; the datum states only that Q is one-dimensional, and the gap is closed as elsewhere in this tier by the registered bridge $\dim \mathcal{S}_Q = 1 \Rightarrow \mathcal{S}_Q \neq 0 \Rightarrow Q$ nonvanishing. So $B = C_u$.

The scalar passage. Assume $Z \neq 0$ (for $Z = 0$ both sides of (72) vanish) and put $x := \|J_{P,Q,n}(Z)\|/\|Z\|$. Dividing (74) by $\|Z\|^2$ gives $x^2\|u_Q\| = \|Z^\dagger \cdot Z\|/\|Z\|^2$, so (75) confines

$$x^2 \;\in\; \Big[\tfrac{1-Ae}{1+Be},\; \tfrac{1+Ae}{1-Be}\Big].$$

Take e_{col} no larger than the two external thresholds, $1/(2\max\{B,1\})$ and $1/\max\{A+B,1\}$. Then the denominators are bounded away from 0 and the two ends simplify to $x^2 \ge 1-(A+B)e$ and $x^2 \le 1 + 2(A+B)e$. Finally $\sqrt{1-t} \ge 1-t$ for $t \in [0,1]$ and $\sqrt{1+t} \le 1 + t/2$ for $t \ge 0$ convert these into

$$1-(A+B)e \;\le\; x \;\le\; 1+(A+B)e, \tag{76}$$

which is (72) for the column corner with $C_{\mathrm{col}} = A+B$. The asymmetric factor 2 in the upper bound for x^2 is exactly what the sharper square-root inequality absorbs, so the same constant serves both sides.

Row transfer. For $Z \in M_n \otimes \mathcal{S}_{Q,P}$ put $W := Z^\dagger \in M_n \otimes \mathcal{S}_{P,Q}$. The self-adjoint operator-space axiom gives $\|W\| = \|Z\|$ at every level; `def-canonical-corner-identifications` *defines* the row map by $J_{Q,P,n}(Z) = J_{P,Q,n}(W)^\dagger$; and Hilbert-space adjunction preserves the operator norm. Applying the column case to W therefore returns both row inequalities with the same constants, so $C_J = C_{\mathrm{col}}$ and $e_J = e_{\mathrm{col}}$ serve the root.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb4-canonical-gram`: 9 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Registry imports: `lem-compcb-rectangular-product`, `lem-compcb-compressed-unit-norm` `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the exported tree visibly consumes the first two, the remaining two being available but unused in the validated derivation. This is the base of the canonical layer of the H-CB tier. Canonical closeness (§29) uses it once, to convert $\|J_{P,Q,n}(Z)\|$ into $\|Z\|$; the canonical inverse estimate (§30) uses its lower half twice, once for the bound $\|J_n^{-1}\| \le \frac{4}{3}$ and once for the norm equivalence that makes the Neumann target complete. Through those two it reaches the canonical-identity clause of the parent `conj-hcb` (§31), which lists it among its sixteen dependencies. Nothing here bears on `op-classical`, which is open.

29 Canonical Ha closeness

Lemma 67 (`lem-hcb4-canonical-closeness`)**.** *There are universal $C_{\mathrm{sp}} < \infty$ and $e_{\mathrm{sp}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{sp}}$ satisfies*

$$\max\{\|(\mathrm{Ha}^Q_{P,Q})_n - J_{P,Q,n}\|,\ \|(\mathrm{Ha}^Q_{Q,P})_n - J_{Q,P,n}\|\} \ \le\ C_{\mathrm{sp}}\, e \qquad \textit{for every } n \ge 1. \tag{77}$$

Transcription note. *Both differences are taken between maps defined on the same amplified corner — $M_n \otimes \mathcal{S}_{P,Q}$ in the first case, $M_n \otimes \mathcal{S}_{Q,P}$ in the second — and the outer norm is the induced one: the supremum over Z of the operator norm of the difference, divided by $\|Z\|$. The estimate is* unconditional*: unlike the inverse and lower-modulus results of §§25 –27 it assumes nothing about the level-one behaviour of any map.*

Registry contract (verbatim).

```
Canonical Ha closeness: there are universal C_sp < infinity and e_sp > 0 such that every H-CB datum
with e <= e_sp satisfies max{||(Ha^Q_{P,Q})_n-J_{P,Q,n}||,||(Ha^Q_{Q,P})_n-J_{Q,P,n}||} <= C_sp*e
for every n >= 1.
```

Proof (prose account of the af-validated tree). The tree has three movements: a scalar normalisation, the column estimate, and the same adjoint transfer used in §28. The organising observation is that the *entire* discrepancy between the special Ha map and the canonical map is carried by one number — the amount by which the compressed unit u_Q fails to be a unit vector for the column-Hilbert norm q_Q.

The normalised compressed unit. There are universal $A < \infty$ and $e_0 > 0$ such that, for $e \le e_0$, the scalar

$$\alpha := q_Q(u_Q) \tag{78}$$

is strictly positive, the normalised element $q_0 = \alpha^{-1} u_Q$ of `def-canonical-corner-identifications` satisfies $q_Q(q_0) = 1$ by absolute homogeneity, and $|\alpha - 1| \le Ae$. Two estimates combine. First, `lem-compcb-corner-algebra` (§16) applied with $P = Q$ — legitimate because Q is nonvanishing, by the registered one-dimensional-to-nonvanishing bridge — makes $\mathcal{S}_Q$ an extended γ-C^*-algebra with $\gamma = C_{\mathrm{ca}} e$ and unit u_Q, so the ε-Banach C^* norm axioms give $|\|u_Q\| - 1| \le \gamma$; after imposing $e \le \min\{e_{\mathrm{ca}}, 1/(2\max\{C_{\mathrm{ca}}, 1\})\}$ one has $\frac{1}{2} \le \|u_Q\| \le \frac{3}{2}$ and in particular $u_Q \ne 0$. Second, `lem-hcb-column-hilbert-squared` (§17) at $n = 1$, with its variable corner specialised to Q and $X = u_Q \in \mathcal{S}_Q$, compares the inner product with the norm: $\langle u_Q, u_Q\rangle = \alpha^2$, so $|\alpha^2 - \|u_Q\|^2| \le C_{\mathrm{col}} e \|u_Q\|^2$, and for $C_{\mathrm{col}} e \le \frac{1}{2}$ the inequality $|\sqrt{1+t} - 1| \le |t|$ on $|t| \le \frac{1}{2}$ gives $\alpha > 0$ and $|\alpha/\|u_Q\| - 1| \le C_{\mathrm{col}} e$. Adding the two through $\|u_Q\| \le \frac{3}{2}$,

$$|\alpha - 1| \le |\alpha - \|u_Q\|| + |\|u_Q\| - 1| \le \big(\tfrac{3}{2} C_{\mathrm{col}} + C_{\mathrm{ca}}\big) e,$$

so $A = 2C_{\mathrm{col}} + C_{\mathrm{ca}}$ works, with e_0 the minimum of the thresholds just used.

The column estimate. Fix $n \ge 1$ and $Z \in M_n \otimes \mathcal{S}_{P,Q}$ and put $H := \big(\mathrm{Ha}^Q_{P,Q}\big)_n(Z)$. The comparison is made *on the adjoint side*, where the column-action lemma is directly applicable. By `lem-hcb2-amplified-adjointness` (§12), $H^\dagger = \big(\mathrm{Ha}^Q_{Q,P}\big)_n(Z^\dagger)$, and by `def-canonical-corner-identification` $J_{P,Q,n}(Z)^\dagger = J_{Q,P,n}(Z^\dagger)$; so it suffices to compare those two. Applying `lem-hcb1-column-action` (§20) with corner variables (Q, P), matrix symbol $Z^\dagger$ and an arbitrary column X, and using $\|Z^\dagger\| = \|Z\|$,

$$q_Q\big(H^\dagger X - Z^\dagger \cdot X\big) \ \le\ C_{\mathrm{act}}\, e\, \|Z\|\, q_P(X). \tag{79}$$

The remaining term is exactly scalar. Put $d := J_{P,Q,n}(Z)^\dagger X \in \mathbb{C}^n$. Entrywise, the column-Hilbert displays and the defining formula for J give $[Z^\dagger \cdot X]_i = \sum_k Z^\dagger_{ki} \cdot X_k = d_i\, u_Q$; under the canonical identification by q_0 that vector *is* αd, whereas $J_{P,Q,n}(Z)^\dagger X$ is d. Hence

$$q_Q\big(Z^\dagger \cdot X - J_{P,Q,n}(Z)^\dagger X\big) \;=\; |\alpha - 1|\, \|d\|_2 \;\le\; A\, e\, (1 + C_J e)\, \|Z\|\, q_P(X), \tag{80}$$

the last step bounding $\|d\|_2$ by $\|J_{P,Q,n}(Z)\|\, q_P(X)$ and then invoking `lem-hcb4-canonical-gram` (§28) — its only use in this proof. Shrinking e below e_0, the two external thresholds, e_J and $1/\max\{C_J, 1\}$ makes the factor $(1 + C_J e)$ at most 2, so adding (79) and (80) by the triangle inequality and taking the supremum over nonzero X gives $\|H^\dagger - J_{P,Q,n}(Z)^\dagger\| \le C_* e \|Z\|$ with $C_* = C_{\mathrm{act}} + 2A$; Hilbert-space adjunction preserves the operator norm, so $\|H - J_{P,Q,n}(Z)\|$ obeys the same bound. No constant and no threshold depends on n.

Row transfer and completion. For $W \in M_n \otimes \mathcal{S}_{Q,P}$, adjointness gives $\big(\mathrm{Ha}^Q_{Q,P}\big)_n(W) = \big(\big(\mathrm{Ha}^Q_{P,Q}\big)_n(W^\dagger)\big)^\dagger$ and the definition gives $J_{Q,P,n}(W) = J_{P,Q,n}(W^\dagger)^\dagger$; with $\|W^\dagger\| = \|W\|$ and adjoint norm invariance, the column estimate transfers verbatim. Taking suprema over nonzero Z and W turns both pointwise bounds into map-norm bounds, so (77) holds with $C_{\mathrm{sp}} = C_*$ and e_{sp} the positive minimum of the thresholds collected above.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb4-canonical-closeness`: 11 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Registry imports: `lem-hcb-column-hilbert-squared`, `lem-hcb1-column-action`, `lem-hcb2-amplified-adjointness`, `lem-hcb4-canonical-gram`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the exported tree visibly consumes the first five. It is the "canonical-identity closeness" clause of the parent `conj-hcb` (§31) — one of the eight suppliers that parent's tree cites directly — and it supplies the hypothesis $\|A_n - J_n\| \le C_{\mathrm{sp}} e$ of the Neumann step in §30. Nothing here bears on `op-classical`, which is open.

30 The canonical Ha inverse estimate

Lemma 68 (`lem-hcb4-canonical-inverse`). *There are universal $C_{\mathrm{sp,inv}} < \infty$ and $e_{\mathrm{sp,inv}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{sp,inv}}$ has the special maps $\mathrm{Ha}^Q_{P,Q}$ and $\mathrm{Ha}^Q_{Q,P}$ completely bijective, with their amplified inverses differing from the corresponding canonical inverses by at most*

$$\big\|\big((\mathrm{Ha}^Q_{P,Q})_n\big)^{-1} - J^{-1}_{P,Q,n}\big\| \;\le\; C_{\mathrm{sp,inv}}\, e, \tag{81}$$

and likewise for the row pair $\big((\mathrm{Ha}^Q_{Q,P})_n, J_{Q,P,n}\big)$. Transcription note. *"Completely bijective" means bijective at* every *amplification level $n \ge 1$, not merely at level one; and the estimate is uniform in n. Unlike the diagonal and off-diagonal inverse statements of §§26–27, this lemma is* unconditional*: no level-one hypothesis is assumed, because the comparison map J_n is bijective by construction and the smallness of e does the rest.*

Registry contract (verbatim).

```
Canonical Ha inverse estimate: there are universal C_sp,inv < infinity and e_sp,inv > 0 such that
every H-CB datum with e <= e_sp,inv has the special maps Ha^Q_{P,Q} and Ha^Q_{Q,P} completely
bijective, with their amplified inverses differing from the corresponding canonical inverses by at
most C_sp,inv*e.
```

Proof (prose account of the af-validated tree). The mathematics is a perturbation of an invertible map by a small one; the length of the tree is due almost entirely to justifying that the target space is Banach, which is where the Neumann series lives.

The ledger and the canonical inverse. Let C_J, e_J come from `lem-hcb4-canonical-gram` (§28) and $C_{\mathrm{sp}}, e_{\mathrm{sp}}$ from `lem-hcb4-canonical-closeness` (§29), and put

$$e_{\mathrm{sp,inv}} := \min\Big\{e_J,\ e_{\mathrm{sp}},\ \tfrac{1}{4(C_J + C_{\mathrm{sp}} + 1)}\Big\}, \qquad C_{\mathrm{sp,inv}} := 4\big(C_{\mathrm{sp}} + 1\big). \tag{82}$$

The last entry of the threshold forces both $C_J e \le \frac14$ and $C_{\mathrm{sp}} e \le \frac14$ simultaneously. For either special ordered corner and every n, the canonical map J_n is the amplified coefficient-space identification of `def-canonical-corner-identifications` — the column map, or the row map defined as its adjoint — hence an algebraic linear bijection onto the indicated operator corner, with the corresponding canonical inverse. Its inverse is bounded: the lower half of the Gram estimate reads $\|J_n Z\| \ge (1 - C_J e)\|Z\|$, and applying it to $Z = J_n^{-1} Y$ with $C_J e \le \frac14$ gives

$$\big\|J_n^{-1}\big\| \;\le\; \tfrac43. \tag{83}$$

The quantitative Neumann step. The abstract step used twice below is: if $J : X \to Y$ is a bounded bijection onto a *Banach* space Y with $\|J^{-1}\| \le \frac43$, and $A : X \to Y$ satisfies $\|A - J\| \le C_{\mathrm{sp}} e$ with $C_{\mathrm{sp}} e \le \frac14$, then A is bijective and $\|A^{-1} - J^{-1}\| \le \frac83 C_{\mathrm{sp}} e$. Put $T := (A - J)J^{-1} : Y \to Y$, so that $A = (I_Y + T)J$ and $\|T\| \le \|A - J\|\|J^{-1}\| \le \frac43 C_{\mathrm{sp}} e \le \frac13$. Since Y is Banach so is $B(Y)$, and the partial sums $S_N = \sum_{k=0}^N (-T)^k$ are Cauchy there by $\sum_{k\ge0} \|T\|^k \le (1 - \|T\|)^{-1}$; passing to the limit in the finite geometric identities $(I_Y + T)S_N = S_N(I_Y + T) = I_Y - (-T)^{N+1}$ and using $\|T\|^{N+1} \to 0$ identifies the limit as the two-sided inverse, with $\|(I_Y + T)^{-1}\| \le \frac32$. Then $A^{-1} = J^{-1}(I_Y + T)^{-1}$ and $(I_Y + T)^{-1} - I_Y = -(I_Y + T)^{-1} T$ give

$$\big\|A^{-1} - J^{-1}\big\| \;\le\; \tfrac43 \cdot \tfrac32 \cdot \tfrac43\, C_{\mathrm{sp}} e \;=\; \tfrac83\, C_{\mathrm{sp}}\, e. \tag{84}$$

Why the targets are Banach. This is not a formality, and the tree does not treat it as one. In an H-CB datum the ambient $\mathcal{A}$ is complete by `def-extended-epsilon-cstar-algebra`, and `def-compressed-corner` states that $\mathrm{Co}_{P,Q}$ is an exact bounded idempotent with $\mathcal{S}_{P,Q} = \operatorname{Im}\mathrm{Co}_{P,Q} = \operatorname{Ker}(1-\mathrm{Co}_{P,Q})$; a kernel of a continuous map is closed, and a closed subspace of a Banach space is Banach, so $\mathcal{S}_{P,Q}$ is complete in the *inherited* norm. The norm actually carried by the target is the Euclidean one, and the two are equivalent by the Gram estimate at $n=1$: there $J_{P,Q,1}(Z)c = Zc$, so $\|J_{P,Q,1}(Z)\| = \|Z\|_{\mathrm{Euc}}$ and $(1-C_J e)\|Z\| \le \|Z\|_{\mathrm{Euc}} \le (1+C_J e)\|Z\|$ with $C_J e \le \frac{1}{4}$. Hence $(\mathcal{S}_{P,Q}, \|\cdot\|_{\mathrm{Euc}})$ is complete, its finite Hilbertian sum $\mathbb{C}^n \otimes \mathcal{S}_{P,Q}$ is complete, and since $B(E,F)$ is Banach whenever F is, both actual targets $B(\mathbb{C}^n, \mathbb{C}^n \otimes \mathcal{S}_{P,Q})$ and $B(\mathbb{C}^n \otimes \mathcal{S}_{P,Q}, \mathbb{C}^n)$ — the column and row targets of `def-ha-map` and `def-canonical-corner-identifications` — are Banach.

Assembly. Fix $n \ge 1$ and either special corner, and let A_n be $\big(\mathrm{Ha}^{Q}_{P,Q}\big)_n$ or $\big(\mathrm{Ha}^{Q}_{Q,P}\big)_n$. Canonical closeness gives $\|A_n - J_n\| \le C_{\mathrm{sp}} e$, (83) gives the bijection J_n with $\|J_n^{-1}\| \le \frac{4}{3}$, and (82) gives $C_{\mathrm{sp}} e \le \frac{1}{4}$; so the Neumann step applies and $\|A_n^{-1} - J_n^{-1}\| \le \frac{8}{3} C_{\mathrm{sp}} e \le 4(C_{\mathrm{sp}}+1)e = C_{\mathrm{sp,inv}} e$. The constants do not depend on n or on the corner, so both special maps are completely bijective with the uniform estimate (81).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb4-canonical-inverse`: 19 nodes, all `validated`, root `validated`, taint clean. Six of the nineteen exported nodes form the completeness branch.

Role in the chain. Registry imports: `lem-hcb4-canonical-gram`, `lem-hcb4-canonical-closeness`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the exported tree visibly consumes the first two. Its `defs` line records `def-ha-map`, `def-hcb-datum`, `def-canonical-corner-identificati` `def-compressed-corner` (the range/kernel identity) and `def-extended-epsilon-cstar-algebra` (completeness of $\mathcal{A}$); the last two are consumed essentially in the validated closed-corner completeness branch. Downstream, this lemma is one of the sixteen declared dependencies of `conj-hcb` (§31); the parent's own exported tree cites eight suppliers directly and this is not among them, so the parent consumes it through the registry rather than in its validated derivation. Nothing here bears on `op-classical`, which is open.

31 The H-CB parent contract

Lemma 69 (`conj-hcb`). *There are universal $C_H < \infty$ and $e_H > 0$ such that, whenever $e = \delta + \varepsilon \leq e_H$, Q is a level-one one-dimensional δ-projection in an extended ε-C^*-algebra $\mathcal{A}$, and P, R, S are δ-projections, the maps $1_{M_n} \otimes \mathrm{Ha}^Q_{P,R}$ — read, under the* COL-HILB *identification, as operators from $\mathbb{C}^n \otimes \mathcal{S}_{R,Q}$ to $\mathbb{C}^n \otimes \mathcal{S}_{P,Q}$ — satisfy for every n the adjoint equality, a product defect at most $C_H e\|Z\|\|W\|$, and the uniform unit, upper-norm, homomorphism and canonical-identity closeness estimates required by `lem_extension`; moreover*

1. *if the level-one lower modulus of $\mathrm{Ha}^Q_{P,P}$ is at least $\frac{1}{4}$, then every amplification has lower modulus at least $1 - C_H e$;*
2. *if $\mathrm{Ha}^Q_{P,P}$ is* also *bijective at level one, then every amplification is bijective with inverse norm at most $1 + C_H e$;*
3. *the analogous off-diagonal inverse bound for $\mathrm{Ha}^Q_{P,R}$ is asserted* only *when $\mathrm{Ha}^Q_{P,R}$ is bijective at level one and $\mathrm{Ha}^Q_{R,R}$ satisfies the diagonal lower-modulus hypothesis;*

all constants independent of n, $\dim \mathcal{A}$, block count and block dimensions. Transcription note. *`lem_extension` is a label in the pinned source, not a registry identifier: it names the consumer whose required estimate list this contract is calibrated to. The three clauses above are conditional as written; nothing in this document establishes their level-one hypotheses for any particular datum.*

Registry contract (verbatim).

```
H-CB: there are universal C_H < infinity and e_H > 0 such that, whenever e=delta+epsilon <= e_H, Q
is a level-one one-dimensional delta-projection in an extended epsilon-C*-algebra A, and P,R,S are
delta-projections, the maps 1_{M_n} tensor Ha^Q_{P,R}, under the COL-HILB identification with
operators on C^n tensor S_{R,Q} and C^n tensor S_{P,Q}, satisfy for every n the adjoint equality,
product defect at most C_H*e*||Z||||W||, and the uniform unit, upper-norm, homomorphism, and
canonical-identity closeness estimates required by lem_extension; moreover, if the level-one lower
modulus of Ha^Q_{P,P} is at least 1/4, then every amplification has lower modulus at least 1-C_H*e,
and if Ha^Q_{P,P} is also bijective at level one then every amplification is bijective with inverse
norm at most 1+C_H*e; the analogous off-diagonal inverse bound for Ha^Q_{P,R} is asserted only when
Ha^Q_{P,R} is bijective at level one and Ha^Q_{R,R} satisfies that diagonal lower-modulus
hypothesis; all constants independent of n, dim A, block count, and block dimensions.
```

Proof (prose account of the af-validated tree). The tree is a *composition*: eleven nodes, one for the constant ledger, one per asserted clause, one for uniformity. No new analysis is done at this level, and that is the point — every estimate has already been proved matrix-uniformly in §§12–27 and in the canonical tier (§§28–30), so the parent's only work is to exhibit one constant pair that serves them all simultaneously.

The ledger. Let the named coefficients and thresholds be those supplied by `lem-hcb2-amplified-adjointness` (e_{adj}), `lem-hcb2-product-defect` ($C_{\mathrm{prod}}, e_{\mathrm{prod}}$), `lem-hcb3-diagonal-unit` ($C_{\mathrm{unit}}, e_{\mathrm{unit}}$), `lem-hcb3-diagonal-upp` ($C_{\mathrm{up}}, e_{\mathrm{up}}$), `lem-hcb3-diagonal-lower-modulus` ($C_{\mathrm{diag}}, e_{\mathrm{diag}}$), `lem-hcb3-diagonal-inverse` ($C_{\mathrm{inv}}, e_{\mathrm{inv}}$), `lem-hcb3-offdiagonal-inverse` ($C_{\mathrm{rect}}, e_{\mathrm{rect}}$) and `lem-hcb4-canonical-closeness` ($C_{\mathrm{sp}}, e_{\mathrm{sp}}$). Put

$$C_H := \max\{1, C_{\mathrm{prod}}, C_{\mathrm{unit}}, C_{\mathrm{up}}, C_{\mathrm{diag}}, C_{\mathrm{inv}}, C_{\mathrm{rect}}, C_{\mathrm{sp}}\}, \qquad e_H := \min\{1, e_{\mathrm{adj}}, e_{\mathrm{prod}}, e_{\mathrm{unit}}, e_{\mathrm{up}}, e_{\mathrm{diag}}, e_{\mathrm{inv}}, e_{\mathrm{rect}}, e_{\mathrm{sp}}\}. \tag{85}$$

A finite maximum of finite universal constants is finite and universal; a finite minimum of positive universal thresholds is positive and universal. Consequently $e \le e_H$ meets *every* imported threshold at once, and each imported error coefficient is dominated by C_H — which is the only step where the clauses interact.

The clauses. Each is now the corresponding import, read at $e \le e_H$ and with its coefficient enlarged to C_H. For $Z \in M_n \otimes \mathcal{S}_{P,R}$ the adjoint clause is the exact identity $\big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z)^\dagger = \big(\mathrm{Ha}^{Q}_{R,P}\big)_n(Z^\dagger)$; for $Z \in M_n \otimes \mathcal{S}_{P,S}$ and $W \in M_n \otimes \mathcal{S}_{S,R}$ the product-defect clause is

$$\big\|\big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z \cdot W) - \big(\mathrm{Ha}^{Q}_{P,S}\big)_n(Z)\,\big(\mathrm{Ha}^{Q}_{S,R}\big)_n(W)\big\| \;\le\; C_H\, e\, \|Z\| \|W\|, \tag{86}$$

which is simultaneously the asserted uniform amplified homomorphism estimate; the unit clause is $\|\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(I_n \otimes u_P) - I\| \le C_H e$; the upper-norm clause is $\|\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(Z)\| \le (1 + C_H e)\|Z\|$; and the canonical-identity clause is

$$\max\{\|\big(\mathrm{Ha}^{Q}_{P,Q}\big)_n - J_{P,Q,n}\|,\ \|\big(\mathrm{Ha}^{Q}_{Q,P}\big)_n - J_{Q,P,n}\|\} \;\le\; C_H\, e,$$

with J the maps fixed by `def-canonical-corner-identifications`. The three conditional clauses are, in order, `lem-hcb3-diagonal-lower-modulus`, `lem-hcb3-diagonal-inverse` (§26) and `lem-hcb3-offdiagon` (§27), each carrying its level-one hypotheses unchanged into the parent statement.

Uniformity. The witnesses (85) are finite maxima and minima of universal quantities only. No entry depends on n, on $\dim \mathcal{A}$, on the block count or on the block dimensions, and no clause involves an entrywise sum over the amplification index; the hostile verifier looked for an n-growth family and found none.

Status. `proved`; `af: validated` (T0). Workspace `proofs/conj-hcb`: 11 nodes, all `validated`, root `validated`, taint clean. The identifier keeps its historical `conj-` slug; registry ids are stable.
Contract amendment of record. The conjecture as originally transcribed asserted the inverse estimate *unconditionally.* That clause is false: an exact $\mathbb{C} \oplus \mathbb{C}$ counterexample refutes it, and the hostile verifier confirmed the refutation as genuine. The contract above carries the verifier's replacement clause verbatim. The verifier also recorded that this does not weaken what the EXT tier consumes, because at the point of use the relevant map h_{11} is first shown to be a level-one $O(e)$-isomorphism, so both conditional hypotheses are met at a universal smallness threshold. Three further corrections were wording-level: the compressed-unit input splits into a general upper bound and a nonvanishing-only two-sided bound; the phrase "established separately" means a universal $1 - O(e)$ lower bound threshold-shrunk to $\frac{1}{4}$, not bare bijectivity; and bijectivity of the canonical corners comes from the Neumann condition, not from two-sided norm bounds alone.
Role in the chain. The registry shard lists sixteen dependencies — the whole COMP-fed H-CB tier including `lem-hcb0-compressed-associator`, `lem-hcb-column-hilbert-squared`, `lem-hcb1-variational-` `lem-hcb1-column-action`, `lem-compcb-corner-algebra`, `lem-hcb3-uniform-square-lower` and the canonical tier `lem-hcb4-canonical-gram` (§28), `lem-hcb4-canonical-closeness` (§29), `lem-hcb4-canonical` (§30); the exported tree cites eight of them directly, the others entering through those eight. All sixteen are `proved` with `af: validated`, and all sixteen are now reproduced in this document. Downstream, `conj-hcb` is the principal input to `conj-extcb` (§37), itself `proved` with `af: validated` and reproduced below; the conditional clauses above are consumed there, after their level-one hypotheses have been established. The registry shard additionally records the paper-proof's explicit relative witnesses $C_H = 4000c$ and $e_H = 1/(10000c)$, with c the maximum of the sanctioned COMP-CB and corrected COL-HILB constants; the validated tree does not use them, deriving (85) instead. Nothing here bears on `op-classical`, which is open.

32 One-dimensional corner calculus

The EXT tier works with the corners of a *closed EXT-CB datum* (`def-extcb-datum`) and asks for one map on a matrix algebra of one larger size. The two results of this section are its arithmetic base: through a one-dimensional δ-projection the compressed product is multiplicative on norms up to $1 \pm O(e)$, and consequently a corner between two one-dimensional projections is at most a line. Both are level-one statements; the amplified content of the tier is carried elsewhere.

Lemma 70 (`lem-extcb-one-dimensional-product`). *There are universal $C_{PQR} < \infty$ and $e_{PQR} > 0$ such that, for $e = \delta + \varepsilon \le e_{PQR}$, if Q is one-dimensional then*

$$\big|\, \|X \cdot Y\| - \|X\|\|Y\| \,\big| \;\le\; C_{PQR}\, e\, \|X\|\|Y\| \qquad \text{for } X \in \mathcal{S}_{P,Q},\; Y \in \mathcal{S}_{Q,R}. \tag{87}$$

Transcription note. *One-dimensionality is required of the* middle *index Q — the index shared by the two corners — and of nothing else; P and R are arbitrary δ-projections. "One-dimensional" means* $\dim \mathcal{S}_Q = 1$ *(`def-one-dimensional-delta-projection`), and the statement is level-one.*

Registry contract (verbatim).

```
Level-one one-dimensional corner product: there are universal C_PQR < infinity and e_PQR > 0 such
that, for e=delta+epsilon <= e_PQR, if Q is one-dimensional then abs(||X dot Y||-||X||||Y||) <=
C_PQR*e*||X||||Y|| for X in S_{P,Q} and Y in S_{Q,R}.
```

Proof (prose account of the af-validated tree). The tree proves the *squared* comparison and takes a square root at the end; the squared form is what the algebra produces, because the only handle on X is $X^\dagger \cdot X$.

A uniform ledger. One-dimensionality gives $\mathcal{S}_Q \neq 0$, and vanishing of the corner characterises the small alternative of `def-delta-projection`; hence Q is nonvanishing and `lem-compcb-corner-algebra` makes $\mathcal{S}_Q$ an extended $C_{\mathrm{ca}}e$-C^*-algebra with unit $u_Q = \mathrm{Co}_Q(Q)$ and $|\|u_Q\| - 1| \le C_{\mathrm{ca}}e$. The dichotomy bounds $\|P\|, \|Q\|, \|R\|$ by a universal B; the compressed-product display supplies a universal C_{dot} with $\|A \cdot B - AB\| \le C_{\mathrm{dot}}e\|A\|\|B\|$ on compatible pairs. One consequence is used repeatedly: u_Q is a left unit on $\mathcal{S}_{Q,R}$ up to $O(e)$,

$$\|u_Q Y - Y\| \;\le\; C_L\, e\, \|Y\|, \tag{88}$$

obtained by combining $\|u_Q - Q\| = O(e)$ (from $\mathrm{Co}_Q(A) = Q(AQ) + O(e)\|A\|$ and $\|Q^2 - Q\| \le \delta$) with $\|QY - Y\| = O(e)\|Y\|$ (from $Y = \mathrm{Co}_{Q,R}(Y) = Q(YR) + O(e)\|Y\|$ and one associator move). Write C_0 for the maximum of these constants and shrink the threshold below $1/(2C_0)$, so that $\|u_Q\| \ge \frac{1}{2}$.

Scalarisation. Fix $X \in \mathcal{S}_{P,Q}$. The compression adjoint identity puts $X^\dagger$ in $\mathcal{S}_{Q,P}$, so $a := X^\dagger \cdot X$ lies in $\mathcal{S}_Q$; since $\dim \mathcal{S}_Q = 1$ and $u_Q \neq 0$, the unit spans, and $a = h\, u_Q$ for a unique scalar h, real because $a^\dagger = a$ and $u_Q^\dagger = u_Q$. Comparing a with the ambient square and using the ε-C^* axioms, $|\|a\| - \|X\|^2| \le (C_0 + 1)e\|X\|^2$; and $\|a\| = |h|\, \|u_Q\|$ together with $|\|u_Q\| - 1| \le C_0 e$ and $\|u_Q\| \ge \frac{1}{2}$ converts this into

$$\big|\, |h| - \|X\|^2 \,\big| \;\le\; C_1\, e\, \|X\|^2, \qquad |h| \;\le\; C_h \|X\|^2. \tag{89}$$

The squared comparison. Put $r = \|X\|$, $s = \|Y\|$, $W = X \cdot Y$ and $Z = XY$ for the ambient product. Three estimates combine. First, $\|W - Z\| \le C_0 e\, rs$ and $|\|Z^\dagger Z\| - \|Z\|^2| \le \varepsilon\|Z\|^2$ give

$||W\|^2 - \|Z^\dagger Z\|| \le C_3 e\, r^2 s^2$. Second, exact anti-multiplicativity of the involution, two applications of the ε-associator axiom, the replacement of $X^\dagger X$ by $a = h\, u_Q$ and then (88) give

$$\left\|(XY)^\dagger(XY) - h\, Y^\dagger Y\right\| \;\le\; C_4\, e\, r^2 s^2. \tag{90}$$

Third, $||Y^\dagger Y\| - s^2| \le \varepsilon s^2$ and (89) give $||h\, Y^\dagger Y\| - r^2 s^2| \le C_5 e\, r^2 s^2$. The reverse triangle inequality then yields $||W\|^2 - r^2 s^2| \le C_2 e\, r^2 s^2$ with $C_2 = C_3 + C_4 + C_5$.

Square root. If $rs = 0$ then $X = 0$ or $Y = 0$, and bilinearity of the compressed product makes $W = 0$, so (87) holds exactly. Otherwise $\|W\| + rs \ge rs > 0$ and the factorisation $|\alpha - \beta| = |\alpha^2 - \beta^2|/(\alpha + \beta)$ for nonnegative α, β turns the squared comparison into (87) with $C_{PQR} = C_2$ and e_{PQR} the minimum of the finitely many universal thresholds.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb-one-dimensional-product`: 25 nodes, all `validated`, root `validated`, taint clean.

Lemma 71 (`lem-extcb-one-dimensional-corner-dimension`). *For sufficiently small universal $\delta + \varepsilon$, if P and Q are one-dimensional δ-projections then* $\dim \mathcal{S}_{P,Q} \le 1$.

Registry contract (verbatim).

```
Level-one one-dimensional corner dimension: for sufficiently small universal delta+epsilon, if P
and Q are one-dimensional delta-projections then dim S_{P,Q} <= 1.
```

Proof (prose account of the af-validated tree). Set $e_0 = \min\{e_{PQR},\, 1/(2(C_{PQR} + 1))\}$ and let $e \le e_0$. For nonzero $X, Y \in \mathcal{S}_{P,Q}$ the compression adjoint identity puts $X^\dagger$ in $\mathcal{S}_{Q,P}$ with $\|X^\dagger\| = \|X\|$, and $X^\dagger \cdot Y$ lies in $\mathcal{S}_Q$. Apply Lemma 70 to the triple (Q, P, Q) — whose middle index is P, one-dimensional by hypothesis — to get

$$\|X^\dagger \cdot Y\| \;\ge\; (1 - C_{PQR} e)\, \|X\| \|Y\| \;\ge\; \tfrac{1}{2} \|X\| \|Y\| \;>\; 0. \tag{91}$$

This is the whole analytic content; the rest is linear algebra. If $\mathcal{S}_{P,Q} = 0$ there is nothing to prove. Otherwise fix a nonzero X and let $T_X : \mathcal{S}_{P,Q} \to \mathcal{S}_Q$ be $T_X(Y) = X^\dagger \cdot Y$, linear because the compressed product is bilinear. By (91), $T_X(Y) \ne 0$ whenever $Y \ne 0$, so $\operatorname{Ker} T_X = 0$; an injective linear map into $\mathcal{S}_Q$, which is one-dimensional by hypothesis, forces $\dim \mathcal{S}_{P,Q} \le \dim \mathcal{S}_Q = 1$.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb-one-dimensional-corner-dimension`: 8 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Both lemmas import `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the second imports the first. Together they are the approximate-C^* substitute for the exact fact that a corner through a rank-one projection is a line: the product statement is the result the pinned source invokes for transitivity of equivalence of one-dimensional δ-projections (`def-one-dimensional-delta-projection`), and the dimension statement is consumed by `lem-extcb1-cross-co` (§34), which is where that transitivity is actually used.

33 Corner-dimension additivity

Lemma 72 (`lem-extcb-corner-dimension-additivity`). *For two finite-dimensional commutative C^*-algebras with projection bases and non-unital sufficiently accurate inclusions v, w, the compressed corner $\mathcal{S}_{v(I),w(I)}$ is linearly bijective to*

$$\bigoplus_{j,k} \mathcal{S}_{v(\Pi_j),w(\Sigma_k)}. \tag{92}$$

Transcription note. *"Projection basis" is* **`def-projection-basis`** *($\Pi_j^\dagger = \Pi_j$, $\Pi_j\Pi_k = \delta_{jk}\Pi_k$, $\sum_j \Pi_j = I$, and likewise $\{\Sigma_k\}$); "inclusion" is* **`def-extended-delta-inclusion`**, *and* non-unital *means the unit of the source need not be carried to the unit of the ambient algebra — $v(I)$ is only an approximate projection there. "Sufficiently accurate" is a universal ceiling on the inclusion defect plus the ambient ε; the tree fixes it as the minimum of finitely many universal thresholds, independent of the two basis cardinalities. The conclusion is a* linear *bijection, bounded with bounded inverse; no isometry and no bound uniform in the number of blocks is asserted, and none is needed downstream, where only the dimension count is consumed.*

Registry contract (verbatim).

```
Level-one corner-dimension additivity: for two finite-dimensional commutative C*-algebras with
projection bases and non-unital sufficiently accurate inclusions v,w, the compressed corner
S_{v(I),w(I)} is linearly bijective to the direct sum over j,k of S_{v(Pi_j),w(Sigma_k)}.
```

Proof (prose account of the af-validated tree). The tree proves one binary splitting lemma, transports it across the second index by the involution, and then runs two finite inductions. Write $P_j = v(\Pi_j)$, $Q_k = w(\Sigma_k)$, $P_{[1,j]} = \sum_{r\le j} P_r$ and $Q_{[1,k]} = \sum_{s\le k} Q_s$.

Images of a projection basis. Each Π_j and each partial sum $\sum_{r<j}\Pi_r$ is an exact Hermitian projection of norm one in the source, and distinct basis projections annihilate each other. Since v is linear, star-preserving, δ-multiplicative and two-sidedly norm-bounded, every image is Hermitian with norm at most $1+\delta$ and $\|P_j^2 - P_j\| \le \delta$; the partial-sum identities $P_{[1,j]} = P_{[1,j-1]} + P_j$ and $P_{[1,p]} = v(I)$ are *exact*, by linearity alone, while orthogonality survives only approximately,

$$\|P_{[1,j-1]}P_j\| \le \delta, \qquad \|P_j P_{[1,j-1]}\| \le \delta, \tag{93}$$

the second from the first by star-preservation. The same holds for w and the Q_k. This mixture — exact additivity, approximate orthogonality — is what the splitting lemma is built to consume.

Binary splitting. Let $R = R_0 + R_1$ and T be uniformly bounded approximate projections with R_0, R_1 approximately orthogonal, and let e dominate the ambient associativity and product errors, all projection defects and both orthogonality defects. Define

$$A(X_0, X_1) = \mathrm{Co}_{R,T}(X_0 + X_1), \qquad B(X) = \big(\mathrm{Co}_{R_0,T}X,\ \mathrm{Co}_{R_1,T}X\big),$$

bounded and linear because every compression is a bounded idempotent whose range is the displayed corner. The calculus of `def-compressed-corner` gives, uniformly, $X = U(XV) + O(e)\|X\|$ for $X \in \mathcal{S}_{U,V}$. Hence for $X_i \in \mathcal{S}_{R_i,T}$, expanding $R = R_0 + R_1$ reproduces X_i from the matching summand and leaves $O(e)\|X_i\|$ from the other — one reassociation plus $R_{1-i}R_i = O(e)$ — so $\mathrm{Co}_{R,T}X_i = X_i + O(e)\|X_i\|$; applying $\mathrm{Co}_{R_a,T}$ and using $R_aR_i = \delta_{ai}R_i + O(e)$ gives $\mathrm{Co}_{R_a,T}\mathrm{Co}_{R,T}X_i =$

$\delta_{ai}X_i+O(e)\|X_i\|$, i.e. $\|BA-I\| \le Ce$ for the max norm on the sum. In the other order, for $X \in \mathcal{S}_{R,T}$ the same replacement gives $X = R(XT) + O(e)\|X\|$ and $\mathrm{Co}_{R_i,T}X = R_i(XT) + O(e)\|X\|$; adding the two and using $R_0 + R_1 = R$ *exactly* gives $\mathrm{Co}_{R_0,T}X + \mathrm{Co}_{R_1,T}X = X + O(e)\|X\|$, and applying the bounded $\mathrm{Co}_{R,T}$, which fixes X, gives $\|AB - I\| \le Ce$. Shrink the universal threshold so that $Ce < 1$: the Neumann series invert BA and AB, so $(BA)^{-1}B$ is a left inverse of A and $B(AB)^{-1}$ a right inverse; a map with both is a bounded linear bijection.

The second index. By the compression adjoint identity and isometry of the involution, $X \mapsto X^\dagger$ is a conjugate-linear isometric bijection $\mathcal{S}_{U,V} \to \mathcal{S}_{V,U}$. Conjugating the first-index assembly for a split $T = T_0 + T_1$ by this map on domain and codomain gives a *complex*-linear bounded bijection — two conjugate-linear maps around a complex-linear one — and the adjoint identity identifies it with the canonical second-index assembly $\mathcal{S}_{R,T_0} \oplus \mathcal{S}_{R,T_1} \to \mathcal{S}_{R,T}$. No second calculation is needed.

Two inductions. Fix j. The base $k = 1$ is the identity, since $Q_{[1,1]} = Q_1$ exactly; for $k \ge 2$ the second-index splitting applies to $Q_{[1,k]} = Q_{[1,k-1]} + Q_k$, whose hypotheses are supplied uniformly by (93). Finite induction over the q basis projections, composing each step with the preceding bijection and using $Q_{[1,q]} = w(I)$, produces a bounded linear bijection $\mathcal{S}_{P_j,w(I)} \to \bigoplus_k \mathcal{S}_{P_j,Q_k}$. The first-index splitting run along $P_{[1,j]} = P_{[1,j-1]} + P_j$ produces, in the same way, $\mathcal{S}_{v(I),w(I)} \to \bigoplus_j \mathcal{S}_{P_j,w(I)}$. Composing the second with the direct sum of the first family and reassociating the finite double sum — an isometric relabelling — gives (92). Each direct sum of finitely many bounded bijections is bounded with bounded inverse, the bound being a maximum over a finite index set.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb-corner-dimension-additivity`: 39 nodes, all `validated`, root `validated`, taint clean. Several of those nodes are *re-derivations*: where a child had been written to consume a still-pending sibling, the tree carries a sharper child that discharges the same step from validated material only. That is a validation-ordering artifact of the bottom-up protocol, not additional mathematics; the mathematical content is the four movements above.

Role in the chain. Imports `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`. It is the counting engine of the EXT tier: a corner over a matrix-unit basis decomposes into its blocks, so a dimension over a composite projection is the sum of the dimensions over its parts. Its single registry consumer is `lem-extcb1-cross-corner-dimension` (§34), which applies it to the diagonal subalgebra of M_r and to the scalars, and reads off $\dim \mathcal{S}_{v(I_r),Q} = r\,d$.

34 Dimension selection in an EXT-CB datum

A closed EXT-CB datum (`def-extcb-datum`) carries δ-projections P, Q with $\|P+Q-I\| \le \delta$, an extended δ-isomorphism $v : M_r \to \mathcal{S}_P$, $\dim \mathcal{S}_Q = 1$ and $\mathcal{S}_{P,Q} \ne 0$. Nothing in the datum says how large the cross corner is. The two lemmas of this section settle it: the corner has dimension exactly r, and the diagonal Q-corner exactly 1 — the shape needed to build one map on M_{r+1}. The first lemma is the technical enabler; the second is the count.

Lemma 73 (`lem-extcb1-close-corner-dimension`)**.** *There is a universal $e_{\mathrm{close}} > 0$ such that, in an EXT-CB datum with $e \le e_{\mathrm{close}}$, the compression ranges $\mathcal{S}_{v(I_r),Q}$ and $\mathcal{S}_{P,Q}$ have the same dimension.* Transcription note. *$v(I_r)$ is the image of the* source *unit; it is an approximate projection close to P, not equal to it, and the two corners it and P cut out are literally different subspaces. Only their dimensions are claimed equal.*

Registry contract (verbatim).

```
Close-compression range invariance: there is a universal e_close > 0 such that, in an EXT-CB datum
with e <= e_close, the compression ranges S_{v(I_r),Q} and S_{P,Q} have the same dimension.
```

Proof (prose account of the af-validated tree). Write $R = v(I_r)$, $E = \mathrm{Co}_{R,Q}$ and $F = \mathrm{Co}_{P,Q}$.

R is close to P, and is itself an approximate projection. Bijectivity of v onto $\mathcal{S}_P$ makes that corner nonzero, so P is nonvanishing; $\dim \mathcal{S}_Q = 1$ makes Q nonvanishing; the nonvanishing alternative of `def-delta-projection` then bounds $\|P\|, \|Q\|$ by a universal M_1. By `lem-compcb-corner-algebra` the corner $\mathcal{S}_P$ is an extended $C_{\mathrm{ca}}e$-C^*-algebra with unit $u_P = \mathrm{Co}_P(P)$, and the unit clause for the extended δ-isomorphism v gives $\|R - u_P\| \le \delta$; combining with $\|u_P - P(PP)\| = O(e)\|P\|$ and $\|P(PP) - P\| \le \big((1+\varepsilon)\|P\| + 1\big)\delta$ yields

$$\|R - P\| \;\le\; C_1 e. \tag{94}$$

Star-preservation gives $R^\dagger = R$, δ-multiplicativity gives $\|R \cdot R - R\| \le \delta$ inside $\mathcal{S}_P$, and the compressed-versus-ambient comparison converts this into $\|R^2 - R\| \le C_2 e$ in $\mathcal{A}$: R is a $(C_R e)$-projection there.

Two close bounded idempotents. With the common adjusted defect $\delta' = C_R e$, the pairs (R, Q) and (P, Q) both satisfy the hypotheses of `lem-compcb-amplified-compression-identities` at $n = 1$, so $E^2 = E$ and $F^2 = F$. Each compression is $O(e)$-close to the corresponding left-times-right multiplication, and $\big(L_R R_Q - L_P R_Q\big)(X) = (R-P)(XQ)$ has norm at most $(1+\varepsilon)^2 \|R-P\|\|Q\|\|X\|$; with (94) this gives universal C_0, M_0 with

$$\|E - F\| \;\le\; C_0 e, \qquad \|E\| \;\le\; M_0. \tag{95}$$

Range transport. If E, F are bounded idempotents on a Banach space with $\|E\| \le M$ and $\|E - F\|(2M+1) < 1$, put $W = FE + (I-F)(I-E)$. Idempotence gives the two exact identities

$$W - I = (F - E)(2E - I), \qquad WE = FW = FE, \tag{96}$$

so $\|W - I\| \le \|F - E\|(2\|E\| + 1) < 1$ and W is invertible by its Neumann series. The relation $WE = FW$ sends $\mathrm{Im}\, E$ into $\mathrm{Im}\, F$ and, read as $EW^{-1} = W^{-1}F$, sends $\mathrm{Im}\, F$ back; hence W restricts to a linear bijection $\mathrm{Im}\, E \to \mathrm{Im}\, F$ and the two ranges have equal dimension. Taking e_{close}

below the thresholds of the two preceding steps and below $1/[2C_0(2M_0+1)]$ makes (95) satisfy that hypothesis, and $\operatorname{Im}\mathrm{Co}_{R,Q} = \mathcal{S}_{R,Q}$, $\operatorname{Im}\mathrm{Co}_{P,Q} = \mathcal{S}_{P,Q}$ finish it.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb1-close-corner-dimension`: 16 nodes, all `validated`, root `validated`, taint clean. This lemma exists because a hostile verdict on the EXT-CB transcription demanded that the level-one close-idempotent normalisation be made explicit, and that the selection threshold cover it.

Lemma 74 (`lem-extcb1-cross-corner-dimension`)**.** *There is a universal $e_{\mathrm{sel}} > 0$ such that every EXT-CB datum with $e \le e_{\mathrm{sel}}$ satisfies*

$$\big(\dim \mathcal{S}_{P,Q},\ \dim \mathcal{S}_{Q,Q}\big) = (r, 1). \tag{97}$$

Registry contract (verbatim).

```
EXT-CB cross-corner dimension: there is a universal e_sel > 0 such that every EXT-CB datum with e
<= e_sel satisfies (dim S_{P,Q},dim S_{Q,Q})=(r,1).
```

Proof (prose account of the af-validated tree). The count is produced by transporting the diagonal matrix units through v, showing that each transported unit is a one-dimensional projection equivalent to all the others, and then summing.

Two inclusions. Let $D_r \subseteq M_r$ be the diagonal subalgebra with projection basis $\Pi_a = E_{aa}$, let v_D be the restriction of v to D_r, put $R_a = v(E_{aa})$, and let $w : \mathbb{C} \to \mathcal{A}$ be $w(\lambda) = \lambda Q$, whose source has the one-element projection basis $\{1\}$. Restriction preserves linearity, star-preservation, δ-multiplicativity and the two-sided norm bounds, and the compressed product on $\mathcal{S}_P$ differs from the ambient one by $O(e)$ on bounded elements, so v_D is a non-unital $O(e)$-inclusion into $\mathcal{A}$ with $v_D(I) = v(I_r)$, uniformly in r. For w: nonvanishing of Q gives $|\|Q\| - 1| \le C_{\mathrm{nv}} e$, the operator-space axioms give the exact cross-norm identity $\|X \otimes Q\| = \|X\|\|Q\|$ at every level, and $\|(XY) \otimes Q - (X \otimes Q)(Y \otimes Q)\| = \|(XY) \otimes (Q - Q^2)\| \le \delta \|X\|\|Y\|$; so every amplification of w is an $O(e)$-inclusion as well.

The transported matrix units. For $a, b \le r$ let $c_{ab}(X) = E_{aa} X E_{bb}$ on M_r and put $T_{ab} = v\, c_{ab}\, v^{-1} \mathrm{Co}_P$. Since Co_P is an idempotent with range $\mathcal{S}_P = \operatorname{Im} v$, one has $\mathrm{Co}_P v = v$ and $v^{-1}\mathrm{Co}_P v = \mathrm{id}$; as $c_{ab}^2 = c_{ab}$ with one-dimensional range, T_{ab} is an *exact* bounded idempotent of rank one, with range $\mathbb{C}\, v(E_{ab})$. The tree then compares it with the corresponding compression: uniformly in a, b and r,

$$\|\mathrm{Co}_{R_a,R_b} - T_{ab}\| \ \le\ C_m e. \tag{98}$$

Three fixed-length estimates give this — each R_a is a Hermitian $C_1 e$-projection (from $\|E_{aa}\| = 1$, δ-multiplicativity and the compressed-versus-ambient comparison); the subordination bound $\|R_a(XR_b) - R_a\big((\mathrm{Co}_P X)R_b\big)\| \le C_s e\|X\|$, a finite telescope using $R_i = \mathrm{Co}_P(R_i)$ and $\mathrm{Co}_P(Y) = P(YP) + O(e)\|Y\|$; and the transported-source comparison $\|T_{ab}X - R_a\big((\mathrm{Co}_P X)R_b\big)\| \le C_3 e\|X\|$, from two applications of δ-multiplicativity in M_r and the corner-algebra bounds — together with $\|\mathrm{Co}_{R_a,R_b} X - R_a(XR_b)\| \le C_4 e\|X\|$. Since all source matrix factors have norm one, no constant depends on r.

Rank invariance. If E, F are bounded idempotents with $\|E - F\| < 1$ then $x \in \operatorname{Im} E$ with $Fx = 0$ gives $x = Ex = (E - F)x$, hence $x = 0$; interchanging E and F gives the reverse, so the two ranges have equal dimension when either is finite-dimensional. Shrinking e so that $C_m e < 1$ and applying this to (98) gives $\dim \mathcal{S}_{R_a,R_b} = \operatorname{rank} T_{ab} = 1$ for all a, b: taking $a = b$, each

R_a is a one-dimensional projection; taking $a \neq b$, the R_a are pairwise equivalent in the sense of `def-one-dimensional-delta-projection`.

Dichotomy, additivity, selection. Put $d_a = \dim \mathcal{S}_{R_a,Q}$. By `lem-extcb-one-dimensional-corner-dimension` (§32), applied to the one-dimensional projections R_a and Q with the enlarged common accuracy, $d_a \leq 1$, so $d_a \in \{0, 1\}$. The values coincide: if some $d_b = 1$ then R_b is equivalent to Q, every R_a is equivalent to R_b, and transitivity of equivalence — the registered property whose source proof is the one-dimensional product lemma of §32 — gives $d_a = 1$ for all a; otherwise all $d_a = 0$. Write d for the common value. Now `lem-extcb-corner-dimension-additivity` (§33), applied to D_r and $\mathbb{C}$ with the inclusions v_D, w, gives a linear bijection $\mathcal{S}_{v(I_r),Q} \to \bigoplus_{a=1}^{r} \mathcal{S}_{R_a,Q}$ and hence $\dim \mathcal{S}_{v(I_r),Q} = \sum_a d_a = r\,d$; Lemma 73 turns this into $\dim \mathcal{S}_{P,Q} = r\,d$. The datum assumes $\mathcal{S}_{P,Q} \neq 0$ and $r \geq 1$, so $d \neq 0$, i.e. $d = 1$ and $\dim \mathcal{S}_{P,Q} = r$. Finally $\mathcal{S}_{Q,Q} = \mathcal{S}_Q$ has dimension 1 by hypothesis, and e_{sel} is the minimum of the finitely many universal thresholds used, giving (97).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb1-cross-corner-dimension`: 30 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. The close-corner lemma imports `lem-compcb-amplified-compression-identities`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the cross-corner lemma imports both of the preceding sections and the close-corner lemma. The pair is the *selection* step of the EXT tier: it is what licenses treating $\mathcal{S}_{P,Q}$ as an r-dimensional column and $\mathcal{S}_Q$ as a line, i.e. the block shape that the four-corner merge (§36) assembles into a single map on M_{r+1}. Its consumer is `conj-extcb` (§37), now `proved` with `af: validated`; the close-corner lemma is there because a hostile verdict on that parent demanded it.

35 The four-corner assembled norm estimate

Lemma 75 (`lem-extcb-four-corner-norm`). *There are universal $K_{\mathrm{norm}} < \infty$ and $e_{\mathrm{norm}} > 0$ such that, for every four-corner merging datum with $e = \rho + \varepsilon \le e_{\mathrm{norm}}$, every $n \ge 1$ and every $X \in M_n \otimes \mathcal{B}$, the assembled map $\gamma_n = \alpha_n \Gamma_n \mu_n$, where*

$$\mu_n(X) = \big((I_n \otimes \Pi_j)\, X\, (I_n \otimes \Pi_k)\big)_{j,k}, \qquad \Gamma_n = \bigoplus_{j,k} \gamma_{jk,n}, \qquad \alpha_n\big((Y_{jk})\big) = \sum_{j,k} Y_{jk},$$

satisfies

$$(1 - K_{\mathrm{norm}} e)\|X\| \;\le\; \|\gamma_n(X)\| \;\le\; (1 + K_{\mathrm{norm}} e)\|X\|. \tag{99}$$

Transcription note. *ρ is the* datum's *common defect bound of* ***`def-four-corner-merging-datum`*** *— involution compatibility, compressed-product compatibility, diagonal-unit approximation, two-sided norm control, and the quantitative complementarity $\|P_1 + P_2 - I\| \le \rho$ of the target projections — and not the signed-picture scale of the same name in* ***`CONVENTIONS.md`****. The four maps γ_{jk} are* fixed *at level one and reused at every amplification: γ_n is not a separately chosen family.*

Registry contract (verbatim).

```
Four-corner assembled norm estimate: there are universal K_norm < infinity and e_norm > 0 such
that, for every four-corner merging datum with e = rho+epsilon <= e_norm, every n >= 1, and every X
in M_n tensor B, the assembled map gamma_n = alpha_n Gamma_n mu_n (where mu_n(X) = ((I_n tensor
Pi_j) X (I_n tensor Pi_k))_{j,k}, Gamma_n is the direct sum of the four fixed amplifications
gamma_{jk,n}, and alpha_n((Y_jk)) = sum_{jk} Y_jk) satisfies (1-K_norm*e)||X|| <= ||gamma_n(X)|| <=
(1+K_norm*e)||X||.
```

Proof (prose account of the af-validated tree). The estimate is proved in three stages, and the order matters: crude two-sided bounds first, an approximate multiplicativity of the *square* second, and only then a scalar bootstrap that turns the crude constants into $1 \pm O(e)$. A direct triangle-inequality attack on (99) does not work, because the four blocks can cancel.

Stage 1: block conditioning. Put $\pi_j = I_n \otimes \Pi_j$ and $X_{jk} = \pi_j X \pi_k$. The source projections are exactly complementary, so $X = \sum_{jk} X_{jk}$, $\|X_{jk}\| \le \|X\|$ and

$$\max_{jk}\|X_{jk}\| \;\le\; \|X\| \;\le\; \sum_{jk}\|X_{jk}\| \;\le\; 4\max_{jk}\|X_{jk}\|. \tag{100}$$

On the target side put $p_j = I_n \otimes P_j$ and $C_{jk} = 1_{M_n} \otimes \mathrm{Co}_{P_j,P_k}$. The compression construction commutes with amplification — the operator $L_{P_j} R_{P_k} + R_{P_k} L_{P_j} - 1$ amplifies entrywise, every partial sum of the defining power series satisfies $f_N(T_n) = 1_{M_n} \otimes f_N(T)$, and matrix entries are bounded coordinate maps — so $C_{jk} = \mathrm{Co}_{p_j,p_k}$ is genuinely the compression built inside $M_n \otimes \mathcal{A}$ and $\operatorname{Im} C_{jk} = M_n \otimes \mathcal{S}_{P_j,P_k}$. Quantitative complementarity gives $\|p_j^2 - p_j\| \le \rho$ and $\|p_1 + p_2 - I\| \le \rho$, whence $\|p_j\| \le 2$ and, for $j \ne l$, $\|p_j p_l\| \le 4e$. Consequently C_{jk} fixes its own corner and almost annihilates the other three: $\|C_{jk}(Y_{lm})\| \le L_c e \|Y_{lm}\|$ for $(j,k) \ne (l,m)$, with universal L_c. With $\beta(Y) = (C_{jk} Y)_{jk}$ and the max norm on the fourfold sum this reads $\|\beta\alpha - I\| \le 3L_c e$ and $\|\beta\| \le M_c$. Feeding (100) and the datum's two-sided norm clause through this extraction gives universal $c_{\mathrm{blk}} > 0$ and $B_{\mathrm{blk}} < \infty$ — concretely $c_{\mathrm{blk}} = 1/(16M_c)$ and $B_{\mathrm{blk}} = 5$ — with

$$c_{\mathrm{blk}}\|X\| \;\le\; \|\gamma_n(X)\| \;\le\; B_{\mathrm{blk}}\|X\|, \qquad \text{uniformly in } n. \tag{101}$$

Stage 2: the square defect. For U in the jk target corner and V in the lm one, two cases occur. If $k = l$ the compressed-product estimate applies at matrix level: $\|UV - U \cdot V\| \leq c_c e \|U\|\|V\|$. If $k \neq l$ the product is *small*: replacing U, V by their compressed representatives $p_j(Up_k)$ and $p_l(Vp_m)$ and moving the middle factors together by exactly three ε-associator steps exposes the factor $p_k p_l$, so $\|UV\| \leq K_{\mathrm{inc}} e \|U\|\|V\|$. Expanding $\gamma_n(X^\dagger X)$ and $\gamma_n(X)^\dagger \gamma_n(X)$ over the blocks produces sixteen terms, eight compatible and eight incompatible; the compatible ones are matched using the datum's involution and compressed-product clauses, the incompatible ones are simply small. Hence

$$\big\|\gamma_n(X^\dagger X) - \gamma_n(X)^\dagger \gamma_n(X)\big\| \;\leq\; D_{\mathrm{sq}} e \|X\|^2, \qquad D_{\mathrm{sq}} = 8\big(4 + 2c_c + 2K_{\mathrm{inc}}\big). \tag{102}$$

Stage 3: the scalar bootstrap. Fix n and let a_n and b_n be the infimum and supremum of $\|\gamma_n(X)\|/\|X\|$ over $X \neq 0$; by (101), $0 < c_{\mathrm{blk}} \leq a_n \leq b_n \leq 5$. The source is an exact C^*-algebra, so $\|X^\dagger X\| = \|X\|^2$. For the upper bound, the target lower C^* inequality and (102) give $(1-\varepsilon)\|\gamma_n(X)\|^2 \leq \|\gamma_n(X^\dagger X)\| + D_{\mathrm{sq}} e \|X\|^2 \leq (b_n + D_{\mathrm{sq}} e)\|X\|^2$, so a maximising sequence yields $(1-\varepsilon)b_n^2 \leq b_n + D_{\mathrm{sq}} e$; if $b_n > 1$ then $b_n - 1 \leq b_n(b_n - 1) \leq \varepsilon b_n^2 + D_{\mathrm{sq}} e \leq (25 + D_{\mathrm{sq}})e$. For the lower bound, the same identity read in the other direction gives $a_n \leq (1+\varepsilon)a_n^2 + D_{\mathrm{sq}} e$, and if $a_n < 1$ then $c_{\mathrm{blk}}(1 - a_n) \leq a_n(1 - a_n) \leq (25 + D_{\mathrm{sq}})e$. Therefore

$$b_n \leq 1 + (25 + D_{\mathrm{sq}})e, \qquad a_n \geq 1 - \frac{25 + D_{\mathrm{sq}}}{c_{\mathrm{blk}}}\, e,$$

which is (99) with $K_{\mathrm{norm}} = \max\{25 + D_{\mathrm{sq}}, (25 + D_{\mathrm{sq}})/c_{\mathrm{blk}}\}$ and e_{norm} the minimum of the universal ceilings of Stages 1 and 2 (and, harmlessly, $1/(2K_{\mathrm{norm}})$). The crude constant c_{blk} survives only in K_{norm}: it is what prevents the bootstrap from being vacuous, not what the final estimate looks like.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb-four-corner-norm`: 18 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`. This shard exists because of a recorded *third-stall* classification: three successive orchestrations of `lem-extcb-four-corner-merge` stalled on exactly this claim, so it was factored out of that tree — its statement extracted mechanically from the blocking challenges, with the assembly maps written out from an already-validated node — and validated on its own. Its single registry consumer is `lem-extcb-four-corner-merge` (§36), which uses it as the entire norm-control half of the merge.

36 The complete four-corner merge

Lemma 76 (`lem-extcb-four-corner-merge`). *There are universal $C_{\mathrm{merge}} < \infty$ and $a_{\mathrm{merge}} > 0$ such that four fixed bijective level-one corner maps satisfying* ***`def-four-corner-merging-datum`*** *with common defect ρ and $\rho + \varepsilon \le a_{\mathrm{merge}}$ combine into one extended $C_{\mathrm{merge}}(\rho + \varepsilon)$-isomorphism.* Transcription note. *The smallness hypothesis constrains the* total *defect $\rho + \varepsilon$, not ρ alone; that is a verdict-driven amendment of the transcribed contract, and the record of why it cannot be dropped is reproduced below. "Extended isomorphism" is* ***`def-extended-delta-inclusion`****: every amplification is a δ-homomorphism with two-sided $(1 \pm \delta)$ norm bounds, and the level-one map is bijective.*

Registry contract (verbatim).

```
Complete four-corner merge: there are universal C_merge < infinity and a_merge > 0 such that four
fixed bijective level-one corner maps satisfying def-four-corner-merging-datum with common defect
rho and rho+epsilon <= a_merge combine into one extended C_merge*(rho+epsilon)-isomorphism.
```

Proof (prose account of the af-validated tree). Write $\pi_j = I_n \otimes \Pi_j$, $X_{jk} = \pi_j X \pi_k$, and let $\mu_n, \Gamma_n, \alpha_n$ and $\gamma_n = \alpha_n \Gamma_n \mu_n$ be the assembly maps of §35. Set $e = \rho + \varepsilon$.

One map, not a family. Exact source complementarity gives $\pi_j \pi_k = \delta_{jk} \pi_j$, $\pi_1 + \pi_2 = I$ and $X = \sum_{jk} X_{jk}$; since the four $\gamma_{jk,n}$ are the amplifications of the same four level-one maps,

$$\gamma_n = 1_{M_n} \otimes \gamma_1, \qquad \gamma_1(X) = \sum_{j,k} \gamma_{jk}(\Pi_j X \Pi_k). \tag{103}$$

This is the structural point of the lemma: the conclusion is about *one* linear map and its amplifications, so nothing may be chosen level by level. Two clauses follow at once. Adjoints of source corners transpose indices, $(\pi_j X \pi_k)^\dagger = \pi_k X^\dagger \pi_j$, and the datum's involution clause matches them, so $\gamma_n(X^\dagger) = \gamma_n(X)^\dagger$ *exactly.* The source unit has only diagonal blocks, so $\gamma_n(I) - I$ splits into the two diagonal-unit errors and $1_{M_n} \otimes (P_1 + P_2 - I)$, whose norm equals its level-one value by the direct-sum norm axiom; hence

$$\|\gamma_n(I) - I\| \ \le\ 3\rho, \qquad \text{uniformly in } n. \tag{104}$$

Corner separation and the multiplicative defect. As in §35, compression commutes with amplification, so all matrix-level corner estimates are available with level-one constants; and from $p_k p_l = p_k(s - I) + (p_k I - p_k) - (p_k^2 - p_k)$ with $s = p_1 + p_2$ one gets $\|p_k p_l\| \le 5e$ for $k \ne l$ once $e \le \frac{1}{4}$. Hence a product of corner elements is compressed-product-close when the middle indices match ($\|UV - U \cdot V\| \le c_{\mathrm{prod}} e \|U\| \|V\|$) and is itself small when they do not ($\|UV\| \le c_{\mathrm{cross}} e \|U\| \|V\|$, by three associator moves that expose $p_k p_l$). Expanding $\gamma_n(XY)$ and $\gamma_n(X)\gamma_n(Y)$ over the blocks gives sixteen terms — eight with matching middle index, compared through the datum's compressed-product clause, and eight without, which are small — so

$$\|\gamma_n(XY) - \gamma_n(X)\gamma_n(Y)\| \ \le\ D_{\mathrm{mult}}\, e\, \|X\| \|Y\|, \qquad D_{\mathrm{mult}} = 8\big[1 + 2c_{\mathrm{prod}} + 2c_{\mathrm{cross}}\big], \tag{105}$$

independently of n and of all dimensions.

Norm control. The datum and the assembly (103) are exactly the hypotheses of `lem-extcb-four-corner-norm` (§35), which supplies the two-sided bound $(1 - K_{\mathrm{norm}} e)\|X\| \le \|\gamma_n(X)\| \le (1 + K_{\mathrm{norm}} e)\|X\|$ at every level simultaneously.

Coverage, and bijectivity at level one. Injectivity of γ_1 is immediate from the lower bound once $K_{\mathrm{norm}}e \le \frac{1}{2}$. Surjectivity is the delicate half, and it is obtained without assuming that the target corner sum is direct: with $\beta(Y) = (\mathrm{Co}_{P_j,P_k}Y)_{jk}$, each compression is $O(e)$-close to $Y \mapsto P_j(YP_k)$, so $\alpha_1\beta$ is $O(e)$-close to $Y \mapsto s(Ys)$ with $s = P_1+P_2$ — an identity of bilinearity alone, no associativity used — and $\|s - I\| \le \rho$ together with the product and approximate-unit axioms gives

$$\|\alpha_1\beta - I\| \;\le\; C_{\mathrm{cov}}e. \tag{106}$$

Choosing e with $C_{\mathrm{cov}}e < 1$ makes $\alpha_1\beta$ invertible by its Neumann series, so $Y = \alpha_1\big(\beta((\alpha_1\beta)^{-1}Y)\big)$ for every Y and α_1 is onto. Now μ_1 is a bijection by exact source complementarity, Γ_1 is a bijection because all four fixed level-one maps are bijective by hypothesis, and α_1 is onto; therefore $\gamma_1 = \alpha_1\Gamma_1\mu_1$ is onto, hence bijective, and every $1_{M_n} \otimes \gamma_1$ is algebraically bijective with inverse $1_{M_n} \otimes \gamma_1^{-1}$.

Conclusion. Let a_{merge} be the minimum of the thresholds above — e_{corner}, e_{norm}, e_{cov}, $1/(2K_{\mathrm{norm}})$ and the universal compression radius — and put $C_{\mathrm{merge}} = \max\{3, D_{\mathrm{mult}}, K_{\mathrm{norm}}\}$. Then (104) and (105) make every amplification a $C_{\mathrm{merge}}e$-homomorphism, the norm lemma supplies the two-sided $(1\pm C_{\mathrm{merge}}e)$ bounds, and level-one bijectivity supplies the isomorphism clause; `def-extended-delta-inclusion` then returns exactly the asserted extended $C_{\mathrm{merge}}(\rho+\varepsilon)$-isomorphism.

Why the hypothesis is on the total defect. The transcribed contract bounded only ρ, leaving ε free. That version is *false*, and the tree records an exact counterexample rather than a repair. Take $\mathcal{A} = \mathbb{C}^3$ with coordinatewise involution, the multiplication $(a,b,c)(a',b',c') = (aa',bb',0)$ and designated unit $I = (1,1,0)$: at every matrix level this is an extended 1-C^*-algebra, the lower C^* inequality being vacuous at $\varepsilon = 1$. The elements $P_1 = (1,0,0)$ and $P_2 = (0,1,0)$ are exact projections with $P_1 + P_2 = I$, so $\rho = 0$; the registered θ formula gives $\mathrm{Co}_{P_j} = E_j$ and $\mathrm{Co}_{P_j,P_k} = 0$ for $j \ne k$, so the four target corners are $\mathbb{C}P_1, 0, 0, \mathbb{C}P_2$. With $\mathcal{B} = \mathbb{C}^2$ and its exact structure, the two diagonal identities and the two zero maps form a datum with common defect $\rho = 0$ whose four maps are bijective complete isometries; yet the assembled map is $\gamma(a,b) = (a,b,0)$, which is not surjective, and no linear bijection $\mathbb{C}^2 \to \mathbb{C}^3$ exists at all. So ρ may be arbitrarily small while the conclusion fails, and only a ceiling on $\rho+\varepsilon$ repairs it. The prover amended the root accordingly and the fresh verifiers validated the amended statement; the registry contract carries that amended root verbatim, and downstream consumption supplies a small ambient ε, so the conditional form is the consumable one.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb-four-corner-merge`: 22 nodes `validated` plus 4 `archived`, root `validated`, taint clean. The archived nodes are dependency-gated duplicates superseded by children that discharge the same step from validated material only.

Role in the chain. Imports `lem-compcb-corner-algebra`, `lem-hcb3-uniform-square-lower` and `lem-extcb-four-corner-norm` (§35). It is the assembly primitive of the EXT tier: given the block shape selected in §34 and four corner maps matched to it, the merge produces the single map that the tier's conclusion is about. Its consumer is `conj-extcb` (§37), now `proved` with `af: validated`: this merge is the last step of that proof. Nothing here bears on `op-classical`, which is open.

37 The EXT-CB parent contract

Lemma 77 (`conj-extcb`). *There are universal $C_{\mathrm{ext}} < \infty$ and $e_{\mathrm{ext}} > 0$ such that if $e = \delta+\varepsilon \le e_{\mathrm{ext}}$, P, Q are δ-projections in an extended ε-C^*-algebra $\mathcal{A}$ with $\|P + Q - I\| \le \delta$, $v : M_r \to \mathcal{S}_P$ is an extended δ-isomorphism, $\dim \mathcal{S}_Q = 1$ at level one, and $\mathcal{S}_{P,Q} \neq 0$, then there is one map $v_+ : M_{r+1} \to \mathcal{A}$ whose every amplification is a $C_{\mathrm{ext}}e$-isomorphism; the same level-one unitary and the same four corner maps carry all amplification levels, with constants independent of r, n and* $\dim \mathcal{A}$. Transcription note. *The hypothesis list is exactly **`def-extcb-datum`**; the conclusion's "isomorphism" is **`def-extended-delta-inclusion`** read at accuracy $C_{\mathrm{ext}}e$. The clause* $\dim \mathcal{S}_Q = 1$ *is used* only *at level one: no amplification of Q is anywhere called one-dimensional. The final clause is not decoration — it is the content of "CB": one unitary and four fixed level-one maps serve every n, so nothing is chosen level by level.*

Registry contract (verbatim).

```
EXT-CB: there are universal C_ext < infinity and e_ext > 0 such that if e=delta+epsilon <= e_ext,
P,Q are delta-projections in an extended epsilon-C*-algebra A with ||P+Q-I|| <= delta, v:M_r->S_P
is an extended delta-isomorphism, dim S_Q=1 at level one, and S_{P,Q} is nonzero, then there is one
map v_+:M_{r+1}->A whose every amplification is a C_ext*e-isomorphism; the same level-one unitary
and the same four corner maps carry all amplification levels, with constants independent of r, n,
and dim A.
```

Proof (prose account of the af-validated tree). Put $P_1 = P$, $P_2 = Q$, $H_j = \mathcal{S}_{P_j,Q}$, $K_1 = \mathbb{C}^r$, $K_2 = \mathbb{C}$ and $h_{jk} = \mathrm{Ha}^{Q}_{P_j,P_k}$. The hypotheses are a closed EXT-CB datum, so for e below the universal e_{sel} of `lem-extcb1-cross-corner-dimension` (§34) the cross corners have dimensions $\dim H_1 = r$ and $\dim H_2 = 1$. Fix once and for all a unitary $U_2 : K_2 \to H_2$.

A dimension-free exact-target correction. The tree proves in-workspace, from no external theorem, the following. *There are universal $a_{\mathrm{corr}} > 0$ and $C_{\mathrm{corr}} < \infty$ such that if $\mathcal{B}$ is a finite-dimensional C^*-algebra, H a finite-dimensional Hilbert space, and $T : \mathcal{B} \to B(H)$ is linear, $\dagger$-preserving and satisfies $\|T_n(XY) - T_n(X)T_n(Y)\| \le a\|X\|\|Y\|$ and $\|T_n(I) - I\| \le a$ at every n, with $0 \le a \le a_{\mathrm{corr}}$, then one unital $\dagger$-homomorphism $\mu : \mathcal{B} \to B(H)$ satisfies $\|\mu_n - T_n\| \le C_{\mathrm{corr}}a$ at every n — the same level-one μ at every amplification.* It is a Newton iteration on the multiplicative defect.

At $\|X\| = 1$, $\dagger$-preservation and the defect estimate give $\|T_n(X)\|^2 \le \|T_n(X^\dagger X)\| + a$, so $M_n = \|T_n\|$ obeys $M_n^2 \le M_n + a$ and $\|T\|_{\mathrm{cb}} \le 2$ for $a \le 1$. Fixing a state ϕ on $\mathcal{B}$, $S_0(x) = T(x) + \phi(x)(I - T(I))$ is $\dagger$-preserving and *exactly* unital, with $\|S_0 - T\|_{\mathrm{cb}} \le a$, $\|S_0\|_{\mathrm{cb}} \le 3$ and multiplicative defect $b_0 \le c_u a$, $c_u = 7$. Normalised Haar measure on the compact unitary group of $\mathcal{B}$ gives $D = \int U^\dagger \otimes U\, dU$, in finite dimension a finite convex combination $\sum_s p_s U_s^\dagger \otimes U_s$ of projective norm one, with $XD = DX$ and $\mathrm{mult}(D) = I$ by invariance. For a defect G of a map S, the correction $w'(x) = \sum_s S(p_s U_s^\dagger) G(U_s, x)$ then obeys $\|w'\|_{\mathrm{cb}} \le \|S\|_{\mathrm{cb}}\|G\|$, and at level n the same formula is $1_{M_n} \otimes w'$ entrywise, because the diagonal identity moves arbitrary matrix entries. This is the step that makes every correction *one level-one map* with n-free, dimension-free bounds.

Now the Newton step. Let S be exactly unital, $\dagger$-preserving, $\|S\|_{\mathrm{cb}} \le 4$, with defect G of uniform amplified bilinear norm b; put $w''(x) = w'(x^\dagger)^\dagger$, $w = (w' + w'')/2$ and $S^+ = S + w$. Since $G(y, I) = G(I, y) = 0$ one has $w(I) = 0$, so S^+ is still exactly unital and $\dagger$-preserving, with $\|w\|_{\mathrm{cb}} \le 4b$. Exact associativity in $B(H)$ gives $S(x)G(y, z) - G(xy, z) + G(x, yz) - G(x, y)S(z) = 0$, and expanding $S(x)w(y) - w(xy) + w(x)S(y)$ with $XD = DX$, $\mathrm{mult}(D) = I$ and $S(I) = I$ cancels

G exactly to first order, every surviving term carrying two defect factors; a triangle estimate gives $\|G_{S^+}\| \le K_N b^2$ at every amplification, for one numerical universal K_N, while $\|S^+ - S\|_{\mathrm{cb}} \le 4b$. Choosing a_{corr} with $K_N c_u a_{\mathrm{corr}} \le \frac{1}{2}$ and $8c_u a_{\mathrm{corr}} < 1$ and iterating from S_0, the recursion $b_{k+1} \le K_N b_k^2$ gives $b_k \le b_0 2^{-k}$, hence $\|S_k - S_0\|_{\mathrm{cb}} \le \sum_{j<k} 4b_j \le 8b_0 \le 8c_u a < 1$ and $\|S_k\|_{\mathrm{cb}} < 4$, so every iteration is licensed — including the degenerate case $b_0 = 0$, where every increment vanishes. The sequence converges in cb norm to one unital †-preserving μ; continuity of multiplication and $b_k \to 0$ give $\mu(xy) = \mu(x)\mu(y)$, and cb convergence means every μ_n is the amplification of that same level-one μ. Finally $\|\mu - T\|_{\mathrm{cb}} \le a + 8c_u a$, i.e. $C_{\mathrm{corr}} = 1 + 8c_u = 57$.

One spatial four-corner system. Let $T = h_{11} \circ v$. For $X, Y \in M_n \otimes M_r$, `def-extended-delta-inclusion` gives $\|v_n(XY) - v_n(X) \cdot v_n(Y)\| \le \delta \|X\| \|Y\|$, $\|v_n(X)\| \le (1+\delta)\|X\|$, exact †-preservation and the compressed unit estimate in $\mathcal{S}_P$; feeding these through the upper-norm, product, unit and adjoint clauses of `conj-hcb` (§31) yields

$$\|T_n(XY) - T_n(X)T_n(Y)\| \le \big[(1 + C_H e)\delta + C_H e(1+\delta)^2\big]\|X\|\|Y\|, \qquad \|T_n(I) - I\| \le A_0 e,$$

so below one universal ceiling both defects are at most $A_0 e$ with $A_0 = 4(C_H + 1)$, uniformly in n, r and $\dim \mathcal{A}$. The correction now applies, in two cases that exhaust $e \ge 0$: if $e = 0$ then $\delta = \varepsilon = 0$, both defects vanish, T is itself an exact unital †-homomorphism and one takes $\mu_{11} := T$; if $e > 0$, shrink the threshold so that $a = A_0 e < a_{\mathrm{corr}}$ and correct. Either way there is one level-one unital †-homomorphism $\mu_{11} : M_r \to B(H_1)$ with

$$\|\mu_{11,n} - h_{11,n} v_n\| \;\le\; \kappa e \quad \text{for every } n, \qquad \kappa = C_{\mathrm{corr}} A_0. \tag{107}$$

It is unital and nonzero, so its kernel is a two-sided ideal in the simple algebra M_r and vanishes; domain and codomain both have dimension r^2 by the level-one count, so it is onto. A unital †-isomorphism between full matrix algebras preserves matrix units, hence is spatial: there is one unitary $U_1 : \mathbb{C}^r \to H_1$ with $\mu_{11}(A) = U_1 A U_1^\dagger$. With the already fixed U_2, set $\mu_{jk}(A) = U_j A U_k^\dagger$. These four fixed level-one maps are exact, completely isometric, adjoint-compatible and multiplicative at every amplification via $1_{M_n} \otimes U_j$, with no level-dependent choice.

The conditional H-CB clauses, in the required order. Shrink the threshold so that $\kappa e < 1$ and $C_H e < \frac{1}{4}$. Since μ_{11} is invertible and $\|T - \mu_{11}\| < 1$, Neumann inversion makes T bijective at level one; v is bijective, so h_{11} is, and for $Z = v(A)$ one has $\|h_{11}(Z)\|/\|Z\| \ge (1 - \kappa e)/(1+\delta) > \frac{1}{4}$. The canonical-identity closeness clause of `conj-hcb` makes each of h_{12}, h_{21}, h_{22} a strict-norm perturbation of its canonical level-one bijection, hence level-one bijective by Neumann inversion, and gives h_{22} level-one lower modulus above $\frac{1}{4}$. *Only now* are the conditional clauses invoked: the diagonal clause for h_{11} and h_{22}, then the off-diagonal clause for h_{12} anchored at h_{22} and for h_{21} anchored at h_{11}. Every $h_{jk,n}$ is then bijective with

$$(1 - C_H e)\|Z\| \le \|h_{jk,n}(Z)\| \le (1 + C_H e)\|Z\|, \qquad \|h_{jk,n}^{-1}\| \le 1 + C_H e, \tag{108}$$

and $h_{jk,n}^{-1} = 1_{M_n} \otimes h_{jk}^{-1}$ by uniqueness of inverses, so no inverse is chosen afresh at level n. The order is the delicate point, and the hostile verifier checked it: the parent's inverse clauses are conditional (§31), and their level-one hypotheses are established *before* use, never assumed.

Four fixed corner maps form a merging datum. Put $\gamma_{11} = v$ and $\gamma_{jk} = h_{jk}^{-1} \circ \mu_{jk}$ for $(j,k) \ne (1,1)$, and $d_{jk,n} = h_{jk,n}\gamma_{jk,n} - \mu_{jk,n}$. Off $(1,1)$ the composition is exact, so $d_{jk,n} = 0$; at $(1,1)$, (107) gives $\|d_{11,n}\| \le \kappa e$. *Involution* is exact: off $(1,1)$, exact H-CB adjointness and spatial adjoint compatibility give $h_{kj,n}\big(\gamma_{jk,n}(X)^\dagger\big) = h_{kj,n}\gamma_{kj,n}(X^\dagger)$ and injectivity of $h_{kj,n}$ cancels it, while $\gamma_{11,n} = v_n$ preserves † exactly. For the *product*, (108) and isometry of μ give $\|\gamma_{jk,n}(X)\| \le$

$(1+C_He)(1+\kappa e)\|X\| \le \frac{25}{16}\|X\| \le 2\|X\|$; with $D = \gamma_{jl,n}(XY) - \gamma_{jk,n}(X)\cdot\gamma_{kl,n}(Y)$, the product-defect clause of `conj-hcb` supplies E with $\|E\| \le 4C_He\|X\|\|Y\|$ and the exact spatial multiplication $\mu_{jl,n}(XY) = \mu_{jk,n}(X)\mu_{kl,n}(Y)$ turns the comparison into

$$h_{jl,n}(D) = d_{jl,n}(XY) - \mu_{jk,n}(X)d_{kl,n}(Y) - d_{jk,n}(X)\mu_{kl,n}(Y) - d_{jk,n}(X)d_{kl,n}(Y) - E,$$

whence $\|h_{jl,n}(D)\| \le [3\kappa + \kappa^2 e + 4C_H]e\|X\|\|Y\| \le 4(C_H+\kappa)e\|X\|\|Y\|$ and, applying $\|h_{jl,n}^{-1}\| \le \frac{5}{4}$, $\|D\| \le 5(C_H+\kappa)e\|X\|\|Y\|$. For the *unit and norm* clauses, spatiality gives $\mu_{jj,n}(I) = I$, so the H-CB diagonal-unit estimate, $\|u_{P_j} - P_j\| \le C_He$ and (108) give $\|\gamma_{jj,n}(I) - I_n \otimes P_j\| \le 3(C_H+\kappa)e$, while isometry of μ and the d-bound give $(1-(C_H+\kappa)e)\|X\| \le \|\gamma_{jk,n}(X)\| \le (1+2(C_H+\kappa)e)\|X\|$. Finally $\gamma_{11} = v$ is bijective by hypothesis and every other γ_{jk} composes two bijections. Since the source block projections in $M_{r+1} = B(K_1 \oplus K_2)$ are exactly complementary and $\|P_1+P_2-I\| \le \delta \le e$, every clause of `def-four-corner-merging-datum` holds with the one common defect $\rho = D_0e$, $D_0 = 5(C_H+\kappa)$ enlarged so that $D_0 \ge 1$, for the same four fixed level-one maps at every amplification.

Merge, and the root. Let e_{ext} be the positive minimum of the thresholds above, of $a_{\text{merge}}/(D_0+1)$, and of the finitely many Neumann and $\frac{1}{4}$ ceilings. Then $\rho + \varepsilon \le (D_0+1)e \le a_{\text{merge}}$, so `lem-extcb-four-corner-merge` (§36) — whose norm half is `lem-extcb-four-corner-norm` (§35) — applies and returns the single sum map $v_+ : M_{r+1} \to \mathcal{A}$ as an extended $C_{\text{merge}}(\rho+\varepsilon)$-isomorphism, hence an extended $C_{\text{ext}}e$-isomorphism with $C_{\text{ext}} = C_{\text{merge}}(D_0+1)$. Its n-th map is $1_{M_n} \otimes v_+$, assembled from $1_{M_n} \otimes U_1$, $1_{M_n} \otimes U_2$ and the same four γ_{jk}; every constant used is universal, so none depends on r, on n, on $\dim\mathcal{A}$ or on the block data. That is the root.

Status. `proved`; `af: validated` (T0). Workspace `proofs/conj-extcb`: 40 nodes `validated` plus 6 `archived`, root `validated`, taint clean. The archived nodes are dependency-gated duplicates superseded by siblings that discharge the same step from validated material only: they are not part of the validated spine, and nothing above rests on them. The identifier keeps its historical `conj-` slug; registry ids are stable.

Correction of record (proof-level, not contract-level). The registry shard records one verifier-mandated correction: the level-one close-idempotent normalisation in the selection step must be explicit — the added-dimension corner is stated for $\widehat{P} = v(I_r)$, with $\|\widehat{P} - P\| = O(e)$ and range identification by close idempotents — and the selection threshold must cover it, at a universal threshold with no r- or level-dependence. That is what `lem-extcb1-close-corner-dimension` (§34) supplies; the contract itself was not amended.

Role in the chain. The registry shard lists ten dependencies: `conj-hcb` (§31), the EXT front end (§§32–36), `lem-compcb-corner-algebra` (§16) and `lem-hcb3-uniform-square-lower` (§24); all ten are `proved` with `af: validated` and all ten are reproduced in this document. The exported tree cites four directly — `conj-hcb`, `lem-extcb1-cross-corner-dimension`, `lem-extcb-four-corner-merge`, `lem-extcb-four-corner-norm` — the rest entering through the selection lemma. The registry body additionally records the paper-proof's witness in the expanded form $C_{\text{ext}} = C_{\text{merge}}[1 + 5C_H + 20C_{\text{app}}(C_H+1)]$; the validated tree does not use it, deriving $C_{\text{merge}}(D_0+1)$ instead. Downstream, `conj-extcb` with `conj-hcb` discharges both gap nodes of the extended assembly, whose registry carrier `lem-thmainext-conditional` is `proved-mod-audit` with `af: none` — **open** here and **not** reproduced — and the unconditional end-to-end constant ledger for that route remains separate open work. Nothing here bears on `op-classical`, which is open.

38 The Route F finish edge

Lemma 78 (`lem-routef-prh-finish`). *Let $A : \ell^\infty(k) \to \ell^\infty(n)$ and $M : \ell^\infty(n) \to \ell^\infty(k)$ be positive unital maps and let $Q : \ell^\infty(n) \to \ell^\infty(n)$ be row-stochastic. If $K \ge 1$,*

$$0 \;\le\; \eta \;\le\; \min\bigl\{(24K)^{-1},\, 1\bigr\}, \tag{109}$$

and

$$\|Q - AM\|_{\infty\to\infty} \;\le\; K\eta, \qquad \|MA - I\|_{\infty\to\infty} \;\le\; \frac{3K\eta}{1-3K\eta}, \tag{110}$$

then there is a stochastic idempotent E (`def-stochastic`) with

$$\|Q - E\|_{\infty\to\infty} \;\le\; \bigl(K + 4\sqrt{2K}\bigr)\sqrt{\eta}. \tag{111}$$

Transcription note. *Both displays in* (110) *are* hypotheses*: the pair (A, M) — the datum of `def-positive-approximate-retract` — is given, and nothing in this statement, or anywhere in this document, manufactures it from Q. The denominator in* (110) *is positive precisely because of the window* (109)*. The constant in* (111) *is inherited from the hypothesis constant K and is universal only to the extent that K is; no result reproduced here fixes a value of K.*

Registry contract (verbatim).

```
Route F PRH finish: let A:l-infinity(k)->l-infinity(n) and M:l-infinity(n)->l-infinity(k) be
positive unital maps and let Q be row-stochastic; if K >= 1, 0 <= eta <= min{(24*K)^(-1),1},
||Q-AM||_{infinity->infinity} <= K*eta, and ||MA-I||_{infinity->infinity} <= 3*K*eta/(1-3*K*eta),
then there is a stochastic idempotent E with ||Q-E||_{infinity->infinity} <=
(K+4*sqrt(2*K))*sqrt(eta).
```

Proof (prose account of the af-validated tree). The tree has three steps — a scalar preparation, one import of `lem-prh` (§4) guarded by a dimension split, and a triangle-inequality finish — and a synthesis node that assembles them.

Scalar preparation. Put $x = 3K\eta$ and

$$\varepsilon := \frac{3K\eta}{1-3K\eta} = \frac{x}{1-x}. \tag{112}$$

The window (109) with $K \ge 1$ gives $0 \le x \le \frac{1}{8}$, hence $1 - x \ge \frac{7}{8} > 0$, so (112) is well defined and

$$0 \;\le\; \varepsilon \;\le\; \tfrac{8}{7}x \;\le\; \tfrac{1}{7} \;<\; \tfrac{1}{2}, \qquad \varepsilon \;\le\; \tfrac{24}{7}K\eta \;\le\; 4K\eta.$$

Separately, $0 \le \eta \le 1$ gives $\eta^2 \le \eta$, i.e. $\eta \le \sqrt{\eta}$, and multiplying by $K \ge 1$,

$$K\eta \;\le\; K\sqrt{\eta}. \tag{113}$$

The two bounds on ε are the whole point of the preparation: the first puts ε inside the PRH window $[0, \frac{1}{2})$, the second is what converts a $\sqrt{\varepsilon}$ rate into a $\sqrt{\eta}$ rate.

The PRH import, with an exhaustive dimension split. By (112) the second hypothesis of (110) reads exactly $\|MA - I_k\|_{\infty\to\infty} \le \varepsilon$ with $0 \le \varepsilon < \frac{1}{2}$, which is the hypothesis of `lem-prh`. That lemma, however, is stated for $k, n \ge 1$, and the contract above assumes no dimension bound; the tree therefore splits, and the split is exhaustive. If $k, n \ge 1$, then `def-positive-approximate-retract` identifies A and M as positive unital maps with probability-vector rows, every premise of `lem-prh` is met, and it returns a stochastic idempotent $E : \ell^\infty(n) \to \ell^\infty(n)$ with

$$\|AM - E\|_{\infty\to\infty} \;\le\; 2\sqrt{2\varepsilon}. \tag{114}$$

The mixed degenerate cases cannot occur: if $k = 0 < n$ then each of the n rows of A is the empty vector, of coordinate sum 0 rather than 1, contradicting the probability-row requirement, and the same argument applied to M excludes $n = 0 < k$. There remains $k = n = 0$, where $\ell^\infty(0)$ is the zero space, its unique endomorphism is the empty 0×0 matrix, which is entrywise nonnegative, fixes the (empty) all-ones vector and is idempotent — a stochastic idempotent by `def-stochastic` — and satisfies $AM = E$, so (114) holds with error 0. In every admissible dimension, then, such an E exists, and `lem-prh` is never invoked outside its stated domain.

The finish. Writing $Q - E = (Q - AM) + (AM - E)$, the triangle inequality for $\|\cdot\|_{\infty\to\infty}$ with the first hypothesis of (110) and (114) gives $\|Q - E\|_{\infty\to\infty} \le K\eta + 2\sqrt{2\varepsilon}$. Monotonicity and multiplicativity of the square root on the nonnegative reals turn $\varepsilon \le 4K\eta$ into $2\sqrt{2\varepsilon} \le 2\sqrt{8K\eta} = 4\sqrt{2K}\sqrt{\eta}$, and with (113),

$$\|Q - E\|_{\infty\to\infty} \;\le\; K\sqrt{\eta} + 4\sqrt{2K}\sqrt{\eta} \;=\; \bigl(K + 4\sqrt{2K}\bigr)\sqrt{\eta},$$

which is (111).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-prh-finish`: 22 nodes created and 22 `validated`, root `validated`, taint clean; three challenges raised and three resolved, four node amendments. The workspace registers two definition imports, `def-positive-approximate-retract` and `def-stochastic`, and two external imports carrying the validated contract of `lem-prh`.

Correction of record (proof-level, not contract-level). All three challenges were the same defect, raised three times at three heights of the tree: the import node asserted a direct application of `lem-prh` while neither the root contract nor either definition supplies that lemma's premise $k, n \ge 1$, so the invocation was, as written, outside the external's stated domain — and the $k = n = 0$ branch is admissible under the root's hypotheses. The repair was not to weaken anything but to make the dichotomy explicit and to discharge the degenerate branch by hand, which is the form reproduced above. The degenerate case is a formality; the protocol nonetheless refused the import until it was discharged, which is exactly the guard one wants on a hypothesis-carrying external. The registry contract was not amended.

What this closes, and what it does not. This is the validated conditional PRH finish edge within Route F and it is now `af`-validated, but it is an *implication*. Its antecedent — a positive unital pair (A, M) with $\|Q - AM\|_{\infty\to\infty} \le K\eta$ and $\|MA - I\|_{\infty\to\infty} \le 3K\eta/(1 - 3K\eta)$ for a dimension-free K — is not established unconditionally anywhere in this document. Two of the registry rows that produce it, `lem-routef-f2-positive-unital-compression` (the positive unital pair) and `lem-routef-f3-retract-defect` (the retract-defect bound, whose conclusion is literally the second hypothesis here), are now `proved` with `af: validated` and are reproduced in §43, as are the seam and ledger rows upstream of them (§41, §42). But the F2 row is itself an implication, and the rows that would discharge *its* antecedent — the factorization row `lem-routef-k-ledger`, which also fixes the dimension-free K, and the assembly carrier `lem-thmainext-conditional` — are `proved-mod-audit` with `af: none`: **open** at this document's rigour bar and **not** reproduced here (§53). Consequently nothing here yields an unconditional $\sqrt{\eta}$ bound for a row-stochastic Q, and `op-classical` is **open**. The shape of (111) is the shape the target asks for; the antecedent is the whole remaining problem on this route.

Role in the chain. Registry `defs`: `def-positive-approximate-retract`, `def-stochastic`. Registry `deps`: `lem-prh` (§4), `proved` with `af: validated` and reproduced above — the single mathematical import, and the only place the square-root rate is produced. The registry records no result depending on this one; `lem-routef-f2-positive-unital-compression` and `lem-routef-f3-retract-defect` (§43) name it in their shard bodies as the consumer they feed, which is the direction of the eventual chain, not a current dependency edge.

39 Quantitative inverse-function control

Lemma 79 (`lem-stage1-quantitative-inverse-function`). *Let X, Y be Banach spaces, let $V : X \to Y$ be a Banach-space isomorphism, let $B_r(x_0) \subset X$ be the open ball of radius $r > 0$ about x_0, and let $f : B_r(x_0) \to Y$ be C^1 with*

$$\|V^{-1}Df(x) - I_X\| \;\le\; c \;<\; 1 \qquad \textit{for every } x \in B_r(x_0). \tag{115}$$

Then f is injective,

$$(1-c)\|x_1 - x_2\| \;\le\; \|V^{-1}\big(f(x_1) - f(x_2)\big)\| \;\le\; (1+c)\|x_1 - x_2\| \qquad (x_1, x_2 \in B_r(x_0)), \tag{116}$$

and the image contains a definite ball transported by V:

$$f(x_0) + V\big(B_{(1-c)r}(0)\big) \;\subseteq\; f\big(B_r(x_0)\big). \tag{117}$$

Transcription note. *The comparison operator V is* fixed *and the bound* (115) *is uniform on the whole ball; nothing here is stated at, or centred on, $Df(x_0)$. The covering radius $(1-c)r$ is measured in the* source *and then pushed forward, so* (117) *guarantees an image of the V-transported ball, which is in general not a norm ball of Y. All three conclusions are quantitative in the single scalar c; no smallness beyond $c < 1$ is assumed.*

Registry contract (verbatim).

```
Quantitative inverse-function control: if V:X->Y is a Banach-space isomorphism and f:B_r(x_0)->Y is
C^1 with ||V^(-1)Df(x)-I|| <= c < 1, then f is injective, (1-c)||x_1-x_2|| <=
||V^(-1)(f(x_1)-f(x_2))|| <= (1+c)||x_1-x_2||, and f(B_r(x_0)) contains f(x_0)+V(B_{(1-c)r}(0)).
```

Proof (prose account of the af-validated tree). The tree is self-contained: it imports no definition shard and no external theorem, so the two classical ingredients it needs — a Banach-valued fundamental theorem of calculus and the contraction-mapping principle — are proved inside the workspace rather than cited. The argument runs in two independent halves, the secant estimate and the covering, joined only by a common normalisation.

Normalisation. Put $F(x) = V^{-1}\big(f(x) - f(x_0)\big)$ on $B_r(x_0)$. Then F is C^1 with $DF(x) = V^{-1}Df(x)$ and $F(x_0) = 0$, so the hypothesis (115) reads exactly $\|DF(x) - I_X\| \le c$: the map has been turned into a perturbation of the identity of X, and the target space no longer appears except through V^{-1}.

The secant-error estimate. The core step is: if a C^1 map F on the open ball satisfies $\|DF - I_X\| \le c$ throughout, then

$$\|F(x_1) - F(x_2) - (x_1 - x_2)\| \;\le\; c\|x_1 - x_2\| \qquad (x_1, x_2 \in B_r(x_0)). \tag{118}$$

Fix x_1, x_2, put $d = x_1 - x_2$ and $\gamma(t) = x_2 + td$; convexity of the ball keeps $\gamma([0,1])$ inside the domain, which is where openness and convexity — and nothing else about the domain — are used. For $q(t) = F(\gamma(t)) - F(x_2) - td$ the chain rule gives $q'(t) = \big(DF(\gamma(t)) - I_X\big)d$, hence $\|q'(t)\| \le c\|d\|$ on $[0,1]$. The tree then proves, for a C^1 curve $q : [0,1] \to X$, that $q(1) - q(0) = \int_0^1 q'(t)\,dt$: continuity of q' makes its tagged Riemann sums converge in norm to the Bochner/Riemann integral while the matching sums of increments telescope to $q(1) - q(0)$; the norm inequality for that integral gives $\|q(1) - q(0)\| \le \int_0^1 \|q'(t)\|\,dt$. With $q(1) - q(0) = F(x_1) - F(x_2) - d$ this is (118).

From the secant error to (116) *and injectivity.* The remaining step is a two-line normed-space deduction, isolated in the tree as its own node: if $\|a-d\| \le c\|d\|$ then the reverse and ordinary triangle inequalities give $(1-c)\|d\| \le \|a\| \le (1+c)\|d\|$, and when $c<1$ the vanishing of a forces $d=0$. Applying this with $d = x_1 - x_2$ and $a = F(x_1) - F(x_2) = V^{-1}(f(x_1) - f(x_2))$ yields (116); the strict positivity of $1-c$ then yields injectivity of f.

The centred covering. Let $g(h) = V^{-1}\big(f(x_0+h) - f(x_0)\big)$ on $B_r(0)$, so $g(0)=0$, $\|Dg(h) - I_X\| \le c$, and the segment argument above applies verbatim to give $\|(g(h)-g(k)) - (h-k)\| \le c\|h-k\|$. Fix z with $\|z\| < (1-c)r$ and choose ρ with $0<\rho<r$ and $\|z\| < (1-c)\rho$ — the strict inequality is what buys the room for the next line. On the closed ball $K = \{h \in X : \|h\| \le \rho\}$, which is complete because X is, set

$$T(h) \;=\; z + h - g(h).$$

The secant bound gives $\|T(h) - T(k)\| \le c\|h-k\|$, and taking $k=0$ with $g(0)=0$ gives $\|h-g(h)\| \le c\|h\|$, whence $\|T(h)\| \le \|z\| + c\rho < \rho$. So T is a c-contraction of the complete metric space K into itself. The tree proves the fixed-point principle in place: the iterates $h_{n+1} = T(h_n)$ satisfy $d(h_{n+1}, h_n) \le c^n d(h_1, h_0)$, the geometric bound $d(h_m, h_n) \le c^n d(h_1,h_0)/(1-c)$ makes them Cauchy, completeness supplies a limit h, and $d(T(h), h) \le c\, d(h, h_n) + d(h_{n+1}, h) \to 0$ identifies it as a fixed point. Then $h \in K \subset B_r(0)$ and $T(h) = h$ says precisely $g(h) = z$.

The translation back. For *this* g the defining identity is $Vg(h) = f(x_0+h) - f(x_0)$. Hence $g(h) = z$ gives $f(x_0+h) = f(x_0) + Vz$, and every $y \in f(x_0) + V(B_{(1-c)r}(0))$ is $f(x_0+h)$ with $x_0 + h \in B_r(x_0)$. That is (117), and with it the root. The fixed point lives in X; Y enters only through V^{-1}.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-stage1-quantitative-inverse-function`: 14 nodes created and 14 `validated`, root `validated`, taint clean; three challenges raised and three resolved, two node amendments. No definition imports and no external imports are registered in the workspace.

Corrections of record (proof-level, not contract-level). The tree reached its verdict only after three hostile challenges, and the shape of the final nodes is their residue. (i) A *dependencies* challenge at the Newton–Leibniz node: as first written it was a leaf that silently borrowed the curve q, its formula and its derivative bound from a *pending* sibling node, which the protocol forbids; it was repaired by a child node carrying the abstract estimate and its instantiation together. (ii) A *type-error* challenge at the fixed-point node: the universal statement was phrased for an arbitrary complete metric space but written with norm differences $\|h_{n+1} - h_n\|$, which is ill-typed there; it was amended to the metric d. (iii) A *scope* challenge at the covering node, with a concrete counterexample: as first written, that node asserted the translation "$g(h) = z$ implies $y = f(x_0+h)$" for an *arbitrary* map g satisfying only $g(0)=0$ and the secant bound. Take $X = Y = \mathbb{R}$, $V = I$, $x_0 = 0$, $r = 1$, $c = \frac{1}{2}$ and $f(x) = \frac{3}{2}x$; then $g(h) = h$ is admissible in that sense, and for $z = h = \frac{1}{4}$ one has $y = f(0) + Vz = \frac{1}{4}$ while $f(0+h) = \frac{3}{8}$. The node was amended to state the covering for an arbitrary admissible g but the translation *only* for the specific g of the normalisation, and a further node makes that specialisation explicit. The registry contract itself was not amended.

Role in the chain. The registry shard lists no `deps` and no `defs`: this is a leaf of the knowledge DAG, an item of standard Banach-space calculus re-established here at this document's rigour bar rather than cited, because no matching local source sits in `refs/`. Its one registered registry consumer is `lem-stage1-exact-unit-rectification` (§40), whose workspace imports this validated contract as a byte-matched external — see the caveat recorded there about how that import is actually used. Both results belong to the Stage-1 front of the FUDW decomposition, which is otherwise not reproduced in this document. Nothing here bears on `op-classical`, which is open.

40 Dimension-free exact-unit rectification

Lemma 80 (`lem-stage1-exact-unit-rectification`). *There are universal constants $C_{\text{unit}} < \infty$ and $\varepsilon_{\text{unit}} > 0$ such that every finite-dimensional ε_X-C^*-algebra $\mathcal{A}$ (`def-epsilon-cstar-algebra`) with $\varepsilon_X \le \varepsilon_{\text{unit}}$ admits, on the same involutive normed space, an exact unit J and a bilinear product $\boldsymbol{\cdot}$ with*

$$\|J - I_X\| \;\le\; C_{\text{unit}}\,\varepsilon_X, \qquad \|x \boldsymbol{\cdot} y - xy\| \;\le\; C_{\text{unit}}\,\varepsilon_X\|x\|\|y\| \quad (x, y \in \mathcal{A}). \tag{119}$$

Transcription note. *"On the same involutive normed space" is a real restriction and not a formality: the underlying vector space, its norm and its involution are untouched, and only the multiplication and the distinguished unit are replaced. "Exact unit" has exactly the meaning of Definition 22; no alternative definition is introduced here. I_X denotes the given approximate unit, and the threshold written $\varepsilon_{\text{unit}}$ here is the contract's `e_unit` — a ceiling on ε_X alone, not on a combined defect. What the statement asserts about $\boldsymbol{\cdot}$ is exactly* (119) *together with the exactness of J:* no *clause claims that $(\mathcal{A}, \boldsymbol{\cdot}, J)$ is again an ε'-C^*-algebra, and the validated tree estimates neither the associativity defect nor the product-norm constant of $\boldsymbol{\cdot}$.*

Registry contract (verbatim).

```
Dimension-free exact-unit rectification: there are universal C_unit < infinity and e_unit > 0 such
that every finite-dimensional epsilon_X-C*-algebra with epsilon_X <= e_unit admits on the same
involutive normed space an exact unit J and product bold-dot with ||J-I_X|| <= C_unit*epsilon_X and
||x bold-dot y-xy|| <= C_unit*epsilon_X*||x||||y||.
```

Proof (prose account of the af-validated tree). The construction is explicit and the constants are exhibited: the tree closes with the universal witnesses

$$\varepsilon_{\text{unit}} = \tfrac{1}{2}, \qquad C_{\text{unit}} = 8. \tag{120}$$

It proceeds in four steps — rescale the unit, produce a Hermitian norming functional, correct the product by three terms of rank one in each argument, and estimate.

A norm-one self-adjoint unit. Write $I = I_X$, $a = \|I\|$ and $J = I/a$. The approximate-unit clause of `def-epsilon-cstar-algebra` gives $|a-1| \le \varepsilon_X$, so $\varepsilon_X \le \frac{1}{2}$ forces $a \ge \frac{1}{2} > 0$ and J is defined. Then $\|J\| = 1$; $J^\dagger = J$ because $I^\dagger = I$ and a is a positive real; and

$$\|J - I\| = |1 - a| \le \varepsilon_X. \tag{121}$$

This is the only place the self-adjointness of J is produced, and the next step consumes it.

A Hermitian norming functional at J. On the one-dimensional subspace $\mathbb{C}J$ define $g(\alpha J) = \alpha$; since $\|\alpha J\| = |\alpha|$ this has norm one, and complex Hahn–Banach extends it to a complex-linear $\psi : \mathcal{A} \to \mathbb{C}$ with $\|\psi\| = 1$ and $\psi(J) = 1$. Put $\psi^\sharp(x) = \overline{\psi(x^\dagger)}$. Because the involution is conjugate-linear, isometric and bijective, $\psi^\sharp$ is again complex-linear of norm one, and $J^\dagger = J$ gives $\psi^\sharp(J) = \overline{\psi(J^\dagger)} = \overline{\psi(J)} = 1$. Hence $\varphi = \frac{1}{2}(\psi + \psi^\sharp)$ is complex-linear with $\|\varphi\| \le 1$, $\varphi(J) = 1$ — so in fact $\|\varphi\| = 1$, since $1 = |\varphi(J)| \le \|\varphi\|\|J\|$ — and is Hermitian:

$$\varphi(x^\dagger) = \overline{\varphi(x)}, \qquad \varphi(J) = 1, \qquad \|\varphi\| = 1. \tag{122}$$

The corrected product. Define, bilinearly,

$$x \boldsymbol{\cdot} y \;:=\; xy + \varphi(x)\big(y - Jy\big) + \varphi(y)\big(x - xJ\big) - \varphi(x)\varphi(y)\big(J - JJ\big). \tag{123}$$

Substituting $x = J$ and using $\varphi(J) = 1$, the first correction term cancels Jy against y and the last two terms cancel each other, giving $J \cdot y = y$; symmetrically $x \cdot J = x$. With $\|J\| = 1$ and $J^\dagger = J$ this is exactly the exact-unit condition of `def-epsilon-cstar-algebra`. The involution is unchanged and remains compatible: Hermiticity (122), the axiom $(xy)^\dagger = y^\dagger x^\dagger$, $J^\dagger = J$ and $(JJ)^\dagger = JJ$ give $(x \cdot y)^\dagger = y^\dagger \cdot x^\dagger$ term by term, so the construction really does live on the same involutive normed space.

The estimate. For any z, the product-norm axiom, (121) and the approximate-unit axiom give

$$\|Jz - z\| \;\le\; \|(J - I)z\| + \|Iz - z\| \;\le\; \big((1+\varepsilon_X)\varepsilon_X + \varepsilon_X\big)\|z\| = (2+\varepsilon_X)\varepsilon_X\|z\|,$$

and likewise $\|zJ - z\| \le (2+\varepsilon_X)\varepsilon_X\|z\|$; taking $z = J$ and using $\|J\| = 1$ also gives $\|J - JJ\| \le (2+\varepsilon_X)\varepsilon_X$. Since $\|\varphi\| = 1$, the three correction terms of (123) are each bounded by $(2+\varepsilon_X)\varepsilon_X\|x\|\|y\|$, so

$$\|x \cdot y - xy\| \;\le\; 3(2+\varepsilon_X)\varepsilon_X\|x\|\|y\| \;\le\; \tfrac{15}{2}\,\varepsilon_X\|x\|\|y\| \;\le\; 8\,\varepsilon_X\|x\|\|y\| \qquad \text{for } \varepsilon_X \le \tfrac{1}{2},$$

while (121) gives $\|J - I_X\| \le \varepsilon_X \le 8\varepsilon_X$. That is (119) with the witnesses (120), and it is the root.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-stage1-exact-unit-rectification`: 6 nodes created and 6 `validated`, root `validated`, taint clean; one challenge raised and resolved. The workspace registers one definition import, `def-epsilon-cstar-algebra`, byte-matched to its pinned source, and two external imports naming the validated contract of `lem-stage1-quantitative-inverse-fu` (§39).

Correction of record (proof-level, not contract-level). The Hahn–Banach step drew a *gap* challenge with a counterexample. As first written, the node asserted only that J has norm one and then symmetrised ψ; but $\psi^\sharp(J) = 1$ needs $J^\dagger = J$, and without it the symmetrisation can destroy the value at J. The verifier's example: $\mathcal{A} = \mathbb{C}$ with $z^\dagger = \overline{z}$, $J = i$ and $\psi(z) = -iz$ — a complex-linear functional of norm one with $\psi(J) = 1$ — gives $\psi^\sharp(z) = iz$, hence $\psi^\sharp(J) = -1$ and $\varphi = 0$, so $\varphi(J) = 0$; indeed no Hermitian complex-linear functional takes the value 1 at i. The repair was a child node deriving $J^\dagger = J$ from the rescaling step and using it explicitly, which is why the account above records self-adjointness where it does. The registry contract was not amended.

Two hypotheses that are carried but not used. First, *finite dimension*: no node of the exported tree invokes it — the rescaling, Hahn–Banach, symmetrisation and norm estimates are all dimension-free — so it is carried from the contract, not consumed by the proof. Second, the *dependency*: the registry lists `lem-stage1-quantitative-inverse-function` as a `dep` and the workspace registers it as a byte-matched external, but no node of the exported tree invokes that contract; the rectification here is elementary. The dependency edge is the one fixed by the governing verdict for the Stage-1 package, and it is recorded rather than exercised. Both observations are about the exported tree, not about the contract, which stands as transcribed.

Role in the chain. Registry `defs`: `def-epsilon-cstar-algebra`. Registry `deps`: `lem-stage1-quantitative-in` (§39), which is `proved` with `af: validated` and reproduced in this document. Downstream, the Stage-1 decomposition rows that would consume an exactly unital model are *not* `af`-validated and are **not** reproduced here; in particular nothing in this document assembles this rectification into a statement about almost-idempotent maps. `op-classical` is **open**.

41 The F0 seam: a stochastic matrix as a unital completely positive map

Route F attacks `op-classical` with the approximate-C^* machinery of the pinned source, and that machinery speaks about unital completely positive maps of a matrix algebra, not about stochastic matrices. The two lemmas of this section are the seam between the two languages: the first says that the classical datum lifts to an admissible input for the machinery, the second says that the lift costs nothing in the one quantity the machinery consumes.

Notation for this section. Fix $n \geq 1$. Write M_n for the complex $n \times n$ matrices, $\mathbb{C}^n = \ell_n^\infty(\mathbb{C})$ for the commutative C^*-algebra of diagonals of M_n, $J : \mathbb{C}^n \to M_n$ for the diagonal inclusion and $D : M_n \to \mathbb{C}^n$, $D([a_{ij}]) = (a_{11}, \ldots, a_{nn})$, for diagonal extraction. The real self-adjoint part of $\mathbb{C}^n$ is written $\ell^\infty(n)$; a row-stochastic Q (`def-stochastic`, Definition 3) acts there, and $Q_\mathbb{C}$ denotes its canonical complex-linear extension to $\mathbb{C}^n$. "UCP" is `def-ucp-map` (Definition 40) throughout, and $\|\cdot\|_{\mathrm{cb}}$ is the completely bounded norm, the supremum over matrix levels $r \geq 1$ of the norms of the amplifications $T_r = \mathrm{id}_{M_r} \otimes T$.

Lemma 81 (`lem-routef-f0-ucp-lift`). *With the notation above, let Q be row-stochastic. Then*

$$\Phi \;:=\; J\, Q_\mathbb{C}\, D \;:\; M_n \longrightarrow M_n \tag{124}$$

is a unital completely positive map. Transcription note. *The typing is the content: D lands in the* complex *diagonal algebra $\mathbb{C}^n$, and it is $Q_\mathbb{C}$, not Q, that acts there. No smallness hypothesis on Q is assumed, and nothing is claimed about Φ beyond unitality and complete positivity.*

Registry contract (verbatim).

```
Route F F0 UCP lift: let n >= 1, let D: M_n -> C^n be diagonal extraction onto the complex diagonal
algebra C^n = l_inf^n(C) and J: C^n -> M_n the diagonal inclusion, let Q: l_inf^n -> l_inf^n be
row-stochastic, and let Q_C: C^n -> C^n be the canonical complex-linear extension of Q; then Phi :=
J Q_C D: M_n -> M_n is a unital completely positive map.
```

Proof (prose account of the af-validated tree). The tree factors the claim through the three composed maps and then closes by composition.

The middle map. By `def-stochastic` the matrix Q has nonnegative entries q_{ij} and satisfies $Q\mathbf{1} = \mathbf{1}$. If $x = (x_j)$ is positive in $\mathbb{C}^n$ then every x_j is a nonnegative real, so $(Q_\mathbb{C} x)_i = \sum_j q_{ij} x_j \geq 0$; and $Q_\mathbb{C}(\mathbf{1}) = \mathbf{1}$. Thus $Q_\mathbb{C}$ is positive and unital, and since its domain $\mathbb{C}^n$ is a commutative C^*-algebra it is automatically completely positive (`def-ucp-map`); hence UCP.

The two diagonal maps. D is unital, and for $r \geq 1$ and a positive block matrix $X = [X_{ab}] \in M_r(M_n)$ the amplification D_r returns the n-tuple of principal compressions $\big[(X_{ab})_{ii}\big]_{a,b=1}^r$, $i = 1, \ldots, n$, each of which is positive in M_r; so D is CP, hence UCP. Dually J is unital, and for $r \geq 1$ a positive element $(A_1, \ldots, A_n)$ of $M_r(\mathbb{C}^n) = \bigoplus_{i=1}^n M_r$ is sent by J_r, after the canonical permutation of tensor factors, to the positive block-diagonal matrix $\mathrm{diag}(A_1, \ldots, A_n) \in M_{rn}$; so J is CP, hence UCP.

Composition. Compositions of UCP maps are UCP (`def-ucp-map`), and (124) is such a composition.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-f0-ucp-lift`: root and 9/9 nodes `validated`, taint clean, one challenge raised and resolved. Two definition imports are registered, `def-stochastic` and `def-ucp-map`; there are no external imports.

Correction of record (contract-level). The single mid-run challenge was a typing defect: the contract as first seeded routed D into the *real* space $\ell^\infty(n)$, while the diagonal of a complex matrix algebra lies in $\mathbb{C}^n$. The repair was an amendment of the contract to the text quoted above — the diagonal algebra made complex and Q replaced by $Q_\mathbb{C}$ — after which the tree was verifier-accepted. The same defect family is repaired in Lemma 86 below, where it was the cause of a first, abandoned elevation run.

Lemma 82 (`lem-routef-f0-defect-identity`). *With the notation above and $\Phi = J\,Q_\mathbb{C}\,D$ as in* (124)*,*

$$\|\Phi^2 - \Phi\|_{\mathrm{cb}} \;=\; \|Q^2 - Q\|_{\infty\to\infty}. \tag{125}$$

Transcription note. *This is an* equality*, not a two-sided estimate with constants: the seam neither loses nor gains defect, and no $\eta \mapsto \varepsilon$ conversion happens here. The left side is a completely bounded norm on M_n, the right side the ℓ^∞ map norm of* `def-almost-idempotent` *(Definition 4); the identity is what licenses reading one as the other.*

Registry contract (verbatim).

```
Route F F0 defect identity: let n >= 1, let D: M_n -> C^n be diagonal extraction onto the complex
diagonal algebra C^n = l_inf^n(C) and J: C^n -> M_n the diagonal inclusion, let Q: l_inf^n ->
l_inf^n be row-stochastic with canonical complex-linear extension Q_C: C^n -> C^n, and put Phi := J
Q_C D; then ||Phi^2 - Phi||_cb = ||Q^2 - Q||_{infinity->infinity}.
```

Proof (prose account of the af-validated tree). Put $L := Q_\mathbb{C}^2 - Q_\mathbb{C}$. The proof is an algebraic reduction, then two matched inequalities at every matrix level, then a single scalar computation.

Reduction. For $x \in \mathbb{C}^n$ the matrix $J(x)$ is diagonal with diagonal x, and D extracts that diagonal, so $DJ = I_{\mathbb{C}^n}$. Hence

$$\Phi^2 - \Phi \;=\; J\,Q_\mathbb{C}(DJ)Q_\mathbb{C}\,D - J\,Q_\mathbb{C}\,D \;=\; J\big(Q_\mathbb{C}^2 - Q_\mathbb{C}\big)D \;=\; JLD. \tag{126}$$

The matrix levels. Fix $r \geq 1$ and identify $M_r(\mathbb{C}^n) = \bigoplus_{j=1}^n M_r$ with the norm $\max_j\|X_j\|$. After the canonical permutation of tensor factors $J_r(X) = \mathrm{diag}(X_1, \ldots, X_n)$, so $\|J_rX\| = \max_j\|X_j\|$ and J_r is isometric; D_r returns the tuple of diagonal $n\times n$ blocks, each a compression, so D_r is contractive; and $D_rJ_r = I$. Submultiplicativity therefore gives $\|J_rL_rD_r\| \leq \|L_r\|$, while testing on inputs of the form J_rX gives $\|J_rL_rD_r(J_rX)\| = \|J_rL_rX\| = \|L_rX\|$ and hence $\|J_rL_rD_r\| \geq \|L_r\|$. So $\|(\Phi^2-\Phi)_r\| = \|L_r\|$ for every r, by (126).

The scalar computation. Let (l_{ij}) be the coefficient matrix of L, which is the real matrix of $Q^2 - Q$. For $X = (X_j) \in M_r(\mathbb{C}^n)$ one has $(L_rX)_i = \sum_j l_{ij}X_j$, so $\|L_rX\| \leq \big(\max_i \sum_j|l_{ij}|\big)\max_j\|X_j\|$. Conversely pick i_0 attaining the maximal row sum and take $X_j = \mathrm{sgn}(l_{i_0j})I_r$ (with $\mathrm{sgn}\,0 = 0$): then $\max_j\|X_j\| \leq 1$ and the i_0-component of L_rX is $\big(\sum_j|l_{i_0j}|\big)I_r$. Hence $\|L_r\| = \max_i\sum_j|l_{ij}|$ for *every* r, and that row maximum is $\|Q^2 - Q\|_{\infty\to\infty}$. Taking the supremum over r, which is the definition of the cb norm, gives (125).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-f0-defect-identity`: root and 12/12 nodes `validated`, taint clean, *zero* challenges — the fresh verifiers confirmed both directions independently, including the zero-row edge case and the matrix-level supremum

identity. Three definition imports are registered, `def-stochastic`, `def-almost-idempotent` and `def-ucp-map`; there are no external imports, and in particular this proof does not use Lemma 81.
Role in the chain. Neither row has registry `deps`: both are leaves. Together they are the entry point of Route F — a row-stochastic Q with defect η in the sense of `def-almost-idempotent` becomes a UCP Φ with $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \leq \eta$, which is exactly the input the ledger of §42 consumes. The registry records their intended consumer as the strengthened `lem-routef-k-ledger`, which is `proved-mod-audit` with `af: none` — **open** at this document's rigour bar and **not** reproduced here — so no chain is closed by this section, and `op-classical` is **open**.

42 The approximate-algebra ledger for an almost-idempotent UCP map

The seam of §41 delivers a unital completely positive map Φ on a matrix algebra whose square is close to itself. This section reproduces the three `af`-validated rows that turn that datum into an *algebra*: an exact idempotent $\widetilde{\Phi}$ obtained by functional calculus, its image equipped with a corrected product, and the bookkeeping that shows the resulting structure is an extended ε-C^*-algebra whose defect is linear in η.

Notation for this section. $\mathcal{H}$ is a nonzero Hilbert space and $\Phi : B(\mathcal{H}) \to B(\mathcal{H})$ is UCP (`def-ucp-map`, Definition 40). All norms written $\|\cdot\|_{\mathrm{cb}}$ are completely bounded norms; I is the identity map, and products of maps are compositions, so that $CB(B(\mathcal{H}))$ is a unital Banach algebra. At amplification level n we write $T = 1_{M_n} \otimes \Phi$ and $D = T^2 - T$. Finally

$$r \;=\; \tfrac{3}{2}\big((1-4\eta)^{-1/2} - 1\big), \qquad M \;=\; 1 + r. \tag{127}$$

Lemma 83 (`lem-kitaev-almost-idemp-audit`). *Let $\mathcal{H} \neq 0$ and let $\Phi : B(\mathcal{H}) \to B(\mathcal{H})$ be UCP with $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta < \frac{1}{4}$. Put*

$$\widetilde{\Phi} \;=\; \tfrac{1}{2}\Big(I + (2\Phi - I)\big(I - 4(\Phi - \Phi^2)\big)^{-1/2}\Big), \qquad \mathcal{A} = \mathrm{Im}(\widetilde{\Phi}), \qquad X \star Y = \widetilde{\Phi}(XY). \tag{128}$$

Then $\widetilde{\Phi}^2 = \widetilde{\Phi}$; both amplified associativity identities Φ_{assoc1} and Φ_{assoc2}, namely

$$\begin{aligned} \big\|T\big(T(T(X)T(Y))T(Z)\big) - T\big(T(X)T(Y)T(Z)\big)\big\| \;&\le\; 10\eta\|X\|\|Y\|\|Z\|, \\ \big\|T\big(T(X)T(T(Y)T(Z))\big) - T\big(T(X)T(Y)T(Z)\big)\big\| \;&\le\; 10\eta\|X\|\|Y\|\|Z\|, \end{aligned} \tag{129}$$

hold at every amplification after the local source type and index corrections recorded below; and for sufficiently small universal η the inherited operator-space norms, involution and unit make $\mathcal{A}$ an extended $\varepsilon_{\mathrm{AI}}(\eta)$-$C^*$-algebra (`def-extended-epsilon-cstar-algebra`, Definition 23) with

$$\varepsilon_{\mathrm{AI}}(\eta) \;=\; \max\big\{\, r,\; 20\eta + 2(M^5 - 1),\; 3r - r^2 \,\big\} \;=\; O(\eta), \tag{130}$$

r, M as in (127)*, dimension-free at every amplification.* Transcription note. *This row is an* audit*: it registers the corrected statement of the pinned source's `th_almost_idemp`, together with the explicit interface* (130)*. It claims nothing about the source's `th_main_ext`, and no optimality of the displayed constants is asserted. The threshold "sufficiently small universal η" is not made explicit by the contract and is not made explicit here.*

Registry contract (verbatim).

```
Corrected th_almost_idemp audit: let H be nonzero and let Phi:B(H)->B(H) be UCP with
||Phi^2-Phi||_cb <= eta < 1/4; for tilde-Phi=(1/2)(I+(2Phi-I)(I-4(Phi-Phi^2))^(-1/2)),
A=Im(tilde-Phi), and X star Y=tilde-Phi(XY), one has tilde-Phi^2=tilde-Phi, both amplified
associativity identities Phi_assoc1 and Phi_assoc2 have error at most 10*eta*||X||||Y||||Z|| after
the local source type/index corrections, and for sufficiently small universal eta the inherited
operator-space norms, involution, and unit make A an extended epsilon_AI(eta)-C*-algebra with
epsilon_AI=max{r,20eta+2(M^5-1),3r-r^2}=O(eta), r=(3/2)((1-4eta)^(-1/2)-1), M=1+r, dimension-free
at every amplification.
```

Proof (prose account of the af-validated tree). The tree has four branches: a spectral package, the first associativity estimate, its adjoint twin, and the packaging into the axioms.

The spectral package. In $CB(B(\mathcal{H}))$ set $F = \Phi - \Phi^2$ and $A_0 = 2\Phi - I$. Both are polynomials in Φ, so they commute, and $A_0^2 = I - 4F$ while $\|4F\|_{\mathrm{cb}} \le 4\eta < 1$. The binomial series $\mathcal{R} = \sum_{k\ge 0} \binom{2k}{k} F^k$ therefore converges in norm to $(I-4F)^{-1/2}$ and commutes with A_0; hence $S = A_0\mathcal{R}$ satisfies $S^2 = I$ and $\widetilde{\Phi} = (I+S)/2$ is idempotent, which is the first clause. Since $\widetilde{\Phi} - \Phi = \frac{1}{2}A_0(\mathcal{R} - I)$, $\|A_0\|_{\mathrm{cb}} \le 2\|\Phi\|_{\mathrm{cb}} + 1 \le 3$ and the series coefficients are nonnegative, $\|\mathcal{R} - I\|_{\mathrm{cb}} \le (1-4\eta)^{-1/2} - 1$, so $\|\widetilde{\Phi} - \Phi\|_{\mathrm{cb}} \le r$. Unitality and $*$-preservation of Φ pass through the real-coefficient series, so $\widetilde{\Phi}(I) = I$ and $\widetilde{\Phi}(X^\dagger) = \widetilde{\Phi}(X)^\dagger$; a bounded idempotent has $\mathrm{Im} = \mathrm{Ker}(I - \widetilde{\Phi})$, so $\mathcal{A}$ is closed, complete, self-adjoint and contains the unit, and every formula commutes with amplification.

The associativity estimate. This is the delicate branch, and the tree proves it without importing a Stinespring theorem. It first constructs, from complete positivity alone, an iterated GNS dilation: Hilbert spaces K_j, isometries $V_j : K_{j-1} \to K_j$ and contractive unital $*$-homomorphisms $u_j : B(K_{j-1}) \to B(K_j)$ with $T(H) = V_1^\dagger u_1(H) V_1$ and the one-step compression identity $u_j(V_j^\dagger U V_j) = V_{j+1}^\dagger u_{j+1}(U) V_{j+1}$ for $U \in B(K_j)$. Writing $u_{b,a} = u_b \circ \cdots \circ u_{a+1}$, $P_1 = V_1 V_1^\dagger$ and $E_X = (I_1 - P_1)u_1(T(X))$, the elementary defect ledger $\|T\| \le 1$, $\|D\| \le \eta$ gives $\|T^2(U) - T(U)\| \le \eta\|U\|$ and $\|T^2(U)T^2(V) - T(U)T(V)\| \le 2\eta\|U\|\|V\|$, whence $G_X = T^2(T(X^\dagger)T(X)) - T(T^2(X^\dagger)T^2(X))$ obeys $\|G_X\| \le 3\eta\|X\|^2$. The "rectangles" $R_k(X) = u_{k+2,1}(E_X)V_{k+2}V_{k+1}$ then satisfy the exact identity $R_k(X)^\dagger R_k(X) = u_{k,0}(G_X)$, so $\|R_k(X)\| \le \sqrt{3\eta}\|X\|$. With $A = T(X)$, $B = T(Y)$, $C = T(Z)$ and $A_1 = T^2(X)$, $B_1 = T^2(Y)$, $C_1 = T^2(Z)$, the operator $W = V_1^\dagger R_1(X^\dagger)^\dagger u_{3,0}(B) V_3 R_0(Z)$ expands *exactly* to $W = T(Q)$ with

$$Q = T\big(T(AB)C\big) - T^2(AB)C_1 - T(A_1B_1C) + T(A_1B_1)C_1, \tag{131}$$

and $\|W\| \le 3\eta\, p$ where $p = \|X\|\|Y\|\|Z\|$. Setting $F_1 = T(T(AB)C) - T(ABC)$, cancellation in (131) gives $Q - F_1 = [T(ABC) - T(A_1B_1C)] + [T(A_1B_1) - T^2(AB)]C_1$, and the perturbation bounds $\|A_1B_1 - AB\| \le 2\eta\|X\|\|Y\|$ and $\|T(A_1B_1) - T^2(AB)\| \le 3\eta\|X\|\|Y\|$ give $\|Q - F_1\| \le 5\eta p$. Contractivity of T transports this to $\|T(Q) - T(F_1)\| \le 5\eta p$, and since F_1 is a difference of two T-values the defect estimate applied to each gives $\|T(F_1) - F_1\| \le 2\eta p$. Hence $\|W - F_1\| \le 7\eta p$ and, with $\|W\| \le 3\eta p$, $\|F_1\| \le 10\eta p$: the first line of (129). The second follows by applying the first to $(Z^\dagger, Y^\dagger, X^\dagger)$ and taking adjoints, T being $*$-preserving.

The axioms. At each level write $P = 1_{M_n} \otimes \widetilde{\Phi}$, so $\|P - T\| \le r$ and $\|P\| \le M$. Bilinearity, range-valuedness, the exact unit and $(X \star Y)^\dagger = Y^\dagger \star X^\dagger$ are immediate from idempotence, unitality and $*$-preservation of P; $\|X \star Y\| \le M\|X\|\|Y\|$ is the submultiplicativity error r. For $X \in \mathrm{Im}(P)$, $\|T(X)\| \ge (1-r)\|X\|$, the Kadison–Schwarz inequality gives $\|T(X^\dagger X)\| \ge \|T(X)\|^2$, and hence $\|P(X^\dagger X)\| \ge (1 - 3r + r^2)\|X\|^2$: the C^* lower-norm error $3r - r^2$. Finally, replacing the five occurrences of P in $P(P(P(X)P(Y))P(Z))$ and in $P(P(X)P(P(Y)P(Z)))$ one at a time by T and using $\|T\| \le 1$, $\|P\| \le M$, $\|P - T\| \le r = M - 1$ bounds each telescoping sum by $r(1 + M + M^2 + M^3 + M^4)p = (M^5 - 1)p$; combined with the two estimates (129) this gives $\|(X \star Y) \star Z - X \star (Y \star Z)\| \le (20\eta + 2(M^5 - 1))p$ for $X, Y, Z \in \mathrm{Im}(P)$. The maximum of the three errors is (130), every input estimate is amplification-uniform, and $M^5 - 1 \le 31r$ for $0 \le r \le 1$ with $r = O(\eta)$, so $\varepsilon_{\mathrm{AI}} \to 0$ and a universal threshold makes it less than one.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-kitaev-almost-idemp-audit`; the registered definition import is `def-extended-epsilon-cstar-algebra`, and the dilation is built inside the tree rather than imported, which is why no Stinespring external appears.

Corrections of record (to the local source, carried by the contract). The audit's corrections are type- and index-level and are named in the contract itself: the operator variables at the source loci belong to the operator algebras $B(K_j)$ and not to the Hilbert spaces K_j, and the final isometry of the auxiliary compression identity is V_{1+k}, not V_1 — with V_1 the identity is ill-typed for $k > 0$, since V_1 has domain K_0. The canonical Stinespring construction remains a source dependency of the original argument; here it is replaced by the in-tree recursive dilation.

Lemma 84 (`lem-routef-functional-calculus-closeness`). *For $0 \le \eta \le \frac{1}{8}$ the exact functional-calculus projector satisfies*

$$\|\widetilde{\Phi} - \Phi\|_{\mathrm{cb}} \;\le\; C_\theta \eta, \qquad C_\theta = 12(\sqrt{2} - 1). \tag{132}$$

Transcription note. *The contract is elliptic by design: "the exact functional-calculus projector" is $\widetilde{\Phi}$ of* (128)*, and the ambient hypotheses — Φ UCP on $B(\mathcal{H})$ with $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta$ — are those of the imported Lemma 83, which is how the validated tree reads them. The statement is a* linear *bound valid on the stated window only; it says nothing for $\eta > \frac{1}{8}$.*

Registry contract (verbatim).

```
Functional-calculus closeness: for 0 <= eta <= 1/8, the exact functional-calculus projector
satisfies ||tilde-Phi-Phi||_cb <= C_theta*eta, where C_theta=12*(sqrt(2)-1).
```

Proof (prose account of the af-validated tree). The tree proves the estimate $\|\widetilde{\Phi} - \Phi\|_{\mathrm{cb}} \le r$ of (127) in its own right and then linearises r. For the first half it records that a UCP map is completely contractive — the matrix Kadison–Schwarz inequality $\Phi_m(X)^\dagger \Phi_m(X) \le \Phi_m(X^\dagger X)$ obtained from positivity of $[X^\dagger; I][X, I]$, combined with $X^\dagger X \le \|X\|^2 I$ — so $\|2\Phi - I\|_{\mathrm{cb}} \le 3$; then the algebraic identity $\widetilde{\Phi} - \Phi = \frac{1}{2}(2\Phi - I)\big((I - 4\Delta)^{-1/2} - I\big)$ with $\Delta = \Phi - \Phi^2$; then, since $\|4\Delta\|_{\mathrm{cb}} \le 4\eta \le \frac{1}{2} < 1$, the absolutely convergent binomial series with nonnegative coefficients gives $\|(I - 4\Delta)^{-1/2} - I\|_{\mathrm{cb}} \le (1 - 4\eta)^{-1/2} - 1$. For the second half, put $t = 4\eta \in [0, \frac{1}{2}]$ and $s = \sqrt{1 - t} \ge 1/\sqrt{2}$; rationalisation gives

$$\frac{(1-t)^{-1/2} - 1}{t} = \frac{1}{s(1+s)} \;\le\; \frac{1}{\frac{1}{\sqrt{2}}(1 + \frac{1}{\sqrt{2}})} = 2(\sqrt{2} - 1),$$

the case $t = 0$ being an equality in the inequality $(1 - t)^{-1/2} - 1 \le 2(\sqrt{2} - 1)t$ itself. Multiplying by t and then by $\frac{3}{2}$ turns $r \le \frac{3}{2} \cdot 2(\sqrt{2} - 1) \cdot 4\eta$ into (132).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-functional-calculus-closeness`; registry `defs`: `def-almost-idempotent`; registry `deps`: `lem-kitaev-almost-idemp-audit` (Lemma 83), from which the projector formula and the ambient hypotheses are imported.

Lemma 85 (`lem-routef-ai-defect-linearization`). *Set $C_\theta = 12(\sqrt{2} - 1)$. There are universal constants $C_A < \infty$ and $\eta_A > 0$, with $C_A = 20 + \frac{211}{8} C_\theta$, such that for every nonzero Hilbert space $\mathcal{H}$, every UCP map $\Phi : B(\mathcal{H}) \to B(\mathcal{H})$ and every η with $0 \le \eta \le \eta_A$ and $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta$, the data $\widetilde{\Phi}, \mathcal{A}, \star$ of* (128) *together with r of* (127) *and $\varepsilon_{\mathrm{AI}}(\eta)$ of* (130) *make $\mathcal{A}$, with its inherited operator-space norms, involution and unit, an extended $\varepsilon_{\mathrm{AI}}(\eta)$-$C^*$-algebra with*

$$\varepsilon_{\mathrm{AI}}(\eta) \;\le\; C_A \eta. \tag{133}$$

Transcription note. *What this row adds to Lemma 83 is the* linearity (133) *and an explicit universal constant; the threshold η_A is asserted to exist and to be universal, and is not exhibited, because it inherits the unexhibited threshold of that lemma.*

Registry contract (verbatim).

```
Approximate-algebra defect linearization: set C_theta=12*(sqrt(2)-1). There are universal C_A <
infinity and eta_A > 0, with C_A=20+(211/8)*C_theta, such that for every nonzero Hilbert space H,
every UCP map Phi:B(H)->B(H), and every eta satisfying 0 <= eta <= eta_A and ||Phi^2-Phi||_cb <=
eta, if tilde-Phi=(1/2)*(I+(2*Phi-I)*(I-4*(Phi-Phi^2))^(-1/2)), A=Im(tilde-Phi), X star
Y=tilde-Phi(XY), r=(3/2)*((1-4*eta)^(-1/2)-1), and
epsilon_AI(eta)=max{r,20*eta+2*((1+r)^5-1),3*r-r^2}, then the inherited operator-space norms,
involution, and unit together with star make A an extended epsilon_AI(eta)-C*-algebra and
epsilon_AI(eta) <= C_A*eta.
```

Proof (prose account of the af-validated tree). Let $\eta_* > 0$ be the universal threshold of Lemma 83, shrunk below $\frac{1}{4}$, and put $\eta_A = \min\{\eta_*, \frac{7}{64}\}$, which is universal and positive. For $0 \le \eta \le \eta_A$ that lemma supplies the extended $\varepsilon_{\mathrm{AI}}(\eta)$-$C^*$-algebra structure and the formula (130), so only the three entries of the maximum have to be bounded. On $0 \le \eta \le \frac{7}{64}$ one has, with $t = \sqrt{1-4\eta} \ge 1/\sqrt{2}$, that $r/\eta = 6/(t(1+t)) \le C_\theta$, hence $0 \le r \le C_\theta \eta$ — the constant being exactly the one fixed by Lemma 84; also $1 - 4\eta \ge \frac{9}{16}$ gives $r \le \frac{1}{2}$. For $0 \le r \le \frac{1}{2}$ the binomial identity $2((1+r)^5 - 1) = 2r(5 + 10r + 10r^2 + 5r^3 + r^4)$ and monotonicity of the parenthesis give $2((1+r)^5-1) \le \frac{211}{8} r$. Therefore

$$20\eta + 2\big((1+r)^5 - 1\big) \;\le\; 20\eta + \tfrac{211}{8} C_\theta \eta \;=\; C_A \eta,$$

while $C_A - C_\theta = 20 + \frac{203}{8} C_\theta > 0$ and $C_A - 3C_\theta = 20 + \frac{187}{8} C_\theta > 0$ give $r \le C_A \eta$ and $3r - r^2 \le 3r \le C_A \eta$. Every entry of the maximum is at most $C_A \eta$, which is (133).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-ai-defect-linearization`: a 13-node tree, taint clean.
Correction of record (contract-level, banking-time). The prover expanded the seeded one-line contract into the self-contained root statement reproduced above, inlining the value of C_θ and the formula (128) from its imported dependencies; the fresh verifiers validated *that* statement, so the registry contract was mechanically replaced by the validated root verbatim at banking. No equivalence judgement was exercised by the orchestrator; the linker's contract-match refusal is what forced the reconciliation.
Role in the chain. Registry `defs`: `def-almost-idempotent`, `def-extended-epsilon-cstar-algebra`. Registry `deps`: `lem-kitaev-almost-idemp-audit` and `lem-routef-functional-calculus-closeness`, both reproduced above and both `af`-validated. The consumer of this ledger is the factorization row `lem-routef-k-ledger`, which is `proved-mod-audit` with `af: none` — **open** at this document's rigour bar and **not** reproduced here. Nothing in this section therefore produces a bound on a stochastic matrix, and `op-classical` is **open**.

43 The F2/F3 block: manufacturing the positive unital retract pair

The finish edge of §38 consumes a pair of positive unital maps A, M between real ℓ^∞ spaces, subject to two estimates. This section reproduces the two `af`-validated rows that manufacture that pair, and the estimates, out of an approximate factorization of the F0 lift Φ of §41 through a finite-dimensional unital C^*-algebra.

Notation for this section. The notation of §41 is in force: M_n, the complex diagonal algebra $\mathbb{C}^n$, its real self-adjoint part $\ell^\infty(n)$, the maps $J, D, Q_\mathbb{C}$ and $\Phi = JQ_\mathbb{C}D$. The intermediate algebra of the contract is written $\mathcal{B}$, and M denotes a map, not a matrix algebra. All completely bounded norms are written $\|\cdot\|_{\mathrm{cb}}$, and $K \geq 1$ is a dimension-independent constant supplied with the hypotheses, not fixed here.

Lemma 86 (`lem-routef-f2-positive-unital-compression`). *Let $K \geq 1$ be dimension-independent, $n \geq 1$, $Q : \ell^\infty(n) \to \ell^\infty(n)$ row-stochastic, $\Phi = JQ_\mathbb{C}D$, let $\mathcal{B}$ be a finite-dimensional unital C^*-algebra and let $\Delta : \mathcal{B} \to M_n$ and $\Upsilon : M_n \to \mathcal{B}$ be UCP. If*

$$0 \;\leq\; \eta \;\leq\; \min\big\{(24K)^{-1},\, 1\big\} \tag{134}$$

and

$$\|\Delta\Upsilon - \Phi\|_{\mathrm{cb}} \leq K\eta, \quad \|\Upsilon\Delta - I_\mathcal{B}\|_{\mathrm{cb}} \leq K\eta, \quad \|\Upsilon(\Delta x\, \Delta y) - xy\| \leq K\eta\|x\|\|y\| \;\; (x, y \in \mathcal{B}), \tag{135}$$

then $\mathcal{B}$ is commutative, and there are $k \geq 1$ and a unital $*$-isomorphism $\iota_\mathbb{C} : \mathbb{C}^k \to \mathcal{B}$ such that $D\Delta\iota_\mathbb{C}$ maps $\mathbb{R}^k$ into $\mathbb{R}^n$, $\iota_\mathbb{C}^{-1}\Upsilon J$ maps $\mathbb{R}^n$ into $\mathbb{R}^k$, and the resulting restrictions and corestrictions

$$A := (D\Delta\iota_\mathbb{C})|_{\mathbb{R}^k} : \ell^\infty(k) \to \ell^\infty(n), \qquad M := (\iota_\mathbb{C}^{-1}\Upsilon J)|_{\mathbb{R}^n} : \ell^\infty(n) \to \ell^\infty(k) \tag{136}$$

are positive unital maps satisfying

$$\|Q - AM\|_{\infty\to\infty} \leq K\eta, \qquad \|QA - A\|_{\infty\to\infty} \leq 2K\eta, \qquad \|Ax\|_\infty \geq (1 - 3K\eta)\|x\|_\infty \;\; (x \in \ell^\infty(k)). \tag{137}$$

Transcription note. *The three displays of* (135) *are* hypotheses*: the factorization datum $(\mathcal{B}, \Delta, \Upsilon)$ is given, and nothing here, or anywhere in this document, manufactures it from Q. Commutativity of $\mathcal{B}$ is* derived*, not assumed. The real/complex typing is load-bearing: Δ, Υ and $\iota_\mathbb{C}$ live over $\mathbb{C}$, and A, M are their restrictions and corestrictions to the real self-adjoint parts.*

Registry contract (verbatim).

```
Route F F2 positive-unital compression: let K >= 1 be a dimension-independent constant, n >= 1, Q:
l_inf^n -> l_inf^n row-stochastic, D: M_n -> C^n diagonal extraction onto the complex diagonal
algebra C^n = l_inf^n(C), J: C^n -> M_n diagonal inclusion, Q_C: C^n -> C^n the canonical
complex-linear extension of Q, and Phi = J Q_C D, B a finite-dimensional unital C*-algebra, and
Delta: B -> M_n, Upsilon: M_n -> B UCP maps; if 0 <= eta <= min{(24K)^{-1},1}, ||Delta Upsilon -
Phi||_cb <= K*eta, ||Upsilon Delta - I_B||_cb <= K*eta, and ||Upsilon(Delta x Delta y) - xy|| <=
K*eta*||x||*||y|| for all x,y in B, then B is commutative and there are k >= 1 and a unital
*-isomorphism iota_C: C^k = l_inf^k(C) -> B such that D Delta iota_C maps R^k into R^n, iota_C^{-1}
Upsilon J maps R^n into R^k, and the resulting restrictions and corestrictions A := (D Delta
iota_C)|_{R^k}: l_inf^k -> l_inf^n and M := (iota_C^{-1} Upsilon J)|_{R^n}: l_inf^n -> l_inf^k are
positive unital maps satisfying ||Q - AM||_{inf->inf} <= K*eta, ||QA - A||_{inf->inf} <= 2K*eta,
and ||Ax||_inf >= (1-3K*eta)*||x||_inf for every x in l_inf^k.
```

Proof (prose account of the af-validated tree). The tree runs in three movements: force commutativity, coordinatise, estimate.

Preliminaries. One shared node derives complete contractivity of a UCP map from 2-positivity: for $S = \mathrm{id}_{M_r} \otimes T$ the block matrix $[a,1]^\dagger[a,1]$ is positive, so $S(a)^\dagger S(a) \le S(a^\dagger a) \le \|a\|^2$, whence $\|T\|_{\mathrm{cb}} \le 1$; in particular $\|\Delta\|_{\mathrm{cb}}, \|\Upsilon\|_{\mathrm{cb}} \le 1$. A second node records the typing facts of the seam: Φ is UCP hence contractive, its range lies in $J(\mathbb{C}^n)$ so any two of its values commute, $DJ = I$, J is isometric, and every positive map is $*$-preserving.

Approximate invariance and approximate commutativity. For $b \in \mathcal{B}$,

$$\Delta b - \Phi\Delta b = \Delta\big(b - \Upsilon\Delta b\big) + (\Delta\Upsilon - \Phi)(\Delta b), \qquad \text{so} \qquad \|\Delta b - \Phi(\Delta b)\| \le 2K\eta\|b\|.$$

Put $a = \Delta x$, $b = \Delta y$ and $a_0 = \Phi a$, $b_0 = \Phi b$. Then $\|a - a_0\| \le 2K\eta\|x\|$, $\|b - b_0\| \le 2K\eta\|y\|$, $\|a\|, \|a_0\| \le \|x\|$, $\|b\|, \|b_0\| \le \|y\|$ and $[a_0, b_0] = 0$, so $\|[a,b]\| \le 8K\eta\|x\|\|y\|$. Feeding this through the third hypothesis of (135) for (x,y) and for (y,x), and using contractivity of Υ,

$$\|xy - yx\| \;\le\; 10K\eta\|x\|\|y\| \qquad (x, y \in \mathcal{B}). \tag{138}$$

The norm-two witness, and commutativity. The tree does *not* cite the finite-dimensional C^*-classification — it is not in the local sources — and proves instead: a noncommutative finite-dimensional unital C^*-algebra carries contractions x, y with $\|xy - yx\| = 2$. Decompose 1 into mutually orthogonal minimal projections $p_1, \dots, p_m$ (a minimal projection p has $p\mathcal{B}p = \mathbb{C}p$, else a proper spectral projection of a non-scalar self-adjoint element of $p\mathcal{B}p$ would sit below p). If every off-diagonal corner $p_i\mathcal{B}p_j$ vanished, then $b = \sum_{i,j} p_i b p_j = \sum_i \lambda_i(b) p_i$ and $\mathcal{B}$ would be commutative; so pick $0 \ne z \in p\mathcal{B}q$ with $p \ne q$ minimal. Then $z^\dagger z \in q\mathcal{B}q$ and $zz^\dagger \in p\mathcal{B}p$, so the C^*-identity gives $z^\dagger z = \|z\|^2 q$ and $zz^\dagger = \|z\|^2 p$; with $v = z/\|z\|$ one has $v^\dagger v = q$, $vv^\dagger = p$ and $v^2 = (v^\dagger)^2 = 0$. The Pauli pair $x = p - q$, $y = v + v^\dagger$ consists of self-adjoint contractions with $xy - yx = 2(v - v^\dagger)$ of norm 2. Under (134) this contradicts (138), since $2 \le 10K\eta \le \frac{10}{24} = \frac{5}{12} < 2$; hence $\mathcal{B}$ is commutative.

Coordinates. A commutative finite-dimensional unital C^*-algebra carries a projection basis $\{\Pi_1, \dots, \Pi_k\}$ — this, byte-matched to its pinned source, is the *only* external the tree is permitted (`def-projection-basis`, Definition 29). Then $k \ge 1$, the Π_j are nonzero and span, and $\iota_{\mathbb{C}}(z) = \sum_j z_j \Pi_j$ is a unital $*$-isomorphism $\mathbb{C}^k \to \mathcal{B}$. The compositions $D\Delta\iota_{\mathbb{C}}$ and $\iota_{\mathbb{C}}^{-1}\Upsilon J$ are UCP, hence positive, unital and $*$-preserving, so they carry self-adjoint parts to self-adjoint parts and (136) defines positive unital maps.

The three estimates. For $z \in \mathbb{R}^n$, $DJ = I$ and $\Phi = JQ_{\mathbb{C}}D$ give $Qz = D\Phi Jz$ while $AMz = D\Delta\Upsilon Jz$; contractivity of D and J then turns the first hypothesis of (135) into $\|Q - AM\|_{\infty\to\infty} \le K\eta$. For $x \in \mathbb{R}^k$, $D\Phi = Q_{\mathbb{C}}D$ and the restriction of $Q_{\mathbb{C}}$ to $\mathbb{R}^n$ give $QAx - Ax = D(\Phi\Delta\iota_{\mathbb{C}}x - \Delta\iota_{\mathbb{C}}x)$, so approximate invariance yields $\|QA - A\|_{\infty\to\infty} \le 2K\eta$. Finally $\|\Delta\iota_{\mathbb{C}}x\| \ge \|\Upsilon\Delta\iota_{\mathbb{C}}x\| \ge (1 - K\eta)\|x\|_\infty$ by the second hypothesis, approximate invariance costs a further $2K\eta\|x\|_\infty$, and $\|\Phi\Delta\iota_{\mathbb{C}}x\| = \|JQ_{\mathbb{C}}Ax\| \le \|Ax\|_\infty$; combining gives the lower modulus in (137).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-f2-positive-unital-compression`: root and 22/22 nodes `validated`, taint clean; one genuine mid-run challenge, resolved. Registered imports: the definitions `def-stochastic`, `def-ucp-map`, `def-projection-basis`, and the single byte-verbatim projection-basis external.

Corrections of record. Two, at different levels. (i) *Contract-level*: the earlier text typed D into the real space $\ell^\infty(n)$ while the diagonal of M_n is complex — the same defect repaired on both F0 seam rows, and the cause of a first elevation run that was abandoned rather than resumed. The contract quoted above is the corrected text; every hypothesis, constant, threshold and estimate is unchanged by the correction. (ii) *Proof-level*: the challenge was the endpoint $\eta = 0$, where a strict inequality chain used in the commutativity node failed. The repair is the endpoint bridge reproduced implicitly above: at $\eta = 0$, (138) reads $\|xy - yx\| = 0$ and commutativity is immediate, so the conclusion holds on the whole closed window (134).

Lemma 87 (`lem-routef-f3-retract-defect`). *Let $K \geq 1$ be dimension-independent, $n, k \geq 1$, let $A : \ell^\infty(k) \to \ell^\infty(n)$ and $M : \ell^\infty(n) \to \ell^\infty(k)$ be positive unital maps, let $Q : \ell^\infty(n) \to \ell^\infty(n)$ be row-stochastic, and let $\eta \geq 0$ satisfy $3K\eta < 1$. If the three estimates* (137) *hold, then*

$$\|MA - I_k\|_{\infty\to\infty} \;\leq\; \frac{3K\eta}{1 - 3K\eta}. \tag{139}$$

Transcription note. *This row is stated entirely in the real picture; its hypotheses are exactly the three conclusions of Lemma 86, but they are* hypotheses *here, and the row does not import that lemma. The denominator is positive precisely because $3K\eta < 1$.*

Registry contract (verbatim).

```
Route F F3 retract defect: let K >= 1 be a dimension-independent constant, n,k >= 1, A: l_inf^k ->
l_inf^n and M: l_inf^n -> l_inf^k positive unital maps, Q: l_inf^n -> l_inf^n row-stochastic, and
eta >= 0 with 3K*eta < 1; if ||Q - AM||_{inf->inf} <= K*eta, ||QA - A||_{inf->inf} <= 2K*eta, and
||Ax||_inf >= (1-3K*eta)*||x||_inf for every x in l_inf^k, then ||MA - I_k||_{inf->inf} <=
3K*eta/(1-3K*eta).
```

Proof (prose account of the af-validated tree). First, A is contractive: for $x \in \ell^\infty(k)$ and $c = \|x\|_\infty$ one has $-c\mathbf{1}_k \leq x \leq c\mathbf{1}_k$ coordinatewise, positivity of A preserves the order and unitality sends $\mathbf{1}_k$ to $\mathbf{1}_n$, so $-c\mathbf{1}_n \leq Ax \leq c\mathbf{1}_n$ and $\|Ax\|_\infty \leq \|x\|_\infty$. Next, linearity gives the exact identity

$$A\big((MA - I_k)x\big) = (AM - Q)(Ax) + (QA - A)x. \tag{140}$$

Since $\|AM - Q\|_{\infty\to\infty} = \|Q - AM\|_{\infty\to\infty}$, the two operator-norm hypotheses and contractivity of A bound the right-hand side of (140) by $K\eta\|x\|_\infty + 2K\eta\|x\|_\infty = 3K\eta\|x\|_\infty$. Put $\alpha = 1 - 3K\eta > 0$ and apply the lower-modulus hypothesis to $y = (MA - I_k)x$: then $\alpha\|y\|_\infty \leq \|Ay\|_\infty \leq 3K\eta\|x\|_\infty$, i.e. $\|(MA - I_k)x\|_\infty \leq (3K\eta/\alpha)\|x\|_\infty$ for every x. Taking the supremum over $\|x\|_\infty \leq 1$ gives (139).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-f3-retract-defect`: a first-pass elevation, root and 11/11 nodes `validated`, taint clean, *zero* challenges. The registered definition import is `def-stochastic`; there are no external imports, and no dimension or norm loss occurs anywhere in the tree.

Role in the chain, and what is still missing. Neither row has registry `deps`. Their conclusions are, literally, the hypothesis list of the finish edge `lem-routef-prh-finish` (§38): (137) supplies $\|Q - AM\|_{\infty\to\infty} \leq K\eta$ and the input of (139), and (139) is verbatim that lemma's second hypothesis, at the same threshold $\eta \leq (24K)^{-1}$. What is *not* supplied anywhere in this document is the antecedent of Lemma 86 itself: the factorization datum $(\mathcal{B}, \Delta, \Upsilon)$ with the three estimates (135) and a dimension-free K. The registry row that would supply it, `lem-routef-k-ledger`, is `proved-mod-audit` with `af: none` — **open** at this document's rigour bar and **not** reproduced here — as is the assembly carrier `lem-thmainext-conditional`. The Route F chain is therefore *not* closed, no unconditional $\sqrt{\eta}$ bound follows, and `op-classical` is **open**.

44 Rectified algebras and graph charts on the approximate unitaries

This section and the three that follow (§45, §46, §47) reproduce eleven `af`-validated Stage-1 results about the unitary set of an approximate C^*-algebra. Throughout, the ambient object is an *exact-unit* ε_r-C^*-algebra $(\mathcal{X}, J, \boldsymbol{\cdot}, \dagger)$ in the sense of Definition 22, and every set and symbol — $\mathcal{H}$, $i\mathcal{H}$, $\overline{\mathcal{U}}_\delta$, $\mathcal{U}_\delta$, $\mathcal{U}$, L_V, ϕ_V with its components $\phi_V^{\parallel}, \phi_V^{\perp}$, and the reserved letters u, h, μ, σ — is the vocabulary of Definition 31, used here and never restated. None of these results bears on `op-classical`, which is **open**.

Lemma 88 (`lem-stage1-rectified-cstar-control`). *There are universal* $C_{\mathrm{rect}} \geq 1$ *and* $e_{\mathrm{rect}} \in (0, 1/C_{\mathrm{rect}}]$ *such that every finite-dimensional* ε_X*-*C^**-algebra with* $0 \leq \varepsilon_X \leq e_{\mathrm{rect}}$ *admits, on the same involutive normed space, a bilinear product* $\boldsymbol{\cdot}$ *and an element* $J = J^\dagger$ *for which* $(\mathcal{X}, J, \boldsymbol{\cdot}, \dagger)$ *satisfies* every *exact-unit* ε_r*-*C^**-algebra axiom of Definition 22, including* $\|J\| = 1$*, where*

$$\varepsilon_r = C_{\mathrm{rect}}\varepsilon_X, \qquad \|J - I_X\| \leq C_{\mathrm{rect}}\varepsilon_X, \qquad \|x \boldsymbol{\cdot} y - xy\| \leq C_{\mathrm{rect}}\varepsilon_X \|x\|\|y\|. \tag{141}$$

Transcription note. *This is strictly stronger than Lemma 80 (§40), which produces an exact unit and a close product but estimates neither the associativity defect nor the product-norm constant of* $\boldsymbol{\cdot}$*. Here the whole axiom list is transferred, at the inflated parameter* $\varepsilon_r = C_{\mathrm{rect}}\varepsilon_X$*; that inflation is the price, and it is what makes the rectified algebra a legitimate input to everything below.*

Registry contract (verbatim).

```
Controlled exact-unit C*-rectification: there are universal C_rect >= 1 and e_rect in (0, 1/C_rect]
such that every finite-dimensional epsilon_X-C*-algebra with 0 <= epsilon_X <= e_rect admits, on
the same involutive normed space, a bilinear product bold-dot and J = J^dagger for which (calX, J,
bold-dot, dagger) satisfies EVERY exact-unit epsilon_r-C*-algebra axiom of
def-epsilon-cstar-algebra, including ||J|| = 1, where epsilon_r = C_rect*epsilon_X, and ||J - I_X||
<= C_rect*epsilon_X, ||x bold-dot y - xy|| <= C_rect*epsilon_X*||x||*||y||.
```

Proof (prose account of the af-validated tree). The witnesses are exhibited: the tree closes with

$$C_{\mathrm{rect}} = 100, \qquad e_{\mathrm{rect}} = \tfrac{1}{100}. \tag{142}$$

A Hermitian norming functional. Write $I = I_X$ and $a = \|I\|$, so $|a - 1| \leq \varepsilon_X$ and $a > 0$. A finite-dimensional Hahn–Banach extension produces a complex-linear f with $\|f\| = 1$ and $f(I) = a$; since $\dagger$ is a conjugate-linear isometric involution and $I^\dagger = I$, the reflection $f^\sharp(x) = \overline{f(x^\dagger)}$ has the same norm and the same value at I, so $\varphi = (f + f^\sharp)/(2a)$ satisfies

$$\varphi(I) = 1, \qquad \|\varphi\| \leq 1/a, \qquad \varphi(x^\dagger) = \overline{\varphi(x)}. \tag{143}$$

The corrected product. With $d_L(y) = y - Iy$, $d_R(x) = x - xI$ and $h = II - I$, put

$$m_0(x, y) = xy + \varphi(x)d_L(y) + \varphi(y)d_R(x) + \varphi(x)\varphi(y)h, \qquad J = I/a, \qquad x \boldsymbol{\cdot} y = a\, m_0(x, y). \tag{144}$$

Because $\varphi(I) = 1$ and $d_L(I) = d_R(I) = -h$, substitution gives $m_0(I, y) = y$ and $m_0(x, I) = x$, so J is an exact two-sided unit for $\boldsymbol{\cdot}$ with $J^\dagger = J$ and $\|J\| = 1$. Term by term, $d_L(y)^\dagger = d_R(y^\dagger)$, $d_R(x)^\dagger = d_L(x^\dagger)$, $h^\dagger = h$ and the last identity of (143) give $(x \boldsymbol{\cdot} y)^\dagger = y^\dagger \boldsymbol{\cdot} x^\dagger$ *exactly*, using only that a is a positive real.

The estimates and the axiom transfer. The approximate-unit axioms give $\|d_L(y)\| \le \varepsilon_X \|y\|$, $\|d_R(x)\| \le \varepsilon_X \|x\|$ and $\|h\| \le a\varepsilon_X$; with $\|\varphi\| \le 1/a$ this bounds $\|m_0(x,y) - xy\|$ by $3\varepsilon_X \|x\|\|y\|/a$, and after rescaling

$$\Delta(x,y) := x \cdot y - xy, \qquad \|\Delta(x,y)\| \le 5\,\varepsilon_X \|x\|\|y\| \qquad (\varepsilon_X \le \tfrac{1}{100}). \tag{145}$$

Every non-associative axiom now transfers by one triangle inequality: $\|x \cdot y\| \le (1+6\varepsilon_X)\|x\|\|y\|$ and $\|x^\dagger \cdot x\| \ge (1-6\varepsilon_X)\|x\|^2$, both comfortably inside the tolerance $100\varepsilon_X$. For the associator, writing m for the original product and $m' = m + \Delta$, the four-term expansion of $m'(m'(x,y),z) - m'(x, m'(y,z))$ is bounded by $[\varepsilon_X + 2(1+\varepsilon_X)\rho + 2(1+6\varepsilon_X)\rho]\|x\|\|y\|\|z\|$ with $\rho = 5\varepsilon_X$, i.e. by $(21\varepsilon_X + 70\varepsilon_X^2)\|x\|\|y\|\|z\| \le 100\varepsilon_X\|x\|\|y\|\|z\|$. Finally $\|J - I\| = |1-a| \le \varepsilon_X$, so (142) closes the root.

Lemma 89 (`lem-stage1-unitary-graph-control`)**.** *There are universal $C_{\mathrm{ch}} \ge 1$ and $\kappa_{\mathrm{ch}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra, every $\delta > 0$ with $C_{\mathrm{ch}}(\varepsilon_r + \delta) \le \kappa_{\mathrm{ch}}$, every $V \in \overline{\mathcal{U}}_\delta$ and every $A^\parallel \in B^{i\mathcal{H}}_{2\delta}(0)$, there is a unique $g_V(A^\parallel) \in B^{\mathcal{H}}_{2\delta}(0)$ with $f_V\big(A^\parallel + g_V(A^\parallel)\big) = 0$, and the corresponding $V \cdot \big(J + A^\parallel + g_V(A^\parallel)\big)$ lies in $\mathcal{U}$, where*

$$f_V(A) = \tfrac{1}{2}\Big(\big((J + A^\dagger) \cdot V^\dagger\big) \cdot \big(V \cdot (J + A)\big) - J\Big); \tag{146}$$

moreover

$$\|g_V(A^\parallel) + \tfrac{1}{2}(V^\dagger \cdot V - J)\| \le C_{\mathrm{ch}}(\varepsilon_r \delta + \delta^2), \quad \|Dg_V(A^\parallel)\| \le C_{\mathrm{ch}}(\varepsilon_r + \delta), \tag{147}$$

$$\|D_{A^\perp} f_V\big(A^\parallel + g_V(A^\parallel)\big) - I_{\mathcal{H}}\| \le C_{\mathrm{ch}}(\varepsilon_r + \delta) < 1, \tag{148}$$

and the resulting C^1 graph charts cover $\mathcal{U}$.

Registry contract (verbatim).

```
Uniform unitary graph control: there are universal C_ch >= 1, kappa_ch in (0, 1/2] such that for
every finite-dimensional exact-unit epsilon_r-C*-algebra, every delta > 0 with C_ch*(epsilon_r +
delta) <= kappa_ch, every V in calUbar_delta, and every A^par in B^{icalH}_{2delta}(0), there is a
unique g_V(A^par) in B^{calH}_{2delta}(0) with f_V(A^par + g_V(A^par)) = 0, and the corresponding V
bold-dot (J + A^par + g_V(A^par)) lies in calU, where f_V(A) = (1/2)*(((J + A^dagger) bold-dot
V^dagger) bold-dot (V bold-dot (J + A)) - J); moreover ||g_V(A^par) + (1/2)*(V^dagger bold-dot V -
J)|| <= C_ch*(epsilon_r*delta + delta^2), ||Dg_V(A^par)|| <= C_ch*(epsilon_r + delta), and
||D_{A^perp} f_V(A^par + g_V(A^par)) - I_{calH}|| <= C_ch*(epsilon_r + delta) < 1; the resulting
C^1 graph charts cover calU.
```

Proof (prose account of the af-validated tree). The closing witnesses are $C_{\mathrm{ch}} = 1024$ and $\kappa_{\mathrm{ch}} = \frac{1}{4}$; write $e = \varepsilon_r$ and $s = e + \delta$, so the guard gives $s \le 1/4096$.

Uniform multiplier control. For $V \in \overline{\mathcal{U}}_\delta$ with $q = V^\dagger \cdot V - J$ one has $\|q\| \le 2\delta$, and the C^*-lower bound gives $\|V\| \le 1 + 2s$. If Y is a right inverse of V then approximate associativity and exact unitality give $\|V^\dagger - Y\| \le 4s\|Y\|$, hence $\|Y\| \le 2$; the two Neumann estimates $\|L_V L_Y - I\| < 1$ and $\|L_{V^\dagger} L_V - I\| \le 4s < 1$ exhibit a right and a left inverse of L_V, which therefore coincide, with $\|L_V^{-1}\| \le 2$ and $\|L_V^{-1} - L_{V^\dagger}\| \le 8s$. For $\|A\| \le 4\delta$ and $X = V \cdot (J + A)$ the product-norm axiom gives $\|L_V^{-1}(L_X - L_V)\| \le 16s < 1$, so L_X is invertible and X has the right inverse $L_X^{-1}J$.

The graph equation. Put $F(P,H) = f_V(P + H)$ for $P \in i\mathcal{H}$, $H \in \mathcal{H}$ of norm $< 2\delta$; exact product reversal identifies $2F = X^\dagger \cdot X - J$, a degree-two real polynomial with values in $\mathcal{H}$, hence C^1. Expanding $2F$ with $V^\dagger \cdot V = J + q$ and reassociating *only* the two linear terms yields $2F = q + 2H + R$ with $\|R\| \le 128(e\delta + \delta^2)$, i.e.

$$\|F(P,H) - \left(\tfrac{q}{2} + H\right)\| \le 64\,(e\delta + \delta^2), \tag{149}$$

and the same expansion differentiated gives $\|D_H F - I_{\mathcal{H}}\| \le 64s$ and $\|D_P F\| \le 64s$.

Solving it. With $c = 64s \le \frac{1}{8}$, apply Lemma 79 (§39) to $H \mapsto F(P,H)$ on $B^{\mathcal{H}}_{2\delta}(0)$ with comparison isomorphism $I_{\mathcal{H}}$: it is injective and its image contains $F(P,0) + B^{\mathcal{H}}_{2\delta(1-c)}(0)$, while (149) and $\|q\| \le 2\delta$ give $\|F(P,0)\| \le \delta(1+c) < 2\delta(1-c)$. So 0 lies in the image and determines the unique $g_V(P)$. Substituting $H = g_V(P)$ in (149) gives the first bound of (147); the lower Lipschitz inequality of the same lemma together with $\|D_P F\| \le c$ makes g_V locally Lipschitz, after which $Dg_V(P) = -(D_H F)^{-1} D_P F$ gives $\|Dg_V(P)\| \le c/(1-c) \le 2c = 128s$ and the continuity of DF and of operator inversion makes g_V a C^1 map.

Unitarity and the cover. For $A = P + g_V(P)$, exact product reversal gives $X_V(P)^\dagger \cdot X_V(P) - J = 2F(P, g_V(P)) = 0$, and the multiplier step supplies the right inverse, so $X_V(P) \in \mathcal{U}$. Injectivity of L_V makes $P \mapsto X_V(P)$ injective with inverse $\phi^{\|}_V$, and conversely any unitary in the coordinate box solves $f_V = 0$ and is caught by uniqueness. Every $U \in \mathcal{U}$ lies in $\overline{\mathcal{U}}_\delta$ and has $f_U(0) = 0$, so $g_U(0) = 0$ and $U = X_U(0)$: the self-centred charts cover $\mathcal{U}$. Finally $64 \le C_{\mathrm{ch}}$, $128 \le C_{\mathrm{ch}}$ and $64s \le \frac{1}{4} < 1$ deliver (147)–(148).

Lemma 90 (`lem-stage1-maurer-cartan-trivialization`)**.** *There are universal $C_{\mathrm{ch}} \ge 1$ and $\kappa_{\mathrm{ch}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ with $C_{\mathrm{ch}}(\varepsilon_r + \delta) \le \kappa_{\mathrm{ch}}$, the graph maps supplied by Lemma 89 satisfy: every tangent space $T_U\mathcal{U}$ is the image of $L_U(I + Dg_U(0)) : i\mathcal{H} \to \mathcal{X}$, and*

$$\omega_U(Z) = \left(L_U^{-1} Z\right)^{\|} \;:\; T_U\mathcal{U} \longrightarrow i\mathcal{H} \tag{150}$$

is a global C^1 bundle trivialization with distortion at most $1 + C_{\mathrm{ch}}\varepsilon_r$, satisfying $\omega_{cU}(cZ) = \omega_U(Z)$ and $\omega_U(iU) = iJ$ for every $c \in U(1)$.

Registry contract (verbatim).

```
Uniform global tangent/Maurer-Cartan control: there are universal C_ch >= 1, kappa_ch in (0, 1/2]
such that for every finite-dimensional exact-unit epsilon_r-C*-algebra and every delta > 0 with
C_ch*(epsilon_r + delta) <= kappa_ch, the graph maps supplied by lem-stage1-unitary-graph-control
satisfy: every tangent space T_U calU is the image of L_U(I + Dg_U(0)): icalH -> calX, and
omega_U(Z) = (L_U^{-1} Z)^par : T_U calU -> icalH is a global C^1 bundle trivialization with
distortion at most 1 + C_ch*epsilon_r, satisfying omega_{cU}(cZ) = omega_U(Z) and omega_U(iU) = iJ
for every c in U(1).
```

Proof (prose account of the af-validated tree). No graph-existence work is repeated: the tree fixes the witnesses of Lemma 89 and enlarges them.

Multiplier bounds at a unitary. For $U \in \mathcal{U}$ one has $U^\dagger \cdot U = J$ and $\|J\| = 1$, so the lower C^*-inequality gives $\|U\| \le (1 - \varepsilon_r)^{-1/2}$ and the associator axiom gives $\|L_{U^\dagger} L_U - I\| \le \varepsilon_r \|U\|^2 \le \varepsilon_r/(1 - \varepsilon_r) < 1$ once $\varepsilon_r \le \frac{1}{4}$. Neumann inversion in finite dimension makes L_U invertible with $\|L_U\|, \|L_U^{-1}\| \le 1 + K\varepsilon_r$ for a universal K; C_{ch} is then chosen at least $\max(C_0, K, 4)$, with C_0, κ_0 the graph witnesses, and $\kappa_{\mathrm{ch}} = \min(\kappa_0, \frac{1}{4})$.

The centred graph. Exact unitality and $U^\dagger \cdot U = J$ give $f_U(0) = 0$, so $g_U(0) = 0$. Comparing the graph germs at two admissible radii — they agree near 0 by pointwise uniqueness — and letting the radius tend to 0 sharpens the derivative bound of (147) to $\|Dg_U(0)\| \le C_0\varepsilon_r$. Differentiating the centred chart $\Psi_U(A) = U \cdot (J + A + g_U(A))$ at 0 gives $D\Psi_U(0) = L_U(I + Dg_U(0))$, whose image is exactly $T_U\mathcal{U}$ because these charts cover $\mathcal{U}$.

The trivialization. Let $P^{\|}(X) = (X - X^\dagger)/2$, of norm at most 1 by isometry of the involution. Since $Dg_U(0)$ maps $i\mathcal{H}$ into $\mathcal{H}$, one has $P^{\|}(A + Dg_U(0)A) = A$, so ω_U of (150) inverts $A \mapsto L_U(I + Dg_U(0))A$ fibrewise; its two norms are at most $\|L_U^{-1}\|$ and $\|L_U\|(1 + \|Dg_U(0)\|)$, both absorbed into $1 + C_{\mathrm{ch}}\varepsilon_r$.

Regularity. $U \mapsto L_U$ is linear and inversion is C^1 on the invertibles, so ω is C^1. For the *inverse* the tree does not argue from chart regularity (see the correction below) but from the defining equation: with $B_U = D_{A^\perp} f_U(0)|_{\mathcal{H}}$ and $C_U = D_{A^\|} f_U(0)|_{i\mathcal{H}}$, both polynomial in U, (148) gives $\|B_U - I_{\mathcal{H}}\| < 1$, and differentiating $f_U(A + g_U(A)) = 0$ at $A = 0$ yields

$$Dg_U(0) = -B_U^{-1} C_U, \tag{151}$$

a C^1 function of U. Hence $(U, A) \mapsto (U, L_U(A + Dg_U(0)A))$ is a C^1 inverse.

Equivariance. $U \mapsto cU$ preserves $\mathcal{U}$ with derivative $Z \mapsto cZ$, and $L_{cU} = cL_U$, so $\omega_{cU}(cZ) = \omega_U(Z)$. The curve $t \mapsto e^{it}U$ lies in $\mathcal{U}$ with velocity iU, and complex bilinearity with $U \cdot J = U$ gives $L_U(iJ) = iU$; as $J^\dagger = J$ the element iJ is anti-Hermitian, so $\omega_U(iU) = iJ$.

Status. All three are `proved` with `af: validated` (T0). Workspaces `proofs/lem-stage1-rectified-cstar-co` (17 nodes created, 17 `validated`, taint clean, root `validated`), `proofs/lem-stage1-unitary-graph-control` (15 live nodes all `validated` plus 3 `archived`, taint clean on all 18), and `proofs/lem-stage1-maurer-cartan-triv` (15 nodes, 15 `validated`, taint clean).

Correction of record (proof-level, not contract-level). The trivialization tree drew a challenge at the regularity node: as first written it inferred C^1 regularity of the fibrewise *inverse* from the merely C^1 graph frame, which is not valid in general. The repair is (151) — an explicit implicit-function formula with invertible normal derivative — and the challenge's counterexample is inapplicable precisely because it has no such formula. No contract was amended.

Role in the chain. Registry `defs`: `def-epsilon-cstar-algebra`, `def-approximate-unitary-space`. Registry `deps`: `lem-stage1-exact-unit-rectification` (§40) and `lem-stage1-quantitative-inverse-functi` (§39) for the rectification; the rectification and the same inverse-function lemma for the graph control; the graph control alone for the trivialization. Downstream they carry §45–§47.

45 The closed polar retraction and its coherence

Lemma 91 (`lem-stage1-polar-retraction`). *There are universal $C_{\rm pol} \geq 1$ and $\kappa_{\rm pol} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ with $C_{\rm pol}(\varepsilon_r + \delta) \leq \kappa_{\rm pol}$, the map*

$$\Pi_\delta(U, H) = U \cdot H \tag{152}$$

is a C^1 diffeomorphism from $\mathcal{U} \times B^{\mathcal{H}}_\delta(J)$ onto an open set S_δ; its inverse (u_δ, h_δ) obeys

$$X = u_\delta(X) \cdot h_\delta(X), \qquad u_\delta(U) = U, \qquad h_\delta(U) = J \quad (U \in \mathcal{U}); \tag{153}$$

and, writing $D = \varepsilon_r\delta + \delta^2$,

$$\mathcal{U}_{\delta - C_{\rm pol}D} \ \subseteq\ S_\delta \ \subseteq\ \mathcal{U}_{\delta + C_{\rm pol}D}. \tag{154}$$

Transcription note. *Only the* shrunken *inner set of* (154) *is claimed; no larger inner radius is asserted. Both $\mathcal{U}_\cdot$ are the strict-inequality sets of Definition 31, so membership always carries the right-inverse clause as well as the defect bound.*

Registry contract (verbatim).

```
Closed C^1 polar retraction: there are universal C_pol >= 1, kappa_pol in (0, 1/2] such that for
every finite-dimensional exact-unit epsilon_r-C*-algebra and delta > 0 with C_pol*(epsilon_r +
delta) <= kappa_pol, Pi_delta(U, H) = U bold-dot H is a C^1 diffeomorphism from calU x
B^{calH}_delta(J) onto an open S_delta, its inverse (u_delta, h_delta) obeys X = u_delta(X)
bold-dot h_delta(X), u_delta(U) = U, h_delta(U) = J, and calU_{delta - C_pol*(epsilon_r*delta +
delta^2)} subseteq S_delta subseteq calU_{delta + C_pol*(epsilon_r*delta + delta^2)}.
```

Proof (prose account of the af-validated tree). The constants $C_{\rm pol}, \kappa_{\rm pol}$ are fixed at the end as a finite maximum and a finite positive minimum of the thresholds used below, so they remain universal.

The coordinate map. Fix a base $V \in \overline{\mathcal{U}}_r$ with $r \leq c\delta$ ($c = 4$ is the only scale used). As in §44, $\|L_{V^\dagger}L_V - I\| \leq \varepsilon_r\|V\|^2 + 2(1+\varepsilon_r)r$, which the guard makes at most $\frac{1}{4}$; hence L_V is injective, therefore bijective in finite dimension, with a universal bound on $\|L_V^{-1}\|$. Writing $A(a) = a + g_V(a)$ for the graph map of Lemma 89 at scale $c\delta$, exact unitality and bilinearity give the *exact* identity

$$p_V(a, q) := L_V^{-1}\big(\Pi_\delta(V \cdot (J + A(a)),\, J + q) - V\big) = A(a) + q + L_V^{-1}\big((V \cdot A(a)) \cdot q\big), \tag{155}$$

with no reassociation. Differentiating (155) and using $\|Dg_V\| \leq C_{\rm ch}(\varepsilon_r + c\delta)$, $\|A(a)\| \leq K\delta$ and the product-norm axiom bounds

$$\sup\|J_0^{-1}\big(Dp_V - J_0\big)\| \ \leq\ C_0(\varepsilon_r + \delta) \ =:\ \theta\ \leq\ \tfrac{1}{4} \tag{156}$$

on the convex box $\|a\| < 2c\delta$, $\|q\| < \frac{3}{2}\delta$, where $J_0(\alpha, \beta) = \alpha + \beta$ is the real-linear isomorphism $i\mathcal{H} \oplus \mathcal{H} \to \mathcal{X}$ (of inverse norm at most 1, because the involution is isometric). At the base point, $p_V(0,0) = g_V(0)$, so (147) gives $\|p_V(0,0) + \frac{1}{2}(V^\dagger \cdot V - J)\| \leq C_0 D$.

Existence of the polar factorisation. For $V \in \mathcal{U}_{\delta - C_{\rm pol}D}$ the base-point estimate reads $\|p_V(0,0)\| < \delta - (C_{\rm pol} - K_b)D$; choosing $C_{\rm pol} > K_b + K_d$ and shrinking the guard so that $\theta < \frac{1}{4}$ makes this at most $(1-\theta)\delta$, so some $\rho < \delta$ satisfies $\|p_V(0,0)\|/(1-\theta) < \rho$. On the closed ρ-ball in the max norm, $F(z) = z - p_V(z)$ is a θ-contraction into itself; its unique fixed point $z = (a, q)$ solves $p_V(a, q) = 0$, i.e.

$$V = \big[V \cdot (J + a + g_V(a))\big] \cdot (J + q), \qquad \|q\| < \delta, \tag{157}$$

with the bracket in $\mathcal{U}$ by Lemma 89. Integrating (156) along segments makes p_V injective on the whole box, so this zero is the only one there.

Global injectivity and the diffeomorphism. Suppose $X = U_i \cdot (J + q_i)$ with $U_i \in \mathcal{U}$ and $\|q_i\| < \delta$ for $i = 1, 2$. Expanding $X^\dagger \cdot X - J$ around $q = 0$ without losing the q factor shows $X \in \overline{\mathcal{U}}_{4\delta}$ and L_X invertible, so one may take the graph chart *at X itself* at scale 4δ: with $A_i = L_X^{-1}(U_i - X)$, the vanishing of $f_X(A_i)$ and uniqueness force $A_i^\perp = g_X(A_i^\parallel)$, and the near-identity estimate for p_X on that box gives $(a_1, q_1) = (a_2, q_2)$, hence $U_1 = U_2$ and $q_1 = q_2$. Thus Π_δ is injective; by (156) its derivative is invertible at every point, so the finite-dimensional C^1 inverse function theorem makes it a local C^1 diffeomorphism, S_δ open, and the local inverses glue to a C^1 inverse (u_δ, h_δ) with $X = u_\delta(X) \cdot h_\delta(X)$. Exact unitality gives $U = U \cdot J$, whence $u_\delta(U) = U$ and $h_\delta(U) = J$.

The radius sandwich. For $X = U \cdot (J + q) \in S_\delta$, the star rule expands $X^\dagger \cdot X - J$ into two cross terms — each equal to q up to an associator of norm $O(\varepsilon_r \delta)$ — plus one quadratic term of norm $O(\delta^2)$, giving $\|X^\dagger \cdot X - J\| < 2\delta + C_{\mathrm{out}} D$; and L_X is a small perturbation of the invertible L_U, so X has a right inverse. Taking $C_{\mathrm{pol}} \geq C_{\mathrm{out}}/2$ gives the outer inclusion of (154). The inner inclusion is (157).

Lemma 92 (`lem-stage1-polar-coherence-naturality`)**.** *For every exact-unit algebra and every two polar data $(\delta_j, S_j, u_j, h_j)$, $j = 1, 2$, for which $\Pi_{\delta_j} : \mathcal{U} \times B^{\mathcal{H}}_{\delta_j}(J) \to S_j$, $(U, H) \mapsto U \cdot H$, is bijective with inverse (u_j, h_j), one has*

$$(u_1, h_1) = (u_2, h_2) \qquad \text{on } S_1 \cap S_2; \tag{158}$$

moreover, for $c \in U(1)$ and $X, cX \in S_j$,

$$u_j(cX) = c\, u_j(X), \qquad h_j(cX) = h_j(X). \tag{159}$$

Transcription note. *This row is* conditional*: the polar data are hypotheses and no free polar witness is produced. It is never used as an existence assertion — existence is Lemma 91 — but only to identify inverse components that have already been produced.*

Registry contract (verbatim).

```
Polar coherence and scalar naturality: for every exact-unit algebra and every two polar data
(delta_j, S_j, u_j, h_j), j = 1, 2, for which Pi_{delta_j}: calU x B^{calH}_{delta_j}(J) -> S_j,
(U, H) |-> U bold-dot H, is bijective with inverse (u_j, h_j), one has (u_1, h_1) = (u_2, h_2) on
S_1 intersect S_2; moreover, for c in U(1) and X, cX in S_j, u_j(cX) = c*u_j(X) and h_j(cX) =
h_j(X).
```

Proof (prose account of the af-validated tree). *Coherence.* Fix $X \in S_1 \cap S_2$ and put $p_j = (u_j(X), h_j(X))$, so $p_j \in \mathcal{U} \times B^{\mathcal{H}}_{\delta_j}(J)$ and $\Pi_{\delta_j}(p_j) = X$. Let k index the larger radius; then $B^{\mathcal{H}}_{\delta_l}(J) \subseteq B^{\mathcal{H}}_{\delta_k}(J)$ for $l \neq k$, so both p_1, p_2 lie in the domain of Π_{δ_k}. Every Π_δ has the *same* formula $(U, H) \mapsto U \cdot H$, so both have Π_{δ_k}-image X, and the assumed bijectivity — hence injectivity — of Π_{δ_k} gives $p_1 = p_2$. That is (158).

Scalar naturality. $\mathcal{U}$ is closed under the circle action: if $U^\dagger \cdot U = J$ and $U \cdot R = J$ then conjugate-linearity of $\dagger$, bilinearity and $\bar{c}c = 1$ give $(cU)^\dagger \cdot (cU) = J$ and $(cU) \cdot (c^{-1}R) = J$, using no associativity and no positivity. Hence for $p = (u_j(X), h_j(X))$ the pair $p_c = (c\, u_j(X), h_j(X))$ again lies in $\mathcal{U} \times B^{\mathcal{H}}_{\delta_j}(J)$, and bilinearity in the first variable gives $\Pi_{\delta_j}(p_c) = c\, X$. Since $q = (u_j(cX), h_j(cX))$ has the same image, injectivity gives $q = p_c$, which is (159).

Status. Both are `proved` with `af: validated` (T0). Workspace `proofs/lem-stage1-polar-retraction`: 29 nodes created, 29 `validated`, root `validated`, taint clean — above the soft node cap, so the argument linker carries a standing REFACTOR advisory for this row even though the run completed cleanly. Workspace `proofs/lem-stage1-polar-coherence-naturality`: 10 nodes, 10 `validated`, taint clean.

Corrections of record (proof-level, not contract-level). Two shaping challenges are visible in the retraction tree. (i) A *scope* challenge at the multiplier node: as first written it inferred a small Neumann error from $r \leq \frac{1}{4}$ alone, with r unrelated to δ; the repair ties the base radius to the scale, $r \leq c\delta$ with c one of the finitely many absolute constants actually used, and every later estimate cites that hypothesis. (ii) An *endpoint* challenge in the existence chain: the corrected chain is $B < \delta - K_d D \leq (1 - \theta)\delta$, with the strictness carried by the first inequality alone, so the equality case the verifier exhibited is harmless. Several nodes were also rewritten to derive their estimates internally rather than from a then-pending sibling. No contract was amended.

Role in the chain. Registry `defs`: `def-approximate-unitary-space` (both) and `def-epsilon-cstar-algebra` (the retraction). Registry `deps`: `lem-stage1-unitary-graph-control` (§44) for the retraction, and the retraction itself for the coherence row. Downstream they carry the approximate group laws (§46) and the path and inversion rows (§47). `op-classical` remains **open**.

46 Quantitative approximate group laws on the unitaries

The unitary set $\mathcal{U}$ of Definition 31 is not a group: $\cdot$ is only approximately associative and $U^\dagger$ is only approximately an inverse. The three results below turn it into an approximate group by *projecting* the naive product and adjoint back to $\mathcal{U}$ through the polar inverse u_δ of Lemma 91. Throughout, all three carry the same pair of guards

$$C_{\mathrm{pol}}(\varepsilon_r + \delta) \le \kappa_{\mathrm{pol}}, \qquad C_{\mathrm{grp}}\varepsilon_r < \delta - C_{\mathrm{pol}}(\varepsilon_r\delta + \delta^2), \tag{160}$$

with universal $C_{\mathrm{grp}}, C_{\mathrm{pol}} \ge 1$ and $\kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$.

Lemma 93 (`lem-stage1-group-domain-membership`). *There exist universal $C_{\mathrm{grp}}, C_{\mathrm{pol}} \ge 1$ and $\kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ satisfying* (160)*, the inverse $u_\delta : S_\delta \to \mathcal{U}$ of the polar map is defined at $U \cdot V$ and at $U^\dagger$ for every $U, V \in \mathcal{U}$; moreover $U \cdot V$ and $U^\dagger$ each have a right inverse.*

Registry contract (verbatim).

```
Group-input polar-domain membership: there exist universal C_grp, C_pol >= 1, kappa_pol in (0, 1/2]
such that for every finite-dimensional exact-unit epsilon_r-C*-algebra and delta > 0 satisfying
C_pol*(epsilon_r + delta) <= kappa_pol and C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta +
delta^2), the inverse u_delta:S_delta -> calU of the polar map is defined at U bold-dot V and
U^dagger for every U, V in calU; moreover, U bold-dot V and U^dagger each have a right inverse.
```

Proof (prose account of the af-validated tree). Take $C_{\mathrm{pol}}, \kappa_{\mathrm{pol}}$ from Lemma 91 and $C_{\mathrm{grp}} = 4$; abbreviate $e = \varepsilon_r$ and $t = \delta - C_{\mathrm{pol}}(e\delta + \delta^2)$. By the inner inclusion (154) it suffices to place $U \cdot V$ and $U^\dagger$ in $\mathcal{U}_t$, right-inverse clause included.

In-scope smallness. Everything below cites one arithmetic node, derived from (160) alone: $C_{\mathrm{pol}} \ge 1$ and $\kappa_{\mathrm{pol}} \le \frac{1}{2}$ give $s := e + \delta \le \frac{1}{2}$ and $\delta \le \frac{1}{2}$, while the second guard with $C_{\mathrm{grp}} = 4$ gives $4e < t$, hence $t > 0$ and $e < t \le \delta(1 - s)$. Since $(s - e)(1 - s)$ is increasing in s on $[e, \frac{1}{2}]$, this forces

$$e < \tfrac{1}{6}. \tag{161}$$

No threshold beyond (161) is available, and none is used.

Multiplier control. For $W \in \mathcal{U}$, $W^\dagger \cdot W = J$ and the C^*-lower bound give $(1 - e)\|W\|^2 \le 1$, while the associator axiom gives $\|L_{W^\dagger}L_W - I\| \le e\|W\|^2 \le e/(1 - e) < \frac{1}{5}$ by (161). Hence L_W is injective, therefore bijective in finite dimension, with $\|L_{W^\dagger}\| \le \frac{7}{5}$ and $\|L_W^{-1}\| \le \frac{7}{4}$.

The product. Put $X = U \cdot V$, so $X^\dagger = V^\dagger \cdot U^\dagger$ by the star rule. Inserting $((V^\dagger \cdot U^\dagger) \cdot U) \cdot V$ and $(V^\dagger \cdot (U^\dagger \cdot U)) \cdot V$ between $X^\dagger \cdot X$ and J costs two associators and lands on $(V^\dagger \cdot J) \cdot V = V^\dagger \cdot V = J$, so

$$\|X^\dagger \cdot X - J\| \;\le\; \frac{2e(1 + e)}{(1 - e)^2} \;\le\; 4e \;<\; t \;<\; 2t. \tag{162}$$

Approximate associativity gives $\|L_X - L_U L_V\| \le e\|U\|\|V\| < \frac{1}{5}$, while $\|(L_U L_V)^{-1}\| \le \frac{49}{16}$; the Neumann perturbation is $< \frac{49}{80} < 1$, so L_X is invertible and $L_X^{-1}J$ is a right inverse of X.

The adjoint. $U^\dagger \cdot U = J$ says U is a right inverse of $U^\dagger$. If R is the right inverse of U supplied by Definition 31 then $R = L_U^{-1}J$ and $\|R\| \le \frac{7}{4}$; one associator gives $\|U^\dagger - R\| \le e\|U\|^2\|R\|$ and hence $\|U \cdot U^\dagger - J\| \le (1 + e)e\|U\|^3\|R\| \le \frac{441}{125}e < 4e < t < 2t$. So both inputs lie in $\mathcal{U}_t \subseteq S_\delta$.

Endpoint discipline. Every estimate above is non-strict and remains meaningful at $e = 0$; the single strict passage is $C_{\mathrm{grp}}e < t$, followed by $t < 2t$ because $t > 0$. That is exactly what membership in the strict set $\mathcal{U}_t$ needs.

Lemma 94 (`lem-stage1-group-closeness`). *There exist universal $C_{\mathrm{grp}}, C_{\mathrm{pol}} \geq 1$ and $\kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ satisfying* (160)*, the inverse $u_\delta : S_\delta \to \mathcal{U}$ of the polar map satisfies*

$$\|u_\delta(U \cdot V) - U \cdot V\| \leq C_{\mathrm{grp}}\varepsilon_r, \qquad \|u_\delta(U^\dagger) - U^\dagger\| \leq C_{\mathrm{grp}}\varepsilon_r \qquad (U, V \in \mathcal{U}). \tag{163}$$

Registry contract (verbatim).

```
Group-input polar closeness: there exist universal C_grp, C_pol >= 1, kappa_pol in (0, 1/2] such
that for every finite-dimensional exact-unit epsilon_r-C*-algebra and delta > 0 satisfying
C_pol*(epsilon_r + delta) <= kappa_pol and C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta +
delta^2), the inverse u_delta:S_delta -> calU of the polar map satisfies ||u_delta(U bold-dot V) -
U bold-dot V|| <= C_grp*epsilon_r and ||u_delta(U^dagger) - U^dagger|| <= C_grp*epsilon_r for every
U, V in calU.
```

Proof (prose account of the af-validated tree). Take $\kappa_{\mathrm{pol}} \leq \frac{1}{16}$ and $C_{\mathrm{grp}} = 5$, and write $\rho = \delta - C_{\mathrm{pol}}(\varepsilon_r\delta + \delta^2)$, so $\varepsilon_r + \delta \leq \frac{1}{16}$ and $\rho > 5\varepsilon_r \geq 0$. This row deliberately does *not* import Lemma 93: the displayed expressions assert that both inputs lie in the domain of u_δ, and the tree re-derives that typing internally — $\|(U \cdot V)^\dagger \cdot (U \cdot V) - J\| \leq 3\varepsilon_r$ with a right inverse obtained from injectivity of $L_{U \cdot V}$ in finite dimension, and $\|U \cdot U^\dagger - J\| \leq 2\varepsilon_r$ with the exact right inverse U — so that $U \cdot V, U^\dagger \in \mathcal{U}_\rho \subseteq S_\delta$. The duplication is the price of removing every edge between the two children.

The estimate itself is one calculation, stated for any $X \in S_\delta$ with $\|X^\dagger \cdot X - J\| \leq 3\varepsilon_r$. Write $u = u_\delta(X)$, $h = h_\delta(X)$ and $a = h - J$, so $X = u \cdot h$, $u \in \mathcal{U}$, $h \in \mathcal{H}$ and $\|a\| < \delta$ by Lemma 91. Two associator comparisons of $(h \cdot u^\dagger) \cdot (u \cdot h)$ against $h \cdot h$ give

$$\|X^\dagger \cdot X - h \cdot h\| \;\leq\; 2\varepsilon_r(1 + \varepsilon_r)\|h\|^2\|u\|^2 \;\leq\; 3\varepsilon_r. \tag{164}$$

Exact unitality and bilinearity give $h \cdot h - J = 2a + a \cdot a$, so $2\|a\| \leq 6\varepsilon_r + (1 + \varepsilon_r)\|a\|^2$; since $\|a\| < \delta \leq \frac{1}{16}$ the quadratic term is at most $\frac{17}{256}\|a\|$ and absorbs, leaving $\|a\| \leq \frac{1536}{495}\varepsilon_r \leq 4\varepsilon_r$. Finally exact right unitality gives $\|u_\delta(X) - X\| = \|u \cdot (J - h)\| \leq (1 + \varepsilon_r)\|u\|\|a\| \leq 5\varepsilon_r$, which is (163).

Lemma 95 (`lem-stage1-approximate-group-laws`). *There exist universal $C_{\mathrm{grp}}, C_{\mathrm{pol}} \geq 1$ and $\kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ satisfying* (160)*, the inverse u_δ of the polar map defines C^1 maps*

$$\mu(U, V) = u_\delta(U \cdot V), \qquad \sigma(U) = u_\delta(U^\dagger) \tag{165}$$

on all of $\mathcal{U}$, with $\mu(J, U) = \mu(U, J) = U$, $\sigma(J) = J$, and

$$\|\mu(U, V) - U \cdot V\| \leq C_{\mathrm{grp}}\varepsilon_r, \qquad \|\sigma(U) - U^\dagger\| \leq C_{\mathrm{grp}}\varepsilon_r, \tag{166}$$

$$\|\mu(\mu(U, V), W) - \mu(U, \mu(V, W))\| \leq C_{\mathrm{grp}}\varepsilon_r, \tag{167}$$

$$\|\mu(\sigma(U), U) - J\| \leq C_{\mathrm{grp}}\varepsilon_r, \qquad \|\mu(U, \sigma(U)) - J\| \leq C_{\mathrm{grp}}\varepsilon_r. \tag{168}$$

Registry contract (verbatim).

```
Quantitative approximate group laws: there exist universal C_grp, C_pol >= 1, kappa_pol in (0, 1/2]
such that for every finite-dimensional exact-unit epsilon_r-C*-algebra and delta > 0 satisfying
```

```
C_pol*(epsilon_r + delta) <= kappa_pol and C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta +
delta^2), the inverse u_delta of the polar map defines C^1 maps mu(U, V) = u_delta(U bold-dot V),
sigma(U) = u_delta(U^dagger) on all of calU, with mu(J, U) = mu(U, J) = U, sigma(J) = J, ||mu(U, V)
- U bold-dot V|| <= C_grp*epsilon_r, ||sigma(U) - U^dagger|| <= C_grp*epsilon_r, ||mu(mu(U, V), W)
- mu(U, mu(V, W))|| <= C_grp*epsilon_r, ||mu(sigma(U), U) - J|| <= C_grp*epsilon_r, and ||mu(U,
sigma(U)) - J|| <= C_grp*epsilon_r.
```

Proof (prose account of the af-validated tree). *Witness synchronisation.* If (G_d, P_d, k_d), (G_c, P_c, k_c) and (P_r, k_r) are witnesses for the two lemmas above and for Lemma 91, put $C_{\text{pol}} = \max(P_d, P_c, P_r)$, $\kappa_{\text{pol}} = \min(k_d, k_c, k_r, \frac{1}{16})$ and $C_{\text{grp}} = \max(G_d, 8G_c, 8)$. Both guards are monotone in these witnesses, so (160) implies each upstream guard, and $\varepsilon_r \le \frac{1}{16}$. Every polar datum in play uses the *same* map Π_δ on the *same* domain, so the sets S_δ coincide and Lemma 92 identifies the inverse components: there is one common u_δ carrying all three upstream conclusions.

Regularity and basepoints. $U \cdot V$ and $U^\dagger$ lie in the open set S_δ by Lemma 93, and u_δ is C^1 there; the product is bilinear and $\dagger$ is real-linear, hence both are C^1 in finite dimension, so (165) defines C^1 maps on all of $\mathcal{U}$. Since $J \in \mathcal{U}$, $J \cdot U = U \cdot J = U$, $J^\dagger = J$ and u_δ fixes $\mathcal{U}$ pointwise ((153)), the three basepoint identities are exact. Estimate (166) is Lemma 94 together with $C_{\text{grp}} \ge 8G_c$.

The three defects. All of $\mathcal{U}$ obeys $\|T\| \le \sqrt{16/15} < \frac{4}{3}$, by the C^*-lower bound applied to $T^\dagger \cdot T = J$. For (167), insert $A \cdot W$, $(U \cdot V) \cdot W$, $U \cdot (V \cdot W)$ and $U \cdot B$ between the two iterated products, where $A = \mu(U, V)$ and $B = \mu(V, W)$: the two outer differences are each $\le G_c\varepsilon_r$, each replacement difference is $\le \frac{17}{12}G_c\varepsilon_r$ by the product-norm axiom, and the sole reassociation costs $\le \frac{64}{27}\varepsilon_r$; the total is $\le \frac{389}{54}G_c\varepsilon_r \le 8G_c\varepsilon_r$. For the first bound of (168), closeness at $(\sigma(U), U)$ plus $\|\sigma(U) \cdot U - U^\dagger \cdot U\| \le \frac{17}{12}G_c\varepsilon_r$ and $U^\dagger \cdot U = J$ give $\le 3G_c\varepsilon_r$. For the second, the same two steps land on $U \cdot U^\dagger$, and a local estimate — $R = L_U^{-1}J$ with $\|R\| \le \frac{3}{2}$, one associator for $\|R - U^\dagger\|$, then $U \cdot U^\dagger - J = U \cdot (U^\dagger - R)$ — gives $\|U \cdot U^\dagger - J\| \le \frac{34}{9}\varepsilon_r \le 4\varepsilon_r$, for a total $\le \frac{77}{12}G_c\varepsilon_r \le 8G_c\varepsilon_r$. All bounds are non-strict and hold at $\varepsilon_r = 0$.

Status. All three are `proved` with `af: validated` (T0). Workspaces `proofs/lem-stage1-group-domain-membe` (10 nodes, 10 `validated`, taint clean), `proofs/lem-stage1-group-closeness` (12 nodes, 12 `validated`, taint clean) and `proofs/lem-stage1-approximate-group-laws` (14 nodes, 14 `validated`, taint clean).

Correction of record (proof-level, not contract-level). The parent's first elevation **aborted on the balloon tripwire** at 60 live nodes. Per that protocol the *proof* was factored — the contract above is byte-unchanged — into the two children reproduced here, chosen so that neither imports the other; the parent then re-elevated on the factored dependencies. Separately, the membership tree's first run aborted STUCK on prover indiscipline about smallness: unjustified thresholds such as $\varepsilon_r \le \frac{1}{8}$ were being assumed. The second run carries the binding rule recorded above — derive (161) once from the guards, cite that node for every later smallness inference, and absorb the worse constants into the proof body rather than sharpening the threshold. No contract was amended.

Role in the chain. Registry `defs`: `def-approximate-unitary-space`, `def-epsilon-cstar-algebra`. Registry `deps`: `lem-stage1-polar-retraction` (§45) for both children; that lemma together with `lem-stage1-polar-coherence-naturality` and the two children for the parent. Downstream the parent supplies the all-$\mathcal{U}$ adjoint domain used in §47. Nothing here bears on `op-classical`, which is **open**.

47 Paths, the inversion derivative, and the smooth atlas

Lemma 96 (`lem-stage1-polar-path-admissibility`). *There exist universal* $C_{\mathrm{path}}, C_{\mathrm{pol}} \geq 1$ *and* $\kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$ *such that, for every finite-dimensional exact-unit* ε_r*-*C^**-algebra, every* $\delta > 0$ *with* $C_{\mathrm{pol}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{pol}}$*, and* $U_0, U_1 \in \mathcal{U}$*: if* $0 \leq q \leq 1$*,* $\|U_1 - U_0\| \leq q$*,* $C_{\mathrm{path}} q \leq \frac{1}{4}$ *and*

$$C_{\mathrm{path}}(q + \varepsilon_r q + q^2) \ < \ \delta - C_{\mathrm{pol}}(\varepsilon_r \delta + \delta^2), \tag{169}$$

then, for $Z_t = (1-t)U_0 + tU_1$*, every* L_{Z_t} *is invertible, every* $Z_t \in \overline{\mathcal{U}}_{C_{\mathrm{path}}(q+\varepsilon_r q+q^2)}$*, and*

$$H(t, U_0, U_1) = u_\delta(Z_t) \tag{170}$$

is jointly continuous in all displayed variables, joins U_0 *to* U_1*, and obeys* $H(t, cU_0, cU_1) = c\, H(t, U_0, U_1)$ *for* $c \in U(1)$*.*

Registry contract (verbatim).

```
Joint projected-straight-path admissibility: there exist universal C_path, C_pol >= 1, kappa_pol in
(0, 1/2] such that, for every finite-dimensional exact-unit epsilon_r-C*-algebra, every delta > 0
with C_pol*(epsilon_r + delta) <= kappa_pol, and U_0, U_1 in calU, if 0 <= q <= 1, ||U_1 - U_0|| <=
q, C_path*q <= 1/4, and C_path*(q + epsilon_r*q + q^2) < delta - C_pol*(epsilon_r*delta + delta^2),
then for Z_t = (1-t)*U_0 + t*U_1 every L_{Z_t} is invertible, every Z_t in calUbar_{C_path*(q +
epsilon_r*q + q^2)}, and H(t, U_0, U_1) = u_delta(Z_t) is jointly continuous in all displayed
variables, joins U_0 to U_1, and obeys H(t, cU_0, cU_1) = c*H(t, U_0, U_1) for c in U(1).
```

Proof (prose account of the af-validated tree). Take C_{pol} from Lemma 91, $\kappa_{\mathrm{pol}} = \min(\kappa^0_{\mathrm{pol}}, C^0_{\mathrm{pol}}/8)$ and $C_{\mathrm{path}} = 1$; the guard then gives $\varepsilon_r \leq \frac{1}{8}$, and for $U \in \mathcal{U}$ the usual chain $\|U\| \leq \sqrt{8/7}$, $\|L_{U^\dagger} L_U - I\| \leq \frac{1}{7}$ makes L_U invertible with $\|L_U^{-1}\| \leq \frac{3}{2}$.

An exact quadratic identity. With $D = U_1 - U_0$, bilinearity, the star rule and $U_i^\dagger \cdot U_i = J$ give, with no error term,

$$Z_t^\dagger \cdot Z_t - J \ = \ t(t-1)\big(D^\dagger \cdot D\big), \tag{171}$$

whence $\|Z_t^\dagger \cdot Z_t - J\| \ \leq \ t(1-t)(1+\varepsilon_r)\|D\|^2 \ \leq \ (1+\varepsilon_r)q^2/4$. Since $L_{Z_t} = L_{U_0} + tL_D$ and $\|L_D\| \leq (1+\varepsilon_r)\|D\|$, the bound $q \leq \frac{1}{4}$ gives $\|tL_{U_0}^{-1} L_D\| \leq \frac{27}{64} < 1$, so every L_{Z_t} is invertible by Neumann and $L_{Z_t}^{-1} J$ is a right inverse of Z_t. As $(1+\varepsilon_r)q^2/4 \leq 2(q + \varepsilon_r q + q^2)$, this is exactly $Z_t \in \overline{\mathcal{U}}_{q+\varepsilon_r q+q^2}$ in the sense of Definition 31.

Into the polar domain. With $\gamma = q + \varepsilon_r q + q^2$ and $\rho = \delta - C_{\mathrm{pol}}(\varepsilon_r \delta + \delta^2)$, hypothesis (169) reads $\gamma < \rho$, so $\|Z_t^\dagger \cdot Z_t - J\| \leq 2\gamma < 2\rho$ and $Z_t \in \mathcal{U}_\rho$; the inner inclusion of (154) puts $Z_t \in S_\delta$ for every t.

The projected path. $(t, U_0, U_1) \mapsto Z_t$ is continuous and u_δ is C^1, hence continuous, on the open set S_δ, so H of (170) is jointly continuous; $Z_0 = U_0$, $Z_1 = U_1$ and $u_\delta(U_i) = U_i$ give the endpoints. For $c \in U(1)$ the rotated pair again lies in $\mathcal{U}$ with the same separation, its segment is cZ_t, and both Z_t and cZ_t lie in S_δ; the scalar-naturality clause (159) of Lemma 92, applied to the polar datum $(\delta, S_\delta, u_\delta, h_\delta)$, gives $u_\delta(cZ_t) = c\, u_\delta(Z_t)$.

Lemma 97 (`lem-stage1-inversion-derivative-control`). *There exist universal* $C_{\mathrm{der}}, C_{\mathrm{ch}}, C_{\mathrm{pol}}, C_{\mathrm{grp}} \geq 1$ *and* $\kappa_{\mathrm{der}}, \kappa_{\mathrm{ch}}, \kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$ *such that for every finite-dimensional exact-unit* ε_r*-*C^**-algebra, every* $\varsigma \in \{+1, -1\}$ *and every* $0 < r \leq \delta$ *satisfying*

$$C_{\mathrm{ch}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{ch}}, \qquad C_{\mathrm{pol}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{pol}}, \qquad C_{\mathrm{grp}}\varepsilon_r < \delta - C_{\mathrm{pol}}(\varepsilon_r \delta + \delta^2), \tag{172}$$

$$C_{\mathrm{der}}(\varepsilon_r + r) \leq \kappa_{\mathrm{der}}, \qquad (1+\varepsilon_r)\big(1 + C_{\mathrm{ch}}(\varepsilon_r + \delta)\big)r + C_{\mathrm{grp}}\varepsilon_r < 2\delta, \tag{173}$$

the globally defined $\sigma(U) = u_\delta(U^\dagger)$ *maps* $\chi_\varsigma\big(B_r^{i\mathcal{H}}(0)\big)$ *into the same* ςJ*-graph chart, where* $\chi_\varsigma(A) = \varsigma J \cdot \big(J + A + g_{\varsigma J}(A)\big)$*, and* $F_\varsigma(A) = \phi^{\|}_{\varsigma J}\big(\sigma(\chi_\varsigma(A))\big)$ *satisfies*

$$\|D(F_\varsigma - \mathrm{id})(A) + 2I_{i\mathcal{H}}\| \;\le\; C_{\mathrm{der}}(\varepsilon_r + r) \qquad \text{for all } A \in B_r^{i\mathcal{H}}(0). \tag{174}$$

Transcription note. *The registry writes the sign as s; it is renamed* ς *here only to keep it apart from the scale variables. The second display of* (173) *is the* chart-retention guard*: it is what makes the coordinate identity* $F_\varsigma(A) = B$ *below legitimate, rather than a comparison of coordinates in two different charts.*

Registry contract (verbatim).

```
Typed inversion derivative with chart retention: there exist universal C_der, C_ch, C_pol, C_grp >=
1 and kappa_der, kappa_ch, kappa_pol in (0, 1/2] such that for every finite-dimensional exact-unit
epsilon_r-C*-algebra, s in {+1, -1}, and 0 < r <= delta satisfying C_ch*(epsilon_r + delta) <=
kappa_ch, C_pol*(epsilon_r + delta) <= kappa_pol, C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta
+ delta^2), C_der*(epsilon_r + r) <= kappa_der, and (1 + epsilon_r)*(1 + C_ch*(epsilon_r +
delta))*r + C_grp*epsilon_r < 2*delta, the globally defined sigma(U) = u_delta(U^dagger) maps
chi_s(B_r^{icalH}(0)) into the same sJ-graph chart, where chi_s(A) = sJ bold-dot (J + A +
g_{sJ}(A)), and F_s(A) = phi_{sJ}^par(sigma(chi_s(A))) satisfies ||D(F_s - id)(A) + 2*I_{icalH}||
<= C_der*(epsilon_r + r) for all A in B_r^{icalH}(0).
```

Proof (prose account of the af-validated tree). Constants are aligned first: $C_{\mathrm{ch}}, \kappa_{\mathrm{ch}}$ from Lemma 89, C_{pol} the larger and κ_{pol} the smaller of the polar and group-law witnesses, C_{grp} from Lemma 95; $C_{\mathrm{der}}, \kappa_{\mathrm{der}}$ are chosen last.

The scalar chart. Write $V = \varsigma J$, $G = g_V$ and $Z_T = T + G(T)$. Exact unitality, bilinearity, the involution laws and $\varsigma^2 = 1$ collapse (146) to $f_V(Z_T) = G(T) + \frac{1}{2}Z_T^\dagger \cdot Z_T$, so $G(0) = 0$ by uniqueness and $G(T) = -\frac{1}{2}Z_T^\dagger \cdot Z_T$; differentiating and absorbing the resulting DG term gives $\|DG(T)\| \le K_g r$ for $\|T\| \le 2r$.

Polar factorisation and the normal term. For $A \in B_r^{i\mathcal{H}}(0)$, $U = \chi_\varsigma(A)$ lies in $\mathcal{U}$; the all-$\mathcal{U}$ adjoint domain of Lemma 95, with the common u_δ identified by Lemma 92, gives C^1 maps $W = \sigma(U)$ and $Q = h_\delta(U^\dagger) - J \in \mathcal{H}$ with $\|Q\| < \delta$, $U^\dagger = W \cdot (J + Q)$ and $\|W - U^\dagger\| \le C_{\mathrm{grp}}\varepsilon_r$. Since $W^\dagger \cdot W = J$, one associator and the C^*-lower bound absorb into

$$\|Q\| \;\le\; K_q C_{\mathrm{grp}}\varepsilon_r. \tag{175}$$

Chart retention. The involution law and the product estimate give $\|U^\dagger - \varsigma J\| \le (1 + \varepsilon_r)(1 + C_{\mathrm{ch}}(\varepsilon_r + \delta))\|A\|$, so the second guard of (173) gives $\|W - \varsigma J\| < 2\delta$. As $L_{\varsigma J} = \varsigma I$ exactly, $C = \phi_{\varsigma J}(W)$ has norm below 2δ and contractive components; unitarity of W makes $f_{\varsigma J}(C) = 0$, so uniqueness in Lemma 89 forces $C^\perp = G(C^\|)$. Thus $W = \chi_\varsigma(B)$ with $B = C^\| = F_\varsigma(A)$ in the *same* chart.

The coordinate identity and its derivative. Substituting the three factorisations and cancelling ς gives the exact identity $J - A + G(A) = (J + B + G(B)) \cdot (J + Q)$, whose anti-Hermitian part reads

$$-A \;=\; B + P^\|\big((B + G(B)) \cdot Q\big), \tag{176}$$

so $\|B\| \le 2r$ once (175) is small. Differentiating (176) in a unit direction ξ, with $P = DF_\varsigma(A)\xi$ and $R = DQ(A)\xi$, gives $-\xi + DG(A)\xi = P + DG(B)P + R + E$ with $E = (P + DG(B)P) \cdot Q + (B + G(B)) \cdot R$; typing by Hermitian and anti-Hermitian parts yields $P + \xi = -P^\|(E)$.

The product bound with (175), $\|B\| \le 2r$ and $\|DG\| \le K_g r$ gives $\|E\| \le \alpha\|P\| + \beta\|R\|$ with $\alpha \le K_1\varepsilon_r$ and $\beta \le K_2 r$; elementary absorption then bounds $\|P\| \le 2$, $\|R\| \le K_3(\varepsilon_r + r)$ and finally $\|P + \xi\| \le K_4(\varepsilon_r + r)$. Choosing κ_{der} small and $C_{\mathrm{der}} \ge K_4$ enforces every absorption condition, and since $D(F_\varsigma - \mathrm{id}) + 2I_{i\mathcal{H}} = DF_\varsigma + I_{i\mathcal{H}}$, this is (174).

Lemma 98 (`lem-stage1-smooth-unitary-atlas`)**.** *For every finite-dimensional exact-unit ε_r-C^*-algebra, if $\mathcal{U}$ is covered by the unique C^1 graph functions g_V of Lemma 89 and $D_{A^\perp} f_V$ is invertible at every graph point, then those same g_V are C^∞, and the same graph charts make $\mathcal{U}$ a smooth embedded manifold; no point or first derivative of a graph or chart is changed.* Transcription note. *The statement is qualitative and conditional: it introduces no smoothness threshold and no new coefficient, and it asserts nothing about* which δ *makes the hypothesis hold — that is* (148).

Registry contract (verbatim).

```
Smooth graph-atlas upgrade: for every finite-dimensional exact-unit epsilon_r-C*-algebra, if calU
is covered by the unique C^1 graph functions g_V of lem-stage1-unitary-graph-control and D_{A^perp}
f_V is invertible at every graph point, then those same g_V are C^infinity, and the same graph
charts make calU a smooth embedded manifold; no point or first derivative of a graph or chart is
changed.
```

Proof (prose account of the af-validated tree). *The defining map is polynomial.* By Definition 22, $\cdot$ is complex-bilinear and $\dagger$ conjugate-linear, hence real-bilinear and real-linear respectively; with the parenthesisation fixed as in (146), the map $\Phi_V(P, Q) = f_V(P + Q)$ is a real polynomial of degree at most two on $i\mathcal{H} \times \mathcal{H}$, so it is C^∞ in finite dimension. The exact involution axioms give $\Phi_V(P, Q) = \frac{1}{2}(B^\dagger \cdot B - J)$ with $B = V \cdot (J + P + Q)$, without reassociation, so Φ_V indeed takes values in $\mathcal{H}$.

Lee's implicit function theorem. Identify $i\mathcal{H}$ and $\mathcal{H}$ linearly with $\mathbb{R}^n$ and $\mathbb{R}^k$. At a graph point $(P, g_V(P))$ one has $\Phi_V = 0$, and the derivative of $Q \mapsto \Phi_V(P, Q)$ is exactly $D_{A^\perp} f_V(P + g_V(P))$, invertible by hypothesis. Applying, with $c = 0$, Theorem C.40 (Implicit Function Theorem) of J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Springer GTM 218, 2012 — registered in the workspace as the byte-matched external `GT-lee-2ed-thm-C.40` against the pinned local source at the recorded locus — produces neighbourhoods O_P, O_Q and a C^∞ map $F : O_P \to O_Q$ whose graph is exactly $\Phi_V^{-1}(0) \cap (O_P \times O_Q)$.

Identification with the given graph. g_V is C^1, hence continuous, takes values in the permitted $\mathcal{H}$-ball, solves $\Phi_V(P', g_V(P')) = 0$ and is the unique such branch; shrinking O_P so that $g_V(O_P) \subseteq O_Q$, the zero-set equivalence forces $g_V = F$ on O_P. Doing this at every P shows the *same* g_V is C^∞ on its original domain; because this is an equality with the pre-existing C^1 function, its values and its first derivative are literally unchanged.

The atlas. In each graph coordinate the parametrisation $P \mapsto (P, g_V(P))$ is now smooth and its inverse on the graph is the real-linear projection to $i\mathcal{H}$; the chart transports ϕ_V of Definition 31 are affine-linear. On an overlap the transition is a target projection composed with a source parametrisation, hence smooth, and symmetrically for its inverse. The cover is the cover already asserted by Lemma 89, and every set, function and coordinate map used is the old one, so $\mathcal{U}$ is a smooth embedded manifold with nothing replaced.

Status. All three are proved with `af: validated` (T0). Workspaces `proofs/lem-stage1-polar-path-admissi` (12 nodes, 12 `validated`, taint clean, no challenges), `proofs/lem-stage1-inversion-derivative-control` (10 nodes, 10 `validated`, taint clean, no challenges) and `proofs/lem-stage1-smooth-unitary-atlas`

(14 nodes, 14 `validated`, taint clean). The last registers three imports: the validated contracts of `lem-stage1-rectified-cstar-control` and `lem-stage1-unitary-graph-control`, and the byte-matched external `GT-lee-2ed-thm-C.40`.

Correction of record (proof-level, not contract-level). The atlas tree drew one genuine challenge: the transition-map node leaned on a *pending* sibling for the C^∞ smoothness of g_V, which the protocol forbids. It was repaired in run by a self-contained smoothness bridge that re-derives the Lee argument and the identification $g_V = F$ inside the node itself. No contract was amended.

Role in the chain. Registry `defs`: `def-approximate-unitary-space`, `def-epsilon-cstar-algebra`. Registry `deps`: `lem-stage1-polar-retraction` and `lem-stage1-polar-coherence-naturality` (§45) for the path row; those two together with `lem-stage1-unitary-graph-control` (§44) and `lem-stage1-approximate-group-laws` (§46) for the inversion row; `lem-stage1-rectified-cstar-control` and `lem-stage1-unitary-graph-control` for the atlas. The Stage-1 polar rows beyond these — the smooth polar inverse, the transport helpers and the constant ledger — are **not** `af`-validated and are **not** reproduced here; nothing in this document assembles the Stage-1 front into a statement about almost-idempotent maps, and `op-classical` is **open**.

48 The smooth polar inverse, smooth operations, and the polar arithmetic

The three rows of this section close the `af`-validated part of the Stage-1 polar block begun in §44. The first two are smoothness upgrades in the style of Lemma 98: they replace no map, no point and no first derivative, but promote the C^1 regularity of the polar inverse and of the approximate-group operations to C^∞ for the embedded smooth structure. The third is pure scalar arithmetic: it packages, once and for all, the guard inequalities that every analytic row of the block demands, so that downstream consumers can instantiate the whole front from a single smallness assumption on ε_X.

Lemma 99 (`lem-stage1-smooth-polar-inverse`)**.** *For every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$: if Lemma 98 gives the smooth embedded atlas and $\Pi_\delta : \mathcal{U} \times B^{\mathcal{H}}_\delta(J) \to S_\delta$ is the bijective C^1 local diffeomorphism of Lemma 91 onto an open set, then the same ambient-bilinear Π_δ is a smooth diffeomorphism and its same set-theoretic inverse (u_δ, h_δ) is smooth; no point or first derivative is changed.* Transcription note. *Like the atlas row, this is qualitative and conditional: both hypotheses are supplied by the two named lemmas, no new threshold is introduced, and the conclusion re-asserts nothing about* which δ *realises them.*

Registry contract (verbatim).

```
Smooth polar-inverse upgrade: for every finite-dimensional exact-unit epsilon_r-C*-algebra and
every delta > 0, if lem-stage1-smooth-unitary-atlas gives the smooth embedded atlas and Pi_delta:
calU x B^{calH}_delta(J) -> S_delta is the bijective C^1 local diffeomorphism of
lem-stage1-polar-retraction onto an open set, then the same ambient-bilinear Pi_delta is a smooth
diffeomorphism and its same set-theoretic inverse (u_delta, h_delta) is smooth; no point or first
derivative is changed.
```

Proof (prose account of the af-validated tree). *The two sides are smooth manifolds.* Lemma 98 equips $\mathcal{U}$ with the smooth embedded graph atlas whose charts and first derivatives are those of the C^1 atlas; $\mathcal{H}$ is a real linear subspace of the finite-dimensional ambient space (Definition 31), so $B^{\mathcal{H}}_\delta(J)$ is an open subset of a finite-dimensional real vector space, and $\mathcal{U} \times B^{\mathcal{H}}_\delta(J)$ carries the product smooth structure; S_δ is open in the ambient space by Lemma 91 and carries its open-subset structure.

Π_δ *is smooth.* By Definition 22 the product $\cdot$ is bilinear, so in real coordinates each component of $(X, H) \mapsto X \cdot H$ is a quadratic polynomial, hence C^∞. In a product chart built from a graph chart of the atlas and the identity chart on the ball, the coordinate representative of Π_δ is this polynomial map composed with the smooth ambient parametrisation of the embedded chart, hence smooth; such charts cover the domain and the values lie in S_δ, so Π_δ is smooth.

Everywhere-invertible derivative. At any point p the C^1 local-diffeomorphism assertion of Lemma 91 provides a local C^1 inverse G; the chain rule applied to $G \circ \Pi_\delta = \mathrm{id}$ and $\Pi_\delta \circ G = \mathrm{id}$ shows $D\Pi_\delta(p)$ is a linear isomorphism.

Smooth local inverses, glued. Fix p, put $q = \Pi_\delta(p)$, and choose smooth charts α about p and β about q. The coordinate map $F = \beta \circ \Pi_\delta \circ \alpha^{-1}$ is smooth with invertible derivative at $\alpha(p)$; Theorem C.34 (Inverse Function Theorem) of J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Springer GTM 218, 2012 — registered in the workspace as the byte-matched external `GT-lee-2ed-thm-C.34` against the pinned local source at the recorded locus — makes F a smooth diffeomorphism between neighbourhoods, so Π_δ has a smooth local inverse near q. By the bijectivity assertion of Lemma 91 the global set-theoretic inverse exists; on each such target neighbourhood it coincides with the unique smooth local inverse there, and these neighbourhoods cover S_δ, so the global inverse is smooth because smoothness is local — exactly the gluing recorded in Corollary C.36(b) of the same source, the byte-matched external `GT-lee-2ed-cor-C.36`. A bijection has only one set-theoretic inverse, so this smooth inverse *is* the pair (u_δ, h_δ) already supplied.

Nothing is changed. The atlas upgrade is identity-on-data through first order, the map is literally the same $\Pi_\delta(U,H) = U \cdot H$ on the same domain and codomain, and the inverse is the same set map; the derivative of a C^1 function in unchanged first-order charts is unique, so proving these same functions C^∞ changes no point value and no first derivative.

Lemma 100 (`lem-stage1-smooth-unitary-operations`). *Under Lemmas 95, 98 and 99, the scalar action $U(1)\times\mathcal{U} \to \mathcal{U}$, $(c,U) \mapsto cU$, and the same maps $\mu : \mathcal{U}\times\mathcal{U} \to \mathcal{U}$, $\mu(U,V) = u_\delta(U \cdot V)$, and $\sigma : \mathcal{U} \to \mathcal{U}$, $\sigma(U) = u_\delta(U^\dagger)$, are smooth as maps into the embedded manifold $\mathcal{U}$; they obey*

$$\mu(cU, dV) = cd\,\mu(U,V), \qquad \sigma(cU) = \bar{c}\,\sigma(U) \qquad (c,d \in U(1)), \tag{177}$$

and no point or first derivative is changed. Transcription note. *The registry* **`deps`** *also list Lemma 92, which the contract exercises through its scalar-naturality clause rather than naming it in the antecedent.*

Registry contract (verbatim).

```
Smooth action/operations upgrade: under lem-stage1-approximate-group-laws,
lem-stage1-smooth-unitary-atlas, and lem-stage1-smooth-polar-inverse, the scalar action U(1) x calU
-> calU, (c, U) |-> cU, and the same maps mu: calU x calU -> calU, mu(U, V) = u_delta(U bold-dot
V), and sigma: calU -> calU, sigma(U) = u_delta(U^dagger), are smooth as maps into the embedded
manifold calU; they obey mu(cU, dV) = c*d*mu(U, V) and sigma(cU) = conj(c)*sigma(U), and no point
or first derivative is changed.
```

Proof (prose account of the af-validated tree). *The circle preserves $\mathcal{U}$.* If $U^\dagger \cdot U = J$ and $U \cdot R = J$, then for $c \in U(1)$ conjugate-linearity of $\dagger$ and bilinearity give $(cU)^\dagger \cdot (cU) = \bar{c}c\,J = J$ and $(cU)\cdot(\bar{c}R) = J$, so $cU \in \mathcal{U}$ with the explicit right inverse $\bar{c}R$. This local derivation is repeated wherever membership is needed below, deliberately never citing a sibling node.

The scalar action is smooth. Since $J \in B^{\mathcal{H}}_\delta(J)$ and $\Pi_\delta(U,J) = U \cdot J = U$ by exact unitality, both U and cU lie in the open set S_δ; the ambient map $(c,U) \mapsto cU$, the restriction of complex scalar multiplication, is smooth into S_δ, and composing with the smooth $u_\delta : S_\delta \to \mathcal{U}$ of Lemma 99 gives a smooth $\mathcal{U}$-valued map. The scalar-naturality clause (159) of Lemma 92 gives $u_\delta(cU) = c\,u_\delta(U) = cU$, so this smooth composite *is* the action itself.

μ and σ are smooth. By Lemma 95 the inputs $U \cdot V$ and $U^\dagger$ lie in S_δ for all $U, V \in \mathcal{U}$ and the displayed formulas are the same globally defined C^1 maps. The ambient product is bilinear and $\dagger$ is conjugate-linear, hence both restrict to smooth S_δ-valued maps on the embedded domain; composing with the smooth u_δ makes μ and σ smooth into $\mathcal{U}$.

Covariance. By the local membership derivation, $cU, dV \in \mathcal{U}$, so Lemma 95 puts $X = U \cdot V$ and $Y = (cU)\cdot(dV)$ in S_δ; bilinearity gives $Y = (cd)X$ with $cd \in U(1)$, and (159) gives $u_\delta(Y) = cd\,u_\delta(X)$, the first identity of (177). Likewise $(cU)^\dagger = \bar{c}U^\dagger$ with $\bar{c} \in U(1)$, both $U^\dagger$ and $(cU)^\dagger$ lie in S_δ, and (159) with scalar $\bar{c}$ gives $\sigma(cU) = \bar{c}\,\sigma(U)$.

Nothing is changed. The atlas and the polar inverse retain their point values and first derivatives by Lemmas 98 and 99; the ambient scalar, product and adjoint maps are untouched; and μ, σ keep the identical formulas of Lemma 95. Uniqueness of derivatives through the unchanged component derivatives leaves every first derivative the pre-existing one.

Lemma 101 (`lem-stage1-polar-scalar-arithmetic`)**.** *For every* $C_{\mathrm{rect}}, C_{\mathrm{ch}}, C_{\mathrm{pol}}, C_{\mathrm{grp}}, C_{\mathrm{path}}, C_{\mathrm{der}} \geq 1$*, every* $e_{\mathrm{rect}} \in (0, 1/C_{\mathrm{rect}}]$ *and every* $\kappa_{\mathrm{ch}}, \kappa_{\mathrm{pol}}, \kappa_{\mathrm{der}} \in (0, \frac{1}{2}]$*, set*

$$\delta_* = \min\Big\{\tfrac{1}{4},\ \tfrac{\kappa_{\mathrm{ch}}}{4C_{\mathrm{ch}}},\ \tfrac{\kappa_{\mathrm{pol}}}{4C_{\mathrm{pol}}}\Big\}, \qquad \varepsilon_*^r = \min\Big\{\tfrac{1}{4},\ \tfrac{\kappa_{\mathrm{ch}}}{4C_{\mathrm{ch}}},\ \tfrac{\kappa_{\mathrm{pol}}}{4C_{\mathrm{pol}}},\ \tfrac{\kappa_{\mathrm{der}}}{8C_{\mathrm{der}}},\ \tfrac{1}{C_{\mathrm{grp}}},\ \tfrac{\delta_*}{12C_{\mathrm{path}}C_{\mathrm{grp}}}\Big\}, \tag{178}$$
$$e_{S1} = \min\Big\{e_{\mathrm{rect}},\ \tfrac{\varepsilon_*^r}{C_{\mathrm{rect}}}\Big\}, \qquad r_{\mathrm{iso}} = \min\Big\{\tfrac{\delta_*}{4},\ \tfrac{\kappa_{\mathrm{der}}}{8C_{\mathrm{der}}}\Big\},$$

and, given $0 \leq \varepsilon_X \leq e_{S1}$*, put* $\varepsilon_r = C_{\mathrm{rect}}\varepsilon_X$*,* $q = C_{\mathrm{grp}}\varepsilon_r$*,* $r_- = \delta_* - C_{\mathrm{pol}}(\varepsilon_r\delta_* + \delta_*^2)$ *and* $\eta = C_{\mathrm{path}}(q + \varepsilon_r q + q^2)$*. Then every such* ε_X *satisfies*

$$C_{\mathrm{ch}}(\varepsilon_r + \delta_*) \leq \kappa_{\mathrm{ch}}, \qquad C_{\mathrm{pol}}(\varepsilon_r + \delta_*) \leq \kappa_{\mathrm{pol}}, \qquad q < r_-, \qquad C_{\mathrm{path}}q \leq \tfrac{1}{4}, \tag{179}$$
$$\eta < r_-, \qquad C_{\mathrm{der}}(\varepsilon_r + r_{\mathrm{iso}}) \leq \kappa_{\mathrm{der}}, \qquad (1+\varepsilon_r)\big(1 + C_{\mathrm{ch}}(\varepsilon_r + \delta_*)\big)r_{\mathrm{iso}} + q < 2\delta_*; \tag{180}$$

moreover $r_- \geq 3\delta_*/4$*,* $\eta \leq \delta_*/4$ *and* $C_{\mathrm{der}}(r_{\mathrm{iso}} + \varepsilon_r) \leq \kappa_{\mathrm{der}}/4 < 1$*.* Transcription note. *The row is universal over every already-selected coefficient and margin tuple and carries* no *analytic content: its ten inputs and four derived scales are exactly the fields of the witness tuple of Definition 39, and the seven displayed guards are the hypotheses of the analytic rows of §§44–47 at the scales* $\delta = \delta_*$ *and* $r = r_{\mathrm{iso}}$*.*

Registry contract (verbatim).

```
Universal Stage-1 polar arithmetic: for every C_rect, C_ch, C_pol, C_grp, C_path, C_der >= 1,
e_rect in (0, 1/C_rect], and kappa_ch, kappa_pol, kappa_der in (0, 1/2], setting delta_* = min{1/4,
kappa_ch/(4*C_ch), kappa_pol/(4*C_pol)}, epsilon_*^r = min{1/4, kappa_ch/(4*C_ch),
kappa_pol/(4*C_pol), kappa_der/(8*C_der), 1/C_grp, delta_*/(12*C_path*C_grp)}, e_S1 = min{e_rect,
epsilon_*^r/C_rect}, r_iso = min{delta_*/4, kappa_der/(8*C_der)}, epsilon_r = C_rect*epsilon_X, q =
C_grp*epsilon_r, r_- = delta_* - C_pol*(epsilon_r*delta_* + delta_*^2), and eta = C_path*(q +
epsilon_r*q + q^2), every 0 <= epsilon_X <= e_S1 satisfies C_ch*(epsilon_r + delta_*) <= kappa_ch,
C_pol*(epsilon_r + delta_*) <= kappa_pol, q < r_-, C_path*q <= 1/4, eta < r_-, C_der*(epsilon_r +
r_iso) <= kappa_der, and (1 + epsilon_r)*(1 + C_ch*(epsilon_r + delta_*))*r_iso + q < 2*delta_*;
moreover r_- >= 3*delta_*/4, eta <= delta_*/4, and C_der*(r_iso + epsilon_r) <= kappa_der/4 < 1.
```

Proof (prose account of the af-validated tree). *Channel and polar guards.* From $e_{S1} \leq \varepsilon_*^r/C_{\mathrm{rect}}$ and $C_{\mathrm{rect}} \geq 1$, $0 \leq \varepsilon_r \leq \varepsilon_*^r$. Since both ε_*^r and δ_* are at most $\kappa_{\mathrm{ch}}/(4C_{\mathrm{ch}})$ and $\kappa_{\mathrm{pol}}/(4C_{\mathrm{pol}})$, the first two guards hold with value at most $\kappa/2$; and since $r_- = \delta_*\big[1 - C_{\mathrm{pol}}(\varepsilon_r + \delta_*)\big]$ with $C_{\mathrm{pol}}(\varepsilon_r + \delta_*) \leq \kappa_{\mathrm{pol}}/2 \leq \frac{1}{4}$, one gets $r_- \geq 3\delta_*/4$.

Path guards. $\varepsilon_r \leq \delta_*/(12C_{\mathrm{path}}C_{\mathrm{grp}})$ gives $0 \leq q \leq \delta_*/(12C_{\mathrm{path}}) \leq \delta_*/12$, hence $C_{\mathrm{path}}q \leq \delta_*/12 \leq \frac{1}{48} < \frac{1}{4}$ (using $\delta_* \leq \frac{1}{4}$) and $q < 3\delta_*/4 \leq r_-$. From $\varepsilon_r \leq \frac{1}{4}$ and $q \leq \frac{1}{48}$, $1 + \varepsilon_r + q \leq \frac{61}{48} < 3$, so $\eta = C_{\mathrm{path}}q\,(1 + \varepsilon_r + q) \leq \delta_*/4 < 3\delta_*/4 \leq r_-$.

Derivative guards. $\varepsilon_r \leq \kappa_{\mathrm{der}}/(8C_{\mathrm{der}})$ and $r_{\mathrm{iso}} \leq \kappa_{\mathrm{der}}/(8C_{\mathrm{der}})$ give $C_{\mathrm{der}}(\varepsilon_r + r_{\mathrm{iso}}) \leq \kappa_{\mathrm{der}}/4$; and $0 < \kappa_{\mathrm{der}} \leq \frac{1}{2}$ makes $\kappa_{\mathrm{der}}/4 \leq \kappa_{\mathrm{der}}$ and $\kappa_{\mathrm{der}}/4 \leq \frac{1}{8} < 1$.

The product guard. The minima give $\varepsilon_r \leq \frac{1}{4}$, $C_{\mathrm{ch}}(\varepsilon_r + \delta_*) \leq \frac{1}{4}$, $r_{\mathrm{iso}} \leq \delta_*/4$ and $q \leq \delta_*/12$, so

$$(1+\varepsilon_r)\big(1 + C_{\mathrm{ch}}(\varepsilon_r + \delta_*)\big)r_{\mathrm{iso}} + q \ \leq\ \tfrac{5}{4}\cdot\tfrac{5}{4}\cdot\tfrac{\delta_*}{4} + \tfrac{\delta_*}{12} \ =\ \tfrac{91}{192}\,\delta_* \ <\ 2\delta_*, \tag{181}$$

strictly, because $\delta_* > 0$.

Status. All three are `proved` with `af: validated` (T0). Workspaces `proofs/lem-stage1-smooth-polar-inver` (21 nodes, 21 `validated`, taint clean, no challenges; registers the byte-matched externals `GT-lee-2ed-thm-C.34` and `GT-lee-2ed-cor-C.36` together with the validated contracts of its two registry deps), `proofs/lem-stage1-smo` (15 nodes, 15 `validated`, taint clean) and `proofs/lem-stage1-polar-scalar-arithmetic` (15 nodes, 15 `validated`, taint clean, no challenges; a leaf with no registry deps).

Corrections of record (proof-level, not contract-level). The operations tree drew two genuine cross-sibling challenges: as first written, the covariance nodes leaned on the *pending* sibling that made the scalar action $\mathcal{U}$-valued, which the protocol forbids. Both were repaired in run by the self-contained local membership derivation reproduced above, so no covariance node cites the action node. Its first run also aborted on a *spurious* prover-overreach trip — the orchestrator's own controller bookkeeping, not the prover, had dirtied `.frontier/`; the guard was exempted for that path and the run repeated. No contract was amended.

Role in the chain. Registry `defs`: `def-approximate-unitary-space` and `def-epsilon-cstar-algebra` for the two upgrades; `def-stage1-polar-witness-data` for the arithmetic row. Registry `deps`: `lem-stage1-polar-retraction` (§45) and `lem-stage1-smooth-unitary-atlas` (§47) for the smooth polar inverse; those two conclusions together with `lem-stage1-polar-coherence-naturality` and `lem-stage1-approximate-group-laws` (§46) for the operations; the arithmetic row has none. The remaining Stage-1 polar rows — the seven parameterized transport helpers and the constant ledger — are **not** `af`-validated and are **not** reproduced here; nothing in this document assembles the Stage-1 front into a statement about almost-idempotent maps, and `op-classical` is **open**.

49 Parameterized transports of the Stage-1 polar layer

This section and the next (§50) reproduce six `af`-validated *transport* rows, which carry six fixed-constant Stage-1 polar lemmas of §44–§47 to the *parameterized* form of Definition 39: throughout, W is a Stage-1 polar witness datum with coefficient fields $C_{\mathrm{rect}}, C_{\mathrm{ch}}, C_{\mathrm{pol}}, C_{\mathrm{grp}}, C_{\mathrm{path}}, C_{\mathrm{der}}$ and margin fields $e_{\mathrm{rect}}, \kappa_{\mathrm{ch}}, \kappa_{\mathrm{pol}}, \kappa_{\mathrm{der}}$ (the four derived scales of W play no role in these six rows). Each row asserts universal *thresholds* — coefficient floors C^0_* and margin caps κ^0_* — such that *every* tuple W whose coefficients dominate the floors and whose margins are dominated by the caps witnesses the parent conclusion at its own fields. All ambient vocabulary ($\mathcal{H}$, $i\mathcal{H}$, $\mathcal{U}$, $\overline{\mathcal{U}}.$, $\mathcal{U}.$, L_V, ϕ_V) is that of Definition 31, never restated.

The common transport skeleton. Every tree in the family shares one skeleton, stated once. (i) Instantiate the universal witnesses of the fixed-constant parent lemma as the thresholds. (ii) Derive $\varepsilon_r \geq 0$ from the product-norm axiom of Definition 22 at $x = y = J$: exact unitality and $\|J\| = 1$ give $1 = \|J \cdot J\| \leq (1 + \varepsilon_r)$. Hence every guard combination in play ($\varepsilon_r + \delta$, $\varepsilon_r\delta + \delta^2$, $q + \varepsilon_r q + q^2$, $\varepsilon_r + r$) is nonnegative. (iii) Transport each scalar guard by monotonicity: a coefficient only *shrinks* from the field of W down to the threshold and a margin only *grows*, so, the multiplied quantities being nonnegative, the assumed field guard implies the parent's witness guard (e.g. $\hat{C}(\varepsilon_r + \delta) \leq C_{\mathrm{ch}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{ch}} \leq \hat{\kappa}$). (iv) Apply the parent lemma. (v) Weaken every conclusion coefficient from the witness back up to the field, again by nonnegativity, using also that the defect sets are monotone in the radius: for $r \leq s$, $\overline{\mathcal{U}}_r \subseteq \overline{\mathcal{U}}_s$ and $\mathcal{U}_r \subseteq \mathcal{U}_s$ (the right-inverse clause is unchanged and the defect bound only relaxes). Existence, uniqueness, regularity, coverage and equivariance clauses carry over verbatim.

Lemma 102 (`lem-stage1-rectified-cstar-transport`). *There exist $C^0_{\mathrm{rect}} \geq 1$ and $e^0_{\mathrm{rect}} \in (0, 1/C^0_{\mathrm{rect}}]$ such that for every witness datum W with $C_{\mathrm{rect}} \geq C^0_{\mathrm{rect}}$ and $0 < e_{\mathrm{rect}} \leq \min\{e^0_{\mathrm{rect}}, 1/C_{\mathrm{rect}}\}$, every finite-dimensional ε_X-C^*-algebra with $0 \leq \varepsilon_X \leq e_{\mathrm{rect}}$ admits, on the same involutive normed space, a bilinear product $\cdot$ and an element $J = J^\dagger$ for which $(\mathcal{X}, J, \cdot, \dagger)$ satisfies every exact-unit ε_r-C^*-algebra axiom of Definition 22, including $\|J\| = 1$, where $\varepsilon_r = C_{\mathrm{rect}}\varepsilon_X$, and for every $x, y \in \mathcal{X}$, $\|J - I_X\| \leq C_{\mathrm{rect}}\varepsilon_X$ and $\|x \cdot y - xy\| \leq C_{\mathrm{rect}}\varepsilon_X\|x\|\|y\|$.*

Registry contract (verbatim).

```
Parameterized rectification transport: there exist C_rect^0 >= 1 and e_rect^0 in (0, 1/C_rect^0]
such that, for every def-stage1-polar-witness-data tuple W with C_rect >= C_rect^0 and 0 < e_rect
<= min{e_rect^0, 1/C_rect}, for every finite-dimensional epsilon_X-C*-algebra (calX, I_X, .,
dagger) with 0 <= epsilon_X <= e_rect, there are on the same involutive normed space a bilinear
product bold-dot and an element J = J^dagger for which (calX, J, bold-dot, dagger) satisfies every
exact-unit epsilon_r-C*-algebra axiom of def-epsilon-cstar-algebra, including ||J|| = 1, where
epsilon_r = C_rect*epsilon_X, and for every x, y in calX, ||J - I_X|| <= C_rect*epsilon_X and ||x
bold-dot y - xy|| <= C_rect*epsilon_X*||x||*||y||.
```

Proof (prose account of the af-validated tree). The skeleton, seeded by Lemma 88 with witnesses (c, e) and thresholds $C^0_{\mathrm{rect}} = c$, $e^0_{\mathrm{rect}} = e$, plus one extra node: *parameter monotonicity of the axiom list itself.* If $0 \leq \alpha \leq \beta$ and $(\mathcal{X}, J, \cdot, \dagger)$ satisfies every exact-unit α-C^*-algebra axiom, then it satisfies every β-axiom: $\|x \cdot y\| \leq (1 + \alpha)\|x\|\|y\| \leq (1 + \beta)\|x\|\|y\|$, the associator bound $\alpha\|x\|\|y\|\|z\|$ is at most $\beta\|x\|\|y\|\|z\|$, and $\|x^\dagger \cdot x\| \geq (1 - \alpha)\|x\|^2 \geq (1 - \beta)\|x\|^2$; every remaining axiom is parameter-free. Since $\varepsilon_X \leq e_{\mathrm{rect}} \leq e$, the parent applies and produces the axioms at

$\alpha = c\varepsilon_X$; monotonicity passes to $\beta = C_{\mathrm{rect}}\varepsilon_X \geq \alpha$, and $c \leq C_{\mathrm{rect}}$ weakens the two closeness estimates.

Lemma 103 (`lem-stage1-unitary-graph-transport`). *There exist $C^0_{\mathrm{ch}} \geq 1$ and $\kappa^0_{\mathrm{ch}} \in (0, \frac{1}{2}]$ such that for every witness datum W with $C_{\mathrm{ch}} \geq C^0_{\mathrm{ch}}$ and $0 < \kappa_{\mathrm{ch}} \leq \kappa^0_{\mathrm{ch}}$, for every finite-dimensional exact-unit ε_r-C^*-algebra, every $\delta > 0$ with $C_{\mathrm{ch}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{ch}}$, every $V \in \overline{\mathcal{U}}_\delta$ and every $A^\parallel \in B^{i\mathcal{H}}_{2\delta}(0)$, there is a unique $g_V(A^\parallel) \in B^{\mathcal{H}}_{2\delta}(0)$ with $f_V(A^\parallel + g_V(A^\parallel)) = 0$ — f_V exactly as in (146) — the element $V \cdot (J + A^\parallel + g_V(A^\parallel))$ lies in $\mathcal{U}$, the two graph estimates (147) and the strict normal-derivative bound (148) hold with the field C_{ch} of W in place of the universal constant, and the resulting C^1 graph charts cover $\mathcal{U}$.*

Registry contract (verbatim).

```
Parameterized unitary-graph transport: there exist C_ch^0 >= 1 and kappa_ch^0 in (0, 1/2] such
that, for every def-stage1-polar-witness-data tuple W with C_ch >= C_ch^0 and 0 < kappa_ch <=
kappa_ch^0, for every finite-dimensional exact-unit epsilon_r-C*-algebra (calX, J, bold-dot,
dagger), every delta > 0 with C_ch*(epsilon_r + delta) <= kappa_ch, every V in calUbar_delta, and
every A^par in B^{icalH}_{2delta}(0), there is a unique g_V(A^par) in B^{calH}_{2delta}(0) such
that f_V(A^par + g_V(A^par)) = 0, where f_V(A) = (1/2)*(((J + A^dagger) bold-dot V^dagger) bold-dot
(V bold-dot (J + A)) - J), the element V bold-dot (J + A^par + g_V(A^par)) lies in calU,
||g_V(A^par) + (1/2)*(V^dagger bold-dot V - J)|| <= C_ch*(epsilon_r*delta + delta^2),
||Dg_V(A^par)|| <= C_ch*(epsilon_r + delta), and ||D_{A^perp} f_V(A^par + g_V(A^par)) - I_{calH}||
<= C_ch*(epsilon_r + delta) < 1, and these C^1 graph charts cover calU.
```

Proof (prose account of the af-validated tree). The skeleton verbatim, seeded by Lemma 89 with witnesses $(\hat{C}, \hat{\kappa})$. With $s = \varepsilon_r + \delta$ and $q = \varepsilon_r\delta + \delta^2 = \delta s$, step (ii) gives $s, q > 0$; step (iii) is $\hat{C}s \leq C_{\mathrm{ch}}s \leq \kappa_{\mathrm{ch}} \leq \hat{\kappa}$; the parent supplies the unique g_V, membership in $\mathcal{U}$, cover, and the three bounds at $\hat{C}$; step (v) weakens them to $C_{\mathrm{ch}}q$, $C_{\mathrm{ch}}s$, $C_{\mathrm{ch}}s$, and the strict clause is $C_{\mathrm{ch}}s \leq \kappa_{\mathrm{ch}} \leq \kappa^0_{\mathrm{ch}} \leq \frac{1}{2} < 1$.

Lemma 104 (`lem-stage1-maurer-cartan-transport`). *There exist $C^0_{\mathrm{ch}} \geq 1$ and $\kappa^0_{\mathrm{ch}} \in (0, \frac{1}{2}]$ such that for every witness datum W with $C_{\mathrm{ch}} \geq C^0_{\mathrm{ch}}$ and $0 < \kappa_{\mathrm{ch}} \leq \kappa^0_{\mathrm{ch}}$, for every finite-dimensional exact-unit ε_r-C^*-algebra, every $\delta > 0$ with $C_{\mathrm{ch}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{ch}}$, and every family $g = (g_U)_{U \in \mathcal{U}}$ of C^1 maps $g_U : B^{i\mathcal{H}}_{2\delta}(0) \to B^{\mathcal{H}}_{2\delta}(0)$ such that, for every U and $A^\parallel$, $g_U(A^\parallel)$ is the* unique *element of $B^{\mathcal{H}}_{2\delta}(0)$ with $f_U(A^\parallel + g_U(A^\parallel)) = 0$: every tangent space $T_U\mathcal{U}$ is the image of $L_U(I + Dg_U(0)) : i\mathcal{H} \to \mathcal{X}$, and $\omega_U(Z) = (L_U^{-1}Z)^\parallel$ is a global C^1 bundle trivialization with distortion at most $1 + C_{\mathrm{ch}}\varepsilon_r$, satisfying $\omega_{cU}(cZ) = \omega_U(Z)$ and $\omega_U(iU) = iJ$ for every $c \in U(1)$.* Transcription note. *Unlike Lemma 90, whose graph maps are the distinguished family supplied by Lemma 89, this row quantifies over* any *family with the displayed uniqueness property.*

Registry contract (verbatim).

```
Parameterized Maurer-Cartan transport: there exist C_ch^0 >= 1 and kappa_ch^0 in (0, 1/2] such
that, for every def-stage1-polar-witness-data tuple W with C_ch >= C_ch^0 and 0 < kappa_ch <=
kappa_ch^0, for every finite-dimensional exact-unit epsilon_r-C*-algebra, every delta > 0 with
C_ch*(epsilon_r + delta) <= kappa_ch, and every family g = (g_U)_{U in calU} of C^1 maps g_U:
B^{icalH}_{2delta}(0) -> B^{calH}_{2delta}(0) such that, for every U in calU and A^par in
B^{icalH}_{2delta}(0), g_U(A^par) is the unique element of B^{calH}_{2delta}(0) satisfying
f_U(A^par + g_U(A^par)) = 0, where f_U(A) = (1/2)*(((J + A^dagger) bold-dot U^dagger) bold-dot (U
bold-dot (J + A)) - J), every tangent space T_U calU is the image of L_U(I + Dg_U(0)): icalH ->
calX, and omega_U(Z) = (L_U^{-1} Z)^par : T_U calU -> icalH is a global C^1 bundle trivialization
```

```
with distortion at most 1 + C_ch*epsilon_r, satisfying omega_{cU}(cZ) = omega_U(Z) and omega_U(iU)
= iJ for every U in calU, Z in T_U calU, and c in U(1).
```

Proof (prose account of the af-validated tree). The skeleton seeded by Lemma 90 with witnesses $(\bar{C}, \bar{\kappa})$, plus a genuine extra obligation: the arbitrary family g cannot simply be *identified* with the parent's distinguished family, because the registered external does not export that family's domain/codomain typing (the tree records this as a conditional bridge only). Instead, the tangent formula is re-derived for g itself. From the guard, $\varepsilon_r < \frac{1}{2}$; if $X^\dagger \cdot X = J$ then the C^*-lower bound, approximate associativity and exact unitality give $\|L_{X^\dagger} L_X Y - Y\| \le \frac{\varepsilon_r}{1-\varepsilon_r}\|Y\|$ with coefficient < 1, so L_X is injective, hence bijective in finite dimension, and X has a right inverse: *every exact zero of the displayed f_U-equation lies in $\mathcal{U}$.* Consequently, for $B \in i\mathcal{H}$ the curve $t \mapsto U \cdot (J + tB + g_U(tB))$ stays in $\mathcal{U}$ (using $g_U(0) = 0$, forced by uniqueness at $f_U(0) = 0$) and differentiates at 0 to $L_U(B + Dg_U(0)B)$, so $\operatorname{im} L_U(I + Dg_U(0)) \subseteq T_U\mathcal{U}$. The parent's fibrewise isomorphism ω_U gives $\dim T_U\mathcal{U} = \dim i\mathcal{H}$ and $\omega_U(L_U(B + Dg_U(0)B)) = B$ (the Hermitian part projects away), so the inclusion is an equality of finite-dimensional spaces. The ω, distortion and equivariance conclusions of the parent mention no graph family and transfer unchanged; step (v) weakens the distortion from $1 + \bar{C}\varepsilon_r$ to $1 + C_{\mathrm{ch}}\varepsilon_r$.

Lemma 105 (`lem-stage1-polar-retraction-transport`)**.** *There exist $C^0_{\mathrm{pol}} \ge 1$ and $\kappa^0_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every witness datum W with $C_{\mathrm{pol}} \ge C^0_{\mathrm{pol}}$ and $0 < \kappa_{\mathrm{pol}} \le \kappa^0_{\mathrm{pol}}$, for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ with $C_{\mathrm{pol}}(\varepsilon_r + \delta) \le \kappa_{\mathrm{pol}}$, the map $\Pi_\delta(U, H) = U \cdot H$ is a C^1 diffeomorphism from $\mathcal{U} \times B^{\mathcal{H}}_\delta(J)$ onto the open set $S_\delta := \Pi_\delta(\mathcal{U} \times B^{\mathcal{H}}_\delta(J))$, its inverse (u_δ, h_δ) obeys the normalisation* (153)*, and the sandwich* (154) *holds with the field C_{pol} of W.*

Registry contract (verbatim).

```
Parameterized polar-retraction transport: there exist C_pol^0 >= 1 and kappa_pol^0 in (0, 1/2] such
that, for every def-stage1-polar-witness-data tuple W with C_pol >= C_pol^0 and 0 < kappa_pol <=
kappa_pol^0, for every finite-dimensional exact-unit epsilon_r-C*-algebra and every delta > 0 with
C_pol*(epsilon_r + delta) <= kappa_pol, the map Pi_delta: calU x B^{calH}_delta(J) -> calX,
Pi_delta(U, H) = U bold-dot H, is a C^1 diffeomorphism onto the open set S_delta := Pi_delta(calU x
B^{calH}_delta(J)), its inverse (u_delta, h_delta): S_delta -> calU x B^{calH}_delta(J) satisfies X
= u_delta(X) bold-dot h_delta(X), u_delta(U) = U, and h_delta(U) = J for every X in S_delta and U
in calU, and calU_{delta - C_pol*(epsilon_r*delta + delta^2)} subseteq S_delta subseteq calU_{delta
+ C_pol*(epsilon_r*delta + delta^2)}.
```

Proof (prose account of the af-validated tree). The skeleton verbatim (the shortest tree of the family), seeded by Lemma 91 with witnesses (C_*, κ_*). The diffeomorphism, openness and normalisation clauses are coefficient-free and transfer unchanged; only the sandwich is weakened: with $A = \varepsilon_r\delta + \delta^2 \ge 0$ and $C_* \le C_{\mathrm{pol}}$, one has $\delta - C_{\mathrm{pol}}A \le \delta - C_*A$ and $\delta + C_*A \le \delta + C_{\mathrm{pol}}A$, so radius monotonicity of $\mathcal{U}$. gives $\mathcal{U}_{\delta - C_{\mathrm{pol}}A} \subseteq \mathcal{U}_{\delta - C_*A} \subseteq S_\delta \subseteq \mathcal{U}_{\delta + C_*A} \subseteq \mathcal{U}_{\delta + C_{\mathrm{pol}}A}$.

Status. All four are `proved` with `af:  validated` (T0). Workspaces `proofs/lem-stage1-rectified-cstar-tra` (7 nodes, 7 `validated`, first-pass, zero challenges), `proofs/lem-stage1-unitary-graph-transport` (9 nodes, 9 `validated`, first-pass, zero challenges), `proofs/lem-stage1-maurer-cartan-transport` (13 nodes, 13 `validated`, two genuine challenges repaired in-run) and `proofs/lem-stage1-polar-retraction-tra` (5 nodes, 5 `validated`, first-pass, zero challenges). All taint clean; all runs tier *routine*.

Correction of record (proof-level, not contract-level). The Maurer–Cartan tree drew two genuine challenges: the arbitrary family g had been identified with the parent's distinguished graph family through a uniqueness step whose antecedent the registered external does not supply. The repair keeps that identification as an explicitly *conditional* bridge and derives the tangent formula for g directly — the every-exact-zero-is-unitary node reproduced above. No contract was amended.
Role in the chain. Registry `defs`: `def-stage1-polar-witness-data` and `def-epsilon-cstar-algebra` (all four) and `def-approximate-unitary-space` (all but the rectification row). Registry `deps`: each transport imports exactly its fixed-constant parent — Lemma 88, Lemma 89 and Lemma 90 (§44), and Lemma 91 (§45) respectively. The path and inversion-derivative transports follow in §50; `op-classical` remains **open**.

50 Parameterized transports of the Stage-1 polar layer, II: paths and the inversion derivative

The two rows below complete the `af`-validated part of the transport family begun in §49; the witness datum W, its coefficient and margin fields, the threshold discipline and the *common transport skeleton* (steps (i)–(v)) are exactly those stated there and are not repeated.

Lemma 106 (`lem-stage1-polar-path-transport`). *There exist $C^0_{\mathrm{path}}, C^0_{\mathrm{pol}} \geq 1$ and $\kappa^0_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every witness datum W with $C_{\mathrm{path}} \geq C^0_{\mathrm{path}}$, $C_{\mathrm{pol}} \geq C^0_{\mathrm{pol}}$ and $0 < \kappa_{\mathrm{pol}} \leq \kappa^0_{\mathrm{pol}}$, for every finite-dimensional exact-unit ε_r-C^*-algebra, every $\delta > 0$ with $C_{\mathrm{pol}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{pol}}$, every $U_0, U_1 \in \mathcal{U}$ and every $q \in [0, 1]$ with $\|U_1 - U_0\| \leq q$, $C_{\mathrm{path}} q \leq \frac{1}{4}$ and $C_{\mathrm{path}}(q + \varepsilon_r q + q^2) < \delta - C_{\mathrm{pol}}(\varepsilon_r \delta + \delta^2)$: every L_{Z_t} is invertible and every $Z_t = (1-t)U_0 + tU_1$ lies in $\overline{\mathcal{U}}_{C_{\mathrm{path}}(q+\varepsilon_r q+q^2)}$ for $t \in [0, 1]$; and, writing u_δ for the unique first component of the inverse of Π_δ (the explicit binder of Lemma 105), $H(t, U_0, U_1) = u_\delta(Z_t)$ is jointly continuous in (t, U_0, U_1), joins U_0 to U_1, and obeys $H(t, cU_0, cU_1) = c\,H(t, U_0, U_1)$ for every $c \in U(1)$.*

Registry contract (verbatim).

```
Parameterized polar-path transport: there exist C_path^0, C_pol^0 >= 1 and kappa_pol^0 in (0, 1/2]
such that, for every def-stage1-polar-witness-data tuple W with C_path >= C_path^0, C_pol >=
C_pol^0, and 0 < kappa_pol <= kappa_pol^0, for every finite-dimensional exact-unit
epsilon_r-C*-algebra, every delta > 0 with C_pol*(epsilon_r + delta) <= kappa_pol, every U_0, U_1
in calU, and every q in [0, 1] satisfying ||U_1 - U_0|| <= q, C_path*q <= 1/4, and C_path*(q +
epsilon_r*q + q^2) < delta - C_pol*(epsilon_r*delta + delta^2), every L_{Z_t} is invertible and
every Z_t = (1-t)*U_0 + t*U_1 lies in calUbar_{C_path*(q + epsilon_r*q + q^2)} for t in [0, 1],
and, writing u_delta for the unique first component of the inverse of Pi_delta: calU x
B^{calH}_delta(J) -> S_delta := Pi_delta(calU x B^{calH}_delta(J)), Pi_delta(U, H) = U bold-dot H,
the map H(t, U_0, U_1) = u_delta(Z_t) is jointly continuous in (t, U_0, U_1), joins U_0 to U_1, and
satisfies H(t, cU_0, cU_1) = c*H(t, U_0, U_1) for every c in U(1).
```

Proof (prose account of the af-validated tree). The skeleton verbatim, seeded by Lemma 96 with witnesses (a, b, k) for $(C_{\mathrm{path}}, C_{\mathrm{pol}}, \kappa_{\mathrm{pol}})$. With $A = q + \varepsilon_r q + q^2 \geq 0$ and $B = \varepsilon_r \delta + \delta^2 \geq 0$, step (iii) transports the smallness guard and $aq \leq C_{\mathrm{path}} q \leq \frac{1}{4}$, and the *strict* loss inequality survives as the chain $aA \leq C_{\mathrm{path}} A < \delta - C_{\mathrm{pol}} B \leq \delta - bB$, the strictness carried by the middle (hypothesised) inequality alone. The parent yields everything with $Z_t \in \overline{\mathcal{U}}_{aA}$; step (v) upgrades the membership to $\overline{\mathcal{U}}_{C_{\mathrm{path}} A}$ by radius monotonicity, and every other conclusion — invertibility, joint continuity, endpoints, circle equivariance — already matches verbatim, the bare-u_δ anaphor of the parent resolving against the explicit polar binder.

Lemma 107 (`lem-stage1-inversion-derivative-transport`). *There exist $C^0_{\mathrm{der}}, C^0_{\mathrm{ch}}, C^0_{\mathrm{pol}}, C^0_{\mathrm{grp}} \geq 1$ and $\kappa^0_{\mathrm{der}}, \kappa^0_{\mathrm{ch}}, \kappa^0_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every witness datum W with $C_{\mathrm{der}} \geq C^0_{\mathrm{der}}$, $C_{\mathrm{ch}} \geq C^0_{\mathrm{ch}}$, $C_{\mathrm{pol}} \geq C^0_{\mathrm{pol}}$, $C_{\mathrm{grp}} \geq C^0_{\mathrm{grp}}$, $0 < \kappa_{\mathrm{der}} \leq \kappa^0_{\mathrm{der}}$, $0 < \kappa_{\mathrm{ch}} \leq \kappa^0_{\mathrm{ch}}$ and $0 < \kappa_{\mathrm{pol}} \leq \kappa^0_{\mathrm{pol}}$, for every finite-dimensional exact-unit ε_r-C^*-algebra, every $\delta > 0$, every $\varsigma \in \{+1, -1\}$ and every $0 < r \leq \delta$ satisfying the five guards* (172)–(173) *at the fields of W: writing u_δ for the unique first component of the inverse of Π_δ (as in Lemma 105) and $g_{\varsigma J}$ for the unique C^1 map $B^{i\mathcal{H}}_{2\delta}(0) \to B^{\mathcal{H}}_{2\delta}(0)$ with $f_{\varsigma J}(A + g_{\varsigma J}(A)) = 0$ throughout, define $\chi_\varsigma(A) = \varsigma J \boldsymbol{\cdot} (J + A + g_{\varsigma J}(A))$ and the global C^1 map $\sigma(U) = u_\delta(U^\dagger)$; then σ maps $\chi_\varsigma(B^{i\mathcal{H}}_r(0))$ into the same ςJ-graph chart, and $F_\varsigma(A) = \phi^{\|}_{\varsigma J}(\sigma(\chi_\varsigma(A)))$ satisfies $\|D(F_\varsigma - \mathrm{id})(A) + 2I_{i\mathcal{H}}\| \leq C_{\mathrm{der}}(\varepsilon_r + r)$ for every $A \in B^{i\mathcal{H}}_r(0)$.* Transcription note. *The* *registry writes the sign as s; it is renamed ς here exactly as in Lemma 97. Where that parent binds u_δ, $g_{\varsigma J}$ and σ as bare anaphors, this row carries* explicit definite descriptions *for all three — the point of the transport.*

Registry contract (verbatim).

```
Parameterized inversion-derivative transport: there exist C_der^0, C_ch^0, C_pol^0, C_grp^0 >= 1
and kappa_der^0, kappa_ch^0, kappa_pol^0 in (0, 1/2] such that, for every
def-stage1-polar-witness-data tuple W with C_der >= C_der^0, C_ch >= C_ch^0, C_pol >= C_pol^0,
C_grp >= C_grp^0, 0 < kappa_der <= kappa_der^0, 0 < kappa_ch <= kappa_ch^0, and 0 < kappa_pol <=
kappa_pol^0, for every finite-dimensional exact-unit epsilon_r-C*-algebra, every delta > 0, every s
in {+1, -1}, and every 0 < r <= delta satisfying C_ch*(epsilon_r + delta) <= kappa_ch,
C_pol*(epsilon_r + delta) <= kappa_pol, C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta +
delta^2), C_der*(epsilon_r + r) <= kappa_der, and (1 + epsilon_r)*(1 + C_ch*(epsilon_r + delta))*r
+ C_grp*epsilon_r < 2*delta, writing u_delta for the unique first component of the inverse of
Pi_delta: calU x B^{calH}_delta(J) -> S_delta := Pi_delta(calU x B^{calH}_delta(J)), Pi_delta(U, H)
= U bold-dot H, and g_{sJ}: B_{2delta}^{icalH}(0) -> B_{2delta}^{calH}(0) for the unique C^1 map
such that, for every A in B_{2delta}^{icalH}(0), f_{sJ}(A + g_{sJ}(A)) = 0, where f_{sJ}(B) =
(1/2)*(((J + B^dagger) bold-dot (sJ)^dagger) bold-dot (sJ bold-dot (J + B)) - J), define chi_s(A) =
sJ bold-dot (J + A + g_{sJ}(A)) and the global C^1 map sigma(U) = u_delta(U^dagger); then sigma
maps chi_s(B_r^{icalH}(0)) into the same sJ-graph chart and, with F_s(A) =
phi_{sJ}^par(sigma(chi_s(A))), one has ||D(F_s - id)(A) + 2*I_{icalH}|| <= C_der*(epsilon_r + r)
for every A in B_r^{icalH}(0).
```

Proof (prose account of the af-validated tree). The skeleton with *joint* thresholds: each coefficient floor is the maximum, and each margin cap the minimum, of the corresponding witnesses of the four parents (Lemmas 97, 91, 89 and 95), so that every field guard transports simultaneously to all four — including the strict group-defect chain $C^q_{\mathrm{grp}}\varepsilon_r \le C_{\mathrm{grp}}\varepsilon_r < \delta - C_{\mathrm{pol}}y \le \delta - C^q_{\mathrm{pol}}y$ (with $y = \varepsilon_r\delta + \delta^2 \ge 0$) and the radius-chart guard, both monotone in the same direction. Then three binder identifications discharge the explicit definite descriptions. *Polar:* Lemma 91 supplies the bijective Π_δ on the identical domain with the identical image, and a bijection has exactly one inverse, so the description selects its first component. *Graph:* $(\varsigma J)^\dagger = \varsigma J$ and $(\varsigma J)\boldsymbol{\cdot}(\varsigma J) = J$ exactly, so $\varsigma J \in \mathcal{U} \subseteq \overline{\mathcal{U}}_\delta$ and Lemma 89 at $V = \varsigma J$ supplies the unique $g_{\varsigma J}$ — its displayed f_V is verbatim the $f_{\varsigma J}$ above. *Inversion:* the group laws (Lemma 95) define $\sigma(U) = u_\delta(U^\dagger)$ on all of $\mathcal{U}$ for this same u_δ; the graph-control conclusions satisfy the hypotheses of the smooth-atlas row (Lemma 98, via the Neumann invertibility of $D_{A^\perp} f_V$ from (148)), and, through the `af`-validated registry rows `lem-stage1-smooth-polar-inverse` (Lemma 99) and `lem-stage1-smooth-unitary-operations` (Lemma 100), the same σ is a smooth — hence C^1 — self-map of $\mathcal{U}$, with no point or first derivative changed. Finally Lemma 97 applies with its own witnesses; after the identifications its χ_ς, σ, F_ς and chart are precisely those above, and step (v) enlarges only the displayed bound, $C^I_{\mathrm{der}}(\varepsilon_r + r) \le C_{\mathrm{der}}(\varepsilon_r + r)$.

Status. Both are `proved` with `af: validated` (T0). Workspaces `proofs/lem-stage1-polar-path-transport` (9 nodes, 9 `validated`, first-pass, zero challenges) and `proofs/lem-stage1-inversion-derivative-transport` (run 2, 2026-07-28: 13 live nodes all `validated` out of 22 created, 9 archived in an in-run repair of an e-binding challenge). Both taint clean; both runs tier *routine*.

Correction of record (deps-level, not contract-level). The inversion-derivative row's first orchestration (2026-07-28, W95 run 1) aborted STUCK with the root pending: verifier-validated audit nodes recorded that the sole allowed external (the parent control lemma) binds u_δ and $g_{\varsigma J}$ as bare anaphors, so three root premises — the polar binder, the graph binder, and global C^1-ness of σ —

were underivable from the allowed inputs. All three are carried verbatim by existing `af`-validated results, so the registry `deps` line was widened by six T0 imports (`lem-stage1-polar-retraction`, `lem-stage1-unitary-graph-control`, `lem-stage1-smooth-unitary-operations` and the latter's three antecedents `lem-stage1-approximate-group-laws`, `lem-stage1-smooth-unitary-atlas`, `lem-stage1-smooth-polar-inverse`), the contract staying **byte-unchanged**; the workspace was re-seeded and run 2 validated cleanly. No contract was amended.

Role in the chain. Registry `defs` (both rows): `def-stage1-polar-witness-data`, `def-approximate-unitary-s` `def-epsilon-cstar-algebra`. Registry `deps`: the path row imports exactly its fixed-constant parent, Lemma 96 (§47); the inversion row imports its parent Lemma 97 (§47) together with the six widened imports listed above. The family's remaining transport row (for the approximate group laws) and the scalar-arithmetic and constant-ledger rows are **not** `af`-validated and are **not** reproduced here; nothing in these two sections assembles the Stage-1 front into a statement about almost-idempotent maps, and `op-classical` remains **open**.

51 Argument-DAG atlas: the Route-F proof chain for `op-classical`

This section is **generated** by `scripts/gen-report-dag.py` directly from the registry shards `argument/lemmas/*.md` (parsed with `scripts/argument.py`'s own parser) and is re-checked by `scripts/check-all.sh`: if a shard's `status`, `af` state or `deps` change and this section is not regenerated, the gate fails. It is therefore a live view of the argument, not a snapshot. Its browsable twin is `argument/DAG.md` (Mermaid, also generated from the same shards, whole registry); this atlas is the print/PDF view, restricted to the landing chain.

What is drawn. Not the whole registry (295 results) — only the sub-DAG that carries the argument landing the north star: the transitive closure of `op-classical` along `deps:` and `routes:` edges, *together with* the Route-F families named by the live strategy `docs/plans/CURRENT.md`. That is **142 results** and 290 edges. The Route-F chain is deliberately included even though it does not yet reach the root through registry edges: that missing composition is the point, and it is drawn as 2 dashed **design-pending (GAP)** links.

Honest reading. Of the 142 results drawn, **84 are rigorous** in this repo's sense (`af: validated`, or `cited` and byte-matched to `refs/`). Everything else — including every node coloured blue (`status: proved`) — is a prose proof that has *not* cleared an independent `af` validation, and the north star itself is `open`. A path of coloured boxes from left to right is *not* a proof; it is a map of what would have to be discharged.

Crosslinking (the anchor scheme). Every drawn result carries the hypertarget `dag:<registry-id>` on its node, and every imported definition carries `dag:<def-id>` on its index row. From anywhere in this lab-book, write `\dagref{lem-prh}` or `\dagdef{def-height}` to jump to the atlas, and `\dagphase{comp}{...}` to jump to a phase page. In the other direction, a node links back (§) to its typeset statement whenever the registry shard names one via its `provenance: report <label>` token — the same join key `scripts/check-provenance.py` uses.

How to read the atlas

The colour of a node is its *rigour rung* (`CLAUDE.md` L0), computed by `scripts/argument.py`'s `proof_class()` — the same function that colours the browsable Mermaid twin `argument/DAG.md`. Colour is never the only channel: each node also carries a glyph and the spelled-out status, so the atlas survives greyscale printing and every form of colour-vision deficiency.

	class	rung	what it means
	■ af-validated (84)	RIGOROUS (L0 rung b)	`af: validated` — root + every node validated by fresh codex verifiers, taint clean. The only rung this repo currently reaches.
	♦ cited (0)	RIGOROUS (L0 rung a)	`status: cited` — byte-verbatim string-matched to a local published source under `refs/`. (No result in this subgraph currently holds this rung.)
	• proved (21)	NOT rigorous	`status: proved` with `af: none/seeded` — a proof written in-repo that has NOT cleared an independent `af` validation. Solid, but not L0.
	◊ seeded (2)	NOT rigorous	`af: seeded` — an `af` workspace exists and the contract is pinned, but no validated tree yet.

	class	rung	what it means
	▲ mod-audit / conj. (24)	NOT rigorous (flagged)	`proved-mod-audit` · `conjecture` · `heuristic` · `numerical` — the workhorse rungs of the inherited campaign. Content, no independent check.
	○ stated (8)	NOT rigorous	`status: stated` — transcribed from an upstream artifact, unchecked here.
	★ OPEN (3)	OPEN	`status: open` or `kind: open-problem` — a live open question. `op-classical` is the north star and is one of these.

Edges. ⟶ a registry `deps:` edge (the dependency *points at* the result that uses it); ⇢ a member of an `routes:` OR-route (*any one* route discharges the shard — here `op-hlc`); ⇢ a **design-pending (GAP) composition**: *not* a registry edge, reserved by the live strategy (`docs/plans/CURRENT.m` and not yet wired. Small dashed grey chips are *boundary copies* of a node that really lives on an earlier band or in another phase; their id is abbreviated to the shortest unambiguous prefix and each is a hyperlink to the real node.

The design-pending (GAP) links

The atlas exists to show *where the chain is still open.* These links are drawn but are **not** registry `deps:` edges — nothing below may be read as an established implication:

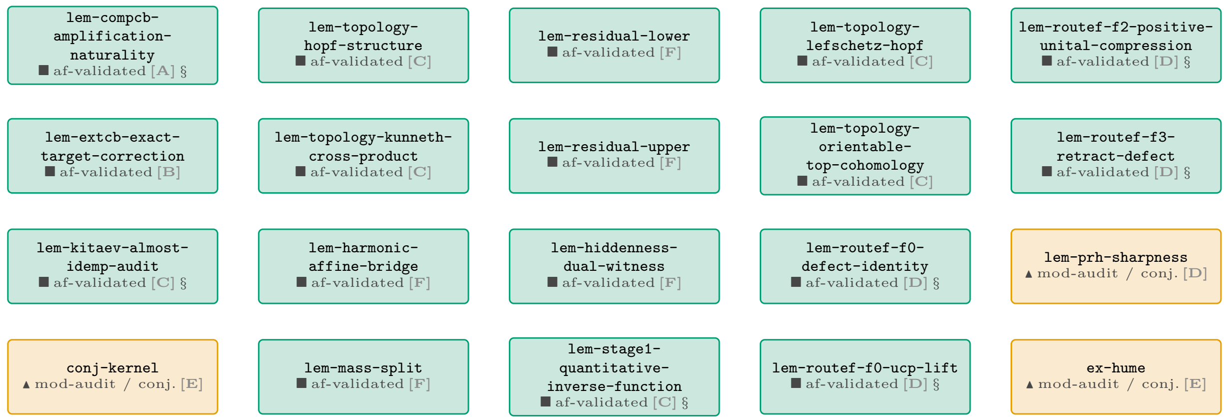
- `lem-routef-k-ledger` ⇢ `op-classical` — phase 5 (bd aism-y81y): F0 codification + root composition; docs/plans/CURRENT.md (v33 §The live campaign frontier: PHASE 4 preliminaries, elevation order and phase-5 root composition)
- `lem-routef-prh-finish` ⇢ `lem-routef-k-ledger` — the registered F2/F3 rows close the stochastic-retract bridge and "compose literally with lem-routef-prh-finish"; docs/plans/CURRENT.md (v33 §UNCHANGED)

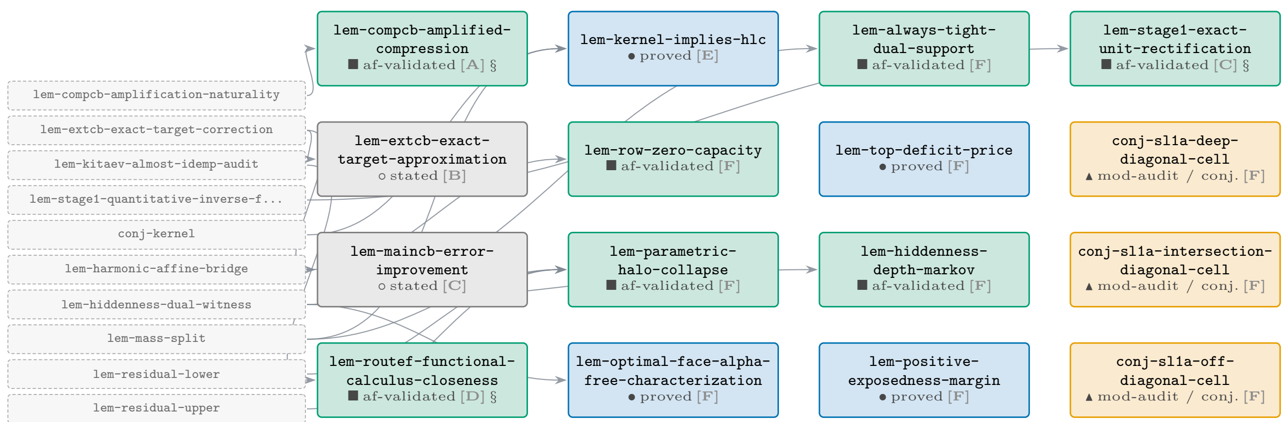
Statistics of the drawn subgraph

quantity	value	note
results drawn	142	of 295 in the registry
registry edges drawn	288	`deps:` 285, OR-route members 3
design-pending (GAP) links	2	drawn dashed; *not* registry edges
rigorous (L0)	**84**	`af: validated` or `status: cited` — 59.2% of the chain
non-rigorous	58	everything else, flagged loudly by colour and glyph
phase COMP/H-CB	27	COMP-CB / H-CB corner tower
phase EXT-CB	10	EXT-CB corner extension
phase Stage-1/topology	41	Stage-1 reset, topology leaves & the Kitaev repair
phase Route F	11	Route F bridges, PRH & the K-ledger
phase root	12	root, classical reduction & the exposed-hull spine
phase trunk	41	signed-geometry trunk (ancestors of the exposed-hull route)
`kind: corollary`	1	
`kind: lemma`	132	
`kind: obstruction`	1	
`kind: open-problem`	4	
`kind: proposition`	2	
`kind: theorem`	2	

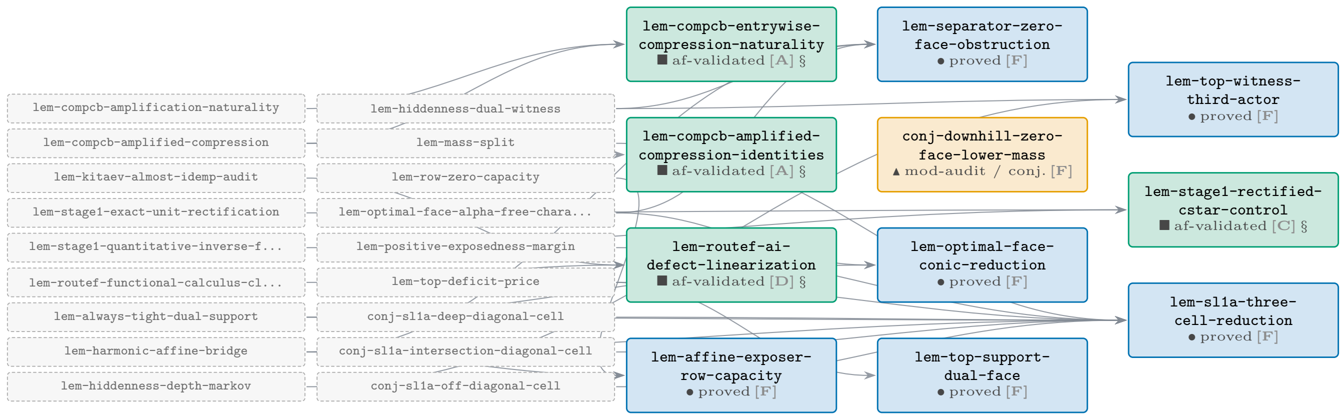
The Route-F proof chain for `op-classical` — complete sub-DAG

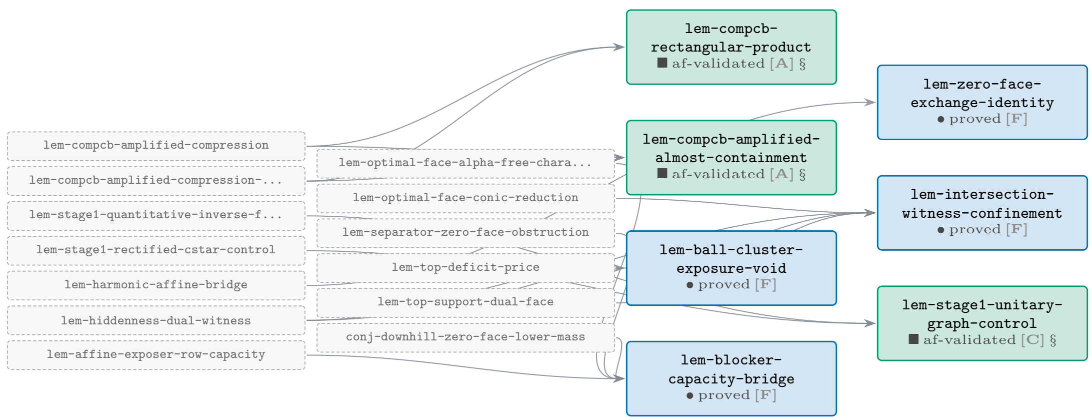
The complete Route-F landing chain: every registry result in the transitive `deps:`/`routes:` closure of `op-classical` together with the Route-F families. Dependencies flow **left to right**: a node sits one layer to the right of its deepest prerequisite, so the north star `op-classical` is on the far right of the last band. Bands read like text — finish a band, drop to the next.



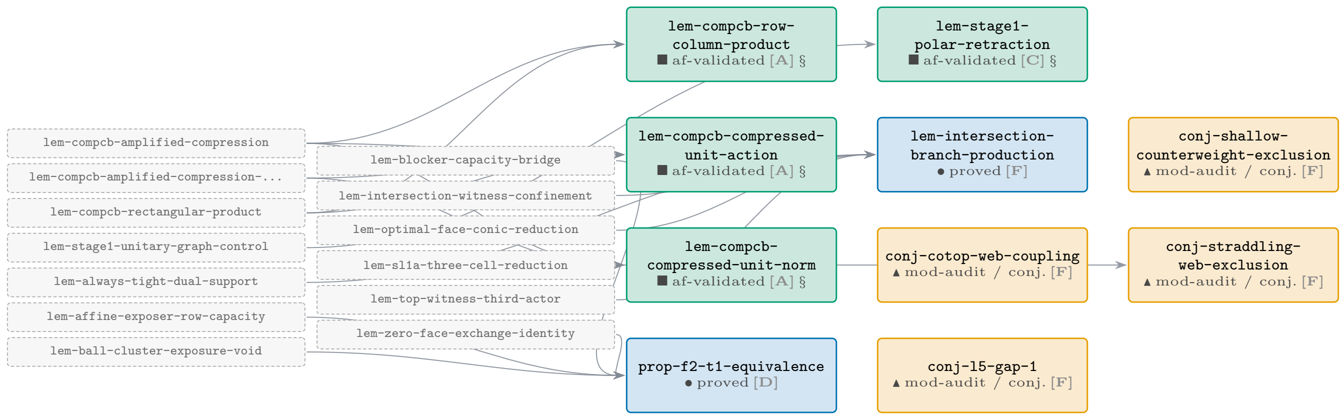


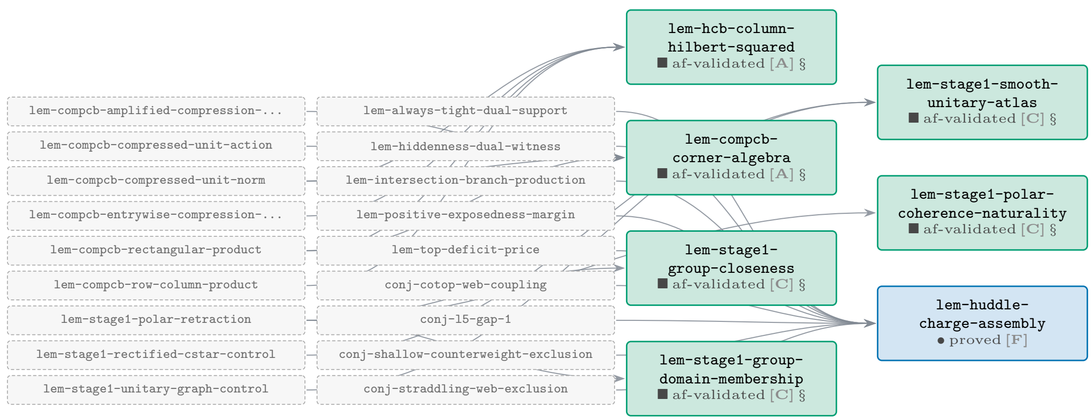
The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*



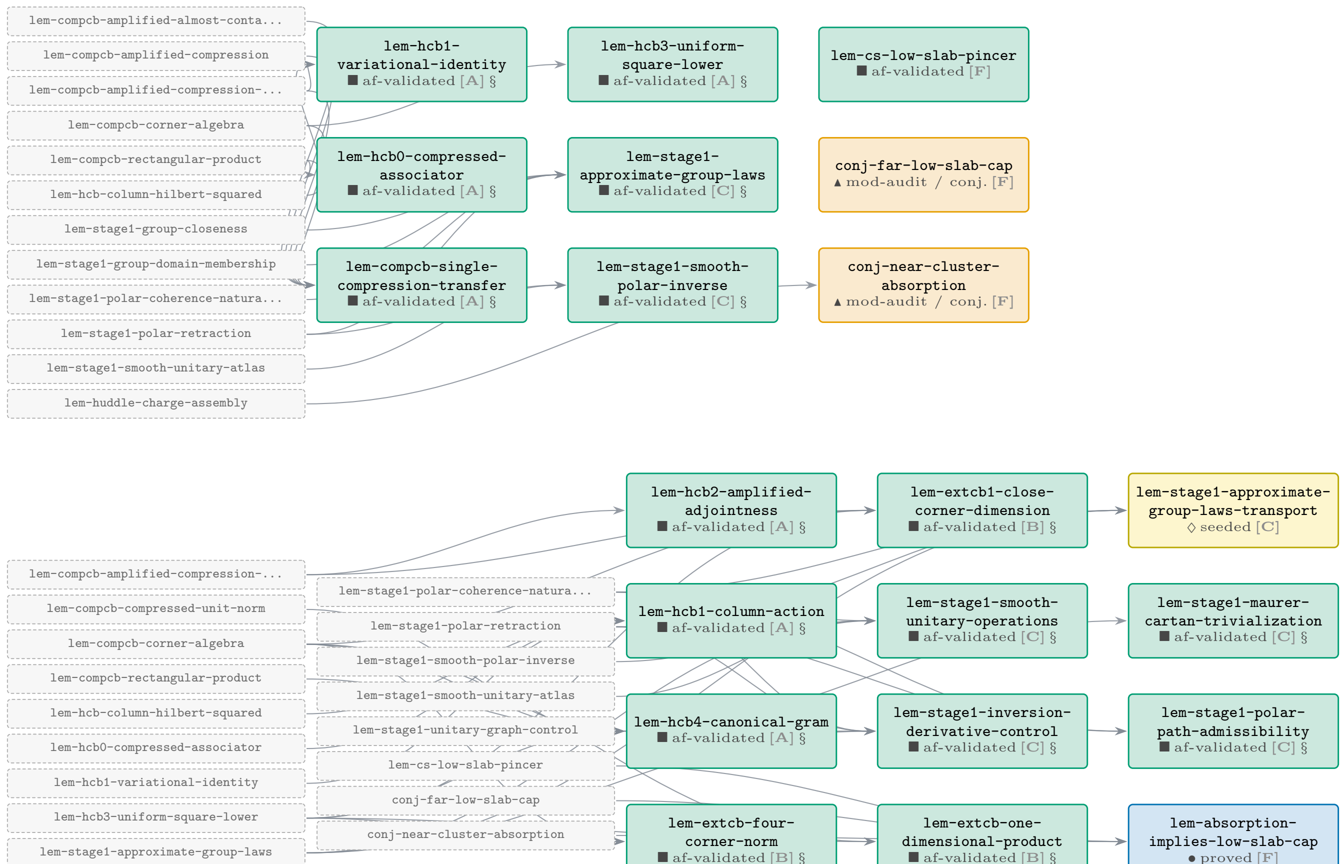


The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*



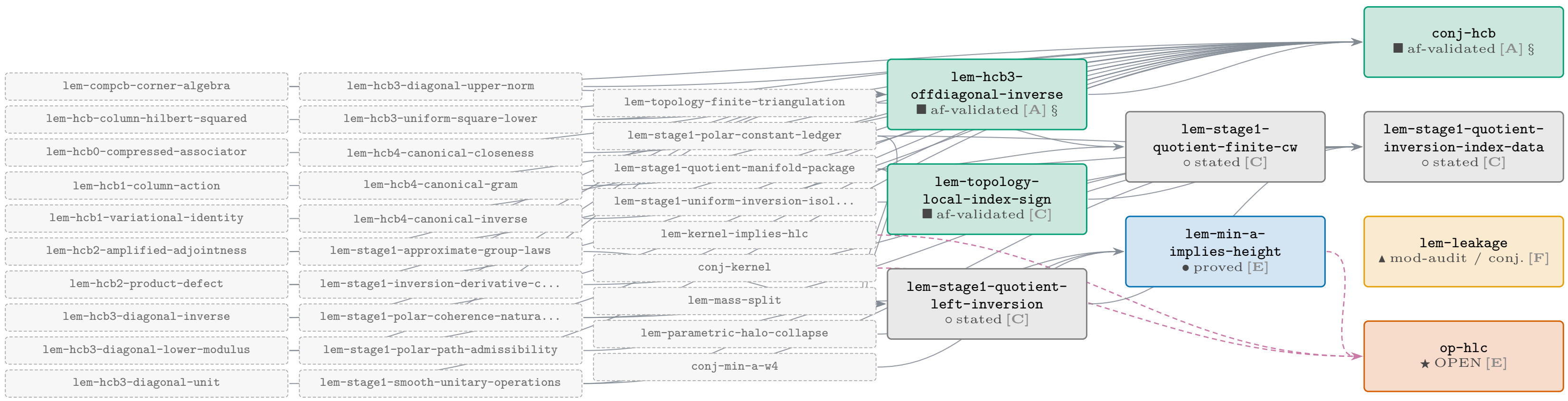


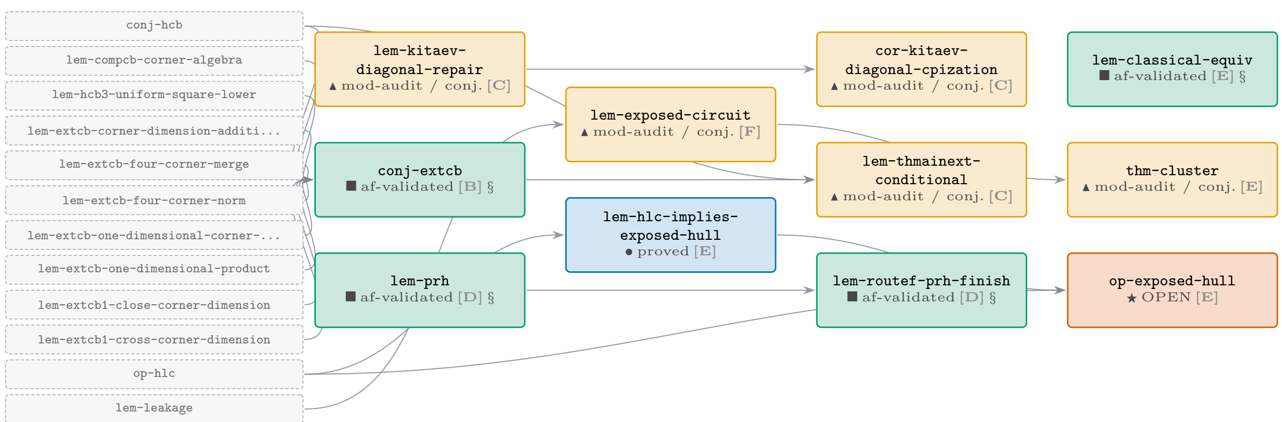
The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*



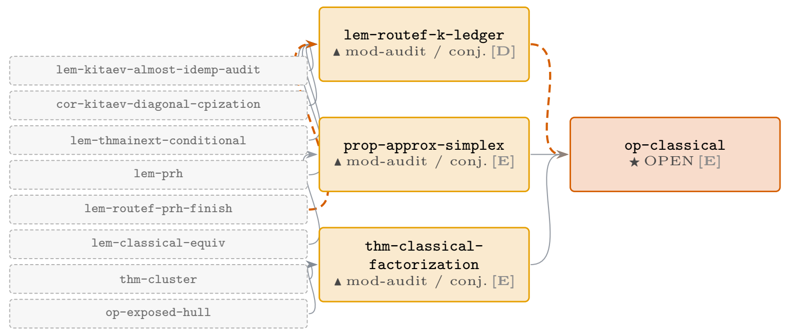
The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*

The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*



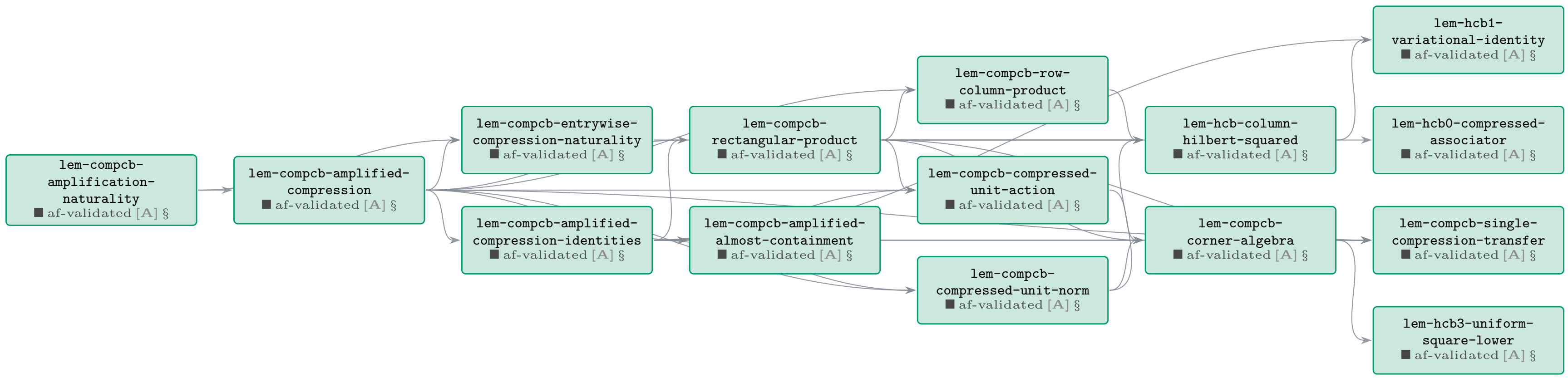


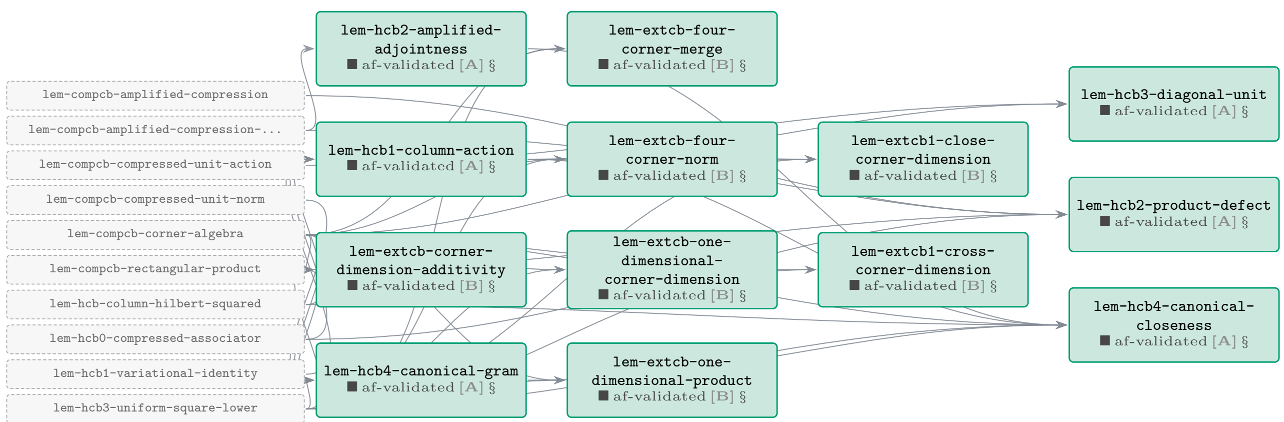
The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*



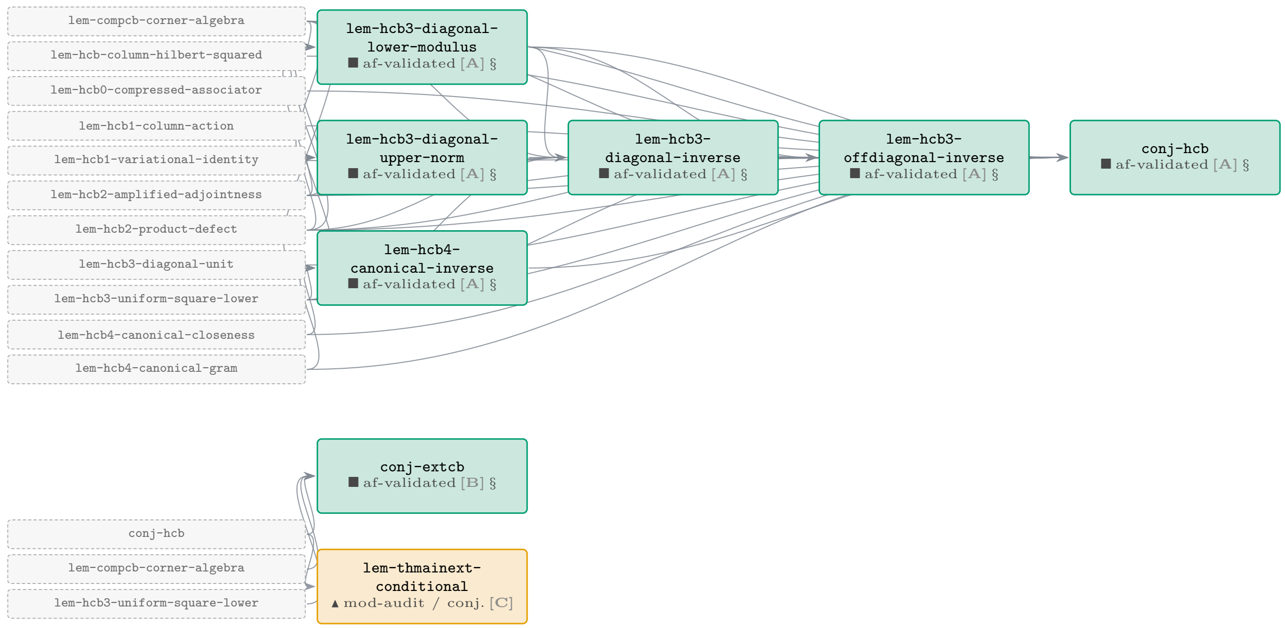
Phase A — COMP-CB / H-CB corner tower

27 results (27 af-validated). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.



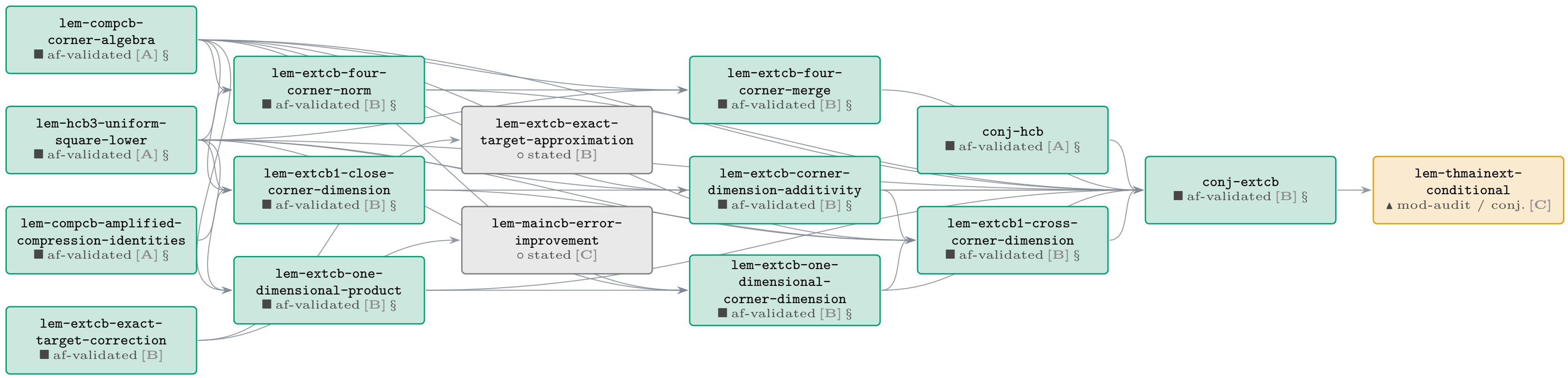


Phase A — COMP-CB / H-CB corner tower *(cont.)*



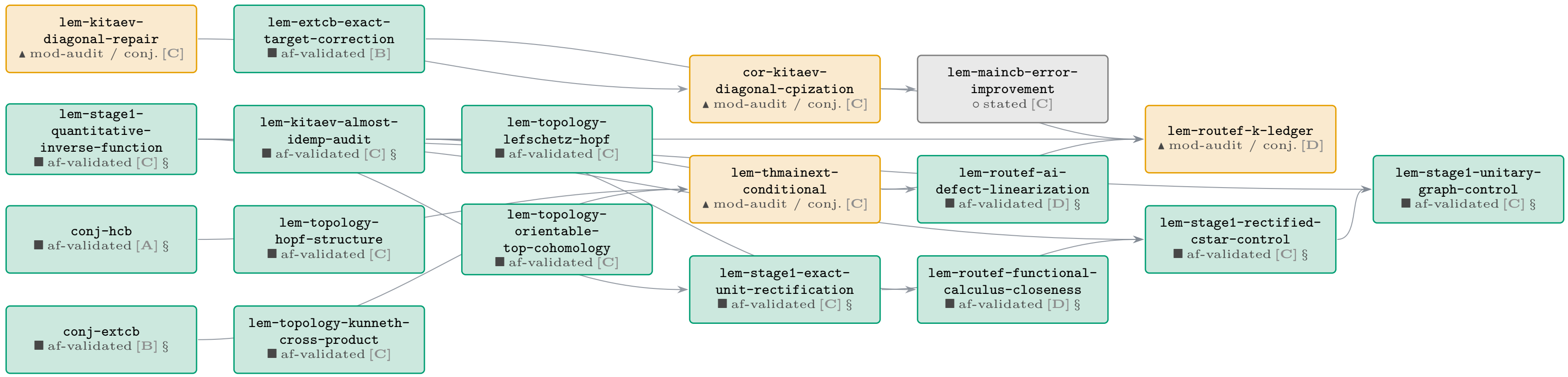
Phase B — EXT-CB corner extension

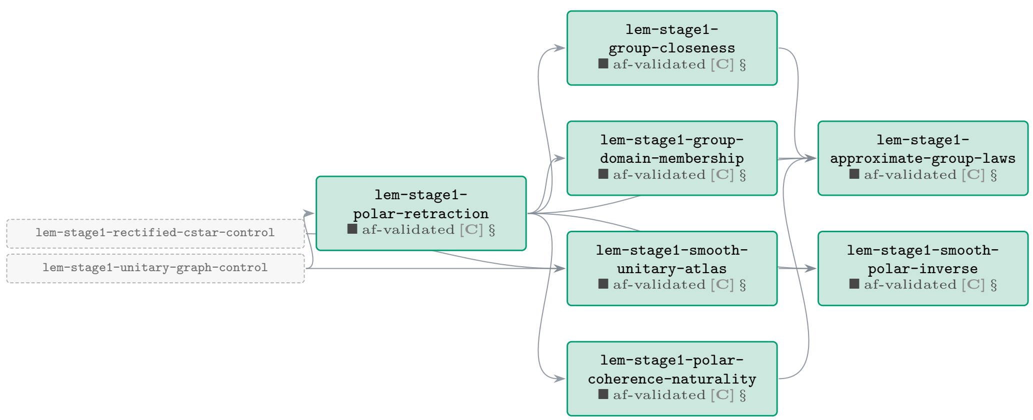
10 results (9 af-validated, 1 stated). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.



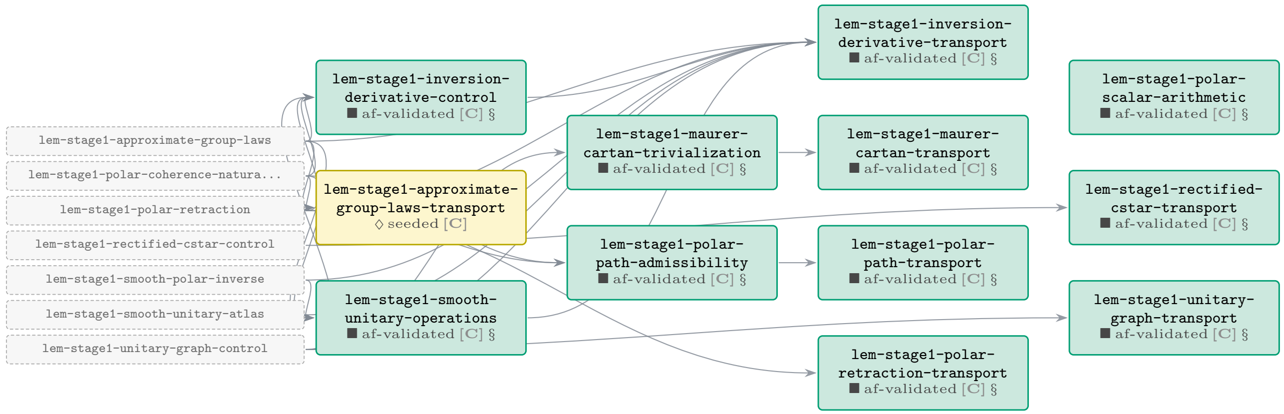
Phase C — Stage-1 reset, topology leaves & the Kitaev repair

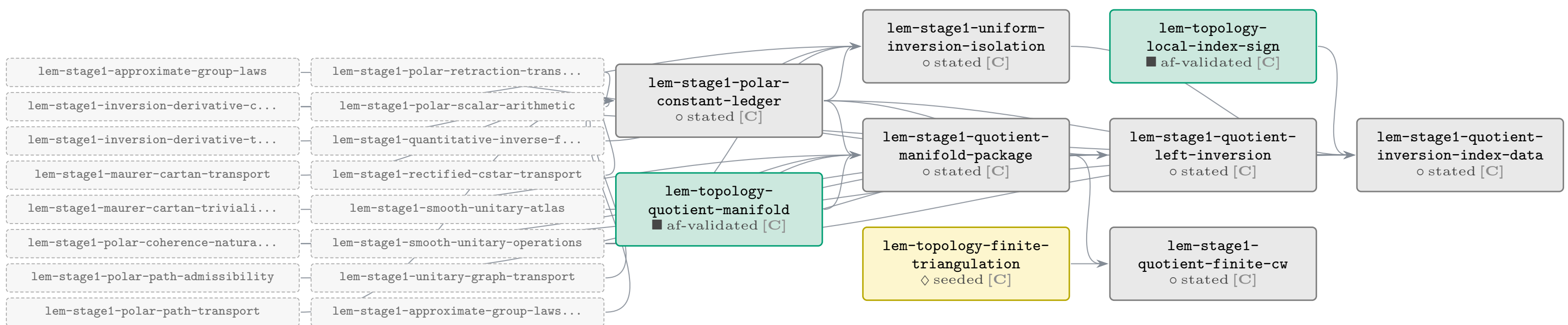
41 results (29 af-validated, 2 seeded, 3 mod-audit / conj., 7 stated). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.





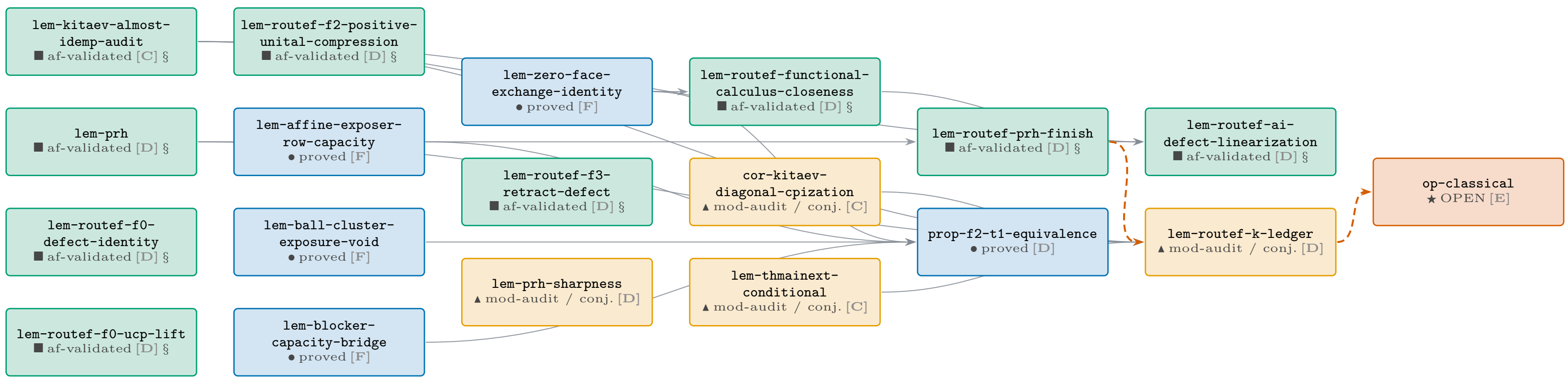
Phase C — Stage-1 reset, topology leaves & the Kitaev repair *(cont.)*





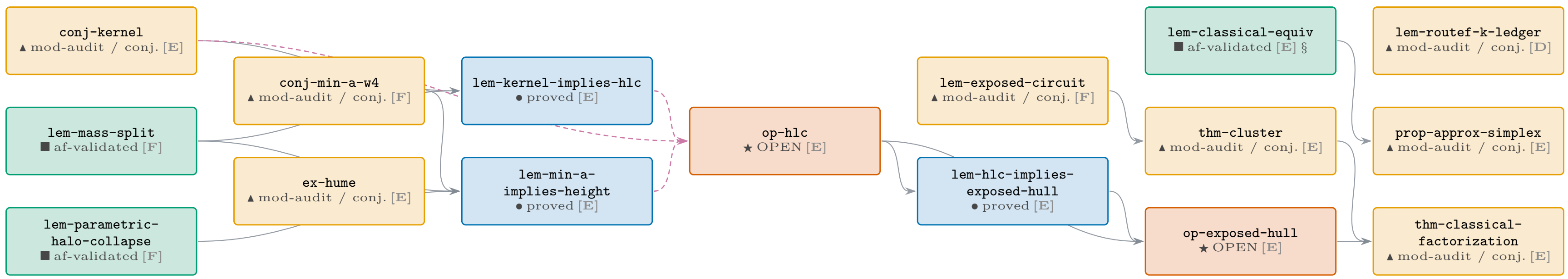
Phase D — Route F bridges, PRH & the K-ledger

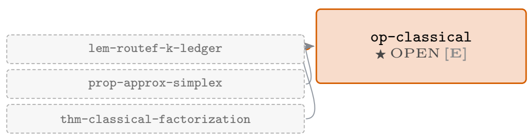
11 results (8 af-validated, 1 proved, 2 mod-audit / conj.). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.



Phase E — root, classical reduction & the exposed-hull spine

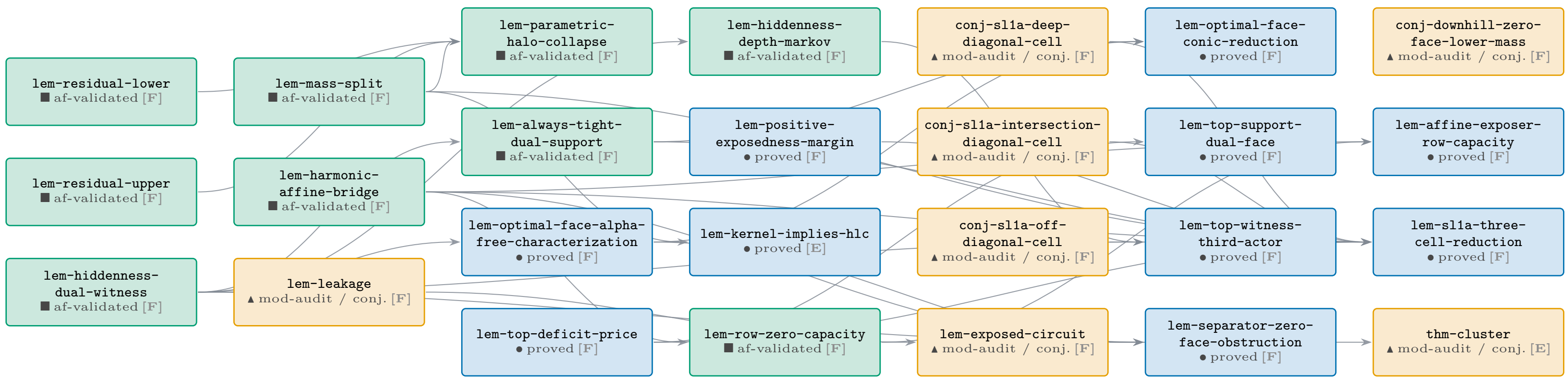
12 results (1 af-validated, 3 proved, 5 mod-audit / conj., 3 OPEN). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.

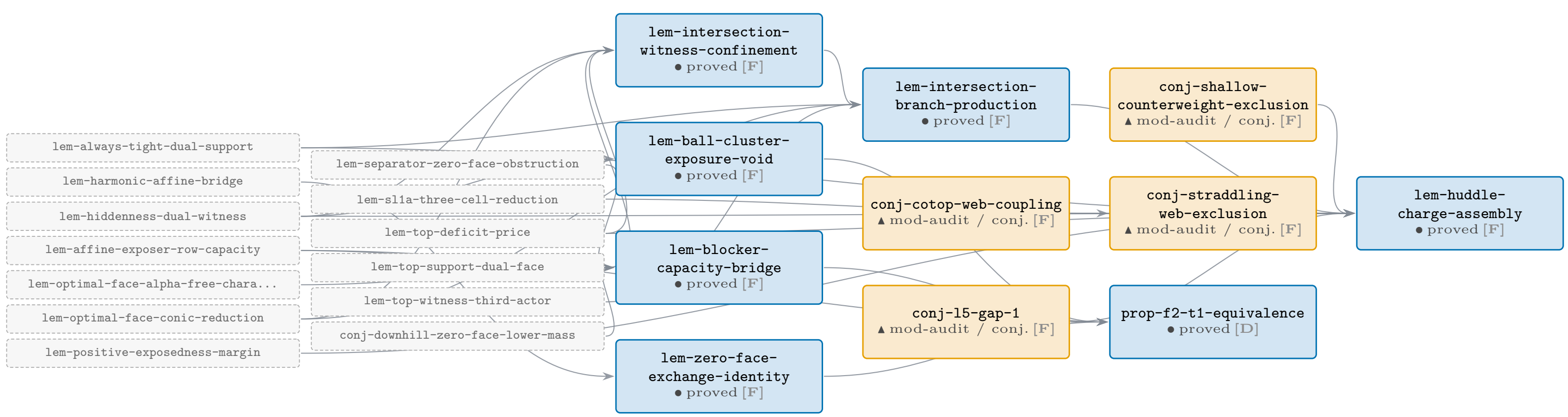




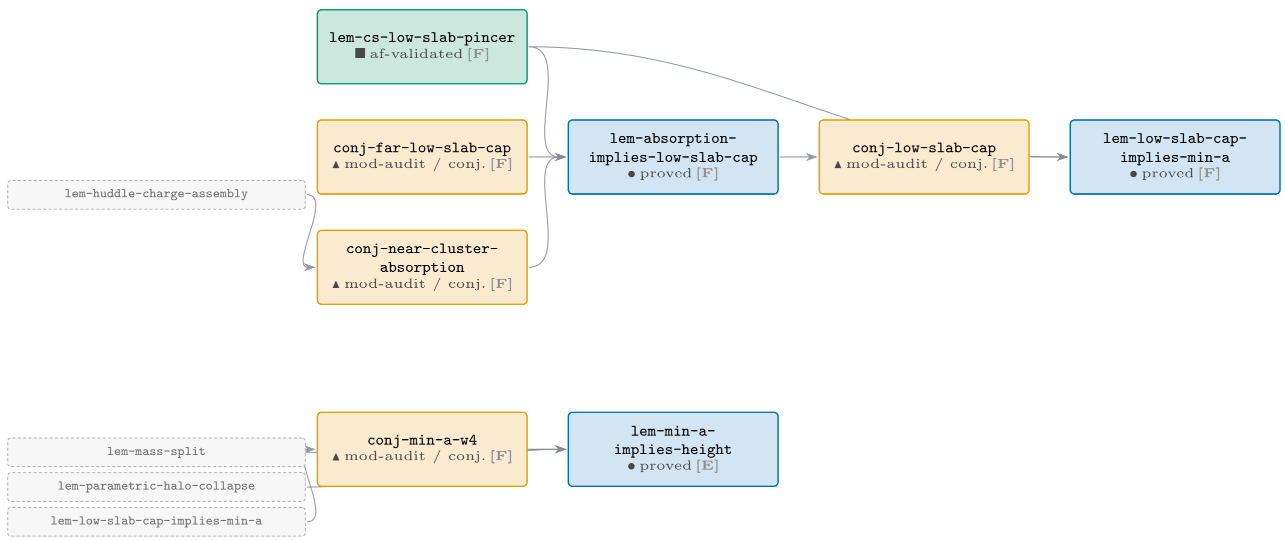
Phase F — signed-geometry trunk (ancestors of the exposed-hull route)

41 results (10 af-validated, 17 proved, 14 mod-audit / conj.). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.





Phase F — signed-geometry trunk (ancestors of the exposed-hull route) *(cont.)*



Node index — every result in the chain

Anchors are `dag:<id>`. The **id** column links to the node in the atlas; **report** links to the typeset statement in this lab-book when one exists. Sorted by id (stable).

id	phase	kind	status	af	in	out	report
`conj-cotop-web-coupling`	trunk	lemma	conjecture	none	0	1	—
`conj-downhill-zero-face-lower-mass`	trunk	lemma	conjecture	none	0	1	—
`conj-extcb`	EXT-CB	lemma	proved	validated	10	1	§ conj:extcb
`conj-far-low-slab-cap`	trunk	lemma	conjecture	none	0	1	—
`conj-hcb`	COMP/H-CB	lemma	proved	validated	16	2	§ conj:hcb
`conj-kernel`	root	open-problem	conjecture	none	0	2	—
`conj-l5-gap-1`	trunk	lemma	conjecture	none	0	1	—
`conj-low-slab-cap`	trunk	lemma	conjecture	none	1	1	—
`conj-min-a-w4`	trunk	lemma	conjecture	none	1	1	—
`conj-near-cluster-absorption`	trunk	lemma	conjecture	none	1	1	—
`conj-shallow-counterweight-exclusion`	trunk	lemma	conjecture	none	0	1	—
`conj-sl1a-deep-diagonal-cell`	trunk	lemma	conjecture	none	0	1	—
`conj-sl1a-intersection-diagonal-cell`	trunk	lemma	conjecture	none	0	1	—
`conj-sl1a-off-diagonal-cell`	trunk	lemma	conjecture	none	0	1	—
`conj-straddling-web-exclusion`	trunk	lemma	conjecture	none	1	1	—
`cor-kitaev-diagonal-cpization`	Stage-1/topology	corollary	proved-mod-audit	none	1	1	—
`ex-hume`	root	obstruction	proved-mod-audit	seeded	0	0	—
`lem-absorption-implies-low-slab-cap`	trunk	lemma	proved	none	3	1	—
`lem-affine-exposer-row-capacity`	trunk	lemma	proved	none	2	2	—
`lem-always-tight-dual-support`	trunk	lemma	proved	validated	1	4	—
`lem-ball-cluster-exposure-void`	trunk	lemma	proved	none	1	1	—
`lem-blocker-capacity-bridge`	trunk	lemma	proved	none	4	1	—
`lem-classical-equiv`	root	lemma	proved	validated	0	1	§ lem:classical-equiv
`lem-compcb-amplification-naturality`	COMP/H-CB	lemma	proved	validated	0	2	§ lem:compcb-amplification-naturality
`lem-compcb-amplified-almost-containment`	COMP/H-CB	lemma	proved	validated	2	1	§ lem:compcb-amplified-almost-containment
`lem-compcb-amplified-compression`	COMP/H-CB	lemma	proved	validated	1	9	§ lem:compcb-amplified-compression
`lem-compcb-amplified-compression-identities`	COMP/H-CB	lemma	proved	validated	1	9	§ lem:compcb-amplified-compression-identities
`lem-compcb-compressed-unit-action`	COMP/H-CB	lemma	proved	validated	3	2	§ lem:compcb-compressed-unit-action
`lem-compcb-compressed-unit-norm`	COMP/H-CB	lemma	proved	validated	2	3	§ lem:compcb-compressed-unit-norm
`lem-compcb-corner-algebra`	COMP/H-CB	lemma	proved	validated	4	18	§ lem:compcb-corner-algebra
`lem-compcb-entrywise-compression-naturality`	COMP/H-CB	lemma	proved	validated	2	1	§ lem:compcb-entrywise-compression-naturality
`lem-compcb-rectangular-product`	COMP/H-CB	lemma	proved	validated	2	8	§ lem:compcb-rectangular-product
`lem-compcb-row-column-product`	COMP/H-CB	lemma	proved	validated	2	1	§ lem:compcb-row-column-product
`lem-compcb-single-compression-transfer`	COMP/H-CB	lemma	proved	validated	5	0	§ lem:compcb-single-compression-transfer
`lem-cs-low-slab-pincer`	trunk	lemma	proved	validated	0	2	—
`lem-exposed-circuit`	trunk	lemma	proved-mod-audit	none	1	1	—
`lem-extcb-corner-dimension-additivity`	EXT-CB	lemma	proved	validated	2	2	§ lem:extcb-corner-dimension-additivity
`lem-extcb-exact-target-approximation`	EXT-CB	lemma	stated	none	1	0	—
`lem-extcb-exact-target-correction`	EXT-CB	lemma	proved	validated	0	2	—
`lem-extcb-four-corner-merge`	EXT-CB	lemma	proved	validated	3	1	§ lem:extcb-four-corner-merge
`lem-extcb-four-corner-norm`	EXT-CB	lemma	proved	validated	2	2	§ lem:extcb-four-corner-norm

id	phase	kind	status	af	in	out	report
lem-extcb-one-dimensional-corner-dimension	EXT-CB	lemma	proved	validated	3	2	§ lem:extcb-one-dimensional-corner-dimension
lem-extcb-one-dimensional-product	EXT-CB	lemma	proved	validated	2	2	§ lem:extcb-one-dimensional-product
lem-extcb1-close-corner-dimension	EXT-CB	lemma	proved	validated	3	2	§ lem:extcb1-close-corner-dimension
lem-extcb1-cross-corner-dimension	EXT-CB	lemma	proved	validated	5	1	§ lem:extcb1-cross-corner-dimension
lem-harmonic-affine-bridge	trunk	lemma	proved	validated	0	5	—
lem-hcb-column-hilbert-squared	COMP/H-CB	lemma	proved	validated	4	4	§ lem:hcb-column-hilbert-squared
lem-hcb0-compressed-associator	COMP/H-CB	lemma	proved	validated	1	3	§ lem:hcb0-compressed-associator
lem-hcb1-column-action	COMP/H-CB	lemma	proved	validated	3	4	§ lem:hcb1-column-action
lem-hcb1-variational-identity	COMP/H-CB	lemma	proved	validated	2	2	§ lem:hcb1-variational-identity
lem-hcb2-amplified-adjointness	COMP/H-CB	lemma	proved	validated	1	6	§ lem:hcb2-amplified-adjointness
lem-hcb2-product-defect	COMP/H-CB	lemma	proved	validated	4	5	§ lem:hcb2-product-defect
lem-hcb3-diagonal-inverse	COMP/H-CB	lemma	proved	validated	5	2	§ lem:hcb3-diagonal-inverse
lem-hcb3-diagonal-lower-modulus	COMP/H-CB	lemma	proved	validated	4	3	§ lem:hcb3-diagonal-lower-modulus
lem-hcb3-diagonal-unit	COMP/H-CB	lemma	proved	validated	2	1	§ lem:hcb3-diagonal-unit
lem-hcb3-diagonal-upper-norm	COMP/H-CB	lemma	proved	validated	3	1	§ lem:hcb3-diagonal-upper-norm
lem-hcb3-offdiagonal-inverse	COMP/H-CB	lemma	proved	validated	6	1	§ lem:hcb3-offdiagonal-inverse
lem-hcb3-uniform-square-lower	COMP/H-CB	lemma	proved	validated	1	15	§ lem:hcb3-uniform-square-lower
lem-hcb4-canonical-closeness	COMP/H-CB	lemma	proved	validated	6	2	§ lem:hcb4-canonical-closeness
lem-hcb4-canonical-gram	COMP/H-CB	lemma	proved	validated	4	3	§ lem:hcb4-canonical-gram
lem-hcb4-canonical-inverse	COMP/H-CB	lemma	proved	validated	4	1	§ lem:hcb4-canonical-inverse
lem-hiddenness-depth-markov	trunk	lemma	proved	validated	1	1	—
lem-hiddenness-dual-witness	trunk	lemma	proved	validated	0	8	—
lem-hlc-implies-exposed-hull	root	lemma	proved	none	1	1	—
lem-huddle-charge-assembly	trunk	lemma	proved	none	9	1	—
lem-intersection-branch-production	trunk	lemma	proved	none	4	1	—
lem-intersection-witness-confinement	trunk	lemma	proved	none	4	1	—
lem-kernel-implies-hlc	root	lemma	proved	none	2	1	—
lem-kitaev-almost-idemp-audit	Stage-1/topology	lemma	proved	validated	0	3	§ lem:kitaev-almost-idemp-audit
lem-kitaev-diagonal-repair	Stage-1/topology	lemma	proved-mod-audit	none	0	1	—
lem-leakage	trunk	lemma	proved-mod-audit	none	0	1	—
lem-low-slab-cap-implies-min-a	trunk	lemma	proved	none	2	1	—
lem-maincb-error-improvement	Stage-1/topology	lemma	stated	none	1	0	—
lem-mass-split	trunk	lemma	proved	validated	0	4	—
lem-min-a-implies-height	root	lemma	proved	none	3	1	—
lem-optimal-face-alpha-free-characterization	trunk	lemma	proved	none	1	3	—
lem-optimal-face-conic-reduction	trunk	lemma	proved	none	2	2	—
lem-parametric-halo-collapse	trunk	lemma	proved	validated	3	1	—
lem-positive-exposedness-margin	trunk	lemma	proved	none	0	2	—
lem-prh	Route F	lemma	proved	validated	0	2	§ lem:prh
lem-prh-sharpness	Route F	lemma	proved-mod-audit	none	0	0	—
lem-residual-lower	trunk	lemma	proved	validated	0	1	—
lem-residual-upper	trunk	lemma	proved	validated	0	1	—
lem-routef-ai-defect-linearization	Route F	lemma	proved	validated	2	0	§ lem:routef-ai-defect-linearization
lem-routef-f0-defect-identity	Route F	lemma	proved	validated	0	0	§ lem:routef-f0-defect-identity
lem-routef-f0-ucp-lift	Route F	lemma	proved	validated	0	0	§ lem:routef-f0-ucp-lift
lem-routef-f2-positive-unital-compression	Route F	lemma	proved	validated	0	0	§ lem:routef-f2-positive-unital-compression
lem-routef-f3-retract-defect	Route F	lemma	proved	validated	0	0	§ lem:routef-f3-retract-defect

id	phase	kind	status	af	in	out	report
lem-routef-functional-calculus-closeness	Route F	lemma	proved	validated	1	1	§ lem:routef-functional-calculus-closeness
lem-routef-k-ledger	Route F	lemma	proved-mod-audit	none	5	1	—
lem-routef-prh-finish	Route F	lemma	proved	validated	1	1	§ lem:routef-prh-finish
lem-row-zero-capacity	trunk	lemma	proved	validated	1	1	—
lem-separator-zero-face-obstruction	trunk	lemma	proved	none	3	1	—
lem-sl1a-three-cell-reduction	trunk	lemma	proved	none	8	1	—
lem-stage1-approximate-group-laws	Stage-1/topology	lemma	proved	validated	4	5	§ lem:stage1-approximate-group-laws
lem-stage1-approximate-group-laws-transport	Stage-1/topology	lemma	stated	seeded	3	1	—
lem-stage1-exact-unit-rectification	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-exact-unit-rectification
lem-stage1-group-closeness	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-group-closeness
lem-stage1-group-domain-membership	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-group-domain-membership
lem-stage1-inversion-derivative-control	Stage-1/topology	lemma	proved	validated	4	3	§ lem:stage1-inversion-derivative-control
lem-stage1-inversion-derivative-transport	Stage-1/topology	lemma	proved	validated	7	1	§ lem:stage1-inversion-derivative-transport
lem-stage1-maurer-cartan-transport	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-maurer-cartan-transport
lem-stage1-maurer-cartan-trivialization	Stage-1/topology	lemma	proved	validated	1	2	§ lem:stage1-maurer-cartan-trivialization
lem-stage1-polar-coherence-naturality	Stage-1/topology	lemma	proved	validated	1	7	§ lem:stage1-polar-coherence-naturality
lem-stage1-polar-constant-ledger	Stage-1/topology	lemma	stated	none	8	4	—
lem-stage1-polar-path-admissibility	Stage-1/topology	lemma	proved	validated	2	2	§ lem:stage1-polar-path-admissibility
lem-stage1-polar-path-transport	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-polar-path-transport
lem-stage1-polar-retraction	Stage-1/topology	lemma	proved	validated	1	10	§ lem:stage1-polar-retraction
lem-stage1-polar-retraction-transport	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-polar-retraction-transport
lem-stage1-polar-scalar-arithmetic	Stage-1/topology	lemma	proved	validated	0	1	§ lem:stage1-polar-scalar-arithmetic
lem-stage1-quantitative-inverse-function	Stage-1/topology	lemma	proved	validated	0	4	§ lem:stage1-quantitative-inverse-function
lem-stage1-quotient-finite-cw	Stage-1/topology	lemma	stated	none	2	0	—
lem-stage1-quotient-inversion-index-data	Stage-1/topology	lemma	stated	none	8	0	—
lem-stage1-quotient-left-inversion	Stage-1/topology	lemma	stated	none	6	1	—
lem-stage1-quotient-manifold-package	Stage-1/topology	lemma	stated	none	5	3	—
lem-stage1-rectified-cstar-control	Stage-1/topology	lemma	proved	validated	2	3	§ lem:stage1-rectified-cstar-control
lem-stage1-rectified-cstar-transport	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-rectified-cstar-transport
lem-stage1-smooth-polar-inverse	Stage-1/topology	lemma	proved	validated	2	2	§ lem:stage1-smooth-polar-inverse
lem-stage1-smooth-unitary-atlas	Stage-1/topology	lemma	proved	validated	2	4	§ lem:stage1-smooth-unitary-atlas
lem-stage1-smooth-unitary-operations	Stage-1/topology	lemma	proved	validated	4	5	§ lem:stage1-smooth-unitary-operations
lem-stage1-uniform-inversion-isolation	Stage-1/topology	lemma	stated	none	4	1	—
lem-stage1-unitary-graph-control	Stage-1/topology	lemma	proved	validated	2	6	§ lem:stage1-unitary-graph-control
lem-stage1-unitary-graph-transport	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-unitary-graph-transport
lem-thmainext-conditional	Stage-1/topology	lemma	proved-mod-audit	none	2	1	—
lem-top-deficit-price	trunk	lemma	proved	none	0	4	—
lem-top-support-dual-face	trunk	lemma	proved	none	1	1	—
lem-top-witness-third-actor	trunk	lemma	proved	none	2	1	—
lem-topology-finite-triangulation	Stage-1/topology	lemma	stated	seeded	0	1	—
lem-topology-hopf-structure	Stage-1/topology	lemma	proved	validated	0	0	—
lem-topology-kunneth-cross-product	Stage-1/topology	lemma	proved	validated	0	0	—
lem-topology-lefschetz-hopf	Stage-1/topology	lemma	proved	validated	0	0	—
lem-topology-local-index-sign	Stage-1/topology	lemma	proved	validated	0	1	—
lem-topology-orientable-top-cohomology	Stage-1/topology	lemma	proved	validated	0	0	—
lem-topology-quotient-manifold	Stage-1/topology	lemma	proved	validated	0	1	—
lem-zero-face-exchange-identity	trunk	lemma	proved	none	1	1	—

id	phase	kind	status	af	in	out	report
`op-classical`	root	open-problem	open	none	3	0	—
`op-exposed-hull`	root	open-problem	open	none	2	1	—
`op-hlc`	root	open-problem	open	none	3	2	—
`prop-approx-simplex`	root	proposition	proved-mod-audit	seeded	1	1	—
`prop-f2-t1-equivalence`	Route F	proposition	proved	none	4	0	—
`thm-classical-factorization`	root	theorem	proved-mod-audit	seeded	2	1	—
`thm-cluster`	root	theorem	proved-mod-audit	seeded	1	1	—

Definition index — the vocabulary the chain imports

One row per `definitions/` shard imported by a drawn result (L2: a term is defined exactly once and only referenced elsewhere). Each row carries the anchor `dag:<def-id>`, so the definitions section can link here with `\dagdef{def-...}` and this table links back to the results that use it.

definition	status	uses	phases that import it
`def-actor-hull`	draft	7	trunk (7)
`def-almost-idempotent`	locked	5	Route F (3), root (2)
`def-approximate-unitary-space`	locked	24	Stage-1/topology (24)
`def-canonical-corner-identifications`	locked	4	COMP/H-CB (4)
`def-co-top`	draft	7	trunk (7)
`def-column-hilbert-corner`	locked	2	COMP/H-CB (2)
`def-compressed-associator`	locked	2	COMP/H-CB (2)
`def-compressed-corner`	locked	17	COMP/H-CB (11), EXT-CB (6)
`def-delta-projection`	locked	15	COMP/H-CB (11), EXT-CB (4)
`def-dual-witness`	draft	1	trunk (1)
`def-epsilon-cstar-algebra`	locked	29	EXT-CB (3), Stage-1/topology (26)
`def-exposed`	locked	38	Route F (1), root (4), trunk (33)
`def-extcb-datum`	locked	3	EXT-CB (3)
`def-extended-delta-inclusion`	locked	5	COMP/H-CB (1), EXT-CB (3), Stage-1/topology (1)
`def-extended-epsilon-cstar-algebra`	locked	23	COMP/H-CB (13), EXT-CB (5), Stage-1/topology (3), Route F (2)
`def-fd-cstar-diagonal`	locked	4	EXT-CB (1), Stage-1/topology (3)
`def-four-corner-merging-datum`	locked	2	EXT-CB (2)
`def-h-space-left-inversion`	locked	2	Stage-1/topology (2)
`def-ha-map`	locked	13	COMP/H-CB (12), EXT-CB (1)
`def-hcb-datum`	locked	15	COMP/H-CB (15)
`def-height`	locked	31	Route F (1), root (5), trunk (25)
`def-invisible-mass`	locked	7	root (3), trunk (4)
`def-lefschetz-fixed-point-data`	locked	3	Stage-1/topology (3)
`def-near-cluster`	draft	3	trunk (3)
`def-near-positive-projection`	locked	2	root (2)
`def-negative-mass`	locked	40	Route F (1), root (6), trunk (33)
`def-one-dimensional-delta-projection`	locked	3	COMP/H-CB (1), EXT-CB (2)
`def-positive-approximate-retract`	locked	3	Route F (3)
`def-projection-basis`	locked	3	EXT-CB (2), Route F (1)
`def-selected-corner`	draft	4	trunk (4)
`def-signed-idempotent`	locked	45	Route F (1), root (7), trunk (37)
`def-slab`	draft	2	trunk (2)
`def-stage1-polar-witness-data`	locked	9	Stage-1/topology (9)
`def-stochastic`	locked	16	Route F (8), root (7), trunk (1)
`def-theta-idempotent-approximation`	draft	2	COMP/H-CB (2)
`def-top-support-functional`	draft	6	trunk (6)
`def-ucp-map`	locked	3	Route F (3)
`def-visible-set`	locked	37	Route F (1), root (5), trunk (31)
`def-zero-face`	draft	1	trunk (1)

Definitions in `definitions/` not imported by any result on this chain (6): `def-maincb-partition-state`, `def-maincb-raw-call`, `def-maincb-reset-state`, `def-operator-space`, `def-pivot`, `def-top-deficit`.

52 The campaign in numbers

52.1 The campaign at a glance

This section is machine-generated from the two repositories' own records: the `fr` controller log, the append-only `af` proof ledgers, the registry and definition shards, the run bundles, the work log, the issue tracker, and the git history. Table 5 places the progenitor campaign (`almost-idempotent-positive-maps`, where the classical portfolio was born) beside this one. Nothing below is a mathematical claim; nothing below promotes any result up the rigour ladder.

Table 5: The two campaigns side by side. The progenitor column is a snapshot of `../almost-idempotent-positive-maps`; that repository has no `fr` controller, so its exploration record lives only in its work log and git history.

Quantity	Progenitor	This repository
First commit	2026-06-02	2026-07-02
Last commit in snapshot	2026-06-13	2026-07-28
Commits	174	836
Days with a commit	11	19
Work-log entries	22	67
`fr` controller cycles	— (no controller)	800
Registry results	59	295
Definition shards	25	45
`af` workspaces	16	119
`af`-validated roots (T0)	16	107
Adversarial proof nodes built	132	1 894
Ledger entries (append-only)	822	11 375
Numerical run bundles	–	38

"T0" is this project's top-of-ladder tag for a result whose `af` proof tree has a `validated` root under the fresh-prover/fresh-verifier protocol. It is not a Lean proof.

52.2 What is counted, and what cannot be

Counted, exactly. Every figure in this section is a count of an artifact that exists in one of the two repositories at the snapshot instant: a JSON record in `.frontier/log.jsonl`, a ledger entry under `proofs/*/ledger/`, a front-matter field in an `argument/lemmas/*.md` or `definitions/*.md` shard, a directory under `runs/` or `docs/`, a heading in `docs/worklog.md` or `FINDINGS.md`, a record in `.beads/issues.jsonl`, or a commit in `git log`. No figure is modelled, interpolated, or extrapolated.

Not recoverable: LLM token consumption. The single most-asked statistic about a campaign like this one — how many tokens, and at what cost — is *not* present in the record and is not reconstructed here. Neither repository ever logged token counts, context sizes, wall-clock durations, or billing data; the `af` ledger records who claimed which node and when, not how much text was produced to do it, and the `fr` log records a worker's `model:role` pair, not its usage. What *is* recoverable is the *job* count — how many distinct prover/verifier invocations were made — and that is reported exactly in Section 52.7. Multiplying a job count by a guessed per-job token figure would produce a number with no evidentiary basis, so it is omitted rather than invented (CLAUDE.MD L0/L3: a number without a re-run command is not a finding).

Other honest gaps.

- The progenitor repository predates the `fr` controller, so it has no cycle/arm/outcome record at all. Its exploration history is reconstructible only from 22 work-log entries and 174 commits, at coarser granularity than this repository's 800 controller cycles.
- The progenitor repository's issue tracker is stored in an embedded Dolt database rather than an exported `issues.jsonl`, so its issue counts are not in this snapshot.
- `fr` evidence records carry a self-reported `verdict` field whose value is `claimed` on every one of the 379 records that has one. It is a protocol marker, not an adjudication, and is therefore not tabulated as an outcome.
- Wave numbering (`W1`–`W95`) is a human convention in the work log and wave documents; it is not a controller-enforced counter, and a few numbers were skipped or reused. Counts of wave *documents* are exact; the wave *index* is indicative.
- Cross-repository carry-over (what this campaign inherited from the progenitor) is described qualitatively in Section 52.9; it is a judgement about content, not a count.

52.3 The campaign calendar

The two campaigns ran as short, dense bursts rather than as steady background work. This repository put 836 commits into 19 calendar days with a commit (out of 27 days elapsed), and the progenitor put 174 into 11.

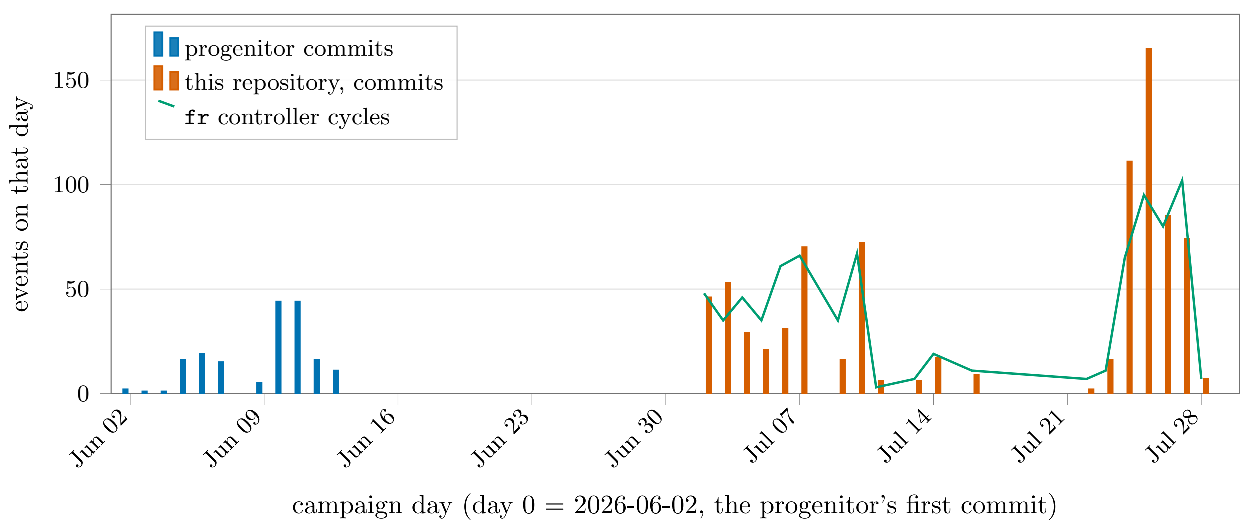


Figure 1: The campaign calendar. Bars are commits per calendar day in each repository; the line is `fr` controller cycles per day in this repository. The three-week gap between the two bar families is the interregnum between the progenitor's last commit (2026-06-13) and this repository's stand-up (2026-07-02).

Table 6: The five heaviest days in this repository, by commit count.

Day	Commits	`fr` cycles	banked	`af` roots validated
2026-07-25	165	95	2	22
2026-07-24	111	65	0	14
2026-07-26	85	80	6	13
2026-07-27	74	102	23	23
2026-07-10	72	67	1	6

"banked" counts `fr` log records with outcome `banked`; the last column counts `af` proof trees whose root node was validated that day.

52.4 The rigour ladder

The project's whole point is that a claim's *status* is tracked honestly and separately from its plausibility. The registry therefore carries a status census, and the only rung that counts as rigorous in-repo is an `af`-validated tree (or a byte-matched citation). Of 295 registry results, 107 carry `af:validated` — 36.3% of the registry. The remainder are flagged non-rigorous in the open.

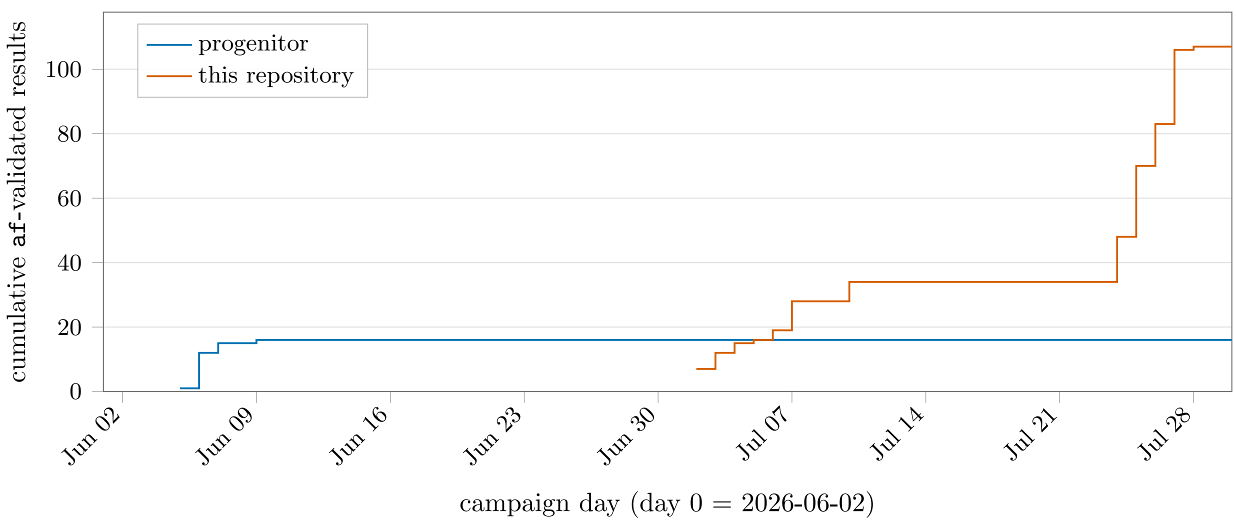


Figure 2: The rigour ladder over time: cumulative count of proof trees whose root node reached `validated` under the fresh-prover/fresh-verifier protocol. Each step is one result crossing from *claimed* to T0. Timestamps are read from the `node_validated` entries of the append-only `af` ledgers.

Table 7: Status census of the result registry.

Registry `status:`	Results	Share
`proved`	220	74.6%
`conjecture`	31	10.5%
`proved-mod-audit`	23	7.8%
`stated`	11	3.7%
`open`	3	1.0%
`disproved`	2	0.7%
`heuristic`	2	0.7%
`numerical`	2	0.7%
`obstruction`	1	0.3%

`proved` here means "the shard's proof has been carried out and reviewed"; it becomes *rigorous* only in combination with `af: validated` (next table). `proved-mod-audit` is the inherited-campaign tag: a paper proof that has not cleared an independent pass in this repository.

Table 8: Formalisation state of the registry: how many results have an `af` workspace, and how many reached a validated root.

`af:` field	Results	Share
`none`	176	59.7%
`validated`	107	36.3%
`seeded`	12	4.1%

`seeded` means a workspace exists with the registry contract as its root conjecture but the tree is not (yet) closed.

Table 9: The vocabulary layer: one definition per shard, provenance-gated.

Definition shards	Count
`original`	26
`cited`	13
`consensus`	6
`status: locked`	34
`status: draft`	11

`cited` definitions byte-match a local source under `refs/`; `consensus` and `original` are adopted/introduced here and lock only on recorded sign-off.

52.5 The exploration controller: arms and outcomes

Research directions are registered as `fr` *arms*; every harvested result is logged as one pull with an outcome, and a Stop-hook circuit-breaker forces a stalled arm to yield. Across 800 cycles, 453 carried an arm label and 347 were orientation or bookkeeping records. The controller recorded 378 explicit allocation decisions: `EXPLOIT` $\times$ 361, `EXPLORE` $\times$ 14, `PIVOT` $\times$ 3.

banked
refuted
lem-topology-hopf-structure af-VALIDATED (81st rigorous result; 13 nodes 13/13 clean) + lem-extcb-exact-target-c
lem-topology-orientable-top-cohomology af-VALIDATED (79th rigorous result; 14 nodes taint clean | Thm 3.26 + fie

Figure 3: Attention allocation across research directions. Each bar is one `fr` arm; segments are the recorded outcomes of the pulls on it. Records with no arm (orientation and bookkeeping cycles) are excluded.

Arm	Pulls	banked	progress	died	refuted	SESSION 28 FINAL: T0 76 -> 83 (six topology + GAP-EA correction banked, al
FH	239	29	134	1	0	
B	113	1	108	4	0	
R	52	31	14	1	0	
G	21	2	19	0	0	
A	14	0	14	0	0	
D	8	0	8	0	0	
F	3	0	3	0	0	
E	2	0	1	0	1	
XE	1	0	0	0	0	

Outcome vocabulary: `banked` = a result harvested and recorded; `progress` = movement without a bankable result; `died` / `refuted` = the direction was killed, with a certificate; `discovery` = an off-goal find routed to the portfolio.

Aggregated over all cycles the outcome mix is 344 `orient`, 301 `progress`, 75 `dispatch`, 63 `banked`, 6 `died`, 3 `discovery`, 1 `SESSION 28 FINAL: T0 76 -> 83 (six topology + GAP-EA correction banked, all oracle-verified); both user escalations ratified in-chat + executed same-session; Cairns->Munkres re-source completed with full manifest provenance (user-supplied scan, per-page OCR, visual confirmation); finite-triangulation parked on aism-j5t9 (provision the C^r-triangulation definition external). All pushed to origin + dolt. Breaker honored: next cycle EXPLORE XE.`, 1 `lem-topology-hopf-structure af-VALIDATED (81st rigorous result; 13 nodes 13/13 clean) + lem-extcb-exact-target-correction af-VALIDATED (82nd; 6 nodes 6/6 clean, first-pass, zero challenges | the designed transcription landed EXACTLY on budget) | GAP-EA, the first of the four v4.1 GAP families, is DISCHARGED at L0. T0 80 -> 82. Session 28 totals: +6 T0 (5 of 7 topology + the GAP-EA correction row), 2 trees parked with` precise diagnoses (1 af-bookkeeping thrash, 1 genuine contract/def scope gap -> user escalation), 3 orchestration-infrastructure laws discovered+banked. Breaker honored: next cycle EXPLOREs off FH., 1 lem-topology-lefschetz-hopf af-VALIDATED (77th rigorous result; 2 nodes taint clean, first-pass canary) + lem-topology-kunneth-cross-product af-VALIDATED (78th; 7 nodes taint clean) | first two of the seven Stage-1 topology elevations banked, oracles af-lem-topology-* registered, fr verify PASS on both exports, registry flips committed, T0 76 -> 78. Session findings: parallel orchestrations mutually abort (sequential law banked to bd memory); 4 balloon-aborts + 1 stuck (external named but not attached as node dep) queued for repair., 1 lem-topology-local-index-sign af-VALIDATED (83rd rigorous result; 23 nodes 23/23 clean) under the user-ratified narrowed contract | the in-place root amendment retained all 21 previously validated nodes; oracle fr verify PASS. T0 82 -> 83; 6 of 7 topology rows T0. finite-triangulation root challenge classified: L1 ground-truth stop (Cairns delegates the class-one/allowable definitions to Veblen-Whitehead 1932, NOT in refs/) | user decision being asked in-chat (narrow to Cairns's literal statement vs acquire V-W)., 1 lem-topology-orientable-top-cohomology af-VALIDATED (79th rigorous result; 14 nodes taint clean | Thm 3.26 + field-UCT + Cor 3.39 composed) + lem-topology-quotient-manifold af-VALIDATED (80th; 4 nodes taint clean | the external-attachment verifier challenge resolved by the re-run prover) | 4 of 7 topology elevations banked, oracles registered, fr verify PASS both, T0 78 -> 80. GAP-EA rows LANDED per DESIGN-GAP-EA option (a): lem-extcb-exact-target-correction (validated node-1.2 contract verbatim, proved/af:none) + lem-extcb-exact-target-approximation (v4.1 bridge, stated), UNWIRED-whitelisted, gate green. Next: seed+orchestrate the correction row; finishing runs for finite-triangulation (14/17), local-index-sign (18/22), hopf re-run., 1 null, 1 obstruction, 1 refuted. Evidence records are tiered 134 T1, 131 T2, 114 T0.

52.6 Adversarial elevations

A result reaches the top in-repo rung only through an af workspace: the registry contract becomes the root conjecture, a fresh prover agent builds a proof tree, and every node is validated by a *separate* fresh verifier told that finding a gap is a success. The orchestrator never judges a step. 119 workspaces were opened; 107 reached a validated root and 12 remain seeded or open. Between them the workers built 1 894 proof nodes, raised 322 challenges (of which 314 were resolved), and forced 245 node amendments.

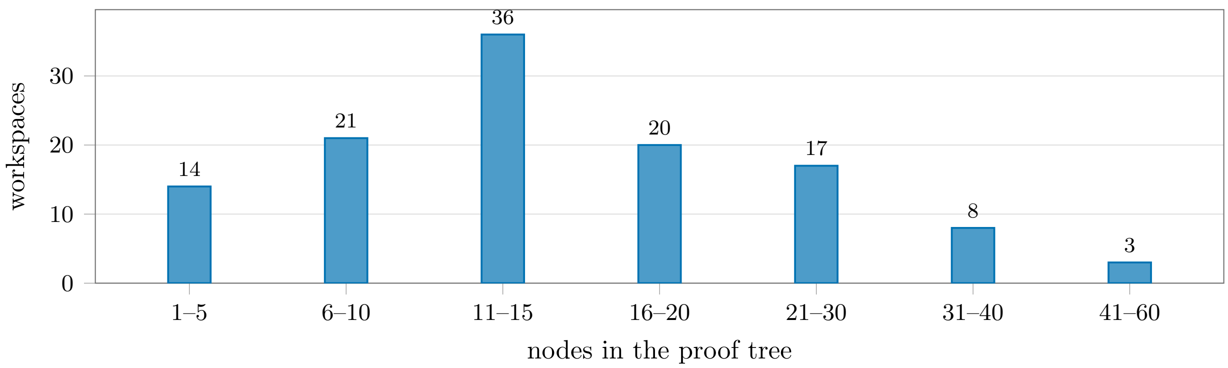


Figure 4: Size distribution of the 119 adversarial proof trees. The median tree has 14 nodes; the largest, `lem-residual-upper`, has 52.

The most-contested tree was `conj-extcb` (24 challenges raised); the longest-running was `conj-extcb`, which needed 20 prover/verifier rounds before its root closed.

Table 11: The append-only `af` ledger, by entry type, summed over every workspace in this repository.

Ledger entry type	Count
`nodes_claimed`	3 046
`nodes_released`	3 046
`node_created`	1 894
`node_validated`	1 804
`def_added`	533
`challenge_raised`	322
`challenge_resolved`	314
`node_amended`	245
`proof_initialized`	119
`node_archived`	51
`approach_tried`	1

`nodes_claimed` is the operational unit of work: one claim event is one worker taking one node. It is the closest thing the record has to a job counter (Section 52.7).

Table 12: Every `af`-validated elevation in this repository, in the order its root node closed.

#	Result	Nodes	Val.	Chal.	Amd.	Defs	Rnds	Jobs	Closed
1	`lem-classical-equiv`	29	21	6	4	4	7	37	2026-07-02
2	`obs-height-collapse`	19	19	2	1	5	5	29	2026-07-02
3	`lem-mass-split`	9	9	1	0	2	3	13	2026-07-02
4	`lem-residual-lower`	36	32	9	2	0	11	56	2026-07-02
5	`lem-residual-upper`	52	49	11	1	0	5	79	2026-07-02
6	`lem-halo-collapse`	20	20	5	3	5	11	33	2026-07-02
7	`lem-factorization`	12	11	1	1	2	4	15	2026-07-02
8	`lem-zerosum-triangle`	10	10	0	0	0	2	13	2026-07-03

(continued)

(Table 12, continued)

#	Result	Nodes	Val.	Chal.	Amd.	Defs	Rnds	Jobs	Closed
9	`lem-weighted-min`	8	8	1	1	0	4	11	2026-07-03
10	`lem-fan-payment`	15	15	1	0	0	6	21	2026-07-03
11	`lem-negpart-subadditive`	16	16	2	3	0	6	25	2026-07-03
12	`lem-fan-payment-restricted`	27	27	2	1	0	9	39	2026-07-03
13	`lem-pivot-removing-move`	9	9	0	2	1	2	14	2026-07-04
14	`lem-collateral-import`	32	31	2	0	1	5	41	2026-07-04
15	`lem-cross-pivot-cancellation`	23	23	3	2	1	5	33	2026-07-04
16	`lem-import-reduction`	10	10	0	0	1	2	13	2026-07-05
17	`lem-parametric-halo-collapse`	14	14	0	0	5	1	19	2026-07-06
18	`lem-genuine-disintegration`	19	19	3	1	6	8	30	2026-07-06
19	`lem-top-concentration`	19	18	4	6	6	8	34	2026-07-06
20	`lem-hiddenness-dual-witness`	11	11	0	0	4	2	15	2026-07-07
21	`lem-cs-low-slab-pincer`	14	14	2	1	2	5	22	2026-07-07
22	`lem-harmonic-affine-bridge`	18	18	0	0	1	1	23	2026-07-07
23	`lem-row-far-dual-certificate`	16	16	3	2	4	8	28	2026-07-07
24	`lem-top-slab-companion`	22	22	1	1	6	5	29	2026-07-07
25	`lem-hiddenness-depth-markov`	32	31	4	3	5	8	47	2026-07-07
26	`lem-depth-d-halo-collapse`	11	11	0	0	5	2	14	2026-07-07
27	`lem-row-zero-capacity`	18	18	0	0	2	2	24	2026-07-07
28	`lem-always-tight-dual-support`	18	18	3	0	4	2	27	2026-07-07
29	`lem-starvation-completion-obstruction`	7	7	0	0	2	2	9	2026-07-10
30	`lem-hx-signed-variation-ledger`	11	11	1	1	2	4	16	2026-07-10
31	`lem-hx-financing-floor`	14	12	5	3	2	8	32	2026-07-10
32	`lem-hx-forced-exterior-coupling`	12	12	0	0	2	3	17	2026-07-10
33	`lem-hx-robust-scalar-starvation`	12	12	0	2	2	2	18	2026-07-10
34	`lem-hx-transverse-moment-identity`	14	14	0	0	1	2	19	2026-07-10
35	`lem-prh`	14	14	2	0	2	6	24	2026-07-24
36	`lem-compcb-amplification-naturality`	12	12	0	0	1	3	16	2026-07-24
37	`lem-compcb-amplified-compression`	13	13	1	5	4	5	19	2026-07-24
38	`lem-compcb-amplified-compression-identities`	7	7	2	1	4	4	15	2026-07-24
39	`lem-compcb-rectangular-product`	12	12	1	0	4	5	19	2026-07-24
40	`lem-compcb-compressed-unit-norm`	15	12	6	4	6	9	33	2026-07-24
41	`lem-compcb-amplified-almost-containment`	13	13	1	1	6	5	20	2026-07-24
42	`lem-compcb-compressed-unit-action`	20	20	5	3	6	8	37	2026-07-24
43	`lem-compcb-corner-algebra`	14	14	2	10	6	4	22	2026-07-24
44	`lem-compcb-single-compression-transfer`	33	32	8	2	7	11	57	2026-07-24
45	`lem-hcb0-compressed-associator`	20	20	2	0	5	6	29	2026-07-24
46	`lem-compcb-entrywise-compression-naturality`	4	4	0	2	6	1	5	2026-07-24
47	`lem-compcb-row-column-product`	20	18	5	3	8	4	32	2026-07-24
48	`lem-hcb-column-hilbert-squared`	50	50	22	17	7	13	108	2026-07-24
49	`lem-hcb1-variational-identity`	8	8	0	0	5	1	9	2026-07-25
50	`lem-hcb1-column-action`	38	38	12	12	9	11	73	2026-07-25
51	`lem-hcb2-amplified-adjointness`	21	21	9	4	9	4	46	2026-07-25
52	`lem-hcb2-product-defect`	12	12	0	0	9	2	15	2026-07-25
53	`lem-hcb3-diagonal-unit`	18	18	3	9	9	8	31	2026-07-25
54	`lem-hcb3-diagonal-upper-norm`	30	30	11	7	9	13	61	2026-07-25
55	`lem-hcb3-uniform-square-lower`	18	18	6	4	11	15	37	2026-07-25
56	`lem-hcb3-diagonal-lower-modulus`	29	29	12	10	11	8	64	2026-07-25
57	`lem-hcb3-diagonal-inverse`	5	5	0	0	11	1	6	2026-07-25
58	`lem-hcb3-offdiagonal-inverse`	16	16	1	0	11	6	23	2026-07-25
59	`lem-hcb4-canonical-gram`	9	9	0	4	12	4	13	2026-07-25
60	`lem-hcb4-canonical-closeness`	11	11	0	0	12	3	15	2026-07-25
61	`lem-hcb4-canonical-inverse`	19	19	3	3	13	6	30	2026-07-25
62	`conj-hcb`	11	11	0	0	15	1	12	2026-07-25
63	`lem-extcb-one-dimensional-product`	25	25	1	0	16	6	35	2026-07-25

(continued)

(Table 12, continued)

#	Result	Nodes	Val.	Chal.	Amd.	Defs	Rnds	Jobs	Closed
64	`lem-extcb-corner-dimension-additivity`	39	39	9	3	18	9	76	2026-07-25
65	`lem-extcb-four-corner-norm`	18	18	6	2	19	11	35	2026-07-25
66	`lem-extcb-four-corner-merge`	26	22	5	9	21	10	44	2026-07-25
67	`lem-extcb-one-dimensional-corner-dimension`	8	8	0	0	20	2	11	2026-07-25
68	`lem-extcb1-close-corner-dimension`	16	16	2	0	21	5	24	2026-07-25
69	`lem-extcb1-cross-corner-dimension`	30	30	8	2	21	7	54	2026-07-25
70	`conj-extcb`	46	40	24	15	21	20	115	2026-07-25
71	`lem-stage1-quantitative-inverse-function`	14	14	3	2	0	4	26	2026-07-26
72	`lem-stage1-exact-unit-rectification`	6	6	1	0	1	4	9	2026-07-26
73	`lem-routef-prh-finish`	22	22	3	4	2	8	34	2026-07-26
74	`lem-kitaev-almost-idemp-audit`	24	24	5	0	1	12	39	2026-07-26
75	`lem-routef-functional-calculus-closeness`	11	11	0	0	1	3	17	2026-07-26
76	`lem-routef-ai-defect-linearization`	13	13	1	1	2	5	19	2026-07-26
77	`lem-topology-lefschetz-hopf`	2	2	0	1	1	1	5	2026-07-26
78	`lem-topology-kunneth-cross-product`	7	7	1	1	0	4	11	2026-07-26
79	`lem-topology-orientable-top-cohomology`	14	14	1	2	0	5	20	2026-07-26
80	`lem-topology-quotient-manifold`	4	4	3	0	0	5	13	2026-07-26
81	`lem-topology-hopf-structure`	13	13	1	2	1	5	20	2026-07-26
82	`lem-extcb-exact-target-correction`	6	6	0	0	4	3	9	2026-07-26
83	`lem-topology-local-index-sign`	23	23	9	10	1	7	55	2026-07-26
84	`lem-routef-f0-ucp-lift`	9	9	1	1	2	3	16	2026-07-27
85	`lem-routef-f0-defect-identity`	12	12	0	0	3	4	19	2026-07-27
86	`lem-routef-f2-positive-unital-compression`	22	22	2	1	3	9	32	2026-07-27
87	`lem-routef-f3-retract-defect`	11	11	0	0	1	4	15	2026-07-27
88	`lem-stage1-rectified-cstar-control`	17	17	5	0	1	9	33	2026-07-27
89	`lem-stage1-unitary-graph-control`	18	15	3	2	2	8	29	2026-07-27
90	`lem-stage1-maurer-cartan-trivialization`	15	15	2	0	2	9	24	2026-07-27
91	`lem-stage1-polar-retraction`	29	29	6	8	2	9	51	2026-07-27
92	`lem-stage1-polar-coherence-naturality`	10	10	0	0	1	4	13	2026-07-27
93	`lem-stage1-group-domain-membership`	10	10	3	1	4	8	17	2026-07-27
94	`lem-stage1-group-closeness`	12	12	0	0	2	3	15	2026-07-27
95	`lem-stage1-approximate-group-laws`	14	14	1	8	2	5	19	2026-07-27
96	`lem-stage1-polar-path-admissibility`	12	12	0	0	2	2	16	2026-07-27
97	`lem-stage1-inversion-derivative-control`	10	10	0	0	2	1	11	2026-07-27
98	`lem-stage1-smooth-unitary-atlas`	14	14	1	0	2	6	21	2026-07-27
99	`lem-stage1-smooth-polar-inverse`	21	21	0	1	2	3	28	2026-07-27
100	`lem-stage1-smooth-unitary-operations`	15	15	2	2	2	3	25	2026-07-27
101	`lem-stage1-polar-scalar-arithmetic`	15	15	0	0	1	2	20	2026-07-27
102	`lem-stage1-rectified-cstar-transport`	7	7	0	1	2	2	8	2026-07-27
103	`lem-stage1-unitary-graph-transport`	9	9	0	0	3	3	12	2026-07-27
104	`lem-stage1-maurer-cartan-transport`	13	13	2	2	3	9	19	2026-07-27
105	`lem-stage1-polar-retraction-transport`	5	5	0	0	3	1	6	2026-07-27
106	`lem-stage1-polar-path-transport`	9	9	0	0	3	2	12	2026-07-27
107	`lem-stage1-inversion-derivative-transport`	22	13	1	1	3	5	23	2026-07-28

"Val." counts validated nodes; "Chal." counts challenges raised by verifiers against nodes of that tree; "Amd." counts node statements the prover had to change in response; "Defs" counts definitions imported into the workspace as byte-matched externals; "Rnds" is the highest prover/verifier round index reached; "Jobs" is the number of node-claim events, i.e. worker invocations against that tree; "Closed" is the day the root node was validated. Fewer validated nodes than created nodes means nodes were archived or superseded during the run.

The 12 workspaces that were seeded but never closed a root are `conj-degenerate-transport`, `conj-nsc`, `conj-rh`, `conj-sc`, `ex-hume`, `lem-hiddenness-alpha-slab-leakage`, `lem-negative-pivot-import`, `lem-stage1-approximate-group-laws-transport`, `lem-topology-finite-triangulation`, `prop-approx-simplex`, `thm-classical-factorization`, `thm-cluster`. A seeded-and-open workspace is a recorded state, not a failure: it means a contract exists and the tree is still being built or was superseded.

52.7 Worker and job census

Two independent records count delegated work. The `fr` log names the workers involved in each harvested cycle as `model:role` pairs; that is a curated, human-entered field and covers 410 worker mentions across 33 distinct model/role pairs. The `af` ledgers count *invocations* mechanically: every `nodes_claimed` entry is one worker taking one node, and the claim's owner string encodes whether it was the initial prover build, a prover fix round, or a verifier pass. That gives 3 046 claim events in this repository and 216 in the progenitor — 3 262 delegated adversarial jobs in total, each one a fresh model invocation.

Table 13: Mechanically counted `af` worker invocations in this repository, by role. Provers and verifiers are always separate fresh agents; the orchestrator never appears here because it never claims a node.

`af` claim owner	Claim events
`verifier`	2 212
`prover (initial build)`	473
`prover (fix round)`	361

Table 14: Worker mix as recorded in the `fr` controller log (curated field; one cycle may name several workers).

Model	Role	Cycles
`codex`	`prover`	168
`codex`	`verifier`	138
`gpt-5.6-sol`	`verifier`	19
`opus`	`prover`	11
`gpt-5.6-sol`	`prover`	10
`codex`	`refuter`	7
`fable`	`author`	7
`codex`	`auditor`	6
`codex`	`applier`	5
`opus`	`author`	5
`codex`	`transcriber`	4
`codex-gpt-5.6-sol`	`prover`	3
`codex`	`adversary`	2
`codex`	`strategist`	2
`codex-gpt-5.6-sol`	`auditor`	2
`codex-gpt-5.6-sol`	`verifier`	2
`fable`	`prover`	2
`opus`	`strategist`	2
15 further model/role pairs		15

`codex` entries are `gpt-5.6-sol` runs under the project's model/effort policy; a few rows carry the fuller `codex-gpt-5.6-sol` spelling from before the label settled. `opus`, `sonnet`, and `fable` are Claude models acting as orchestrator-side authors, strategists, and reviewers.

The progenitor's 216 claim events are counted but not decomposed: that campaign predates the `v-/pf-/prover-build` owner convention and used ad-hoc worker names (`agent-A`, `prover-direct-sum`, `vtb-2.1`, ...), so only its total is comparable with the column above.

What these numbers are not. A claim event is a job, not a token budget. Job counts are exact and mechanically derived; token, cost, and wall-clock figures were never recorded by either repository and are not estimated here (Section 52.2).

52.8 Evidence, artifacts, and dead routes

Numerical work is quarantined by construction: every experiment is a run bundle with a re-run command, a schema, an index row, and a checkable invariant, and is never rigorous. 38 such bundles exist. The inherited campaign record (`docs/ingest/`) carries a further 60 experiment files, including the 67,000-instance exact enumeration that supports the linear law and the (EX) conjecture — evidence, and proof of nothing.

Table 15: Document and tooling production in this repository.

Artifact class	Count
`docs/waves/` — wave records	107
`docs/plans/` — strategy and proof-sketch versions	62
`docs/audits/`, `lit-review/`, `recon/`, `tooling-feedback/`	5
`docs/ingest/` — the inherited classical portfolio	7
of which experiment files	60
`runs/` — numerical run bundles	38
producing scripts inside bundles	128
`report/sections/` — lab-book shards	53
lines of LaTeX in them	6 832
`refs/` — pinned ground-truth sources	10
`scripts/` — tooling files	28
lines of tooling code	9 077
of which validation gates	7
lines in the gates	1 893
gate unit-test files / lines	7 / 1 305

The negative record is kept as deliberately as the positive one. `FINDINGS.md` carries 46 dated entries with 16 explicit do-not-re-walk markers, indexed by mechanism; `docs/LEARNINGS.md` records 3 retracted in-repo claims (a retraction is a success of the machinery, not an embarrassment).

Table 16: The dead-route census, read off the hand-built mechanism index in `FINDINGS.md`.

Mechanism tag in the findings index	Entries
`CERTIFICATE-OR-CENSUS`	14
`CONSTANT-OR-SCALE`	11
`DEAD-ROUTE`	10
`RETIRED-ROUTE`	1
`WALL`	10

DEAD-ROUTE is a permanent kill with a certificate; WALL is a structural limit found, not a route kill; RETIRED-ROUTE is superseded rather than refuted. The distinction is load-bearing: only the first forbids re-walking.

Table 17: The issue tracker at the snapshot instant (210 issues plus 9 persistent memory records).

Issue status	Count
`closed`	149
`open`	53
`in_progress`	6
`deferred`	2

52.9 Progenitor and successor

The classical portfolio was not born here. It began as a side-quest inside `almost-idempotent-positive-maps`, whose own target was the non-commutative problem; the classical case was its proving ground. That campaign ran 2026-06-02 to 2026-06-13 (174 commits, 22 work-log entries) and reached 59 registry results with 16 `af`-validated roots across 16 workspaces and 132 proof nodes. This repository was stood up 2026-07-02 — 19 days later — specifically to re-establish the classical result from scratch under a stricter ladder.

What carried over. Three things, and each was re-tagged *down* on arrival:

- **The reduction chain.** The progenitor reduced the classical stability statement, through audited and audit-pending steps, to a single open kernel (the "(EX)" conjecture). The chain re-entered this repository as `stated` or `proved-mod-audit` — objects of study, not oracles — and 23 of the 295 registry results here still carry that inherited tag.
- **The numerical record.** The 67,000-instance exact enumeration and the seed experiments were copied wholesale into `docs/ingest/` (60 files) and re-homed as run bundles. Their status did not change: `numerical`, L3, never rigorous.
- **One validated result.** Exactly one classical result had cleared an `af` validation upstream — `lem-classical-equiv`, the signed↔stochastic bridge. It was nevertheless re-validated here from scratch (29 nodes, 6 challenges) and that in-repo run, not the upstream one, is what this repository's ladder rests on.

In throughput terms the successor is roughly 4.8 times the progenitor in commits, 5.0 times in registry size, and 6.7 times in `af`-validated results, over 2.2 times the elapsed span.

52.10 Colophon: how this section stays true

Everything in Section 52 is generated by `scripts/gen-report-stats.py` into `report/generated/stats/`. The shard in `report/sections/` contains no numbers; it only \inputs the generated body.

The staleness problem. Statistics about a repository depend on things that change with every commit — git history, the controller log, the proof ledgers. A byte-for-byte freshness gate over such a generator would be permanently red, because the very commit that lands the regenerated tables changes their inputs again. The generator therefore runs in two halves with a committed snapshot between them:

`python3 scripts/gen-report-stats.py --extract`
: mines the live sources into `report/generated/stats/campaign-extract.json` *and* re-renders.

`python3 scripts/gen-report-stats.py`
: renders the `.tex` files from that committed snapshot, deterministically.

`python3 scripts/gen-report-stats.py --check`
: the gate: re-renders, byte-compares, and prints a non-fatal drift advisory.

Staleness semantics. Only the render step is gated, so the check is stable across ordinary commits and fails exactly when it should: a hand-edited generated file, or a renderer change without a re-render. The *numbers*, by contrast, are as of **2026-07-28T07:06:04Z** and go stale silently by design — refreshing them is a deliberate act. `--check` therefore also compares a few cheap live counts against the snapshot and prints how far behind it is, without blocking a commit.

Progenitor data. The `almost-idempotent-positive-maps` block is mined live when that repository sits beside this one and is otherwise carried forward from the committed snapshot, so the gate is green on machines that do not have the sibling. The current block was mined live at the last extraction.

Provenance of the figures. 800 controller records, 11 375 `af` ledger entries, 295 registry shards, 45 definition shards, 38 run bundles, 67 work-log entries, 210 issue records, and 1 010 commits were read to produce this section.

53 Status and outlook

What is validated here

The preceding sections reproduce forty-seven registry results, each `proved` with `af: validated`: the signed–stochastic bridge `lem-classical-equiv` (§3); positive-retract hardening `lem-prh` (§4); the COMP tier — `lem-compcb-amplification-naturality`, `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-identities`, `lem-compcb-entrywise-compression-naturality`, `lem-compcb-rectangular-product`, `lem-compcb-amplified-almost-containment`, `lem-compcb-compressed-un` `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-product`, `lem-compcb-corner-algebra`, `lem-compcb-single-compression-transfer`; the H-CB tier — `lem-hcb0-compressed-associator`, `lem-hcb-column-hilbert-squared`, `lem-hcb1-variational-identity`, `lem-hcb1-column-action`, `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-hcb3-diagonal-unit`, `lem-hcb3-diagonal` `lem-hcb3-uniform-square-lower`, `lem-hcb3-diagonal-lower-modulus`, `lem-hcb3-diagonal-inverse`, `lem-hcb3-offdiagonal-inverse` — its canonical layer — `lem-hcb4-canonical-gram`, `lem-hcb4-canonical-clos` `lem-hcb4-canonical-inverse` — together with the tier's parent contract `conj-hcb` (§31); and the front end of the EXT tier — `lem-extcb-one-dimensional-product`, `lem-extcb-one-dimensional-corner-dimen` `lem-extcb-corner-dimension-additivity`, `lem-extcb1-close-corner-dimension`, `lem-extcb1-cross-corner` `lem-extcb-four-corner-norm`, `lem-extcb-four-corner-merge`; and, closing that tier, its parent contract `conj-extcb` (§37); the Route F finish edge `lem-routef-prh-finish` (§38); and the Stage-1 rectification pair — `lem-stage1-quantitative-inverse-function` (§39) and `lem-stage1-exact-unit-rectific` (§40); and the Route F upstream block — `lem-routef-f0-ucp-lift` and `lem-routef-f0-defect-identity` (§41), `lem-kitaev-almost-idemp-audit`, `lem-routef-functional-calculus-closeness` and `lem-routef-ai-d` (§42), `lem-routef-f2-positive-unital-compression` and `lem-routef-f3-retract-defect` (§43). The registry-index snapshot from which the table below is transcribed contains **73** results with that status pair. It predates the seven Route F upstream rows just listed, none of which appears in it, so the set difference is the thirty-three rows tabulated below.

The headline also needs qualifications. The inverse and lower-modulus results — `lem-hcb3-diagonal-lower-mo` `lem-hcb3-diagonal-inverse`, `lem-hcb3-offdiagonal-inverse` and the corresponding clauses of `conj-hcb` — are validated as *conditional* statements. Their level-one hypotheses are established in exactly one place in this document and for exactly one datum: §37 establishes them for the EXT-CB datum, in the order the tree fixes, before invoking the clauses. Nothing else here supplies them for any datum. Their conditional shape is itself a correction of record: the parent's inverse clause was unconditional until an exact $\mathbb{C}\oplus\mathbb{C}$ counterexample refuted it. `lem-extcb-four-corner-merge` (§36) carries a smallness hypothesis on the *total* defect $\rho+\varepsilon$; the transcribed version that constrained ρ alone is false, and the refuting example is reproduced there. Both corrections were produced by hostile verification, and both are reasons to distrust unconditional restatements of this material.

What is next, and what it is not

With §31 the H-CB tier is closed, and with §37 so is the EXT tier: `conj-extcb` is `proved` with `af: validated` and is reproduced above as Lemma 77. What it delivers is one extension step — from M_r to M_{r+1}, at one universal accuracy, with one unitary and four fixed corner maps serving every amplification level. It is not an induction, and nothing here iterates it.

Beyond the EXT tier lies the assembly, and that is where the live route now stops. Its registry carrier `lem-thmainext-conditional` is `proved-mod-audit` with `af: none` — **open** at this document's rigour bar, and **not** reproduced here. The MAIN-CB assembly rows and the K-ledger factorization rows — the latter being what would discharge the antecedent of §43 — are explicitly *not* in the pipeline and must not be treated as pending detail. They are **quarantined** pending a focused repair and a fresh hostile review, along with the named gap interfaces (exact-target approximation, the polar contract, the MAIN structure, the ledger domains, and the final bridge); and several external topology sources are not yet in `refs/`, which is a provisioning risk, not a proof step. The unconditional end-to-end constant ledger for Route F remains separate open work.

The honest summary is therefore: the base, the middle and both tier parents of the live route have been made rigorous here, and the assembly above them is quarantined. `op-classical` is **open**.

Validated registry results not reproduced here

The table is the exact set difference between the supplied index rows carrying `proved/validated` and the reproduced ids listed above. Its ordering follows the generated index.

Registry id	Status	Note
`lem-always-tight-dual-support`	`proved, validated af:`	off the live route; registry record retained
`lem-collateral-import`	`proved, validated af:`	off the live route; registry record retained
`lem-cross-pivot-cancellation`	`proved, validated af:`	off the live route; registry record retained
`lem-cs-low-slab-pincer`	`proved, validated af:`	off the live route; registry record retained
`lem-depth-d-halo-collapse`	`proved, validated af:`	off the live route; registry record retained
`lem-factorization`	`proved, validated af:`	off the live route; registry record retained
`lem-fan-payment`	`proved, validated af:`	off the live route; registry record retained
`lem-fan-payment-restricted`	`proved, validated af:`	off the live route; registry record retained
`lem-genuine-disintegration`	`proved, validated af:`	off the live route; registry record retained
`lem-halo-collapse`	`proved, validated af:`	off the live route; registry record retained
`lem-harmonic-affine-bridge`	`proved, validated af:`	off the live route; registry record retained
`lem-hiddenness-depth-markov`	`proved, validated af:`	off the live route; registry record retained
`lem-hiddenness-dual-witness`	`proved, validated af:`	off the live route; registry record retained
`lem-hx-financing-floor`	`proved, validated af:`	off the live route; registry record retained
`lem-hx-forced-exterior-coupling`	`proved, validated af:`	off the live route; registry record retained
`lem-hx-robust-scalar-starvation`	`proved, validated af:`	off the live route; registry record retained
`lem-hx-signed-variation-ledger`	`proved, validated af:`	off the live route; registry record retained

Registry id	Status	Note
`lem-hx-transverse-moment-identity`	`proved`, `af: validated`	off the live route; registry record retained
`lem-import-reduction`	`proved`, `af: validated`	off the live route; registry record retained
`lem-mass-split`	`proved`, `af: validated`	off the live route; registry record retained
`lem-negpart-subadditive`	`proved`, `af: validated`	off the live route; registry record retained
`lem-parametric-halo-collapse`	`proved`, `af: validated`	off the live route; registry record retained
`lem-pivot-removing-move`	`proved`, `af: validated`	off the live route; registry record retained
`lem-residual-lower`	`proved`, `af: validated`	off the live route; registry record retained
`lem-residual-upper`	`proved`, `af: validated`	off the live route; registry record retained
`lem-row-far-dual-certificate`	`proved`, `af: validated`	off the live route; registry record retained
`lem-row-zero-capacity`	`proved`, `af: validated`	off the live route; registry record retained
`lem-starvation-completion-obstruction`	`proved`, `af: validated`	off the live route; registry record retained
`lem-top-concentration`	`proved`, `af: validated`	off the live route; registry record retained
`lem-top-slab-companion`	`proved`, `af: validated`	off the live route; registry record retained
`lem-weighted-min`	`proved`, `af: validated`	off the live route; registry record retained
`lem-zerosum-triangle`	`proved`, `af: validated`	off the live route; registry record retained
`obs-height-collapse`	`proved`, `af: validated`	off the live route; registry record retained

Each id and status pair in the table is transcribed from `repo-inputs/argument-INDEX.md`. In particular, `conj-extcb` is absent from the table because it is *reproduced* above (§37), not because of its status: the index records it as `proved` with `af: validated`.